

Stellingen

behorende bij het proefschrift getiteld:

Experimental study of heat transfer in turbulent flows over wall-mounted cubes

van Erwin R. Meinders

- 1 Het effect van lokale stromingsverschijnselen, zoals vortex structuren, op de lokale convectieve warmteoverdracht van kubussen in een kanaalstroming is niet terug te vinden in de kubus-gemiddelde warmteoverdrachtscoëfficiënt.
(uit: dit proefschrift)
- 2 Het gebruik van een akoestisch detectie systeem vergroot de veiligheid bij het boorproces door poreuze steenformaties tijdens exploratie en productie van aardgas en aardolie.
(uit: E.R. Meinders. *Sound propagation in drilling fluids. An investigation to determine the limitations of the Wellhead Sonar System. Shell Research/Universiteit Twente, afstudeerverslag*)
- 3 Ministers zijn marionetten van de samenleving terwijl ze geacht worden de touwtjes zelf in handen te hebben.
- 4 Genetische manipulatie om het geslacht van embryo's te beïnvloeden kan in het meest ongunstige geval leiden tot een kinderloze maatschappij en dus tot het uitsterven van het menselijke ras.
- 5 Recente voorbeelden van toenemende excessieve geweldsuitbarstingen wijzen op een tanende maatschappelijke tolerantie.
- 6 In het Nederlandse rechtssysteem staat de strafmaat voor het toebrengen van zwaar lichamelijk letsel of doodslag in geen verhouding tot die voor bijvoorbeeld belastingfraude of handelen in aandelen met voorkennis.
- 7 Ondergeschikte maar desalniettemin direct gerelateerde doelen zoals internationale waardering en scoringsdrang verdrukken de primaire taken van een universiteit, te weten het opleiden van studenten en promovendi en het verrichten van onderzoek, vaak naar een ondergeschikte plaats.
- 8 Het ziekelijke fenomeen oranjekeorts is duidelijk een gevolg van de mondialisering en individualisering van de Nederlandse samenleving.
- 9 Economische welvaart brengt individualisering van de maatschappij met zich mee.
- 10 Uit het grote aantal excessieve verbale en fysieke gewelddadigheden tijdens een vriendschappelijk duel blijkt dat de voetbalsport onherroepelijk verbonden is met primaire gevoelens van de mens.
- 11 Met het uitsterven van dierlijke en plantaardige organismen verdwijnen unieke genetische bouwstenen waardoor de biodiversiteit, en dus de kans op het vinden van geneesmiddelen, negatief wordt beïnvloed.

- 12 Dat achterkantfactoren vaak bepalend zijn voor het verloop en het resultaat van organisatieveranderingsprocessen is hoogst kwalijk te noemen.
- 13 De inburgering van een computer-gedomineerde tijdsbesteding leidt tot de verloedering van de communicatieve vaardigheden van kinderen.
- 14 Het gebruik van illegale stimulantia in de sportwereld doet afbreuk aan de term sportief.
- 15 De aanduiding *bananenrepubliek* heeft zowel betrekking op de kromheid van het aldaar heersende regime alsmede op haar belangrijkste export product.

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flows over wall-mounted cubes

Experimental study of heat transfer in turbulent flows over wall-mounted cubes

Proefschrift



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Part I

Introduction

CHAPTER 1

Introduction to the problem

1.1 Background

For centuries, turbulent flows over three-dimensional bluff bodies have been challenging issues in fluid dynamic research. Leonardo da Vinci was already fascinated by the vortex structure in flows over prismatic obstacles. His sketched observations of flow visualisation experiments are well known and are frequently used to illustrate the complexity of such kind of bluff-body flows. One of his sketches is shown in Fig. 1.1, taken over from Frisch [52]. The sketch represents the flow past a prismatic vertical wall-mounted object. The drawing clearly shows the turbulent nature of the flow, particularly in the wake of the obstacle. Da Vinci was probably the first who identified the so-called horseshoe vortex, which is induced in front of the obstacle and wraps around its base.

When such a bluff body is internally heated, it will be convectively cooled by the fluid flow. The rate of convective heat transfer, usually expressed in terms of the heat transfer coefficient, is proportional to the local fluid velocity and the temperature difference between the body surface and the fluid. However, in complex situations the heat transfer rate is also dependent on the immediate flow structure around the obstacles. Such vortical structures will lead to steep spatial gradients in the distributions of the local time-averaged convective heat transfer. In addition to this, flow instabilities may lead to periodic flow phenomena (like the well-known vortex shedding at the sides of the obstacle) which affect the time-dependent convective heat transfer.

Turbulent flow around and heat transfer to or from bluff bodies can be found in various practical situations encountered in engineering. For instance in wind-engineering applications, possible aerodynamic interaction between two bluff bodies, such as the pylons of bridges or tall buildings in closely populated areas, is of crucial importance for their dynamical and mechanical characteristics, since flow-induced excitations can result in severe material damage. It is therefore of major importance to understand the flow and vortical structures around these bluff bodies. Further, examples can be found in steam and gas turbines, steam generators and heat exchangers, where the structural integrity of the equipment depends on the local flow structures. An illustrating example is the convective heat transfer from the high-temperature combustion gases to the blades in a turbine which, if insufficient, may lead to overheating of the blade material and, therefore, to irreversible damage of the turbine blades which shortens the life time of the machinery. To prevent the turbine blades from overheating, cold air is usually recirculated through cooling passages inside the blade. These internal passages are equipped with wall-mounted protruding ribs to enhance the overall convective heat transfer. For optimum design of the cooling passages in turbine blades, detailed knowledge of the local wall

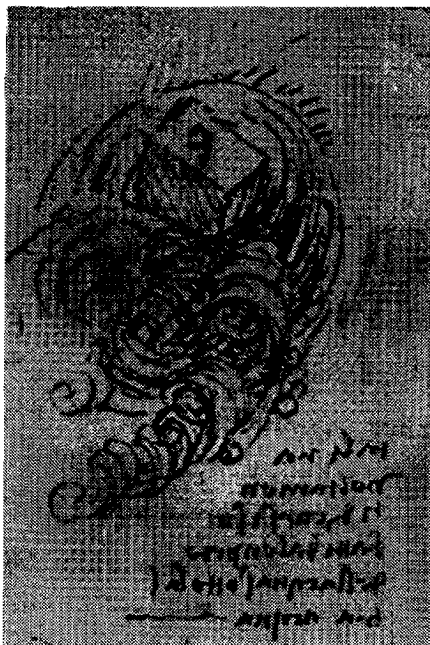


FIGURE 1.1: Sketched observations of the flow past a prismatic obstacle (from Leonardo da Vinci, ms. F., p 18v, bibliotheque de l'Institut de France, shown in Frisch, 1996).

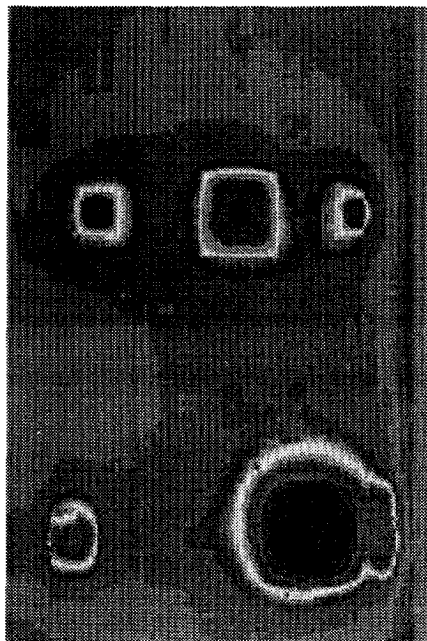


FIGURE 1.2: Grey value representation of an infrared image of a printed circuit board. Black areas are identified as hot spots.

heat transfer and the associated local flow properties is required. Several studies were focussed on this subject [17, 31, 70, 71, 72, 73]. In the present work, we are concerned with another important application, i.e. the thermal performance of electronic printed circuit boards. The research described in this thesis was sponsored by Philips, Nederlandse Philips Bedrijven B.V., in order to get answers to questions related to local overheating problems of integrated circuits (chips) and electronic components. Wrong predictions of the heat transfer may lead to critical local hot spots which obviously reduce the lifetime of an electronic apparatus.

1.2 Electronics cooling application

Severe local overheating of crucial components, leading to component damage, is an important cause of technical failure of electronic equipment [87]. In order to increase the life-time and the product reliability of an apparatus, the thermal performance of such systems during operation needs to be well known. It is the task of an engineer to adequately design the electronic system according to the thermal requirements. Despite extensive research in this topic for decades, the problem of local overheating is still actual. This is mainly due to the miniaturisation of electronic circuitry

and the ever increasing power dissipation in printed circuit boards which impose more and more stringent demands for an efficient thermal management. To keep the manufacturing costs and the noise level of the apparatus as low as possible, designers solve the problem with optimum design rather than with additional cooling capacity provided by extra fans. The various topics related to thermal management of electronic parts and systems is elaborately discussed in an overview by Lasance [87].

Printed circuit boards are characterised by a complex composition of a range of different bluff-shaped electronic components. The components have several different sizes and geometries which make the usage of one characteristic length scale for the heat transfer characterisation impossible. The complex configuration and the strong concentrated power dissipation within the elements result in a high non-uniform convective and radiative heat transfer. An example of a typical surface temperature distribution of a powered printed circuit board, recorded with an infrared camera, is given in Fig. 1.2. The image shows clearly regions with hot spots (the dark areas).

The convective heat transfer depends strongly on the flow field characteristics. Hence, an accurate solution of the temperature field depends on the accuracy of the flow solution. The flow in the devices is usually classified as a low Reynolds number turbulent flow. While the Reynolds numbers remain low because of the small velocities and length scales encountered, turbulence may be generated locally by the protruding electronic components mounted on the printed circuit boards. Depending on the local Reynolds number, turbulence may decay locally or may persist until the next downstream protrusion where it may be enhanced. In electronic equipment turbulence often remains isolated in small patches in the shear layers around the protruding elements, while the flow above the protrusions remains more or less laminar.

This outline of the problem highlights that both the heat transfer and the flow structures on printed circuit boards are very complex. A reliable prognosis and avoidance of hot spots and critical temperatures in a variety of complex printed circuit boards can be achieved only by employing a technique which involves an accurate evaluation of the local surface heat transfer and temperature on and around the components.

Therefore, the well-known Nusselt-Reynolds number relations, which correlate a Nusselt number (the dimensionless heat transfer coefficient h) with a Reynolds number (a measure for the flow rate), are highly unsuited for the description of the convective heat transfer from densely packed printed circuit boards. These relations are usually applicable for a selective class of heat transfer problems in which only one characteristic length is involved and for which the boundary conditions are well defined. Further, the definition of the local reference temperature, encountered on printed circuit boards and used in the definition of h , is not trivial.

Computational fluid dynamics (CFD) codes are more and more used as design tools to predict the thermal processes encountered in electronic circuitry. CFD codes can easily cope with a high level of complexity (component, board and system level) which makes them attractive to use in the design process of printed circuit boards. With large workstations, an entire electronic cabinet can be considered

with many details. An accurate laminar solution for low-Reynolds number flows can be obtained provided that the mesh is fine enough and that the solvers and discretisations techniques are well chosen. But, as argued earlier, it is unlikely that the flow remains laminar. Therefore, sophisticated turbulence models and/or other solution techniques, like large eddy simulations (LES) and direct numerical simulations (DNS), may be required to obtain accurate and reliable solutions for the transitional and turbulent flows. A drawback for both LES and DNS is that the computational capacity of currently available workstations is far from that needed to accurately solve both the kinematic and the non-uniform heat transfer problems as encountered in electronic devices.

A better prediction and understanding of the heat transfer and kinematic processes encountered on printed circuit boards can be achieved by a balanced combination of both experimental and numerical effort. Although computational research will soon or later be the dominant contribution in the design cycle of an electronic apparatus, experimental research will, in essence, always be needed for the validation of numerical simulations.

1.3 Goals of this study

The previous section highlighted the context in which the research was performed. This thesis will be focussed on the experimental approach of the problem which involves the accurate analysis of the local convective heat transfer in idealised models of realistic electronic printed circuit boards. Some of the major goals of the research are itemised as follows:

- The primary purpose of this experimental study is to investigate the physical mechanisms which lead to the intensification or degradation of the local convective heat transfer. This objective involves a thorough evaluation of the time-averaged convective heat transfer and its associated flow structure. In some cases, additional time-dependent convective heat flux measurements were performed to support the interpretation of the results.
- A next objective was to derive quantifiable correlations between the distributions of the local time-averaged heat transfer coefficient and the local flow field parameters for idealised configurations of (multiple) wall-mounted protrusions in a channel flow. These configurations serve as a diagnostic case for the cooling of realistic electronic circuitry.
- A third major goal of the research was to provide accurate and reliable experimental heat transfer and flow field data for benchmarking of commercial and home-made CFD codes and, in particular, of turbulence models used for the thermal design of printed circuit boards. The experimental data needed to be very accurate to serve as a reference. This implies that significant effort had to be put in a thorough accuracy analysis and in the validation of the obtained experimental results.

1.4 Implementation of approach

The investigation was primarily focussed on the flow structures and the turbulent heat transfer in flows over surface-mounted obstacles in a turbulent channel flow. The electronic components were modelled as a cube for the following reasons:

- A cube is the most basic three-dimensional geometry which involves most generic flow phenomena encountered in real electronic devices: flow separation, recirculation and reattachment. Yet the simplicity of the geometry and the fixed separation lines (at the cube edges) make this flow suitable for experimental analysis. The different types of flow will have a significant impact on the local heat transfer. Although in real electronics applications the sizes of the protruding components are somewhat different (more flat-pack like geometries) the physical phenomena encountered in the cube case are more or less representative. For these reasons, the cube is considered to be a suitable test model for an electronic component.
- Several researchers studied the cube-averaged heat transfer from configurations of cubes (particularly isolated cubes or regular matrices of cubes) in a channel flow. These data served for comparison and verification of the results obtained in this research.
- The method to determine distributions of the time-averaged local convective heat transfer from the obstacles required a sufficiently large obstacle. This determined, among others, the size and the geometry of the adopted element (see the description of the experimental method, part II).

An arrangement of these cubes mounted on a wall is then representative for a printed circuit board geometry. While an isolated wall-mounted cube can be considered as the most basic configuration of such a printed circuit board, a configuration of multiple cubes models a densely populated board and can be used to study the interaction between components.

Four basic configurations were selected to investigate the local convective heat transfer and the local flow structure. All four test-cases provided basic data but each configuration had different and complementary features. Figure 1.3 shows

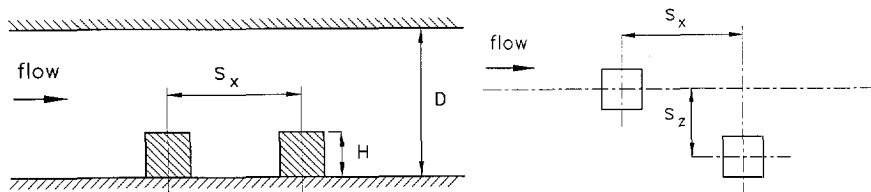


FIGURE 1.3: Sketches of the side and plan view of the tandem configuration of wall-mounted cubes.

the side and top view of a typical configuration of cubes studied in this thesis. The cube height is indicated with H , S_x and S_z denote the streamwise and the

spanwise centre-to-centre distance between the cubes and D is the channel height. The configurations discussed in this thesis are as follows:

- A single cube, placed in a developing turbulent channel flow was considered to be the most basic configuration. For this test case, no thermal and aerodynamic interactions with other heated obstacles were present: this allowed a very detailed analysis of the flow structures and the local convective heat transfer. Further, much research was already devoted to the flow and the vortex structure around a single cube in either a boundary layer flow or a fully developed channel flow with much higher Reynolds numbers. Additional data of the convective heat transfer will, therefore, complement the existing knowledge. The flow field around the single cube contains more or less the most basic features and this case is therefore well suited to study the correlation between surface heat transfer and local flow structure. Experiments were performed for different Reynolds numbers and the effect of the upstream boundary condition was incorporated.
- The in-line row of cubes with a streamwise distance of $S_x/H = 2$ and placed in a developing channel flow was the second test case considered. The experiments were performed with all cubes heated in order to investigate the so-called adiabatic heat transfer. Tests were conducted for different Reynolds numbers. This is the only test case for which mixed convection phenomena were noticed for the lowest Reynolds number studied.
- A tandem arrangement of two cubes was a suitable model to investigate the aerodynamic and thermal interaction between two wall-mounted obstacles. Most of the experimental work was performed for tandem arrangements placed in a fully developed turbulent channel flow at the fixed Reynolds number of $Re_H = 3900$. Some additional measurements were carried out in a developing boundary layer flow for higher Reynolds numbers. Flow field and heat transfer measurements were performed for various streamwise and spanwise distances, S_x/H and S_z/H respectively. The arrangements with $S_z/H = 0$ are denoted as in-line arrangements, those with S_z/H different from 0 are denoted as off-axis tandem arrangements. Additional measurements of the time-dependent convective heat transfer were performed for the in-line tandem.
- The last studied test case is the in-line matrix of wall-mounted cubes with equal streamwise and spanwise distances, i.e. $S_x/H = S_z/H = 4$. The total number of allocated cubes (250 in total) caused a spatially periodic turbulent channel flow far downstream in the matrix. The most important objective for this case was the detailed and accurate description of the flow field and heat transfer characteristics of a cube in the matrix sufficiently far downstream so that the flow could be regarded as fully developed. This situation allowed the use of periodic inlet and outlet flow conditions in the numerical computations. The dataset was used at two ERCOFTAC workshops (June 6-7, 1997, Delft University of Technology, Delft [62], and May 28-29, 1998, UMIST, Manchester [88]) for the validation of turbulence models and numerical solution methods.

These four configurations have been considered separately and the results will be presented in separate chapters.

1.5 Outline of thesis

This thesis is organised in four clusters of chapters. Part I started with introductory words to define the problem and to address the objectives of the investigation (chapter 1). A brief summary of the theoretical background is given in chapter 2. We limit the discussion to the theory needed for the interpretation of the experimental results. Further, a review of relevant literature is given in chapter 2 aimed at indicating the current state of knowledge in this area. Part II is dedicated to the description of the experiment. The test-configuration and the experimental methods to evaluate local heat transfer coefficients are described in chapter 3. Chapter 4 addresses the diagnostic measurement techniques to acquire convective heat transfer and velocity data. This chapter covers, besides other used measurement techniques, in fact two major techniques, namely infrared thermography and the Laser Doppler Anemometry method. Part III of the thesis presents the results for the four test cases investigated. The most basic case, a single cube in a turbulent channel flow, is discussed in chapter 5. Chapter 6 presents the results for the in-line row of 9 cubes. The tandem of two cubes is discussed in chapter 7 and finally, results for the spatially periodic matrix flow are given in chapter 8. The thesis is finished with concluding remarks in part IV. A discussion of the major findings is given in chapter 9 while conclusions and future directions are given in chapter 10. The experimental evaluation of the thermal conductivity of the epoxy substrate, used in the measurement cubes, is given in appendix A.

CHAPTER 2

Theoretical background and literature review

In the past decades, much research was already devoted to the thermal management of electronic systems since product reliability is an important strategy and sales argument. Most of this published material is pragmatic in nature and is mainly focussed on the average convective heat transfer or the thermal resistance of representative electronic components. Entire international conferences are dedicated to the problem of thermal management of electronic systems (e.g. Thermal Management of Electronic Systems (Eurotherm), Semitherm and IThERM (USA), etc.). The approach adapted in this thesis is different and is based on the local flow structure and its relation to the local convective heat transfer for idealised models of printed circuit boards. The chapter starts with a section on general considerations to briefly outline some basic theoretical background information needed for the interpretation of the flow and heat transfer results. Further, the flow over a single wall-mounted cube is discussed to illustrate the generic flow types encountered in the test cases under investigation. The discussion then proceeds with an overview of the most relevant literature in this area, which includes the flow and heat transfer characteristics of single and multiple two-dimensional and three-dimensional objects.

2.1 General considerations

The general flow and energy equations are discussed first, followed by the concepts of convective heat transfer. The statistical description of turbulence is an important issue in this thesis since the observer, the experimenter in this case, deals with continuous signals (velocity, temperature) which are acquired in discretized form. The autocorrelation functions and spectra were mainly used to determine frequencies of vortex shedding in the wake flow of the cubes.

2.1.1 Flow equations

When fluid is considered to be a continuum, the laws of mechanics and thermomechanics can be applied to a control volume to derive the conservation laws of mass, momentum and energy. In Euler's approach, the control volume is fixed relatively to the coordinate system while the fluid is passing through it. Only Newtonian viscous fluids will be considered. The formulas given here were obtained from the well-known literature [66, 166, 63, 52].

Conservation of mass describes the balance between the density change at a fixed point and the change in the mass velocity vector $\rho \mathbf{u}$. The difference of mass flow into and out of the control volume must equal the rate of change in mass contained in the volume (a repetition of the suffix in a term implies a summation over all three values of the suffix):

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_i}{\partial x_i} = 0, \quad (2.1)$$

where u_i are the three instantaneous velocity components, ρ is the density and t and x_i refer to time and the Cartesian coordinates, respectively. For an incompressible fluid ($\rho = \text{constant}$) this conservation law simplifies to:

$$\frac{\partial u_i}{\partial x_i} = 0. \quad (2.2)$$

Conservation of momentum along a streamline is given by the Navier-Stokes equations. For an incompressible flow with constant flow properties, these equations read in index notation:

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + \nu \frac{\partial^2 u_i}{\partial x_j^2} + g_i, \quad (2.3)$$

where p is the hydrostatic pressure, ν is the kinematic viscosity and g_i is the gravitation force. The left-hand side represents the rate of change of momentum along a streamline. The first two terms on the right-hand side are the pressure term, and the rate of momentum transfer by viscous diffusion.

The vorticity is defined as the rotation of the instantaneous velocity field of a fluid element:

$$\omega_i = \epsilon_{ijk} \frac{\partial u_k}{\partial x_j}, \quad (2.4)$$

where ω_i are the vorticity components, ϵ_{ijk} is the 'unit alternating tensor' [25].

Conservation of energy for an incompressible flow with constant flow properties can be represented in terms of the fluid temperature T :

$$\frac{\partial T}{\partial t} + u_j \frac{\partial T}{\partial x_j} = a \frac{\partial^2 T}{\partial x_j^2} + \nu \Phi_v, \quad (2.5)$$

where a is the thermal diffusivity ($a = \lambda / \rho c_p$), ν is the kinematic viscosity and Φ_v is the dissipation function. The left-hand side refers to the rate of energy change along a streamline, i.e. the local change in time and convection. The first term on the right-hand side represents the transfer of energy by conduction and the second term is the irreversible rate of internal energy increase by viscous dissipation. Source terms are not included in the equation.

With Reynolds decomposition, the instantaneous quantities in a turbulent flow can be considered as the superposition of a mean, an average over a period which is much larger than the turbulent fluctuations, and a fluctuating part:

$$u_i = \overline{u_i} + u'_i, \quad (2.6)$$

$$p = \overline{p} + p', \quad (2.7)$$

$$T = \overline{T} + T'. \quad (2.8)$$

The mean conservation laws are obtained by substituting these expressions in the time-dependent Navier Stokes equations. Conservation of mass then becomes:

$$\frac{\partial \overline{u_i}}{\partial x_i} = 0. \quad (2.9)$$

For conservation of momentum:

$$\frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{p}}{\partial x_i} + \nu \frac{\partial^2 \bar{u}_i}{\partial x_j^2} - \frac{\partial \overline{u'_i u'_j}}{\partial x_j}, \quad (2.10)$$

where $\overline{\tau_{ij,t}} = \rho \overline{u'_i u'_j}$ are the turbulent stresses. These stresses are in a time average not zero which means that an extra unknown is introduced for the same set of partial differential equations. This causes the well known closure problem.

When viscous dissipation is omitted, conservation of energy becomes:

$$\frac{\partial \bar{T}}{\partial t} + \bar{u}_j \frac{\partial \bar{T}}{\partial x_j} = \alpha \frac{\partial^2 \bar{T}}{\partial x_j^2} - \frac{\partial \overline{T' u'_j}}{\partial x_j}, \quad (2.11)$$

where $\overline{q_j^{(t)}} = \overline{T' u'_j}$ are the turbulent heat fluxes. These correlations are also unknowns for the given problem.

The turbulent kinetic energy per unit mass (k) is half the sum of the normal stresses and is given as:

$$k = \frac{1}{2} (\overline{u'^2} + \overline{v'^2} + \overline{w'^2}) = \frac{1}{2} (\overline{u'_i u'_i}). \quad (2.12)$$

From the Reynolds-stress transport equations, the turbulent energy equation for a steady flow ($\partial/\partial t = 0$) reads [25]:

$$\bar{u}_j \frac{\partial \overline{u'^2}/2}{\partial x_j} = -\frac{1}{\rho} \overline{u'_i u'_j} \frac{\partial \bar{u}_i}{\partial x_j} - \frac{1}{\rho} \frac{\partial (\overline{p' u'_j} + \overline{u'^2 u'_j})}{\partial x_j} + \overline{\nu u'_i \frac{\partial^2 u'_i}{\partial x_j^2}}. \quad (2.13)$$

The left-hand side represents the convective transport of turbulent kinetic energy by the mean flow. The first term at the right-hand side is the production of turbulent energy due to the mean rate of strain. The second term represents spatial transport of turbulent energy due to the pressure and velocity fluctuations. The last term denotes the viscous part. The latter includes the viscous dissipation: the transfer from turbulent kinetic energy to thermal internal energy.

2.1.2 Convective heat transfer

Convective heat transfer refers to heat exchange between a heated/cooled surface and the fluid flowing over it. Traditionally, the heat transfer coefficient is defined as [42, 83]:

$$h_{ref} = \frac{Q_{conv}}{A(T_{sur} - T_{ref})}, \quad (2.14)$$

where Q_{conv} is the convective heat flow, A is the total area and T_{sur} and T_{ref} are the averaged surface and reference temperatures, respectively. The temperature difference ($T_{sur} - T_{ref}$) is the driving force for the convective transport. The heat transfer coefficient is usually non-dimensionalised with a characteristic length L and the thermal conductivity of the fluid λ , resulting in the Nusselt number:

$$Nu = \frac{h_{ref} L}{\lambda}. \quad (2.15)$$

In some cases, a better insight is gained by the Stanton number St , defined as:

$$St = \frac{h}{\rho c_p U_{ref}}, \quad (2.16)$$

where ρ is the fluid density, c_p is the specific heat and U_{ref} is a reference velocity.

A dimensionless parameter for the flow is the Reynolds number Re_L which is defined as:

$$Re_L = \frac{U_{ref} L}{\nu}, \quad (2.17)$$

where U_{ref} is a reference velocity, L is again the characteristic length and ν is the kinematic viscosity of the flow.

Convective heat transfer is directly related to air flow and, therefore, the Nusselt number is a function of many properties and parameters. Nusselt numbers are usually related to Reynolds numbers according to:

$$Nu = A Re_L^m Pr^n f(geometry), \quad (2.18)$$

where Pr is the Prandtl number ($Pr = \nu/a$, where a is the thermal diffusivity of the fluid), A is a constant, n and m are correlation coefficients and $f(geometry)$ is a geometry-dependent function. Such a correlation is doomed to fail when the situation is not represented by one characteristic length scale and one reference velocity. Further, such a correlation, expressed in the constant A and the coefficients n and m , is dependent on the geometry and thermal and kinematic boundary conditions. This means that congruence scaling rules need to be strictly applied as well as the boundary conditions involved if such a correlation is used to estimate/predict the heat transfer in a given flow situation. $Nu-Re$ relations generally give the correct order of magnitude of the convective heat transfer in simple flow/wall configurations such as flat walls, etc. However, such correlations are usually over-simplistic for complex geometries.

In this thesis it is, therefore, attempted to present the results in their basic form, i.e. in terms of the local heat transfer coefficient for each particular geometry without attempting to involve any similarity or scaling. In some cases, the heat transfer and bulk velocity of the channel flow is scaled to enable a comparison with literature data.

According to Reynolds' analogy, the heat and momentum transfer are caused by the same physical mechanism. The classical relation between the heat and momentum transfer reads:

$$St_x = \frac{1}{2} C_{f,x}, \quad (2.19)$$

where St_x is the local Stanton number, based on the local heat transfer coefficient, and $C_{f,x}$ is the local friction factor, defined as $C_{f,x} = 2\tau_w/\rho U_\infty^2$ where τ_w is the wall shear stress. From this, it appears that the local heat transfer is proportional to:

$$h \sim \frac{\tau_w}{U_\infty}. \quad (2.20)$$

In well-behaved flows without flow separation and recirculations (like, for example, boundary layer flows) this analogy has been successfully applied. When flow recirculation, reattachment and separation are involved, departures from Reynolds' analogy are to be expected.

2.1.3 Statistical description of turbulence

In laboratory experiments, it is common practise to define a time average of the function $f(t)$:

$$\bar{f} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T f(t) dt. \quad (2.21)$$

where T is the period in which the signal is recorded. This integral can be approximated by the sum of f_i , which are values of f at discrete times t_i :

$$\bar{f} = \lim_{N \rightarrow \infty} \frac{\sum_{i=1}^N f_i \Delta t_i}{\sum_{i=1}^N \Delta t_i}, \quad (2.22)$$

where Δt_i is the time interval between two consecutive realisations. In reality, it is impossible to take an infinite number of samples to obtain the time average. There exist tabulated criteria to which the number N should satisfy in order to neglect statistical error. Another often useful definition is the ensemble average which averages the function over different individual events of the experiment each time at a time t_o after the run was started:

$$\bar{f} = \lim_{R \rightarrow \infty} \frac{1}{R} \sum_{r=1}^R f(t_o). \quad (2.23)$$

This average can be interpreted as repeating the experiment R times when the values are recorded at time t_o . The phase average is a special ensemble average. For example, if vortex shedding is present in the flow, each time when a shed vortex passes the monitor point, a sample is taken. Averaging over R samples then gives the average at a certain phase of the shedding cycle. In this manner, it is possible to detect coherent structures in flows provided that a time reference point is available.

It is assumed that the process becomes uncorrelated for large time delays: $f(t)f(t+\tau) \rightarrow 0$ if $\tau \rightarrow \infty$. This implies that the time scale T_u describes the period over which the signal is still correlated with itself (in fact it is a measure for the macroscopic time scale). If the signal is measured in the period T which is much larger than the time scale T_u , the ensemble average can be replaced by the time average [166].

The variance of a signal $f(t)$ is defined as:

$$\sigma_f^2 = \overline{f'^2} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (f(t) - \bar{f})^2 dt. \quad (2.24)$$

σ_f is better known as the standard deviation. Similarly, the third and fourth order moments of $f(t)$ are given by:

$$\overline{f^3} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (f(t) - \bar{f})^3 dt, \quad (2.25)$$

$$\overline{f^4} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T (f(t) - \bar{f})^4 dt. \quad (2.26)$$

The skewness and kurtosis factors, S and K , respectively, are defined as the third and fourth order moments normalised with the standard deviation: $S = \overline{f^3}/\sigma_f^3$ and $K = \overline{f^4}/\sigma_f^4$. The skewness indicates the symmetry or asymmetry of the histogram while the kurtosis is a measure for the peakedness of the distribution. It can be shown for Gaussian distributions that the skewness equals $S = 0$ and the kurtosis $K = 3$. Turbulence is a non-linear phenomenon and, therefore, its statistics will not resemble Gaussian behaviour. For example, strong negative peaks in the velocity distributions will lead to strong negative values of the skewness factor.

2.1.4 Autocorrelation functions and spectra

The temporal or autocorrelation function $\rho(t)$ is a measure for the correlation with itself. It is defined as the auto-covariance function $R(\tau)$ normalised with $R(0)$:

$$\rho(t) = \frac{R(\tau)}{R(0)} = \frac{\int_{-\infty}^{\infty} f(t)f(t+\tau)d\tau}{\overline{f^2}}. \quad (2.27)$$

The integral time scale Tu is defined as the time integration of the autocorrelation function according to:

$$Tu = \int_0^{\infty} \rho(t) dt. \quad (2.28)$$

Another time scale which reveals from the autocorrelation function is the Taylor micro time scale λ_T which is derived from the second time derivative at $t=0$:

$$-\frac{\partial^2 \rho(t)}{\partial t^2} \Big|_{t=0} = \frac{1}{\lambda_T^2}. \quad (2.29)$$

The power spectral density (PSD) is defined as

$$\Phi = \lim_{T \rightarrow \infty} \frac{1}{T} F(\omega) \times F(\omega), \quad (2.30)$$

where $F(\omega)$ is the Fourier transform of the signal $f(t)$ and ω is the angular frequency ($\omega = 2\pi f$). The PSD can also be given in terms of the Fourier transform of the autocorrelation function.

2.2 Generic flow features in bluff-body flows: the cube

The literature review of Schofield [145] showed that a consensus exists for the general flow features around a single wall-mounted cube. The features include the horseshoe vortex, the wake vortex and the top and the side vortices, which are indicated in the three-dimensional 'artist's impression' in Fig. 2.1. These flow features are not only observed for a single cube in turbulent channel flows, but are also representative for configurations of multiple cubes. In a time-averaged sense, the oncoming main flow is separated at the sharp leading side and top edges of the cubes. The separated side shear layer caused a high vorticity recirculation at the side face close to the leading edge with its origin at the channel floor. This side vortex is indicated with **A** in Fig. 2.1. The separation of the top shear layer resulted in a bound recirculation ('top vortex', indicated with **B**) located at the top face and oriented perpendicularly to the direction of the main flow. The flow field in front of the cube is characterised by the horseshoe vortex, indicated with **C**, which originates at the base of the windward stagnation region and extends downstream along the channel floor past the sides of the cube. An arc-shaped vortex was typically found in the wake of the obstacle ('wake vortex'). This wake vortex, indicated with **D**, was confined by the solid walls and the fluid flowing along the side and top faces. The instantaneous

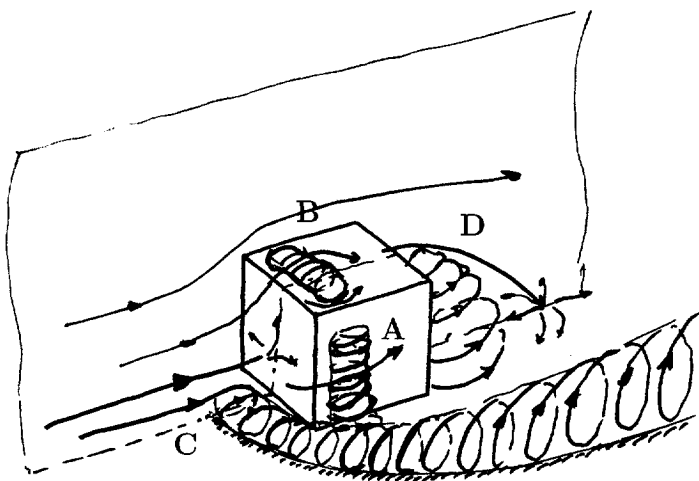


FIGURE 2.1: 'Artist's impression' of the flow structure around the single cube in a turbulent channel flow.

flow field differs significantly from the time-averaged field as sketched in Fig. 2.1. Vortex shedding at the side faces of the cube leads to asymmetric instantaneous flow patterns (only in a time-averaged sense one can speak of a symmetric flow pattern). The shear layer which separates at the leading side edge of the obstacle is subject to Kelvin-Helmholtz instabilities [66]. These instabilities are amplified and force the

flow to alternate between two preferable energy states. In one state, the angle of attack of the main flow is more directed towards the left side of the obstacle, causing a vortex to be shed from the right trailing side edge. In the other state, a vortex is shed at the opposite trailing side. The alternation between both states results in a street of vortices, the so called Von Karman vortex street, which interact with each other. The dynamics of vortex shedding is a well documented phenomenon which occurs behind the side faces of bluff bodies [66, 174, 58].

2.3 Flow field around wall-mounted rectangular protrusions

Hunt and co-workers [68] gave a detailed description of the application of topology to oil-film visualisations to deduce the flow field around single wall-mounted prismatic obstacles. They provided details of the kinematic and topological principles necessary to infer the flow field from the critical points (nodes and saddles) identified on the surfaces which bound the flow. Dependent on the number of saddles and nodal points found, they derived a topologically consistent flow field which contained additional fine details besides the structures given in Fig. 2.1. For example, it was deduced that the primary horseshoe vortex is accompanied by an additional system of smaller counter-rotating vortices upstream of the primary one.

Martinuzzi and co-workers [101, 102] applied the topological concepts to document the mean flow structure around surface-mounted blocks with square cross-sections placed in a fully developed, turbulent channel flow at a Reynolds number of $Re_H = 40000$. The Reynolds number was based on the bulk velocity and the obstacle height. The channel-height-to-cube-height ratio was 2. The purpose of the study was to highlight the differences between nominally and fully three-dimensional flows. The results showed that two distinct flow regimes existed which depended on the obstacle span-to-height ratio. For the larger ratios, the tall obstacles, the mid region was nominally two-dimensional but the alternating saddle and nodal points in front of the obstacle suggested that three-dimensional cellular flow structures caused the transport of fluid over the top of the obstacle. For the smaller ratios, including the single cube, it was found that the flow upstream of the obstacle alternates between two energetic states. In one state, a high-inertia jet of fluid is directed upstream against the pressure gradient. In the other state, this jet rolls up to form a system of four vortices, making the horseshoe vortex system. This hypothesis was supported with laser-light sheet experiments and LDA-velocity measurements [86, 101]. Due to the alternation between the two states, the velocity distributions had bimodal shapes. Despite the vigorous variation of the velocity field, it was found that the alternation did not involve periodicity.

Such a bimodal behaviour was also observed in the horseshoe vortex structure upstream of a wing-body junction by Devenport and Simpson [38]. They observed that the time-averaged flow patterns obtained using oil-film visualisations differed from those observed from velocity measurements. These inconsistencies were attributed to the dynamic, bimodal behaviour of the flow: in a certain region upstream of the wing-body junction, the probability density functions of the streamwise and normal velocity component became significantly bimodal. In an attempt to elucidate the

mechanisms involved, the histograms were decomposed in two major peaks each corresponding to a state. From this decomposed histograms, the time-averaged velocity was calculated for each state. Such a conditional decomposition and analysis of two main velocities was followed by Martinuzzi and co-workers [86, 101]. Time-averaged vector plots obtained in this manner clearly displayed the two different flow states.

Martinuzzi et al. [101, 103, 104, 69] investigated the flow over a cube in a fully developed channel flow using Laser Doppler Anemometry ($Re_H = 40000$, channel-height-to-cube-height ratio was 2). The rotation of the setup allowed the measurement of all three velocity components. Long-term averaged velocities and higher order moments, like the variance, skewness and kurtosis were directly measured. The instantaneous measurement of pairs of the velocity components u and v and pairs of u and w allowed the measurement of correlations, or the Reynolds stresses, between them. The detailed measurements were complemented with oil-film and smoke visualisations. They documented the three-dimensional flow structure around the cube in great detail which displays all features given in Fig. 2.1. Further, they also provided information about the flow dynamics in front of the cube, as well as in the wake of the obstacle. A comparison with these results will be made in this thesis in the paragraphs in question.

The flow over a surface-mounted cube in a uniform flow and in a boundary layer flow was experimentally studied by Castro and Robins [30]. For both cases, the cube was placed on a flat plate in an open-circuit wind tunnel. The cube size was 60 mm. For the uniform oncoming flow condition, the cube was placed 20 mm from the leading edge, giving a boundary layer thickness of less than 1.5 mm which is significantly smaller than the cube height. In the boundary layer type of flow, the boundary layer thickness was 10 times the cube size. Hot-wire anemometry was used to measure profiles of the mean velocities and turbulence intensities. The results showed that shear layer reattachment depends on the relative thickness of the oncoming boundary layer: shear layer reattachment was observed for the boundary layer flow, contrary to the cube in the uniform oncoming flow. Further, the extent of the wake recirculation appeared to be also a strong function of the relative boundary layer thickness of the oncoming flow.

The flow around wall-mounted vertical cylinders and wing-body junctions (a tapered, streamlined cylinder) is also characterised by a horseshoe vortex system, induced at front of the cylinder and deflected along the sides [13, 136, 48, 175, 178, 176]. In addition to the flow field analysis, Fisher and Eibeck [48] studied the effect of the horseshoe vortex on the surface heat transfer from the base plate downstream of the obstacle. The Reynolds numbers considered were in the range $77,000 < Re_D < 130,000$. The Reynolds number was based on the cylinder width and the bulk velocity. The results indicated that the heat transfer was augmented up to 50% at the centreline of the vortex extensions.

The flow structure around multiple-obstacle configurations in a channel flow received little attention in the literature. Okamoto et al. [126] studied the flow field in a matrix of surface-mounted square blocks of width $D = 23$ mm and height $H = 5$ mm placed in a boundary layer flow. The face-to-face distance S (in both the streamwise and the spanwise direction) was varied as $S/H = 2, 3, 5, 7, 10$ and 13 to

investigate the different flow regimes as imposed by the different distances between the elements. The Reynolds number was constant for all configurations studied, $Re_H = U_\infty H / \nu = 990$, where U_∞ denotes the freestream velocity. They used a two-colour Laser Doppler Anemometer to measure the time-mean velocities and turbulence intensities around the protruding obstacles. The study showed that for $S/H < 5$ the flow from the obstacle top edge separated flow did not reattach at the channel floor in the wake of the obstacle but the inter-obstacle space was fully covered by a large flow recirculation. Reattachment at the channel floor of the separated shear layer was detected for $S/H > 7$. Further, the configuration of blocks with spacing $S/H = 7$ had maximum turbulence intensities around the blocks.

The fully developed matrix flow was one of the test cases of the ERCOFTAC workshop held at the Delft University of Technology, June 6-7, 1997 [62] and at UMIST, Manchester, May 28-29, 1998 [88]). One of the reasons for its selection as benchmark flow was the flow periodicity, which eliminates uncertainties associated with the proper definition of the inlet and outlet boundary conditions. Further, new data of both the flow structure and the convective heat transfer were available (the results of chapter 8). In total 9 contributions were submitted, including steady-state RANS (Reynolds Averaged Navier Stokes) computations with different turbulence models and quasi-DNS simulations. Only one contribution involved the local convective heat transfer (Bornoff and Parry [21]). They performed simulations with the commercial CFD code Flotherm [49, 50] and supplied only the face averaged and cube-averaged convective heat fluxes. The results were in reasonable agreement with the experimental data although they did not correctly predict the flow structure around the cubes. The quasi-DNS data, supplied by Verstappen and Veldman [170, 171], were performed for two different grids. The DNS-method was based on a high-order, spectro-consistent discretisation method. The results, including profiles of the mean velocity and turbulent fluctuations, were in good agreement with the experimental data, particularly for the finer grid (100^3 grid points). The turbulence models (RANS) predicted the mean velocities reasonably well while the calculated Reynolds stresses deviated significantly from the experimental data. Further, Mathey et al. [106] performed large eddy simulations (LES). The calculated mean velocities and second order moments were also in very good agreement with the experiments.

2.4 Vortex shedding past rectangular obstacles

Much experimental work was already devoted to the vortex shedding past square or rectangular cross-sectioned cylinders suspended in a uniform flow. These two-dimensional configurations display the mechanisms of vortex shedding well and can serve as reference cases for studies of separated flows.

Okajima [125] found for rectangular two-dimensional cylinders suspended in a uniform flow that the shedding frequency, presented as the dimensionless Strouhal number St , is a strong function of the width-to-height ratio (B/H) in the Reynolds number range of $70 < Re_H < 2 \times 10^4$. For the larger ratios, $B/H > 1$, a second peak appeared in the power density spectrum, caused by motion induced excitations

(MIE, which are the eigen-vibrations of systems exposed to fluid flow). For the square cylinder, only Von Karman type of vortex shedding was observed and St varied between 0.10 and 0.15 for $70 < Re_H < 2 \times 10^4$.

Shirashi and Matsumoto [153] presented a fundamental study of the mechanisms responsible for the vortex-induced oscillations of two-dimensional structures. They found that three different kinds of vortices are responsible for the vortex induced oscillations of bluff sections which are: separated vortices from the leading edge, secondary vortices at and separated vortices from the trailing edge.

The effect of the variation of the distance between a solid wall and a square cylinder on the flow around the cylinder was investigated by Martinuzzi and Wu [177, 105]. The results showed that the vortex shedding frequency increases sharply for gaps less than one obstacle height H . This behaviour is accompanied by the appearance of local wall flow recirculations and a decrease in the obstacle drag coefficient. It was noted that no vortex shedding activity could be detected for separations of less than $0.3H$.

Lyn and co-workers [94, 95] studied the vortex shedding past a square cylinder suspended in a uniform flow which had a turbulence intensity of 2% and a Reynolds number of $Re_D = U_{in}D/\nu = 22,000$. The Strouhal number found was $St = fD/U_{in} = 0.132 \pm 0.004$. In a time-averaged sense, the streamlines were symmetric and showed two strong recirculations in the wake of the cylinder. A method was developed to perform a phase averaging in order to reveal the coherent structures in the wake of the cylinder. An ordinary LDA system was used to measure the velocity field, yielding consecutive velocity data at different stations. This means that the definition of a fixed reference point in time was necessary to synchronise the measurements. This was done by a pressure transducer at one of the side faces of the square cylinder. The measured pressure revealed also a sinusoidal periodicity. By triggering the Laser Doppler system at different time delays in the reference signals, the phase averaging was performed at different phases which led to the detection of large coherent structures in the wake flow.

In an additional study by Lee [89] and Vickery [173], the vortex shedding is discussed in terms of the surface pressure field and drag/lift forces on the cylinder. They did not give values for the dimensionless frequencies.

Balachandar et al. [14] considered the wake recirculation in separated flows of two-dimensional geometries with circular, elliptical and square cross-sections and normal plates. For the Reynolds number range considered, they observed that a strong interaction existed between both separated shear layers for all cases, which resulted in Von Karman vortex streets. They found that the contribution of the Reynolds normal stresses to the force balance of the wake vortex was significant. Further, they made a conditional decomposition to separate the incoherent contributions from the coherent or organised structures.

For the freely suspended two-dimensional square prism, the consensus of experimental data yields a Strouhal number of $St = 0.133$, based on the oncoming free stream velocity and the width of the square cylinder.

Vortex shedding from the lateral faces of a wall-mounted cube is less pronounced due to the suppression by the base channel wall underneath it. Directly behind the

obstacle, the organised vortices are rather strong and most of the kinetic energy is stored at the largest scales (the energy is concentrated around the vortex shedding frequency). Further downstream, about $1-2H$ distances, the large scale structures have more or less decayed, among others due to the viscous interaction with the channel wall, and the energy associated with these coherent structures cascades to the higher frequencies (smaller scales). This results in more intense turbulence in the far obstacle wake. When the oncoming flow is already turbulent, similar shedding phenomena occur but the coherent structures are not very distinct: instead of a periodic alternation, the vortex shedding occurs in a quasi-periodic manner.

Experimental investigation of the vortex shedding from single wall-mounted rectangular obstacles was reported by several authors. Martinuzzi et al. [101, 102] presented Strouhal numbers ($St = fH/u_B$, where u_B is the bulk velocity, f is the predominant frequency and H is the obstacle size) for bluff-bodies of square cross-section with different span-width-to-height ratio's (W/H) in a fully developed channel flow. The channel-height-to-obstacle-height ratio (h/H) was 2. The Reynolds number Re_H , based on the bulk velocity and the obstacle height, was considerably higher than those considered in this study, namely $Re_H = 40,000$. It was observed that the dominant frequency associated with vortex shedding decreased with increasing span. They measured a Strouhal number of $St = 0.145$ for a single wall-mounted cube. For the larger aspect ratios, ($W/H > 1$), a second peak in the power density spectra was also observed which corresponded to the MIE type of oscillations. It was observed that the dominant frequency associated with vortex shedding decreased with increasing span of the obstacle.

Shah and Ferziger [151] presented results of the time-dependent velocity field with use of large eddy simulations (LES). The time-averaged flow field, resolved from the time-dependent solutions, showed the characteristic vortices usually found in the proximity of an isolated cube (i.e. horseshoe, top, side and wake vortices). These time-averaged results were in good agreement with those measured by Martinuzzi et al. [101, 102]. But the instantaneous flow field was very different from the time-averaged one. For example, the arch-shaped wake vortex did not exist in an instantaneous sense. Although instantaneous vortices were observed, they were asymmetric on the sides of the cube.

The aerodynamic interaction between two rectangular cylinders of infinite length in a tandem arrangement was studied by Takeuchi [163] and Takeuchi and Matsumoto [164, 165]. The results showed that vortex shedding and the motion induced excitations of the obstacles (MIE, which are obstacle excitations induced by flow instabilities) both depend on the ratio S_x/H , where S_x is the in-line centre-to-centre distance and H is the obstacle width.

Devarakonda and Humphrey [37] experimentally studied the turbulent flow in the wake of a tandem arrangement of two square cylinders of infinite length using Laser Doppler Anemometry at a Reynolds number of 10,000, based on the averaged inlet velocity and the size of the downstream prism (D). They varied parameters such as the streamwise centre-to-centre distance between the cylinders ($\lambda = S_x/D$) and the cylinder-height ratio ($\delta = d/D = 0.25, 0.5$ and 1.0 ; d is the width of the leading cylinder). Strouhal numbers were measured behind the second prism. A

maximum in the Strouhal number was found between $\delta = 0.25$ and 1.0 for $\lambda \approx 1.0$. For $\delta = 1.0$, a minimum in the Strouhal number occurred between $\lambda = 1.0$ and 4.0 . They also postulated that the leading upstream cylinder may drive the second cylinder although the presented data contained insufficient detail to support this hypothesis.

The investigation of Luo and Teng [93] of the aerodynamic forces acting on the downstream square cylinder of an in-line tandem arrangement showed the existence of a critical spacing of $S/H = 3$ (S is the centre-to-centre distance, H is the width of the cylinder). The Strouhal number was minimum for this distance. For smaller distances, the shear layer which separated from the leading cylinder reattached on the second cylinder and vortex shedding was detected only downstream of the second cylinder. For larger spacings, vortex shedding was detected behind both cylinders.

Similar results were reported by Oka et al. [124] and Ishigai et al. [79] for two circular cylinders, of diameter D , in an in-line tandem arrangement as discussed in the review of Zdravkovich [179].

de Souza et al. [156] experimentally studied the vortex structure in the near wake of a circular cylinder suspended in a turbulent boundary layer flow. They used Hot-wire anemometry to measure velocity time series to which three different identification techniques were applied to identify coherent structures in the flow. Two methods were based on the filtering of the instantaneous velocity data with a spectral and a time-frequency (wavelet) decomposition while the third technique made use of the relation between sudden variations in the velocity and large-scale structures in the flow.

2.5 Time-averaged convective heat transfer from rectangular objects

2.5.1 Local convective heat transfer

The effect of flow separation, recirculation and reattachment on the time-averaged heat transfer was studied in the past for mainly two-dimensional configurations. Less attention was given to the local convective heat transfer for three-dimensional cases because of the difficulty of measuring local heat fluxes in such a situation.

Two-dimensional situations

Several studies in the accessible literature were focussed on distributions of the local heat transfer coefficient for two-dimensional configurations. Igarashi and Yamasaki [76] studied the local convective heat transfer from a wall-mounted rectangular rib of 10 and 20 mm height and 30 mm depth placed transversal in a turbulent boundary layer flow at velocities of the undisturbed flow between 4 and 18 m/s. They determined the local convective heat transfer coefficient (h) from the uniform heat flux, provided by an electrically powered stainless steel sheet, and surface temperatures measured with thermocouples buried in the material. A large variation in the local heat transfer coefficient was observed, in particular at the top face of the rectangular rib, which was attributed to flow separation and reattachment.

The convective heat transfer from an array of multiple wall-mounted ribs in a fully developed channel flow was reported by Aliaga et al. [5, 6]. They determined the local heat transfer coefficients from a uniform heat flux, generated by an electrically powered metal foil, and the surface temperature distribution, measured with an infrared camera. The Reynolds numbers considered, based on the centreline velocity and the hydraulic diameter of the channel, ranged between 0.5×10^6 and 6×10^6 . Results for a similar configuration were recently reported by Lorenz et al. [91]. They also used a heated foil and infrared thermography to determine heat transfer coefficients. The Reynolds numbers in the latter study, also based on the hydraulic diameter and further on the mean velocity, ranged from 10^4 to 10^5 . Both studies illustrated the difficulties in determining accurately the local heat transfer, and indicated possible causes of temperature inaccuracy when infrared thermography is used for the measurement of surface temperature distributions. From these studies it also appeared that, as expected, flow separation and reattachment dictate the local convective heat transfer phenomena to a great extent.

In other work, Igarashi [74, 75] reported on studies of the local convective heat transfer from a vertical rectangular prism placed in a uniform flow with free stream velocities ranging from 6 to 20 m/s. The investigation included variation of the width-to-height ratio and the flow Reynolds number. Again, a substantial variation in the convective heat transfer was found. The local wall heat transfer from ribbed surfaces in stationary and rotating ducts was also studied by Iacovidis et al. [31, 70, 71, 72, 73]. Ribbed ducts are usually applied in cooling passages for hot-temperature gas turbine blades. The results showed also strong variation in the local convective heat transfer.

Other examples include the heat transfer downstream of a step in the surface of a plate [149], the heat transfer in the recirculation region behind a double step [47] and the heat transfer in the recirculation region on a blunt flat plate [131]. The studies showed also that for these two-dimensional geometries flow impingement and separation had a significant impact on the local heat transfer coefficient. Regions of flow separation were typically characterised by low heat transfer coefficients, while regions where the flow reattached exposed maxima in the local heat transfer coefficient.

Three-dimensional objects

To our knowledge, no results have been published on the direct measurement of the distributions of the local convective heat transfer for all faces of a single cube which are exposed to convective cooling. One of the reasons for this deficiency is the difficulty of measuring local convective heat fluxes for three-dimensional objects. A possible way to measure local heat fluxes, besides the method developed in the work described in this thesis, is by using thin metal foils. The latter method, already mentioned in the previous text, is based on the electrical power dissipation due to the Ohmic resistance of the thin metal foil. The resistance is directly proportional to the inverse of the thickness: $R = \rho_R l / A$ [Ω], where ρ_R [Ω m] is the specific resistance of the metal used, l is the length of the sheet and A is the cross-section area of the

sheet. When the thickness is uniform and the electrical field is directed in parallel through the sheet, a uniform dissipation results which leads to a uniform convective heat flux when the surface is convectively cooled. The complicated shape of cubical obstacles prevents appropriate wiring of the foils and special care should be taken at the corners of the obstacles. It is much easier to equip a two-dimensional obstacle with such a metal sheet.

The constant heat flux method was used by Olsen et al. [127] to measure heat transfer coefficients at two faces of a heated cube. The cubical geometry complicated the coverage of the entire cube surface with these metal foils. The heat transfer was also strongly non-uniform in their case, in particular at the side face of the cube where the side vortex, enveloping the entire side face, was responsible for a monotonic increase in the heat transfer coefficient in the streamwise direction.

A different approach was followed by Natarajan and Chyu [123, 33] who investigated the local convective heat transfer from a single wall-mounted cube in a boundary layer flow with the naphthalene sublimation method which is based on a mass-heat transfer analogy. They placed a cube in a boundary layer flow for which the boundary layer thickness δ was about $1/4$ of the cube height H . The Reynolds numbers, based on the cube height and the free stream velocity, ranged between 3.1×10^4 and 1.1×10^5 . They observed significant differences in the mass transfer rates for the different cube faces.

Recently, Chyu and Natarajan [34] performed a comparative study of the local mass transfer characteristics on the base surface of three-dimensional basic geometries (cylinder, cube, diamond, pyramid and hemisphere) placed in a boundary layer flow at a Reynolds number of 17,000 (the Reynolds number was based on the object height and the freestream velocity). The results showed a mass transfer enhancement especially in vicinity of the horseshoe vortex and wake vortices downstream of the protrusions.

2.5.2 Averaged convective heat transfer

Single objects

Much more research was devoted to the element-averaged convective heat transfer from isothermal obstacles. A study of the average convective heat transfer for an isolated single protrusion in channel flow was performed by Roeller et al. [139, 140] for Reynolds numbers between 500 and 10000. The Reynolds number was based on the channel height. The study included variation of obstacle dimensions, geometry and blockage ratio. The results showed that the average heat transfer coefficient increased with increasing blockage ratio mainly due to local flow acceleration.

Nakayama [121] studied the conjugate (conjugate refers to the interaction between the three modes of heat transfer: convection, conduction and radiation) heat transfer from a wall-mounted block in a channel flow both experimentally and numerically for Reynolds numbers between 2200 to 16000 for two different situations: a heated copper block on an adiabatic wall and an adiabatic block on a heated wall. They found that the conjugate effects were strong and that the local convective heat transfer from the channel wall was strongly related to the three-dimensional flow

field around the protrusion.

Morris and Garimella [116] studied the thermal wake downstream of a three-dimensional obstacle in a water tunnel flow for uniform inlet conditions and Reynolds numbers between 500-1000 (laminar regime) and 10000-25000 (turbulent regime). Long-term mean flow patterns were deduced from flow visualisation experiments. It was shown, as expected, that buoyancy affected the thermal wake structure only for the lower Reynolds number cases.

Multiple objects

Moffat and Anderson [7, 8, 9, 10, 113] dealt with a matrix of aluminium cubes and flat packs mounted on an insulating channel wall. The concept of the adiabatic heat transfer coefficient was successfully applied aimed at eliminating the influence of the upstream thermal boundary conditions. Different parameters were varied: the channel height with respect to the obstacle height, the obstacle sizes (cubes and flat packs), the spanwise and the streamwise separation between the components, and the Reynolds number. The studies provided much insight in the convective heat transfer for these kind of multiple-element configurations. It was shown that a fully developed heat transfer regimes was obtained after four or five rows of cubes.

Garimella et al. [117, 53, 54, 55] also focussed on the average convective heat transfer from isothermal copper cuboids on a flat plate in a water channel. Their studies included variation of different geometric parameters (e.g. the channel height, the streamwise and spanwise separation between the obstacles, in-line and staggered arrangements) and the flow Reynolds number. The range of Reynolds numbers considered covered laminar, transitional and turbulent channel flows. They observed that the convective heat transfer is significantly dependent on the streamwise separation, the channel height and the Reynolds number.

2.6 Time-dependent convective heat transfer

The vortex shedding from obstacles suspended in a cross-flow or from wall-mounted protrusions causes large-scale coherent structures in the wake flow of the obstacle in question. These periodic oscillations affect not only the surface heat transfer from the objects itself or, in the case of wall-mounted protrusions, from the base plate on which the obstacle is mounted, but it will also affect the turbulent heat flux in the boundary layer flow.

The coupling between the surface heat transfer from circular cylinders in cross-flow and the vortex shedding has been experimentally investigated by Scholten and Murray [146, 147, 148] for both low and high freestream turbulence flows. Time-dependent heat fluxes were measured around the circumference of the cylinder with a hot film sensor; velocities were measured with a Laser Doppler Anemometry system. Due to the strong aerodynamic coupling with the pressure field, the heat flux at the front of the cylinder varied periodically with the frequency of vortex shedding.

Wroblewski et al. [175, 176, 178] studied the effect of flow instabilities, caused by vortex shedding, on the turbulent heat flux in the wake of wall-mounted cylinders

and wing-body junctions. Wroblewski and Eibeck [175] performed simultaneous measurements of the fluid velocity and temperatures at different stations in the wake of the streamlined cylinders. The Reynolds numbers considered were $Re_D = 24,700, 36,000$ and $970,000$, where Re_D is based on the root diameter of the cylinder. The boundary layer in the wake of the obstacle was heated by means of a uniform heat flux foil, fully covering the base on which the cylinder was mounted (the cylinder was insulated from the base plate in order to prevent its heating). Cross-spectra of the fluctuating velocity and temperature, $\overline{u'\theta'}$, showed peaks at frequencies associated with vortex shedding for the measurements in the middle-wake region. This suggested that the turbulent heat transfer in the boundary layer flow was augmented by vortex shedding. Spectra of the turbulent heat flux in the inner-wake region showed maxima at a much broader frequency range. Since the flow periodicities were dominantly present in the spanwise direction, the turbulent heat fluxes showed anisotropies. While the turbulent heat flux in the flow was enhanced by vortex shedding, an effect on the total surface heat transfer was not observed. They argued that the turbulence mixing in the wake due to flow instabilities and noted increase in the eddy diffusivity was the dominant mechanism for heat transfer enhancement in the boundary layer flow.

The effect of large-scale periodic unsteadiness in the wake of a streamlined cylinder, attributable to vortex shedding, on the heat transfer in the boundary layer was studied by Wroblewski [176]. He used a triple-wire to measure simultaneously the temperature and the velocity in the wake. By using a triple decomposition and conditional sampling technique, the heat fluxes associated with the coherent structures in the wake flow were separated from the turbulent heat fluxes. The results showed that up to 50 % of the spanwise heat flux was attributed to the coherent contribution, while this was about 30 % for the vertical component of the heat flux.

Xie and Wroblewski [178] studied the effect of large-scale periodicities on the surface heat transfer from the base plate in the wake of a circular cylinder. They measured the time-resolved wall heat flux and turbulent heat fluxes in the boundary layer. In addition to the measurement techniques described in [176], they used a thin film sensor to measure the time-dependent wall heat flux. They found that the surface heat flux in the wake of the cylinder was significantly increased by the large-scale coherent structures in the flow (vortex shedding). This effect was largest in the unstable region, directly downstream of the vortex streets. They postulated that the thickening and the thinning of the near-wall thermal boundary layer, induced by the flow periodicity, caused the augmentation of the surface heat transfer. The strong correlation between the ensemble-averaged velocity, temperature, and wall heat flux confirmed this hypothesis. The fluctuations in the convective heat fluxes were up to 25 %.

Swisher et al. [158] used a fast-response heat flux sensor, similar to the one described in Hager et al. [59] and Simmons et al. [155], to measure the time-dependent heat flux in the horseshoe vortex structure in front of a wing-body junction at its centreline. Velocities were acquired simultaneously. A strong correlation was detected between the heat flux and the velocity, particularly at the location where the flow separates from the wall upstream from the main horseshoe vortex.

Suzuki et al. [82, 160, 161, 162] studied the surface heat transfer in a boundary layer flow disturbed by an inserted cylinder. The cylinder was placed normal to the flow direction. They reported on a dissimilarity between the surface heat transfer and the friction factor. At locations where the latter became minimum, a significant enhancement in the local surface heat transfer was found. This is in contradiction with the similarity rule (Reynolds' analogy) which prescribes a proportionality between the heat transfer coefficient and the wall shear stress. They argued that the dissimilarity was directly related to the vortex shedding from the suspended cylinder. Entrainment of cold air occurred at the location where the vortex was directed downward to the surface while hot fluid was ejected from the wall where the flow separated. Temperature effects associated with this so-called washing action caused the mentioned dissimilarity. Their hypothesis was recently supported by time-dependent numerical simulations [77] which showed a dissimilarity between the instantaneous wall friction and the surface heat transfer.

Part II

Description of the experiment

CHAPTER 3

Experimental set-up and measurement methods

The test configuration in which the experimental work was performed is described in this chapter. A closed wind-tunnel was constructed in a way that the channel section in which all the measurement were performed had well-defined two-dimensional flow conditions. The configurations of cubes were mounted on one of the channel walls. Two methods were explored to measure the convective heat transfer from the cubes. A time-averaged method was developed to evaluate distributions of the local heat transfer coefficient from the surface temperature distributions. For this method, an advanced measurement assembly was designed. In addition to the time-averaged method, a transient heat flux technique was developed to measure the time-dependent convective heat transfer from the cubes. The method used a sensor of finite size and had, therefore, a limited spatial resolution. Both methods are evaluated with an accuracy analysis to support the reliability of the convective heat transfer measurements.

3.1 Test configuration

The experimental work was conducted in a blowing-type low-Reynolds number closed wind tunnel. The flow was recirculated by means of two fans. Two heat exchangers were used to thermostat the air to a constant temperature. Honeycombs and fine grids were used to eliminate the large turbulence scales and to straighten the flow. Approximately 2 m upstream of the test-section, the channel contracted to a rectangular channel of height $D = 51$ mm and of 60 cm width. The sufficiently large width-to-height ratio (12) ensured a two-dimensional flow at the centre of the test-section. A sketch of the test-section with its global right-turning Cartesian coordinate system are given in Fig. 3.1. The x - and y -axis refer to the streamwise and the normal direction. The z -axis points out of the image. A photograph of the test-section with the matrix of wall-mounted cubes is shown in Fig. 3.2. The laser probe in front of the test-section was used to measure the flow field around the cubes. Cubes of size $H = 15$ mm were mounted on the lower channel wall, from now on referred to as channel floor. This gives a channel height-to-cube-height ratio of $D/H = 3.4$. The local coordinate system is chosen in the same way as the global coordinate system, but the local origin is now located at the front centreline of each cube (see Fig. 3.3 for a sketch of the cube with its local coordinate system). The test-section was oriented vertically to allow the measurement equipment to access the measurement area around the cubes. This will be discussed in more detail later in this section. Gravity forces acted in the vertical (spanwise, z) direction. For most of the experiments performed, the Reynolds number was sufficiently large to neglect buoyancy effects. For situations where the induced secondary flow was not negligible, this will be explicitly mentioned in the text.

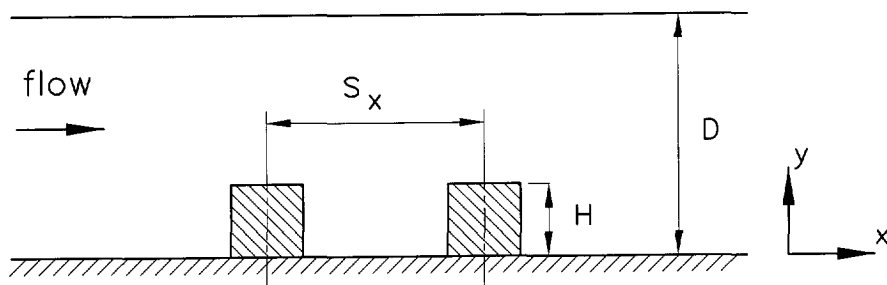


FIGURE 3.1: Sketch of the side view of the test-section with two wall-mounted cubes. The global reference system is also indicated in the figure. The z -coordinate points out of the image (right-turning Cartesian coordinate system).

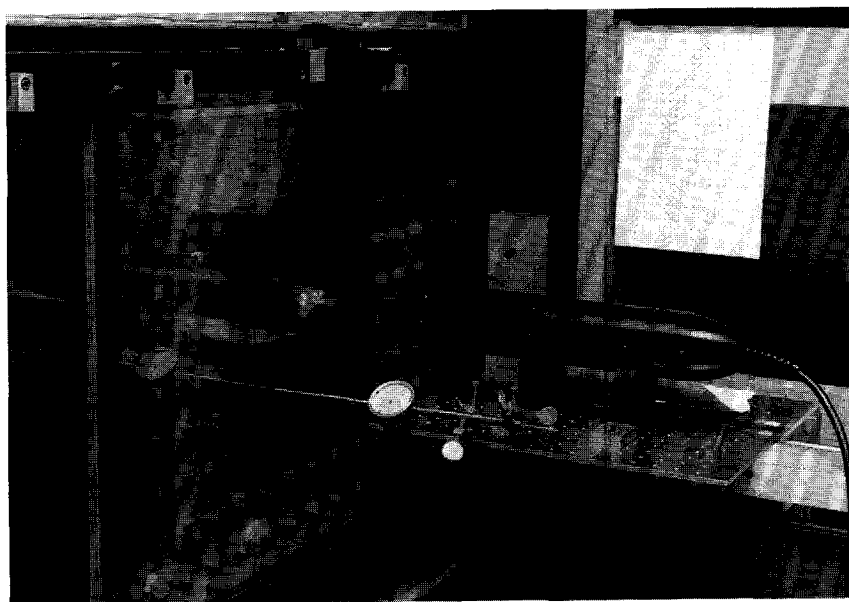


FIGURE 3.2: Photograph of the test-section with the matrix of wall-mounted cubes. The laser probe, mounted on a traverse table in front of the test-section, was used to measure the flow field around the cubes.

At the entrance of the test-section, boundary layers developed along the test-section walls, i.e. the channel roof and floor.¹ The boundary layers which developed on the smaller side walls did not influence the channel flow in the centre-region due

¹Although the channel was placed vertically, the cubes were mounted on one of the vertical walls. For sake of clarity, we shall refer in further text to 'floor' as the channel wall with the cubes and to 'roof' as the opposite channel wall. The other two wall are denoted as 'side' walls.

to the large width-to-height ratio. Tripping strips in the test-section entrance were used to accelerate the transition to a turbulent boundary layer flow. It was found experimentally (from measurements) that the 'zig-zag' strip, commonly applied on aerofoils to stimulate the transition to a turbulent boundary layer flow, was the most suitable strip for our application. For most of the experiments performed, the strip was located at the entrance of the parallel test-section at both the channel roof and floor. This resulted in a fully developed channel flow in the measurement area at $x = 0$, located 2 meter downstream from the entrance. For some measurements, the strip was located somewhat further downstream, leading to a developing turbulent channel flow. The different ways of controlling the inflow conditions resulted in marked differences in the channel flow at the measurement section. For all these regimes, the channel flow was, therefore, well documented with the LDA system (profiles of the mean velocities and turbulence intensities). For the test-cases in question, the oncoming flow conditions are given before the actual discussion starts.

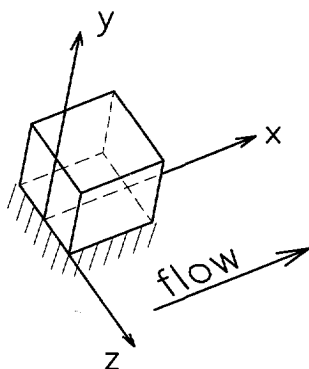


FIGURE 3.3: The local coordinate system of the cube. x , y and z refer to the streamwise, the normal and spanwise direction, respectively.

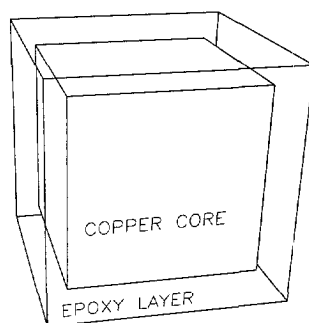


FIGURE 3.4: Schematic of the measurement cube, consisting of the core of copper with the epoxy layer around it.

The cubes were mounted on a base plate of formaldehyde. The thermal conductivity of this insulator was measured with a steady state 'sandwich' technique, described in Appendix A. The results showed a thermal conductivity of $\lambda = 0.33 \text{ W/mK}$. The experimental uncertainty, estimated from a single sample uncertainty analysis and incorporating heat losses, was about 10%. The low thermal conductivity of the base plate prevented excessive conductive heat losses from the heated cubes through this plate.

The experimental work discussed in this thesis was performed for four different configurations of cubes which were already mentioned in the introduction: the single cube, the in-line row of cubes with a streamwise distance of $S_x/H = 2$, the tandem-configuration of two cubes for various values of S_x/H and S_z/H and the regular matrix of cubes with $S_x/H = S_z/H = 4$.

Most of the results described in this thesis were based on the data acquired with two experimental techniques. Infrared thermography was used to measure the

surface temperature distributions of the cubes which served as a basis for the determination of the distributions of the local heat transfer coefficient. Laser Doppler Anemometry was used to document the flow structure quantitatively. Both techniques will be thoroughly discussed in the chapter on experimental techniques. Since both techniques are based on optical principles, a suitable window was needed for the measurement in question. The channel roof (the front wall) was, therefore, made of glass in case of the LDA measurements. For the heat transfer experiments, the channel roof was replaced by a special window (foil). Its transmittance characteristics were compatible with those of the infrared camera used in order to transmit the infrared radiation through the window.

The LDA probe was mounted on a traverse table which could be traversed in all three coordinate directions (see Fig. 3.2). This allowed us to document the flow field around the entire cube. The traverse system had an absolute location accuracy of 0.1 mm provided that a suitable reference point was selected. The reproducibility was better than 0.1 mm. The infrared camera was also mounted on this traverse table, but in a different set-up for the measurement of the surface temperature distributions. The high traverse accuracy was necessary for accurate measurements for both the Laser Doppler Anemometry and IR-thermography.

3.2 Time-averaged measurement method

A major goal of the present work was to investigate distributions of the time-averaged local heat transfer coefficient. A local measure of h requires both a local surface temperature and a local convective heat flux. The most common way is to use passive heat flux sensors which are primarily based upon a known temperature difference across a thermal insulator with a well known thermal resistance (thermal conductivity and layer thickness). A drawback of such kind of sensors is the restricted spatial resolution. Further, a disturbance of both the flow and the thermal boundary condition may occur, particularly if the thermal properties of the sensor differ much from those of the heated surface in question. A good review of the applications, characteristics and drawbacks of passive and active sensors is given by Diller [39]. A different approach is to impose a constant heat flux at the entire surface of the cube. This is done, for example, by electrical powering of a thin sheet of metal of relatively large Ohmic resistance, placed on top of the surface under investigation. The electrical energy will be uniformly dissipated (provided that the metal properties are isotropic and that the thickness is uniform) which results in a constant heat flux through the surface. This approach has major constructive and geometrical disadvantages and was, therefore, not considered to be a suitable candidate.

To measure time-averaged distributions of the heat transfer coefficient with a large spatial resolution, we followed a different approach. This was primarily based on a known temperature difference and a known thermal resistance. A special measurement assembly was designed and constructed. It had the capability of determining the convective heat flux distribution at almost the entire surface of the cube from which the local heat transfer coefficient was derived.

3.2.1 Measurement cube

The measurement assemblies consisted of an internal cubical core of copper covered by a thin epoxy mantle of well defined thickness of 1.5 mm. The schematic of the cube is given in Fig. 3.4, a detailed mechanical drawing is shown in Fig. 3.5. The lower part of the cube was in the form of a bolt which was mounted to the base plate by a nut. The bolt consisted partly of a pass-edge to accurately position the assembly in the base plate. It was important that the cubes were well positioned with respect to the global coordinate system for accurate measurements (both anemometry and thermography). The bolt was also made from the low thermal conductivity epoxy, the nut was made from a different plastic, both for the sake of minimising heat losses.

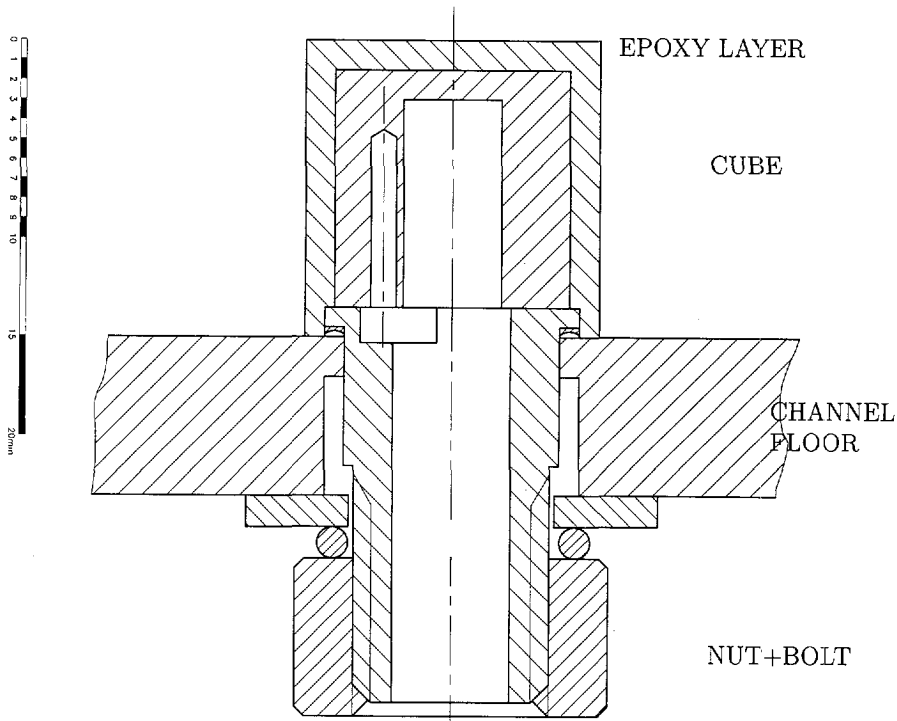


FIGURE 3.5: Detailed cross-section of the composed measurement assembly mounted on the channel floor. The sketch shows also a part of the mounting base and the nut.

It is very crucial that the thickness of the epoxy layer is uniform and well defined to accurately determine heat fluxes through the layer, as will be seen later in this thesis. Heat is generated in the centre of the copper core by a dissipating source. The source consisted of a resistance wire which was twisted spirally around a glass cylinder and electrically insulated by a glass tube. The lead pins to the heat source were made of a low thermally and high electrically conducting alloy to minimise heat losses through the pins and heat dissipation in the pins, respectively. The

thermal conductivity of copper is about 1000 times larger than that of the epoxy substrate (the thermal conductivity of copper is about 390 W/mK , that of the used epoxy 0.24 W/mK). The ratio of the epoxy layer thickness to the copper thickness was almost $1/5$. Hence, the thermal resistance of the epoxy layer is much larger compared to that of the copper core. The temperature change across the copper core is, therefore, negligible compared to that across the epoxy-layer which resulted in an almost uniform copper temperature within the tolerance of 0.05°C . This uniformity holds even for relatively high heat fluxes.

The epoxy layer represents a heat flux sensor: when the uniform copper temperature (measured with a thermocouple) and the surface temperature distribution (measured with thermography) are both known, the conduction problem in the epoxy can be solved. The resulting temperature distribution in the epoxy served to derive distributions of the local heat transfer coefficient. Details of the measurement method are given in the next section. A constant copper temperature, stable in time, was required for the application of the uniform Dirichlet boundary condition. A special power supply was, therefore, designed to heat the cubes individually. This controller used the temperature of the copper core, measured with a thermocouple, and adjusted the power supply such that the set-temperature was obtained. The adaptive feedback loop in combination with the frequency of updating (4-5 seconds per update which was constrained by the scan frequency of the data-logger) ensured a quickly obtained equilibrium state in which the temperature was stable within 0.1°C . The response time of the measurement cubes, i.e. the time after which the temperature of the copper core was constant within the tolerance, was about 5-8 minutes.

Because the epoxy layer was in fact the heat flux sensor, a sufficiently thick layer was required in order to ensure accurate measurements. As we will see later, the major contribution to the uncertainty in the heat transfer coefficient is from the surface temperature measurement. A sufficiently large temperature decay across the epoxy layer, facilitated by the layer thickness, was then required to satisfy the accuracy demands. On the other hand, a too thick layer will cause a too low temperature at the surface of the cube which is not beneficial for infrared thermography. From estimations, an epoxy layer thickness of 1.5 mm appeared to be the optimum selection.

All experiments presented in this thesis were performed at a copper core temperature of 75.0°C except for one series of single cube experiments for the sake of investigating the effect of the thermal boundary condition on the convective heat transfer.

3.2.2 Physical model

The uniform copper temperature of the measurement cube provided the Dirichlet temperature boundary condition for the inner bound of the epoxy layer. The surface temperature distribution of the outer bound of the substrate, acquired from thermography experiments, provided the outer temperature boundary condition. A simple one-dimensional analysis with the neglect of lateral conductive heat fluxes

leads to unacceptable error, especially close to the edges and corners.

Therefore, the temperature distribution in the epoxy substrate was calculated numerically by solving the equation for the three-dimensional conduction problem [78]:

$$\frac{\partial}{\partial x_i} \left(\lambda_{i,j} \frac{\partial T(x, y, z)}{\partial x_j} \right) = 0, \quad (3.1)$$

where $T(x, y, z)$ is the temperature distribution in the epoxy substrate and $\lambda_{i,j}$ is the anisotropic thermal conductivity. Measurements showed an isotropic thermal conductivity of 0.24 W/mK which was spatially and temperature independent. This value was determined with a steady-state and transient measurement technique, both discussed in Appendix A. Equation 3.1 reduces to the ordinary Laplace equation when the thermal conductivity is constant [78]:

$$\nabla^2 T(x, y, z) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0. \quad (3.2)$$

When the temperature distribution in the epoxy layer is known, the spatial temperature gradient in the normal direction at each of the five surfaces of the cube can be calculated yielding the local conductive heat flux:

$$\phi''_{cond}(\eta, \xi) = -\lambda \frac{\partial T(\eta, \xi)}{\partial n}, \quad (3.3)$$

where n is the direction normal to the surface, $T(\eta, \xi)$ is the surface temperature and η and ξ refer to local surface coordinates at the cube faces. The local net radiative heat flux $\phi''_{rad}(\eta, \xi)$ was calculated from the surface temperature distribution, ambient temperatures (channel-walls), surface emissivity and view factors (see below). A heat balance at the surface yields the local convective heat flux $\phi''_{conv}(\eta, \xi)$:

$$\phi''_{conv}(\eta, \xi) = \phi''_{cond}(\eta, \xi) - \phi''_{rad}(\eta, \xi). \quad (3.4)$$

The local heat transfer coefficient h_{ref} is defined as:

$$h_{ref} = \frac{\phi''_{conv}(\eta, \xi)}{T_{sur}(\eta, \xi) - T_{ref}(\eta, \xi)}, \quad (3.5)$$

$T_{sur}(\eta, \xi)$ and $T_{ref}(\eta, \xi)$ denote the surface and the reference temperature, respectively.

The Laplace equation was discretized with the Finite Volume Method [133]. The boundary conditions were incorporated in the coefficients of the discretized diffusion equation. The temperature boundary condition for the back-wall face (the interface between the base plate and the cube) was required to solve the equation explicitly. It was impossible to measure accurately the temperature distribution with sufficient spatial resolution for this face. One can think of thermocouple measurements, but the resolution can never compete with that obtained from the thermography measurements. Therefore, the lower epoxy layer between the base plate and the copper

core was imaginary removed from the computational domain. The temperature distribution of this imaginary plane was composed of the copper temperature for the mid-plane (the square mid-section of size 12×12 mm) and a linear interpolation between the surface temperature of the outer wall and the copper temperature. The remaining numerical domain consisted of the copper core inclusive the epoxy layer covering the top and the four side faces. The dimensions of the computational domain were then $15 \times 15 \times 13.5$ mm. Since Dirichlet boundary conditions were defined for the entire domain, the problem was well defined.

Besides the convective and radiative heat flow from the five faces of the cube exposed to air, heat conduction occurred at the interface between the cube and the lower base plate, the sixth face. This heat loss consisted of conduction through the lead and thermocouple wires and of conduction through the epoxy interface. However, these losses did not affect the measurement: the temperature distribution in the entire cube is the result of the equilibrium of all heat fluxes involved. It is noted that conduction through the base plate preheats the oncoming flow. This will affect the local reference temperature and thus the heat transfer coefficient. This effect is only of minor importance (second order effect) and is, therefore, not taken into account.

The conductive heat losses through the sixth interface can be estimated from an integral approach. The total heat supplied is the total power dissipated in the electrical heat source and is simply the product of the voltage V and the electrical current I , namely $P_{sup} = V \times I$. The total heat Q_{cond} through the five faces exposed to air flow is the sum of the total radiation Q_{rad} and convection Q_{conv} . This sum is in fact the integrated conductive heat flux, as results from the temperature solution of the Laplace equation:

$$Q_{cond} = Q_{conv} + Q_{rad} = \frac{1}{A} \int_A \phi''_{cond}(\eta, \xi) dA, \quad (3.6)$$

where A is the surface of the cube exposed to air flow, used in the numerical domain, and $\phi''_{cond}(\eta, \xi)$ is the local conductive heat flux. This integral is approximated by the next finite summation:

$$Q_{cond} = \frac{1}{M} \sum_{i,j} \phi''_{cond}(\eta, \xi), \quad (3.7)$$

where M is the total number of surface cells, i and j are indices denoting the discrete location at the surface. As noted earlier, the Laplace equation was solved for the numerical domain of $15 \times 15 \times 13.5$ mm since the thermal boundary conditions for the sixth face were unknown. This conductive heat flow Q_{cond} is, therefore, the integrated flux over the reduced area ($15 \times 15 \times 13.5$ mm). The conductive heat flow for the total cube area exposed to air flow ($= 5 \times 15^2$ mm²) was then obtained by assuming that the averaged conductive heat flux Q_{cond}/A , determined for the reduced area applied to the total area which leads to a corrected conductive heat flow: $Q_{cond,c} = (15 \times 15 \times 15) / (15 \times 15 \times 13.5) \times Q_{cond}$. Based on this correction, the heat losses then read:

$$Q_{loss} = P_{sup} - Q_{cond,c}. \quad (3.8)$$

Reference temperature

The reference temperature used in the definition of the local heat transfer coefficient needs special attention. Three different reference temperatures are commonly used: T_{in} , T_{∞} and T_m , which are the temperature of the flow at the channel inlet, the temperature at infinity (often used for boundary layer flows) and the mixed mean temperature. The latter is obtained from an energy balance over an infinitesimal length of the channel, specifically:

$$T_m = T_{in} + \frac{\phi_{tot}}{\dot{m}c_p}, \quad (3.9)$$

where ϕ_{tot} is the total heat added upstream to the flow, $\dot{m}$ is the mass flow rate and c_p is the specific heat of air. For one dataset of convective heat fluxes, the use of different reference temperatures leads to different values of the heat transfer coefficient. Consider a regularly arranged matrix of isothermal cubes (made from aluminium or copper). If all cubes are heated, the mixed mean temperature of the flow will increase in the downstream direction due to heat transfer by convection. Compare this situation with the case that for the similar matrix configuration only the cube in question is powered. The flow field around the cube in the matrix is, when the temperature effects are neglected for simplicity, similar for both cases. If we then define the heat transfer coefficient based on the inlet temperature, we obtain a different value for the heat transfer coefficient. In the case that all cubes were powered, the upstream heated cubes caused the air temperature to be higher in the proximity of the cube in question. This consequently leads to a smaller convective heat flux. If then the inlet temperature is used as reference temperature, we obtain a significantly smaller heat transfer coefficient than in the case that only the cube in question was heated (the temperature of the local environment is then approximately the inlet temperature, leading to a much larger convective heat flux).

This example illustrates that the heat transfer coefficient is a derived property depending on its definition, and that it will have different values dependent on the reference temperatures used. Further, these definitions of h are strongly dependent on the thermal boundary conditions. This is well known, for example, if the convective heat transfer in a pipe flow with a uniform wall heat flux is compared to a similar pipe flow but with a constant wall temperature. The situation becomes complicated if the upstream thermal boundary conditions are not uniform, for example, in the case of discrete not-uniformly distributed heat sources. Heat transfer coefficients based on the aforementioned reference temperatures will all give different values. Further, the results are only applicable in situations with identical thermal boundary conditions. To enable a proper comparison between laboratory experiments and reality, or to allow the application of laboratory experiments to the evaluation of practical heat transfer problems, this ambiguity needs to be eliminated. The goal is then to find a universal descriptor, independent of the thermal boundary conditions upstream.

Isothermal cubes The upstream cubes can be considered as discrete heat sources causing non-uniform distributions of the wall temperature and the surface heat

flux. To address the problem of this non-uniform upstream heating, Moffat et al. [9, 10, 114] introduced the concept of adiabatic heat transfer for this kind of application. An adiabatic situation is, according to the physical meaning of adiabatic, a situation with no heat exchange of any heat transfer mode (radiation, conduction, convection) with the environment. To translate this concept to arrays of cubes, the adiabatic situation can be seen as isothermal and perfectly reflecting cubes (made of polished copper or aluminium) mounted on a perfectly insulating board. Due to the zero emissivity of the surface, radiation can be disregarded. The zero thermal conductivity of the mounting board prevents conductive heat exchange with the environment. When the cube is not powered while the other cubes in the array remain heated, the unpowered cube in question is convectively heated by the air. In the equilibrium state it will reach the mean temperature of the air in the proximity of the cube. Since no heat transfer can take place, this state is then referred to as adiabatic. The constant temperature of the cube in that case is the adiabatic reference temperature which is physically independent on the upstream thermal boundary conditions.

The physical background was provided by Sellars et al. [150]. They considered a steady-state fully developed laminar flow in a pipe. They assumed parabolic velocity profiles in the pipe. Further, they assumed a superposition of individual contributions of the arbitrary heat flux distribution upstream of the location in question, which is allowed since the energy equation is linear. These contributions to the local temperature are better known as Kernel functions. The superposition of these Kernels gives the value for the adiabatic reference temperature. A good review of this theoretical problem is also given in Eckert and Drake [42]. This approach was extended by Moffat and co-workers for the mentioned equidistantly arranged matrix. A superposition of measured thermal wake functions or Kernels resulted in the adiabatic reference temperature for the matrix of cubes.

In laboratory applications, it is impossible to achieve a true adiabatic situation. Board materials will always have a non-zero thermal conductivity and the surface emissivity will always be larger than zero. Thus radiation and conduction can not be ignored in laboratory experiments. This problem was addressed by Moffat and co-workers by minimising the heat losses (conduction and radiation) by considering polished aluminium cubes mounted on a board of balsa wood. The large thermal conductivity of aluminium resulted in a uniform temperature of the entire cube. In the case that the cube in question is unpowered while all others remained heated, the heat losses are in equilibrium with the convective heat flow from the preheated air to the unpowered cube. This heat loss, $Q_{conv,unp}$, is directed towards the cube (the air heats the cube) and it can be expressed as:

$$Q_{conv,unp} = h_{ad}A(T_{unp} - T_{ad}), \quad (3.10)$$

where h_{ad} is the adiabatic heat transfer coefficient, T_{unp} is the unpowered isothermal temperature of the cube and T_{ad} is the adiabatic reference temperature. When the cube is consequently powered while the other cubes remain heated, the temperature of the cube will increase due to its own active contribution. The temperature rise is then relative to the unpowered cube temperature. The convective heat flow,

$Q_{conv,pow}$, then reads:

$$Q_{conv,pow} = h_{ad}A(T_{pow} - T_{unp}), \quad (3.11)$$

where T_{pow} is the isothermal temperature of the cube while it is powered. The first order correction proposed by Moffat and Anderson [9, 10, 114] is only valid for small heat losses. The adiabatic reference temperature was determined from a superposition of thermal wake measurements. Another possibility is to measure the unpowered reference temperature of the cube in question directly when the cube is unpowered while the other cubes remained powered. The simple first order correction then leads to the adiabatic reference temperature.

The advantage of copper or aluminium cubes is that the surface and internal temperatures in both the powered and unpowered situation are uniform. Even in the case of a relatively large heat flux (for large power dissipations), it can be shown with a simple calculation that the temperature is uniform within 0.3°C for the cube size in question.

The protruding shape of the isothermal cubes will also cause complex flow and vortex structures. Although the cube itself is isothermal (with the mentioned tolerances) this is not the case for the fluid in the vicinity of the cube. Since the local convective heat flux is strongly dependent on the local fluid temperature, a marked variation will be present.

Epoxy measurement cubes The measurement cubes used in this study consisted of a copper core with a thin layer of epoxy. A low thermal conductivity of the epoxy substrate was selected to make the method successful (the temperature gradients across the layer give indirectly conductive heat fluxes). A consequence of this approach is that the surface temperature of the cube is not isothermal in the powered situation, nor in the unpowered situation. This epoxy layer makes the experiment different from that of Moffat et al.

Some issues need to be addressed to substantiate the application of the adiabatic concept to the present situation. One of the major questions to be answered is whether or not we should select a local reference temperature in the definition of the local heat transfer coefficient, or whether an averaged temperature suffices. In other words, are the temperature gradients in the adiabatic situation small enough to allow the use of the cube- or surface-averaged reference temperature. The following issues require special attention:

- In addition to strong temperature gradients across the epoxy layer, the low thermal conductivity has other consequences. It will be shown in the following chapters where the results are discussed that the flow field and vortex structure around the cubes caused a strong variation in the surface heat transfer. As a consequence, the cube has large spatial temperature gradients along the surface (due to the low thermal conductivity of epoxy). This temperature distribution can then be considered as a small-scale irregular thermal history, like the heated upstream cubes in the matrix arrangement. In fact the problem should be tackled at both scales: at cube level and system (configuration) level.

To measure the local effects at the cube surface, we should then be able to locally unpower the cube, make it locally adiabatic and then measure the effect of heating of the upstream surface. A difficulty arises for reverse flows since heat is recirculated in the vortices. Questions which arise are: which direction is upstream, what is the net effect on the local reference temperature and so on. Similar difficulties are expected if multiple cube configurations are considered since convected heat will recirculate back to the cube in question.

- In the case that the cube is unpowered while the other cubes in the configuration remain powered, the vortex structures around the cube will cause similar effects: temperature gradients at the cube surface are to be expected, however, less strong. Due to the turbulent flow field around the cube, the heat which is convected from the upstream cube will be quickly diffused. In other words, the vortex and flow structure around the cube does not carry the temperature information of the surface as it would when the cube in question was powered. Therefore, the downstream cube will be exposed to well-mixed fluid at almost uniform temperature.

Despite this well mixing, temperature differences are expected on a larger scale, for example, between the front and the rear face. The copper then acts as a heat sink which spreads the heat uniformly. In the case that a strong temperature gradient exists across the cube, for example, when the temperature of the front face of the cube is significantly higher than that of the rear face, the copper core diminishes the temperature decay. This leads to an underestimation of the temperature of the front face in the unpowered situation. The surface temperature of the rear face is then overestimated.

It was attempted to measure the surface temperature of an unpowered cube, placed in the in-line row of cube with the infrared camera (see chapter 6) but due to the uncertainties encountered when temperatures just above room temperature are measured, the results were not reliable at all.

Based on simple calculations it was concluded that in the unpowered situation, the temperature decay across the epoxy layer is of the same order of magnitude as the variation along the surface is. The decay was estimated to be less than $0.5\text{ }^{\circ}\text{C}$. These local effects will give minor deviations in the reference temperature and its contribution to the experimental uncertainty in the heat transfer coefficient is negligible.

The cubes were painted with a thermally black paint. This paint was used to enhance the thermal radiation from the surfaces in order to increase the accuracy of the infrared surface temperature measurements. Almost all measurement discussed in this thesis were performed in a turbulent channel flow for which the convective heat transfer was significantly larger than the radiative heat transfer. Despite that, the radiation was certainly not negligible.

The cubes were mounted on a board made from formaldehyde. This plastic had a thermal conductivity of 0.33 W/mK which is about 5 times larger than that of balsa wood. The selection of this material was based on the good machining quality

since precise positioning of the cubes was required for accurate thermography and LDA measurements. The value of the thermal conductivity is still sufficiently small to consider the conductive heat losses to be much smaller compared to the rate of convective heat transfer.

Reference temperatures used Based on the foregoing analysis, a cube-averaged adiabatic reference temperature was used in the definition of the local heat transfer coefficient.

For most of the heat transfer results presented in this thesis, only the cube in question was powered while the other cubes remained unpowered. In that case, the local reference temperature of the cube is in fact the isothermal temperature of the entire environment in the situation before starting the experiment: the adiabatic situation. This temperature was then measured and used in the definition of h . When the cube is then subsequently powered, conductive heat losses through the base plate and radiation will both heat the direct environment of the cube and can be convected back to the fluid. Although these second-order effects will increase the reference temperature, they are estimated to be negligible.

The experiments for the in-line array of 9 cubes were performed while all cubes remained powered. The adiabatic reference temperature was then measured in a separate experiment. In that case, only the cube in question was cut from power supply during the operation of all other elements. Due to the relatively small heat fluxes through the walls of the unpowered cube, the temperature decay across the epoxy layer is negligible. Therefore, the surface temperature was approximated with the copper core temperature.

Variation of the epoxy layer thickness

The layer thickness of the epoxy substrate dictates in fact, in combination with the local convective heat transfer, the temperature distribution of its outer boundary. For a very thin epoxy layer, in the limit of zero thickness, the boundary condition approaches the uniform temperature condition. For rather thick layers, the resulting boundary condition is something between a constant temperature and a constant heat flux boundary condition. Although different surface temperatures were encountered, consequently resulting in different local cooling rates, the heat transfer coefficient should, since it is a definition, not be affected by the different temperature distributions and thus different δ 's.

In order to study the effect of the layer thickness on the heat transfer characteristics, the CFD-code FLOTHERM [49, 50] was used. Because of its restriction in the availability of suitable turbulence models, a laminar flow was considered. The simulations were performed for a single wall-mounted cube in a channel flow. The channel height was 5 cm. The cube size was 15 mm for all cases, while the inner copper core was varied in size to obtain different epoxy layer thicknesses. Five thicknesses of the epoxy layer were considered, namely: 0.6, 0.9, 1.5, 2.0 and 2.5 mm. The Reynolds number of the flow was 1000, based on the channel width and the inlet velocity of 0.32 m/s.

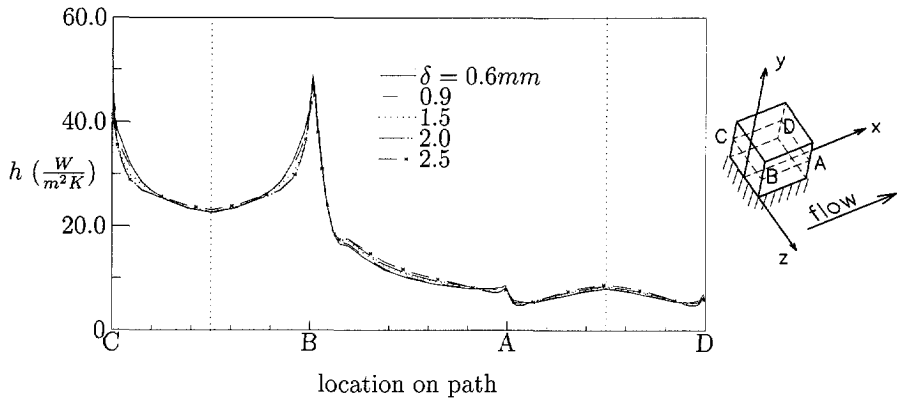


FIGURE 3.6: Distribution of the heat transfer coefficient along path *ABCD*A parametric in the epoxy layer thickness.

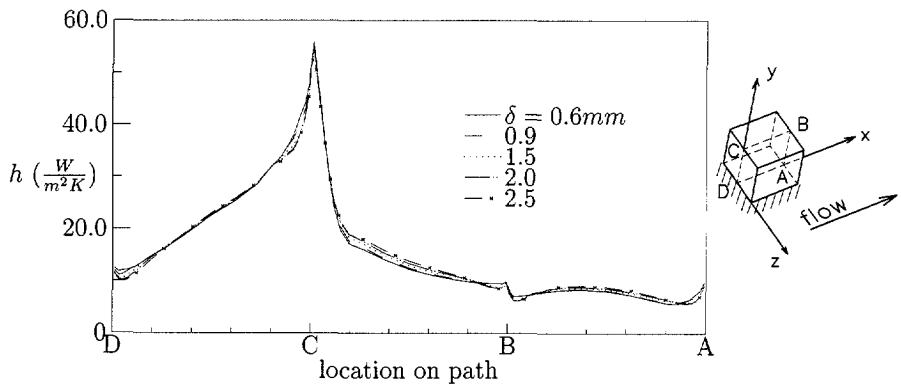


FIGURE 3.7: Distribution of the heat transfer coefficient along path *ABCD* parametric in the epoxy layer thickness.

A sufficiently fine grid was designed to guarantee negligible truncation error. An initial run was performed for the cube with an epoxy layer thickness of 1.5 mm and with the inner core kept at 75 °C. The coupled problem was solved for the cube, i.e. heat conduction in the epoxy substrate, radiation exchange between the heated cube and the environment and convective heat transfer to the surrounding air. For simplicity, the effect of buoyancy was neglected. The total heat through all six faces of the cube, in fact, the heat generation in the isothermal core, was equal to 1.41 W. This total heat production is considered as the reference situation and this was implemented as an evenly distributed heat source per unit volume in the copper core for all other cases. Because of the non-equidistant grid as used in the numerical model, the heat source per unit volume varies as the inverse of the cell volume.

Figures 3.6 and 3.7 show profiles of the heat transfer coefficient along paths

$ABCD$ and $ABCD$, respectively, for the five studied epoxy layer thicknesses δ . Path $ABCD$ denotes a cross-section parallel to the base plate at $y/H = 0.5$, path $ABCD$ denotes a cross-section perpendicular to the base plate at $z/H = 0$. The ambient temperature was chosen as reference temperature. From the figures it is observed that the profiles coincide within discrepancies of less than 5%. It can, therefore, be concluded that the heat transfer coefficient is independent of the epoxy layer thickness. Although the temperature distributions varied significantly with the thickness, the heat transfer coefficient apparently did not.

Although a rather small Reynolds number was chosen to study the behaviour of the epoxy layer thickness on the heat transfer coefficient, it is assumed that these simulations represent reality. In the case of larger Reynolds numbers, representative for the turbulent regimes considered in this study, it is believed that differences become even smaller than for laminar flows since the convective heat transfer is then less dependent on the thermal boundary conditions.

3.2.3 Radiative heat transfer

For most situations considered in this thesis, the radiative heat flux comprises just a small part of the total heat transfer. However, in some situations with low characteristic velocities, i.e. natural and mixed convection and areas with strong recirculations, the radiative heat fluxes were of the same order of magnitude as the convective heat fluxes. The net radiative heat flux from an irradiating surface consists of the partly reflected irradiation and the emitted radiation [108]:

$$\phi''_{rad} = E + (\rho - 1)H, \quad (3.12)$$

where ϕ''_{rad} (W/m^2) is the net radiative heat flux, E (W/m^2) is the emission, H (W/m^2) is the irradiation and ρ is the object reflectivity. The radiative exchange between surfaces can be expressed in an integral equation [108]:

$$\frac{\phi''_{rad}(r)}{\epsilon(r)} - \int_A \left(\frac{1}{\epsilon(r')} - 1 \right) \phi''_{rad}(r') dF_{dA-dA'} + H_o(r) = E_b(r) - \int_A E_b(r') dF_{dA-dA'}, \quad (3.13)$$

where r (m) is the position vector, ϵ is the emissivity, $dF_{dA-dA'}$ is the view factor between the infinitesimal surfaces dA and dA' (m^2), $E_b(r)$ (W/m^2) is the blackbody emissive power and $H_o(r)$ (W/m^2) is the external contribution to the irradiation.

For a finite number N of surfaces, the net radiative heat flux can be approximated by [108]:

$$\sum_{i=1}^N \left(\frac{\delta_{ki}}{\epsilon_i} - F_{k-i} \frac{1 - \epsilon_i}{\epsilon_i} \right) \phi''_{rad,i} = \sum_{i=1}^N (\delta_{ki} - F_{k-i}) \sigma T_i^4, \quad (3.14)$$

where $\phi''_{rad,i}$ is the unknown radiative heat flux at the subsurface A_i (m^2), δ_{ki} is the Kronecker delta ($\delta_{ki}=1$ for $k=i$ and $\delta_{ki}=0$ for $k \neq i$), ϵ_i is the emissivity for surface A_i , T ($^{\circ}C$) is the temperature and F_{k-i} is the view factor between subsurface A_i and A_k :

$$F_{k-i} = \frac{1}{A_k} \int_{A_k} \int_{A_i} \frac{\cos \theta_k \cos \theta_i}{S^2} dA_k dA_i, \quad (3.15)$$

where S (m) is the distance between subsurfaces A_k and A_i and θ is the angle between the normal and the vector S .

It is not desirable to take an infinite number of radiative surfaces with corresponding view factors into account. Besides the fact that this requires temperature and emissivity information, it is much too complex for an experimental evaluation. A drastic simplification will already give satisfactory and accurate predictions of the radiative heat exchange.

According to the analysis of Ortega and Moffat [130], above algebraic analysis can be adapted to derive the four faces approach. This approach maps the total surface under consideration onto four representative surfaces. For each of the surfaces, the mutual view factors were then calculated. For most of the heat transfer experiments performed, the cube under investigation was only powered in order to achieve adiabatic conditions. In that case, the net radiative heat exchange from the cube surface reduces to the simple Stefan-Boltzmann equation:

$$\phi''_{rad}(\eta, \xi) = \sigma \epsilon \left(T_{sur}(\eta, \xi)^4 - T_{amb}^4 \right), \quad (3.16)$$

where $\phi''_{rad}(\eta, \xi)$ is the local net radiative heat flux, σ is the Stefan-Boltzmann constant ($\sigma = 5.67 \cdot 10^{-8}$), ϵ is the surface emissivity, assumed to be constant for both the cube and the environment, and T_{amb} is the ambient temperature. Most of the experiments were performed in the forced convection regime, where the convective heat fluxes were much larger than radiative heat fluxes (approximately one order of magnitude). Some low-Reynolds number experiments were performed for the in-line array of 9 cubes for which all cubes were powered at the same time (the adiabatic reference temperature was then measured in a separate experiment). For this case, the four faces model was used to get a good approximation of the local radiative heat flux.

3.2.4 Experimental uncertainty in the heat transfer coefficient

The experimental results were evaluated with the single sample uncertainty analysis [109, 110]. In order to illustrate the influence of the individual contributions, some terms were simplified. Of course, this simplification was only adopted in this accuracy analysis. The conductive heat flux $\phi''_{cond}|_s$ is approximated with the one-dimensional Fourier's law [29]:

$$\phi''_{cond}|_s = \lambda \frac{\partial T}{\partial n}|_s = \lambda \frac{(T_{co} - T_{sur})}{d}, \quad (3.17)$$

where T_{co} and T_{sur} are the copper and surface temperature respectively, λ is the thermal conductivity of the epoxy substrate and d is the substrate thickness. The thermal conductivity λ was determined experimentally with two different methods, yielding an isotropic value of 0.24 W/mK with an accuracy of 2%. Details of these experiments are given in Appendix A. In order to illustrate the influence of the radiative heat flux in the overall uncertainty, radiation was simplified by the Stefan Boltzmann's law:

$$\phi''_{rad} = \epsilon \sigma (T_{sur}^4 - T_{amb}^4), \quad (3.18)$$

where ϵ is the surface emissivity, σ the Stefan Boltzmann constant ($=5.670 \cdot 10^{-8} \text{ W/m}^2 \text{ K}^4$) and T_{sur} and T_{amb} are respectively the surface and ambient temperatures. The emissivity of the black paint, measured within an experimental uncertainty of 2%, was 0.95. An expression for the single sample uncertainty in h_{ad} , based on a one-dimensional approach, reads:

$$\begin{aligned}
 (\delta h_{ad})^2 = & \left[\frac{\frac{\partial T}{\partial n}|_s}{(T_{sur} - T_{ad})} \right]^2 (\delta \lambda)^2 + \left[\frac{\lambda}{d(T_{sur} - T_{ad})} \right]^2 (\delta T_{co})^2 + \\
 & \left[\frac{4\epsilon\sigma T_{amb}^3}{T_{sur} - T_{ad}} \right]^2 (\delta T_{amb})^2 + \left[\frac{h_{ad}}{T_{sur} - T_{ad}} \right]^2 (\delta T_{ad})^2 + \\
 & \left[\frac{\lambda(T_{co} - T_{sur})}{d^2(T_{sur} - T_{ad})} \right]^2 (\delta d)^2 + \\
 & \left[\frac{\sigma(T_{sur}^4 - T_{amb}^4)}{T_{sur} - T_{ad}} \right]^2 (\delta \epsilon)^2 + \left[\frac{4\epsilon\sigma T_{sur}^3 + \lambda/d + h_{ad}}{T_{sur} - T_{ad}} \right]^2 (\delta T_{sur})^2.
 \end{aligned} \tag{3.19}$$

Minor contributions to the experimental uncertainty in h_{ad} are the errors in the ambient and reference temperatures. These were measured with copper-constantan thermocouples (type T) which had, after appropriate calibration, an experimental uncertainty of 0.1°C . Due to radiative exchanges etc., the accuracy of these temperature measurements was estimated to be within 0.5°C . The copper temperature, also measured with an embedded and calibrated thermocouple, yielded an accuracy of 0.1°C .

The uncertainty in the surface temperature distribution, measured with infrared thermography, had a large impact on the overall accuracy of the heat transfer coefficient. Application of both the in situ calibration and the image restoration technique resulted in an absolute accuracy of 0.4°C in the surface temperature for the mid-plane of the faces, and of about 0.6°C close to the edges because of the effects of image mapping. In the next chapter, a thorough discussion will be given of both the image restoration and the in situ calibration. Other important contributions to the experimental uncertainty in h_{ad} are the errors in the thermal conductivity of the epoxy substrate, the layer thickness δ and possible truncation errors in the numerical solution of the internal temperature distribution. In the next sections, a detailed discussion is given of these contributions to the total accumulated uncertainty in the heat transfer coefficient.

Thickness of the epoxy layer

The accuracy of the heat transfer coefficient is very sensitive to the epoxy layer thickness. Therefore, this layer was machined very accurately to a thickness of 1.5 mm with an uncertainty of 0.01 mm. The thickness of the layer was also effected by thermal expansion of the epoxy substrate during the heat transfer experiments. A one-dimensional (1D) analysis with additional measurements was performed to examine the thermal expansion of the epoxy substrate.

The thermal expansion of a one-dimensional layer of a material of thickness H_0

(m) in the reference situation (ambient temperature) is expressed as:

$$H = H_o(1 + \alpha\Delta T), \quad (3.20)$$

where H is the thickness of the layer at the temperature T , α ($^{\circ}\text{C}^{-1}$) is the thermal expansion coefficient and ΔT ($^{\circ}\text{C}$) is the temperature rise ($T - T_{amb}$). Such an expression can be assumed for both the copper core and the epoxy layer in the 1D approach. The total thickness of the composed material then becomes $L = 2d + D$, where d (m) is the layer thickness and D (m) denotes the rib of the copper. Expressed in thermal expansion, this becomes:

$$L = L_o + (2d_o\alpha_{epoxy}\Delta T_1 + D_o\alpha_{copper}\Delta T_2), \quad (3.21)$$

where α_{epoxy} and α_{copper} are the thermal expansion coefficients of epoxy and copper. To simplify the analysis, the entire material is assumed to be at uniform temperature which implies that the temperature rises ΔT_1 and ΔT_2 are equal.

Thermal expansion experiments were performed by monitoring the instantaneous thickness L of the entire cube as a function of the instantaneous temperature during a heating up cycle from 20°C to 80°C . The thickness increase was measured with an accurate dilatometer which measures the expansion compared to a reference point. The thickness increase $(L - L_o)/L_o$ is shown in Fig. 3.8. The contribution of the copper core to the total increase in length is estimated from the known thermal expansion coefficient of copper, namely $\alpha_{copper} = 1.7 \cdot 10^{-5} \text{ }^{\circ}\text{C}^{-1}$. The remaining expansion was attributed to the two epoxy layers. For one layer, the relative thermal expansion becomes $(d - d_o)/d_o = 5.8 \times 10^{-3}$, at a temperature of 70°C . This is an increase of less than 0.6%. The thermal expansion coefficient of epoxy was determined from the slope of the curves in Fig. 3.8. A value of 2.71×10^{-5} was found with a least square fit, resulting in a thermal expansion coefficient of $\alpha_{epoxy} = 6.8 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$ for epoxy. This is in good agreement with the range of expansion coefficients reported in handbooks $\alpha_{epoxy} = 5.4\text{--}9.0 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$.

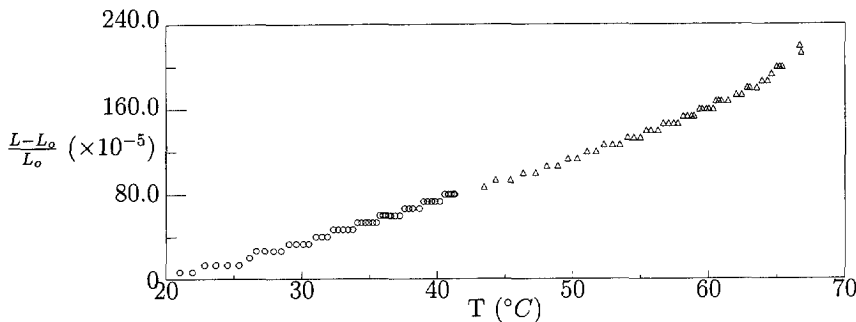


FIGURE 3.8: Thermal expansion of the cube: the relative thickness increase versus the cube temperature. Different symbols refer to different experiments.

The surfaces of the cubes were covered with a black paint to enhance the radiositivity as required for accurate infrared thermography measurements. The thickness of

this layer was uniform and smaller than 0.06 mm at the entire surface. Further, the thermal conductivity of the black paint is much larger than that of the epoxy substrate (about one order of magnitude) which implies that the temperature decay across the paint-layer is negligible. Therefore, the used black paint does not introduce additional error to the epoxy layer thickness and, thus, to the calculated conductive heat fluxes.

Truncation errors

Since the finite volume method (central differencing) was used to solve the Laplace equation numerically, the truncation error in the temperature solution is of second order. Grid refinement was applied to both an artificial test case and real experiments to investigate the truncation errors involved. The test case was examined to study fine grids. The spatial resolution in the measured surface temperature distributions (obtained from the infrared thermography) allowed a selection of maximum 30×30 grid cells on one face of the cube. This resolution was artificially increased by interpolation between pixel points. This, of course, does not add information but allows us to examine the magnitude of the truncation error.

Test case. For the test case, constant outer wall and copper temperatures of 60 and 80 °C were selected as boundary condition. The Laplace equation was solved for different grid sizes. From the temperature solution, the local conductive heat fluxes through the outer surface were calculated. An integration of the local heat flux resulted in the conductive heat flow. Since three-dimensional effects (the lateral fluxes) were expected to be most significantly present in the edge region, a distinction was made between the mid-plane and the edges/corners. The mid-plane refers to the area above the copper core, with dimension 12×12 mm. The edges are the remaining part of the face. Due to the constant temperature boundary condition and the symmetry of the numerical domain, the heat transfer was similar for all four side faces. The conductive heat through the top face will be different from that through the side faces because the lower epoxy layer was imaginary removed from the computational domain. Six equidistant grid sizes were investigated, corresponding to 2, 3, 4, 5, 6 and 7 grid cells in the layer thickness. The results for the test case are given in Figs. 3.9 and 3.10. The different symbols refer to the top and side faces. With an increasing number of cells, the total conduction converges gradually to an asymptoting limit, found for the infinite number of volume cells or the analytical solution. Differences in the calculated conductive heat flow between the course grids and the asymptotic limit are largest for the edges. This is expected since three-dimensional effects (the lateral heat fluxes) are most dominantly present in this part of the epoxy layer. The differences between the calculated conductive heat flow through the edges for the course grid (2 cells along the thickness) and the fine grid (7 cells along the thickness) were about 5% for both the top and side faces. The differences between the grid with 3 cells along the thickness of the epoxy layer, the resolution which is used for most of the results given in this thesis, are of the order of magnitude of 2-3%. Because of the relatively thin epoxy layer compared to

the overall dimensions of the cube, the heat transfer is more or less one-dimensional in the mid-area of the faces. This results in small truncation errors, even for the courser grids (see Figs. 3.9 and 3.10). The differences between the course (2 cells) and fine grid (7 cells) were of the order of magnitude of 0.5% for both faces. This value drops to 0.3% for the grid with 3 cells along the thickness of the epoxy layer.

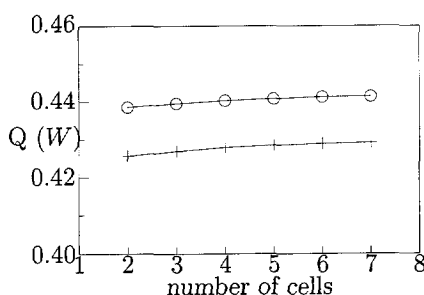


FIGURE 3.9: Conductive heat flow through the mid-plane as a function of the grid size: (+) top face and (o) side face ($T_{co} = 80^\circ\text{C}$ and $T_{sur} = 60^\circ\text{C}$).

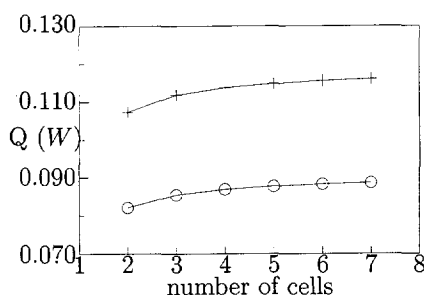


FIGURE 3.10: Conductive heat flow through the edges as a function of the grid size: (+) top face and (o) side face ($T_{co} = 80^\circ\text{C}$ and $T_{sur} = 60^\circ\text{C}$).

Real experiment. The resolution of the surface temperature fields was artificially doubled by linear interpolations between the grid nodes. A resolution of 60×60 surface cells per face was achieved in this way (six cells along the thickness). For a representative case, the single cube in a turbulent channel flow, the results for the two different grid sizes are compared in Fig. 3.11. As expected, differences in the local heat transfer coefficient for the two different sizes are very small in the mid-plane region, while these increase somewhat close to the edges. Values of the truncation error for these regions are in the order of magnitude of 2% and 5%, respectively. It was already argued that the accuracy of the local heat transfer coefficients close to the edges suffered from larger experimental uncertainty than the heat transfer coefficients at the mid-plane do. The larger truncation errors in this area add up to the experimental uncertainty and, therefore, support this conclusion. The total conduction through the five faces exposed to air was calculated for both the regular size of 30×30 grid cells and the finer grid with 60×60 grid cells per face. The results showed that the total conduction was approximately 1-2% smaller for the finer grid. Differences in the averaged heat transfer coefficient were also 1-2%. The artificial increase of the spatial resolution by linear interpolation modifies the surface temperature somewhat and, therefore, introduces an interpolation error. It is not clear if both contributions add up or nullify each other but since the errors are only 1-2%, it is assumed that the 30×30 grid is sufficiently accurate.

From these observations it can be concluded that for the grid size with 30×30 surface cells (3 cells in the layer thickness) the truncation error is within 3%.

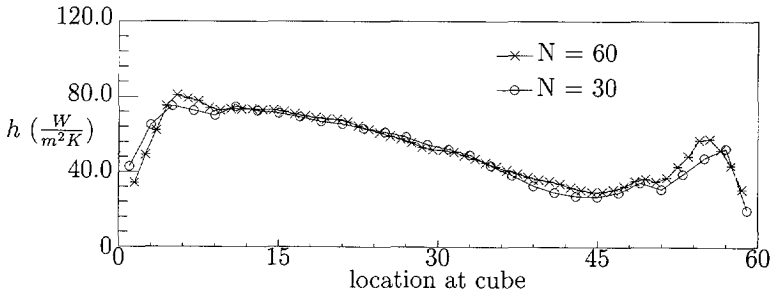


FIGURE 3.11: Distribution of the measured heat transfer coefficient for two grid sizes: (+) 30×30 cells and (o) 60×60 cells.

Therefore, the error in the calculated substrate temperature is of the same order of magnitude as the uncertainty in the copper and the surface temperature.

Conservation of heat fluxes per volume cell. The equilibrium of conductive heat fluxes per volume cell (conservation of energy) was considered to verify the temperature solution of the Laplace equation. The residuals of the heat fluxes were within the rounding error (about 1×10^{-4}) which verified that the code was implemented correctly.

Overall accuracy of the heat transfer coefficient

From the above analysis of the different contributions to the experimental uncertainty it is concluded that the accumulation of errors results in approximately 5% accuracy in the local heat transfer coefficient for the mid-region of the five faces of the cube. The heat transfer coefficients at the edges have a somewhat higher uncertainty of about 10 % due to, among others, mapping inaccuracies.

3.3 Time-dependent heat flux method

It will be shown in the following chapters in which results are discussed that the flow around the configurations of cubes is locally unstable which leads to periodic vortex shedding in the wake of the cube. These large coherent structures have a significant impact on the convective heat transfer. It was assumed that the vortex shedding was directly coupled to the time-dependent convective heat transfer. This has been demonstrated for the flow past obstacles of other geometry, like cylinders in cross-flow [146, 147, 148, 155, 99, 142]. These studies clearly showed that the velocity field and convective heat transfer from the heated cylinder were strongly coupled (the coherence between both time signals showed a strong peak at the vortex shedding frequency in question). To obtain more insight in the physical interaction between the flow and the surface heat transfer in flows over wall-mounted cubes, the time-dependent convective heat transfer was measured with a fast-responding heat flux sensor. Diller [39] has reviewed most of the useful passive and active techniques

and calibration methods. The review summarises the difficulties and problems a researcher has to face to obtain accurate results. The response time of the sensor is one of the most significant requirements for transient heat transfer research. The sensor needs to be flush mounted since layer differences will cause distortion of the flow and, therefore, distortion of the measurement. Further, different thermal properties of the sensor will cause disruption of the thermal boundary layer. This effect is not significantly present when an active sensor is used. In that case, the only thermal boundary which will develop is created by the sensor itself. One of the most recent developments is the heat flux microsensor which basically consists of a deposited insulation layer with a thickness of less than 1 micron with on both sides deposited thermocouple junctions. These junctions are connected serially to form a thermopile aimed at increasing the signal-to-noise ratio. The output signal is then a summation of all small thermocouple voltages. Details of this sensor and the technique involved are discussed in Hager et al. [59, 60], Diller [39], Mancuso and Diller [99] and Simmons et al. [155].

In this work, an active time-dependent heat flux sensor was developed. This sensor consisted of a heated thin metal film deposited on an insulating substrate of glass. The frequency response of the sensor was determined from a calibration with chopped laser beams which illuminated the sensor. Further, the sensor response was calibrated to on fore hand known periodic oscillations in the flow. This was achieved by placing the sensor in the wake of a cylinder. The shedding frequency was then simultaneously measured with a hot-wire anemometer. Our main objective of this heat flux sensor was the measurement of possible coherence between the vortex shedding and the time-dependent convective heat flux. This implied that we were mainly interested in detecting possible characteristic frequencies carried by the time-dependent heat flux signal rather than determining the absolute value of the time-dependent convective heat flux.

3.3.1 Fast-responding heat flux sensor

The heat flux sensor consisted of a rectangular glass substrate of size 10×4 mm. Two vacon pins (a metal alloy) were melted in the glass at a distance of 4 mm apart in a parallel orientation. Lead wires were soldered to these pins for electrical power supply. Vacon was selected since it was well suited to be embedded in glass. Further, vacon has a relatively large electrical conductivity which prevented power dissipation in the lead wires itself. The top layer of the glass substrate with the two vacon pins embedded in parallel was polished to obtain a smooth surface with a roughness of less than 1 nanometer. A layer of platinum with thickness of 0.2 micron was vapour deposited on the entire polished surface with a vacuum deposition technique. Platinum adhered well to the used glass. The two vacon pins were then electrically connected through the platinum film on top of the surface. The active area of the sensor was 4×4 mm. A sketch of this sensor is given in Fig. 3.12. A more or less similar approach was followed by Campbell et al. [26] and Simmons et al. [155].

A voltage drop was maintained across the metal foil. Due to the large electrical

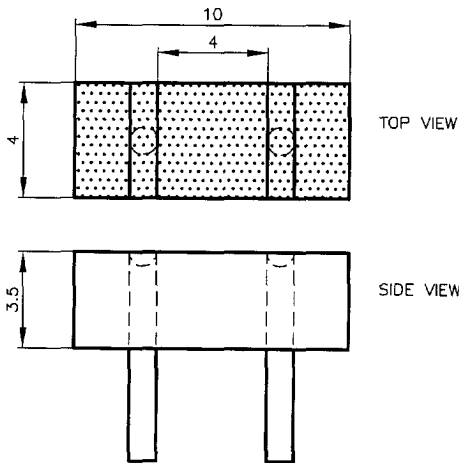


FIGURE 3.12: Sketches of the side and the top view of the heat flux sensor. The active sensor area was 4×4 mm.

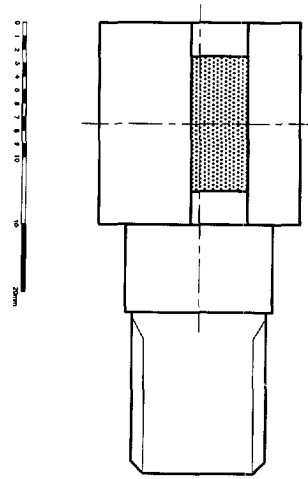


FIGURE 3.13: Sketch of the cube with the flush-mounted heat flux sensor (area with dots-crosshatch). The mounting bolt is also shown.

conductivity of vacon, the two parallel pins acted as a kind of electrical current spreader. This resulted in a parallel electrical field through the platinum film which caused a spatially uniform dissipation of the electrical energy and, consequently, a uniformly generated heat flux in the metal foil. This uniformity in the heat flux distribution is only guaranteed if the electrical field is parallel, the foil is of uniform thickness and the specific resistance of the film material is not a function of the spatial coordinate.

The thin film had a relatively large (positive) temperature-dependent Ohmic resistance $R(T)$ (see Fig. 3.14). Several sensors were made for which the resistances were about $3.9 \, \Omega$ at room temperature. The metal sheet was thin and small enough to assume the temperature to be constant across the entire metal foil (no spatial temperature gradients). The operation of the sensor was based on this constant-temperature assumption. A controller was used to adjust the power supply to maintain the metal sheet at a constant temperature (a conventional hot-wire anemometer controller was used [90]). The convective heat transfer from the heated foil is then directly proportional to the power supply required to maintain the temperature of the foil at the constant set value. The output voltage of the controller was sampled and stored with a signal memory recorder. The sample frequency was between 300 and 500 Hz which was sufficiently large compared to the vortex shedding frequencies in question. The number of data points for one measurement was about 32,000.

The hot-film sensor was built in a cube made from formaldehyde. The thermal conductivity of formaldehyde was experimentally determined to be $0.33 \, \text{W/mK}$ (see appendix A). Care was taken to mount the sensor flush with the surface of the cube to prevent distortion of the flow. The hot-film sensor was embedded in the cube at

actually two different places at the side face, as indicated in Fig. 3.13, to measure the effect of vortex shedding. In the sketch, the sensor is embedded in the back half of the side face. Note that if the cube is 180° rotated, the sensor is located in the front half of the cube.

3.3.2 Time-dependent heat transfer coefficient

The measured resistance across the platinum foil is surface-averaged and corresponds to one film (sensor) temperature, T_{sen} . The heat transfer coefficient $h(t)$ is, therefore, an average over the entire film surface, namely:

$$h(t) = \frac{Q_{conv}(t)}{A(T_{sen}(t) - T_{ref})}, \quad (3.22)$$

where $Q_{conv}(t)$ is the time-dependent convective heat flow, A refers to the sensor area exposed to air ($A = 4 \times 4 \text{ mm}^2$) and T_{ref} is the reference temperature. Since only a small part of the cube is heated (only the sensor area) the local oncoming flow can be assumed at isothermal conditions ($T_{ref} = T_{in}$, the inlet temperature of the flow). The sensor temperature $T_{sen}(t)$ is controlled to a constant value by means of the Wheatstone bridge and is, compared to the variation of the convective heat flux, approximately constant. The only time-dependent quantity in the expression for the instantaneous heat transfer coefficient is then the convective heat flow. The total electrical energy dissipated in the platinum foil is the product of the time-dependent voltage drop $V(t)$ and the electrical current $I(t)$:

$$P(t) = V(t) \times I(t). \quad (3.23)$$

A global energy balance over the platinum foil reads:

$$P(t) = \rho c_p V \frac{\partial T_{sen}(t)}{\partial t} + Q_{conv}(t) + Q_{loss}(t), \quad (3.24)$$

where $\rho c_p V \frac{\partial T}{\partial t}$ is the transient contribution (ρ is the density of platinum, c_p is its specific heat and V is the volume of the platinum foil), $Q_{loss}(t)$ is the time-dependent heat loss which consists of radiation (from the upper face) and conduction (through the glass substrate and vacon pins). The platinum foil had a low emissivity of $\epsilon \approx 0.1$ and radiative heat losses are, therefore, small. The thermal conductivity of glass is in the order of magnitude of 1 W/mK . Due to the slow time response of the underlying system of the glass substrate, i.e. the cube of formaldehyde and the vacon connecting pins, the temperature variations are damped. It is difficult to accurately estimate the time-dependent heat losses. A way of minimising the heat losses is by maintaining the entire cube at the sensor temperature by an additional heat source in the cube itself. The zero temperature difference between the sensor and the cube then minimises the time-dependent heat losses.

In first instance, this sensor was developed to measure absolute values of the time-dependent convective heat flux and heat transfer coefficient. But due to its disappointing accuracy, we decided to concentrate only on the frequency information carried by the time-dependent heat flux signals. Therefore, only a dynamic calibration of the sensor was performed.

3.3.3 Calibration procedures

The electrical resistance dependency $R(T_{sen})$ of the platinum film on the sensor temperature T_{sen} was determined with a temperature-controlled oven. To account for all contributions to the total resistance imposed by the intermediate equipment, the sensor was calibrated while it was connected to the Wheatstone bridge. The temperature dependency of the film resistance, including the connected metal wires, is shown in Fig. 3.14. The linear relation is obvious from the figure. The sensitivity is sufficiently large to estimate the actual sensor temperature while the sensor is operated.

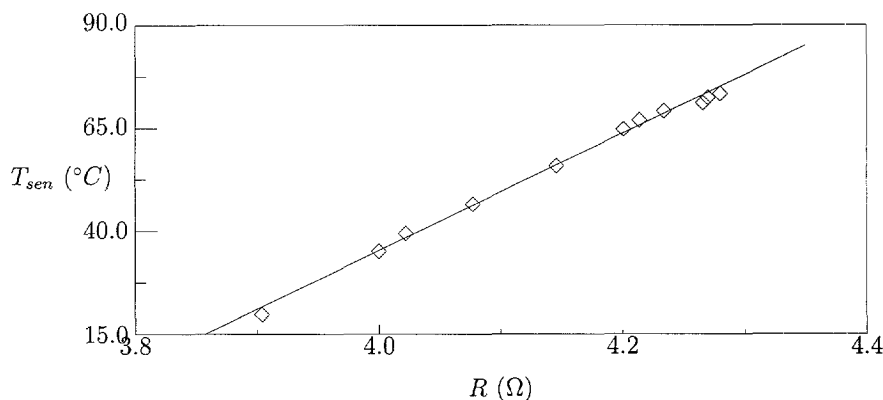


FIGURE 3.14: Temperature dependency of the electrical resistance of the platinum film, including the connecting wires, of the heat flux sensor.

The frequency response of the sensor was calibrated with a chopped laser beam from an Argon-Helium laser. This is a well known technique and has been used by several researchers to obtain the dynamic characteristics of heat flux sensors (see, for example, Holmberg and Diller [65], Hager et al. [59, 60] and Diller [39]). The laser beam illuminated the platinum film with the frequency imposed by the chopper. The resulting saw-tooth like response signal allowed us to estimate the response time of the sensor. From experiments with different chopper frequencies it was concluded that the deposited platinum layer was sufficiently thin to follow frequencies of at least 200 Hz. A second dynamic calibration of the heat flux sensor was performed in the wake flow of a suspended circular cylinder. The sensor was flush mounted in the top face of the cube while the cylinder was oriented horizontally upstream of the cube (in parallel with the base plate). The vortex shedding from the cylinder caused large-scale flow periodicities in its wake. When the sensor is placed in this oscillating flow, the heated area will be periodically cooled by convection. The time-dependent convective heat flux contains then the vortex shedding frequency.

A typical power density spectrum of the time-dependent heat flux is shown in Fig. 3.15 for a bulk velocity of 2.1 m/s. The spectrum clearly shows a peak at about $f = 85$ Hz which is attributed to the vortex shedding. This frequency was estimated to be accurate within 5%. This value included experimental uncertainties as well as

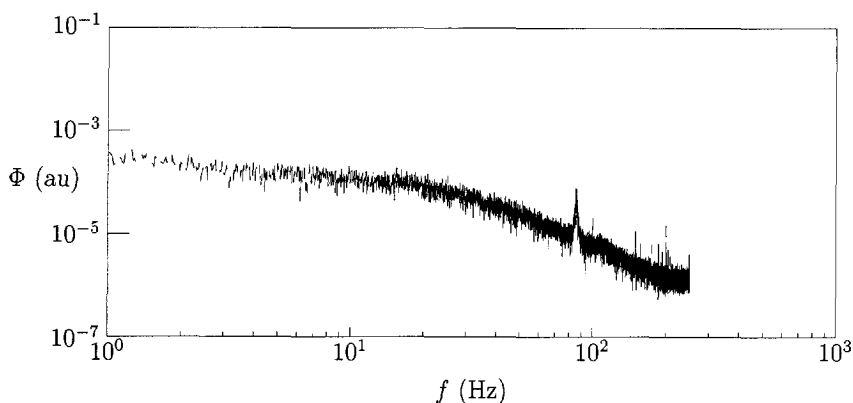


FIGURE 3.15: Power density spectrum of the time-dependent heat flux from the sensor placed in the wake of a suspended circular cylinder.

errors introduced by the location determination of the intensity peak in the power density spectra. The frequency of vortex shedding was determined independently from power density spectra of the velocity time series, measured with a hot-wire anemometer. The diameter of the cylinder was $D = 5.65$ mm. The frequencies were made dimensionless in terms of the Strouhal number ($St = fD/u_B$, where f is the frequency, $D = 5.65$ mm is the known diameter of the cylinder and u_B is the bulk velocity) and are plotted versus the bulk velocity in Fig. 3.16. The solid line

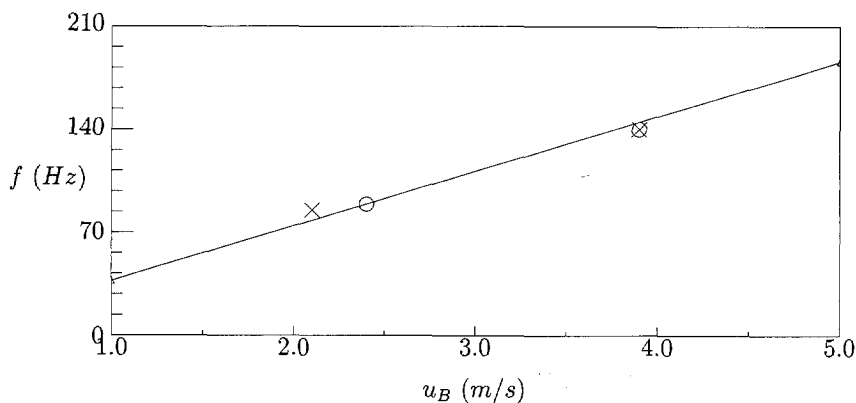


FIGURE 3.16: Dominant frequencies obtained from hot-wire (o) and heat flux spectra (x) for measurements in the wake of a suspended circular cylinder with diameter $D = 5.65$ mm. The solid line refers to the generally accepted value of $St = 0.21$.

in Fig. 3.16 represents the well known relation between the Strouhal number and the vortex shedding frequency for this suspended circular cylinder case ($f = 0.21 u_B/H$). When the inaccuracy of 5% is taken into account, the dominant frequencies obtained from the heat flux and velocity spectra are in good agreement with the calculated values.

The calibration measurements were performed for a range of shedding frequencies, i.e. $10 < f < 200$. The results confirmed a good dynamic response of the used heat flux sensor for frequencies up to at least 200 Hz. This value is an order of magnitude higher than the vortex shedding frequencies present in the wake flow of wall-mounted cubes in the turbulent channel flow for Reynolds numbers between $1000 < Re_H < 5000$ as considered in this study. It is, therefore, concluded that the dynamic response of the sensor is sufficient to accurately detect dominant frequencies in the time-dependent convective heat flux. It is explicitly mentioned that the sensor was not statically calibrated to determine accurate convective heat fluxes with it. Much more effort is then needed to analyse and to estimate the different time-dependent heat losses involved.

CHAPTER 4

Diagnostic measurement techniques

The measurement techniques discussed in this chapter are used to provide the necessary information for the evaluation of the local convective heat transfer from the cubes and of the flow structure around the cubes.

Distributions of the time-averaged surface temperature were determined with infrared thermography (IR). This is a commonly used diagnostic measurement technique applied in a broad range of thermo-fluid problems. But the application of infrared thermography in this study required significant image processing and a sophisticated calibration in order to achieve a high accuracy in the temperature results. The discussion in this chapter will therefore mainly be concentrated on the development of these improvements rather than describing infrared thermography in general. For such a description, reference is made to standard literature. To obtain an objective validation of the time-averaged infrared temperature measurements, the liquid crystal thermography technique was explored. For similar experimental situations, the infrared surface temperature measurements were validated with liquid crystal measurements. The calibration and the application of the liquid crystal method used will be discussed in this chapter.

Flow visualisation was performed with smoke and oil-film visualisation techniques. Since these techniques are commonly used, only a brief description of the used application method will be given. Detailed documentation of the time-averaged vortex structures and flow dynamics around the cubes was performed with the Laser Doppler Anemometry technique (LDA). LDA is a well known and thoroughly discussed non-intrusive measurement technique to measure fluid velocities in a seeded flow. But the LDA method involves several sources of uncertainty which depend, for example, on the system, the geometry, the optics and the situation. Detailed analysis and diagnostics of the results are needed and all errors involved need to be addressed before the experimental results are acceptable and satisfactory.

Further, topological principles were applied to the velocity data obtained with the LDA and visualisation techniques to derive additional flow details.

Finally, local and face-averaged static pressure coefficients were measured with a pressure-cube.

4.1 Liquid Crystal Thermography

The cholesteric liquid crystals (LC) measurements performed in this study served for the validation of the infrared surface temperature measurements. A detailed discussion of the different properties and possible applications of liquid crystals is given in the accessible literature (see, for example, Moffat [114] and Baughn [18]). The used encapsulated cholesteric liquid crystals are substances with both the properties of a liquid and a crystalline state. The molecular ordering of these

liquid crystals changes under the influence of temperature. This reversible change in molecular structure causes wavelength-selective reflection of the incident light. The unambiguous relation between temperature and reflected light enables the observer to identify surface temperatures. The liquid crystals are encapsulated to improve the stability since they are protected from atmospheric contamination. A liquid crystal is characterised by its event temperature and the corresponding band-width being the temperature range in which the crystal changes from colourless to red, green, blue, and back to colourless.

4.1.1 Calibration of the Liquid Crystals

A mixture of five crystals was used which had narrow band-widths of approximately 2°C and event temperatures at respectively 47.5, 55.0, 60.0, 65.0 and 73.0°C [132]. Moffat et al. [45, 46] developed a sophisticated calibration system based on the brightness, hue and intensity of the emitted light which unambiguously related each colour to a surface temperature for a mixture of liquid crystals. Such an approach was also adopted by Stasiek [157], Sun et al. [159], Hollingsworth et al. [64] and Akino et al. [4]. Since we were only interested in an objective validation of the infrared measurements, we adapted a different approach. Only the sharp transition lines between the green and the red appearance, which are isotherms on the surface, were considered. The reason was twofold. First, the transitions were very sharp and easy to recognise. Second, the transitions provide an objective colour interpretation and the real colour, the human sensation, was not relevant.

The LC mixture was calibrated with an a priori known temperature distribution of a heated copper plate, thermally insulated with foam from below and from the sides. This copper plate (calibration rig) is schematically shown in Fig. 4.1. The upper plot in the figure shows the front view. The marked holes represent the locations of the thermocouples. The lower plot in Fig. 4.1 refers to the cross-section at the centre-line of the calibration rig. A temperature gradient across the 10 mm thick copper plate was maintained by means of two heat exchangers, located at each side of the plate. The resulting one-dimensional almost linear temperature decay was monitored with 22 thermocouples distributed along the centre-line, some of them were located off-axis. The thermocouples were mounted just underneath the upper surface. The high thermal conductivity of the copper plate and its relatively small thickness ensured that no temperature gradients existed along the thickness. The off-axis thermocouples were used to ensure the one-dimensionality of the temperature gradient. The thermocouples along the centre-line were used to measure the temperature gradient. A typical temperature profile is given in Fig. 4.2. The calibrated thermocouples were accurate within 0.1°C .

The copper plate was painted with the liquid crystal mixture. The colour patterns were photographed and from these pictures, the temperatures corresponding to the green-red transitions were determined from a linear interpolation between the subsequent temperature readings. The calibration temperatures found with this method were confirmed with a second calibration performed with a pure copper cube. This cube was controlled to a constant temperature within the tolerance of

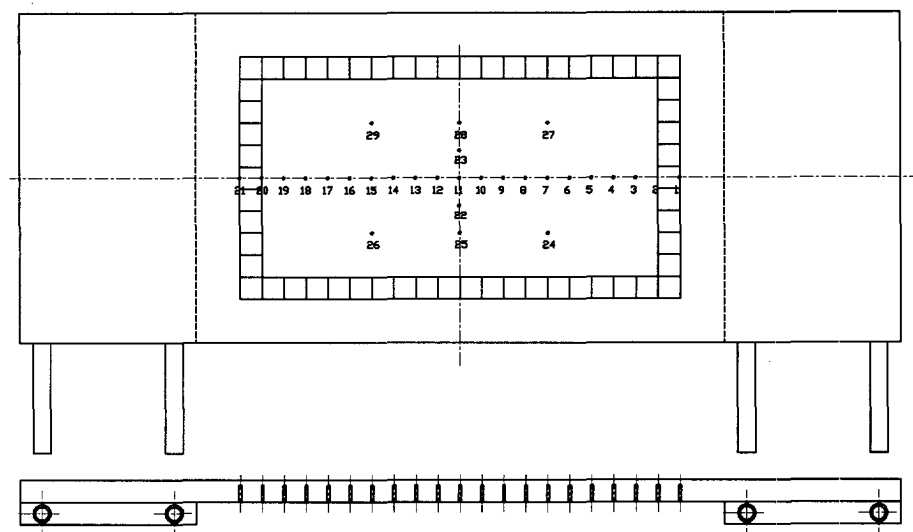


FIGURE 4.1: The copper plate with heat exchangers at both sides used to calibrate the mixture of five thermochromic liquid crystals. The upper plot is a front view of the calibration rig. The marked holes represent the locations of the thermocouples junctions. The lower plot is the view through the cross-section at the centre-line of the set-up.

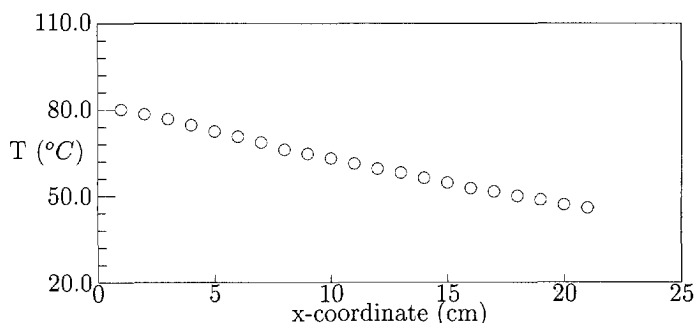


FIGURE 4.2: Temperature profile over the calibration rig. $x = 0$ corresponds to thermocouple junction 1; $x = 21$ corresponds to junction 21.

0.1°C . Care was taken to guarantee similar experimental conditions during calibration and measurement, which included source and angle of illumination, the age of the mixture, and so on. The temperatures at the sharp green-red transitions found with these two methods are 47.5 , 55.8 , 60.5 , 65.2 and 72.8°C for the five subsequent liquid crystals.

The temperature and spatial accuracy of the liquid crystal measurements, evaluated from uncertainty analysis, were estimated to be within $0.3\text{--}0.4^{\circ}\text{C}$ and 0.4 mm , respectively. The spatial resolution is poor but this is of less importance since the

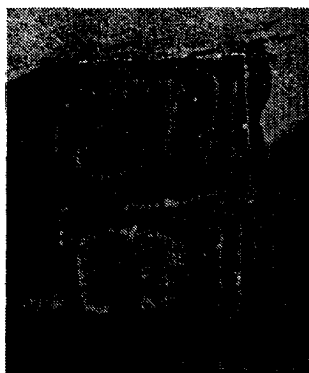


FIGURE 4.3: Liquid Crystal image of two faces (side and top) of the fifth cube in the in-line row of 9 cubes (discussed in chapter 6).

purpose of the liquid crystal measurements was only to validate the temperature results from the infrared technique.

4.1.2 Data acquisition and processing

The surface of the cubes in question was painted with the calibrated mixture of five liquid crystals. Care was taken to guarantee that the experimental conditions during the calibration were similar to those during the real experiments, such as the source and the angle of illumination, the age of the mixture, the intensity of the light source, etc. An infrared filter was used to prevent heating of the surface by the incident thermal radiation. Excessive thermal radiation effects the surface temperature of the faces of the epoxy cube due to the low thermal conductivity of the epoxy substrate. Since this filter effects the incident radiation (light intensity, brightness, etc.), it was applied for both the calibration measurements and real measurements to maintain similar experimental conditions. The colour patterns were then photographed with a fine grain size colour film in order to elucidate the colour pattern at the cube faces. A zoom-ocular was used for the magnification of the object. A typical example of a liquid crystal measurement is given in Fig. 4.3 for a Reynolds number of $Re_H = 5066$. The image shows the temperature distribution at two faces of the cube, namely the side and top face. The cube was located in the in-line row of 9 cubes, as discussed in chapter 6. The method used to calibrate the liquid crystal mixture provided five absolute surface isotherms. For the result shown here, only four isotherms appear at the surface which correspond to the four lowest temperatures (i.e. 47.5, 55.8, 60.5 and 65.2 °C). These photographs were then mapped onto an isotherm plot.

4.2 Infrared thermography

Infrared thermography is commonly used to analyse convective heat transfer phenomena, as encountered in thermo-fluid problems. Good reviews of the different

applications and the technical drawbacks of infrared thermography are hard to find but work of Carlomagno et al. [27, 28] are recommended. Further, background information is given in the Agema training manuals [3].

Some advantages of infrared thermography are that this technique is rather easy to operate and that a two-dimensional temperature field is obtained after proper calibration. Care should be taken when this technique is employed for the measurement of surface temperature fields of small heated three-dimensional protrusions. In particular when temperature distributions with a high spatial resolution and a large depth-of-field are required, the measurements can suffer from large spatial image degradation which may lead to an unacceptable error in the measured surface temperature. This obviously prevents the reliable evaluation of heat transfer phenomena, in particular in laboratory applications. The accuracy of infrared temperature measurements for these kind of applications can be increased by employing an image restoration technique. Furthermore, environmental contributions to the total radiosity received by the infrared sensor are usually difficult to quantify and complicate an accurate analysis. This problem can be faced by employing an in situ calibration which accounts for the various environmental reflections and emissions.

This section gives a detailed analysis of the infrared thermography used in this thesis to measure the steady-state surface temperature distributions of the cubes. The characteristics of the infrared camera used are firstly discussed. Further, details of the mapping procedures and data processing are given. Much attention is given to the in situ calibration and the image restoration techniques, both developed to increase the absolute accuracy of the temperature measurements.

4.2.1 Infrared camera characteristics

The infrared camera used [169] had a two-dimensional scanning mechanism to create 200 image-lines, each consisting of 300 pixels. The low frame-rate of 2 Hz restricted its application to steady-state situations. This camera was suitable for our application since we were only interested in steady-state surface temperature distributions. The wavelength bandwidth of the camera was between 3 and 5 micron. An 8-bits AD converter digitised the received signal in 256 grey values. This dynamic range ensured a thermal resolution of at least 0.2°C for the situations investigated in this research. The images used were accumulations of five images, consecutive in time, stored in a temporary buffer in the camera. The time scales of the heat transfer processes studied were far beyond the time to create these accumulated images (4-5 seconds) and therefore, this low scan frequency did not hamper the measurements.

The calibration curve as supplied by the manufacturer, given in Fig. 4.4, shows the linearity of the system for a relatively large temperature range, namely between -35°C and 200°C . The resolution in the for the present study relevant temperature range ($45\text{--}65^{\circ}\text{C}$) was low and, therefore, a recalibration was performed with a perfect black body for the reduced temperature range of $30\text{--}70^{\circ}\text{C}$. A simple calculation verified that the emissivity of the used black body was almost 1 (0.999, see [108]). The black-body temperature was measured with a PT100 element. The camera response versus the black-body temperature, plotted in Fig. 4.5, shows again the linearity of

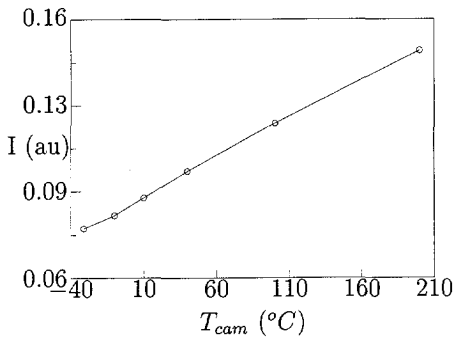


FIGURE 4.4: Default calibration of the infrared camera: camera response versus black-body temperature (supplied by the manufacturer).

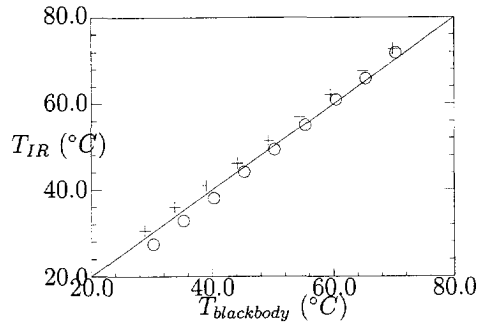


FIGURE 4.5: Calibration of the infrared camera: temperature from camera reading versus black-body temperature without (+) and with (o) a one-point calibration.

the infrared camera. It is important to note that a structural difference is observed between the black-body and infrared camera temperature which becomes larger for higher temperatures. Application of the one-point-calibration option for correction of the thermal reading results only in a local improvement which can also be seen in Fig. 4.5. Only in a small interval around the correction, i.e. for this example 55 °C, the thermal reading approaches the black-body value. However, the remaining part of the calibration is less accurate. An in situ calibration was developed, to be discussed in the next section, to obtain accurate temperature measurements.

A copper object of square cross-section was perpendicularly scanned to measure the sampling density as function of the object-distance. A sampling density of approximately 7 pixels/mm was encountered for an object distance of $O_b = 0.19$ m. No difference in sampling density was measured between the horizontal and vertical scan-direction, thus providing a square sampling grid.

4.2.2 Data acquisition and processing

To measure the surface temperature distribution of the cubes in the closed wind tunnel, the infrared radiation from the cube surfaces needs to be transmitted by the window. This requires a window with transmittance characteristics compatible to those of the infrared camera used. A foil was selected which had a uniform transmittance of 0.84 in the range of 1-10 micron.

The side faces of the cubes, oriented perpendicularly to the mounting base, were scanned at a 45° scan angle because of geometrical restrictions of the closed wind tunnel. This scan facility is drawn in Fig. 4.6. The non-perpendicular scan-direction has two major complications. First, the spatial resolution decreases with the scan-angle. And second, a scan at 45° requires a depth-of-field of at least ± 5 mm for a direct and accurate temperature measurement across the entire face. A typical example of an infrared image scanned with this set-up is given in Fig. 4.7.

The cube surfaces were painted with a thermally black paint to enhance the radiation needed for infrared thermography. For scan angles up to approximately

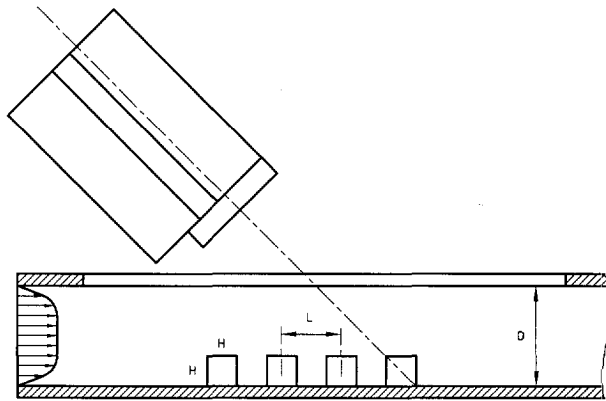


FIGURE 4.6: Schematic of the channel with the wall-mounted cubes where the infrared camera is in position to scan the top and front face of the cube.

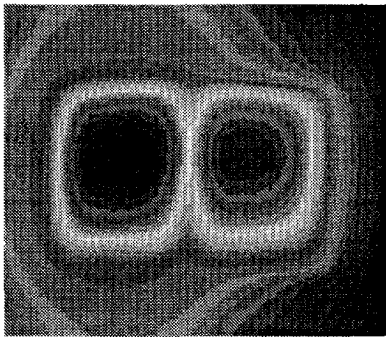


FIGURE 4.7: Infrared image of the a typical two-faced projection (side and top, image is 90° rotated) of the fifth cube in the in-line array arrangement of cubes ($Re_H = 5066$).

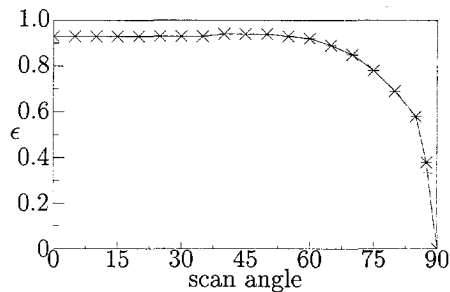


FIGURE 4.8: Angle-dependent surface emissivity of the used black paint.

55-60°, the emissivity appeared to be independent of the scan angle and equals $\epsilon = 0.95$, see Fig. 4.8. This value was verified with measurements performed with two other methods. One method used a vacuum chamber in which a copper cube was hung. The cube was powered internally with an electrical heat source. The vacuum conditions were used to minimise the convective heat transfer from the cube to the environment. Conduction losses were minimised by using thin lead wires. An energy balance over the cube gave an accurate expression for the total radiative exchange in which the surface emissivity of the used paint was the only unknown. The other method was based on a black body experiment. The results involved an experimental uncertainty of 2%. An accurate value of the surface emissivity was not necessary for accurate surface temperature measurements since the in situ calibration, to be discussed later, counter-balanced the emissivity (the calibration

was performed with an identical object which was painted with the same thermally black paint). An accurate value for the emissivity is only required for the calculation of the net radiative heat flux needed to determine the convective heat flux from the equilibrium of heat fluxes at the surface of the cube.

Four different projections were employed to cover the total surface of the three-dimensional cube. A schematic is given in Fig. 4.9. The five different cube faces are denoted as rear (R, the face in the wake of the cube), side (S, the north and south side faces), front (F, the windward face), and top (T). The coordinate system is also given in Fig. 4.9.

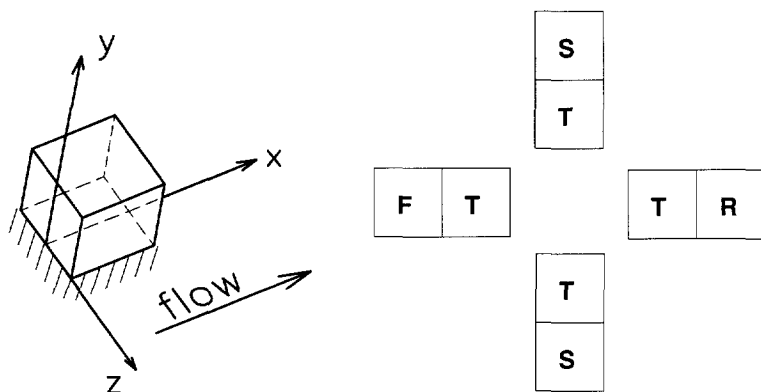


FIGURE 4.9: Sketch of the four different projections employed to cover the total surface of the three-dimensional cube.

Since the cube faces cover only a small part of the entire recorded image, location identification was a necessity. Especially the optical blur prevented a correct identification of the exact position of the object in the infrared image. This problem was solved by a grid cube which had a high precision thermal grid at its surface. This cube was made of copper with precisely located grooves which were filled with black paint. The high thermal conductivity of copper resulted in a uniform copper surface temperature. The pronounced emissivity differences between black paint ($\epsilon \approx 0.95$) and copper ($\epsilon \approx 0.3$) supplied a good contrast. A schematic drawing is shown in figure 4.10. The infrared image in Fig. 4.11 of two faces of the grid-cube clearly shows the good contrast. The cube location in the image was determined from these images of the grid-cube and was applied to all other measurements. The used traverse mechanism had a displacement accuracy of 0.1 mm which ensured a reproducible measurement to allow this one-to-one mapping. It was possible to determine perspective functions from this grid-cube but perspective effects appeared to be minor and, therefore, this is not further discussed here.

With the relative pixel coordinates of the cube faces in the infrared image obtained from a separate experiment with the grid cube, an equidistantly constructed orthogonal grid of $N \times N$ surface cells was then superimposed on the two faces contained by the infrared image to obtain the spatial distribution of the averaged pixel value. In fact the distribution of the pixel intensities is mapped upon a equidis-

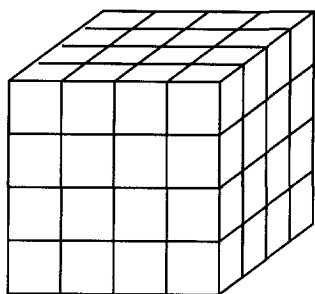


FIGURE 4.10: Schematic of the grid element for location identification.

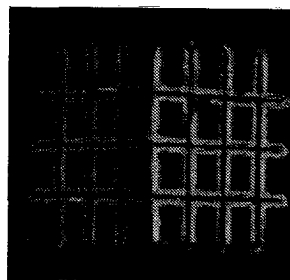


FIGURE 4.11: Infrared image of the grid element.

tant grid of size $N \times N$. The infrared system had a focal matrix of 300×200 pixels. The optimum grid resolution appeared to be 30×30 grid cells per cube face (see chapter 3). For this resolution, the focal length was 0.19 m and the averaged intensity per surface cell was determined from about 5-8 individual pixels. This focal length corresponded to the minimum possible distance to operate the camera due to geometrical restrictions. An increase of the focal length leads consequently to a decrease in image magnification and, thus, to a loss of spatial resolution. For larger grids, for example 40×40 surface cells, the averaged pixel value per surface cell is based on less individual pixels which leads to an increased noise level in the measurements.

The procedure to define the corners is estimated to be accurate within 1 pixel-position. This corresponded to a temperature uncertainty of less than 0.1°C for the mid-plane and less than 0.2°C for the edges.

4.2.3 Intensity-temperature conversion

Global conversion

Infrared thermography is based on the unambiguous relation between the surface emission of electro-magnetic waves of a heated object and its surface temperature. The blackbody spectral emittance is described by Planck's law [108, 154]. From an integration of the spectral intensity distribution over the wavelength interval $0 \leq \lambda \leq \infty$, the total radiant emittance W_b of a black body ($\epsilon = 1$) is obtained, which is known as the Stefan-Boltzmann equation [108]:

$$W_b = \sigma T^4, \quad (4.1)$$

where σ is the Stefan-Boltzmann constant. In a practical situation, a surface will almost never comply with the laws of perfect black-body emission over an extended wavelength region but it may be approached in a certain spectral interval.

The total radiosity received by the thermal detector consists of several contributions from the thermal environment which can be divided in four global main contributions [108, 3]:

- The emission from the target.

The total power emitted by the target is:

$$E_{obj} = \epsilon_{obj} \sigma T_{obj}^4, \quad (4.2)$$

where ϵ_{obj} is the object-emissivity, σ denotes the Stefan-Boltzmann constant and T_{obj} is the object temperature.

- Reflections from the environment.

The total reflected power on the object from the environment:

$$E_{env} = \rho_{obj} \epsilon_{env} \sigma T_{env}^4 = (1 - \epsilon_{obj}) \epsilon_{env} \sigma T_{env}^4, \quad (4.3)$$

where ρ_{obj} is the reflection coefficient of the object, ϵ_{env} is the emissivity of the environment and T_{env} denotes the temperature of the environment.

- The emission from air.

The total emitted power from the air is:

$$E_{air} = \epsilon_{air} \sigma T_{air}^4 = (1 - \tau_{air}) \sigma T_{air}^4, \quad (4.4)$$

where ϵ_{air} is the emissivity of air, τ_{air} is the transmittance of air and T_{air} is the air temperature.

- The internal camera contributions.

Internal camera reflections, emissions, etc., contribute to the total received power:

$$E_{camera} = -\epsilon_{chp} P_{chp} + \epsilon_{opt} P_{opt}, \quad (4.5)$$

where P_{chp} is radiative power from the chopper, P_{opt} is the power from the optics and ϵ_{opt} and ϵ_{chp} are the emissivity of the optics and the chopper, respectively.

Radiative power emitted and reflected by the object is attenuated by air (absorption). This is accounted for by a multiplication with the transmittance of air τ_{air} , specifically:

$$\tau_{air} = e^{-\alpha d}, \quad (4.6)$$

where d is the distance between the target and the sensor and α is the absorptivity of air. For small distances, τ_{air} is almost unity. The camera optics (lens etc.) will also attenuate the transmitted radiation and, therefore, the individual terms have to be multiplied by the transmittance of the optics τ_{opt} . If the object is observed through a window, the transmitted thermal radiation will be attenuated by absorption and reflection. Therefore, the transmittance of the window τ_w should also be taken into account.

The previous emissivities, transmittances and reflection coefficients can be wavelength dependent. The considered expressions denote integrated values over a limited wavelength region (3-5 micron) which is determined by the optical system (lens etc.) and the thermal sensor.

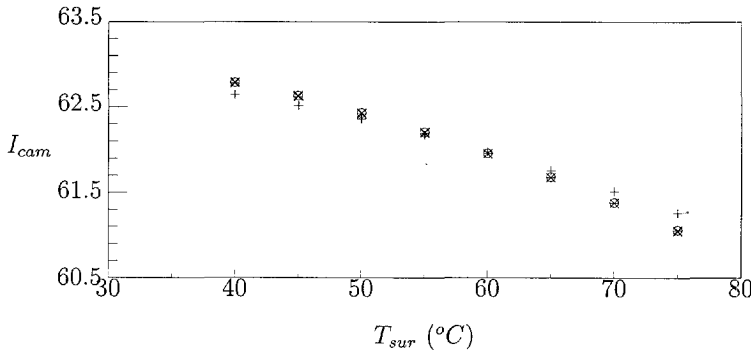


FIGURE 4.12: In situ calibration curves for 4 surface cells at the centreline of the top face: camera readings versus surface temperatures. The different symbols refer to different locations.

The summation of all global contributions finally gives an expression for the object temperature. It is obvious that all contributions are integrated over the entire environment. The camera does not provide a possibility to distinguish between contributions from discrete heat sources or, in other words, to incorporate view factors.

Temperature evaluation from the in situ calibration

Contributions from local hot spots, which are particularly present when the target is surrounded by a discrete distribution of heat sources, for example in multiple-cube configurations, are disregarded by the global approach. This is highlighted when the total radiosity received by the thermal sensor at pixel position (i, j) is expressed in a functional formulation:

$$I_{cam}(i, j) = F(T_{sur}(\xi, \eta), \alpha, f, T_{neig}, T_{amb}, F_{i,j}, \epsilon_{obj}, \epsilon_{neig}, \epsilon_{env}, \tau_w), \quad (4.7)$$

where ξ and η are the surface coordinates at the cube faces, $T_{sur}(\xi, \eta)$ is the local surface temperature of the target, α is the scan angle, f is the focal length, T_{neig} and T_{amb} are the surface temperatures of the neighbouring sources and the ambient, $F_{i,j}$ are view factors between the object and other thermal sources, ϵ_{obj} , ϵ_{neig} and ϵ_{env} are surface emissivities and τ_w is the transmittance of the used foil. This expression shows that most of the radiative contributions are location dependent and an evaluation of the individual contributions is difficult to perform. A possibility is to define view factors for the local thermal sources but this even complicates the situation. The relation between the local surface temperature $T_{sur}(\xi, \eta)$ and the camera reading I_{cam} was determined directly by means of an in situ calibration to incorporate the location dependence of the contributions involved.

The in situ calibration was performed with a cube of only copper with the same dimensions as the epoxy cubes ($15 \times 15 \times 15$ mm). This calibration cube was also equipped with an electrical heater and painted with the same black paint as used for the epoxy cubes. The high thermal conductivity of copper led to a uniform temperature, internally as well as at the surface (the tolerances were within 0.05 $^{\circ}C$,

even for relatively large heat fluxes). The copper temperature was measured with a thermocouple, embedded in the core region of the cube, and controlled by means of a feedback loop to the set-temperature, similarly as for the epoxy cubes. Variation of the copper temperature resulted in different distributions of the radiative emission from the surface of the five faces of the cube exposed to air flow. These intensity distributions were determined according to the mapping and the processing procedures described in the previous section for 9 different surface temperature levels of the copper cube, equidistantly distributed over the temperature range 35-75 °C. This resulted in a matrix of 7200 calibration curves, describing the relation between the camera reading $I_{cam}(i, j)$ and the actual local surface temperature $T_{sur}(\xi, \eta)$ of the copper cube (the five relevant faces of the cube were mapped in 4 projections, each consisting of two faces, each face consisting of 30×30 surface cells (or surface locations), giving a total of 7200 calibration curves). The top face was scanned four times and the obtained temperature fields were subsequently averaged. Fig. 4.12 shows a series of typical calibration curves for four surface cells situated at the centreline of the top face of the cube. It is noted that the camera reading decreases with increasing surface temperature. This was due to the inverted intensity conversion from the received radiosity to the camera reading. The differences in the camera reading between different locations correspond to temperatures up to approximately 5 °C. These differences would not be accounted for when the infrared system was calibrated with only one or a few thermocouples embedded in the cube. For the present case, in which inaccuracies in surface temperatures of maximum 0.5 °C are tolerated, in situ calibration is a necessity.

For the in situ calibration, the epoxy cubes were substituted by the copper calibration cubes. It was guaranteed that the experimental conditions during the in situ calibration and the real experiment were similar. The matrix of calibration curves, measured in a separate experiment, was then directly applicable to determine the local surface temperature of the epoxy cubes.

4.2.4 Image degradation in general

Several sources contribute to temperature inaccuracies in infrared images. Temperature noise, which manifests as deviations in the thermal value of individual pixels, can be minimised by averaging over a sufficiently large number of images. Deviations due to the restricted digitisation resolution of the camera (8-bits converter) is also mentioned as a possible contribution (it was already noted that the temperature resolution was at least 0.2 °C). Further, spatial image degradation is a consequence of optical diffraction and lens aberrations (for instance the defocussing of the object). Spatial averaging over the instantaneous projected area is also classified as possible cause [141].

Noise

It is beyond the context of this thesis to discuss all the different sources that contribute to the noise present in an infrared system. For this, reference is made to the detailed analysis given by Hudson and Wordsworth Hudson [67]. To find the

noise characteristics of the present camera, a uniform temperature field kept at different temperature levels between 30 °C and 80 °C was scanned. This temperature field was created with a black copper plate with a uniform surface emissivity. The variance of the spatial intensities $\bar{I}^2$ is plotted versus the mean field intensity $\bar{I}$ in Fig. 4.13. Above the noise floor, the stable plateau at the lower intensities, the relation between the mean intensity and the variance develops to a more or less linear profile which illustrates the Poisson-limited behaviour of the infrared camera. The noise shown in this figure corresponds to temperatures between roughly 0.1-0.2 °C.

A different way of determining the noise level in the accumulated image is to consider the standard deviation in the separate pixels and subsequently average it over all pixels:

$$\sigma(i, j) = \sqrt{\frac{1}{N-1} \sum_{i,j} (I(i, j) - \bar{I}(i, j))^2}, \quad (4.8)$$

where $I(i, j)$ and $\bar{I}(i, j)$ are respectively the grey-values of the instantaneous images and the accumulation of five images at position (i, j) . $\bar{\sigma}$, which is the average of all pixel positions M , is a measure for the noise level of the accumulated image. The noise-level was calculated for several images for different conditions (i.e. surfaces of uniform temperature, strong gradients, and so on) and in all situations, the noise level corresponded to temperatures of about 0.10-0.12 °C.

Hysteresis

Since a camera with a scanning mechanism with only one thermal sensor was used, spatial blur could possibly be caused by hysteresis effects: the sensor release time exceeds the period of two consecutive pixel recordings. The circular aperture of a large black body was scanned to study possible hysteresis effects. The emitted radiation of the source is uniform across the entire aperture according to the definition of a black body [108]. Cross-sectional slices of the camera response are given in Figs. 4.14 and 4.15 for respectively the horizontal and the vertical direction. The focal length was $M = 0.20$ m. The temperature profile is almost uniform in the centre, while it decays gradually to the ambient temperature at the edges (note that an increase in camera intensity corresponds to a decrease in temperature). The rotation-symmetric profiles illustrate that the scan frequency was sufficiently low to deny hysteresis effects.

Slit response function

When small objects are considered, spatial image degradation becomes critical and a modified recording (consisting of object and environment contributions) is obtained. A common method to measure this effect is based on the placement of a variable slit in front of a large black body. The un-modulated signal is then determined by the radiation from the black body without the slit in front of it. Radiation which passes the variable slit will be modulated. The ratio of the distorted to the undistorted signal is called the modulation (Mod). This modulation is better known as the slit response function (SRF) when it is plotted versus the slit width c.q. object

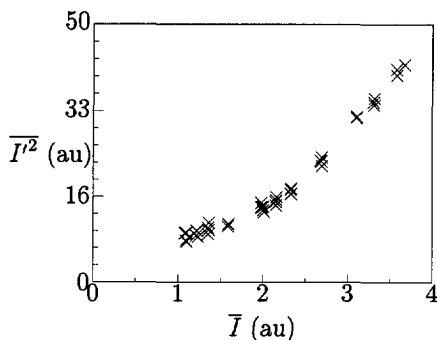


FIGURE 4.13: Variance of the spatial intensity versus the mean intensity for a uniform temperature field.

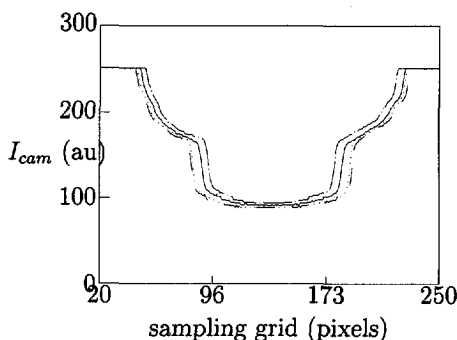


FIGURE 4.14: Horizontal cross-sectional slices of the thermal image of a circular black-body.

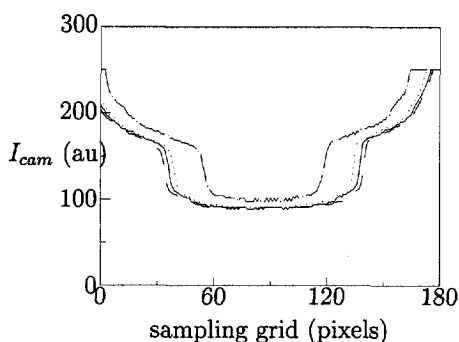


FIGURE 4.15: Vertical cross-sectional slices of the thermal image of a circular black-body.

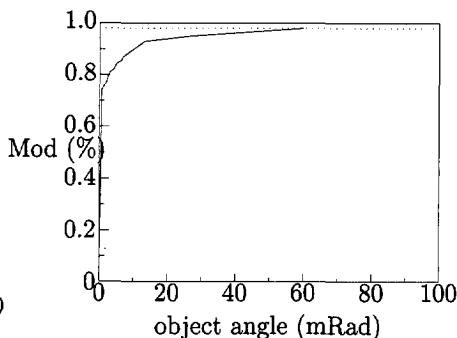


FIGURE 4.16: Slit response function (SRF) of the IR-camera: the modulation Mod versus the object angle. The dotted line represents a modulation of 98%.

angle in mRad (the object angle is defined as the angle between half the slit width and the object distance). A typical SRF of the infrared system used [169] is given in Fig 4.16. A modulation of Mod = 98% is indicated as the dotted line in the figure. For infinite slit widths, the modulation becomes 1 (100%). A zero slit width corresponds to a zero modulation.

Usually, the spatial resolution is specified for the slit width that gives 50% modulation [61]. The spatial resolution is the number of resolving pixels (i.e. the amount of pixel elements in the slit width). The largest acceptable error in the temperature reading can be expressed in terms of a minimum required modulation level. The minimum object size can be calculated when the object distance is known (for small angles, $\tan(\alpha) = \alpha$, and therefore, object angle = object size/object distance).

The SRF, depicted in Fig. 4.16, exhibits a steep gradient for very small object angles. After a modulation of approximately 70%, the modulation improves slowly with increasing object angle. Even for rather large object angles, (up to 80 mRad)

the modulation is still less than 99% which implies an unacceptable error in the temperature reading.

4.2.5 Spatial image degradation

It was argued that the scan angle of 45° required a depth-of-field of at least ± 5 mm. Further, a sufficiently large spatial resolution was needed to calculate the temperature distribution in the epoxy layer accurately. The compromise between these requirements was a focal distance of $M = 0.19$ m. Because of this focal length, spatial image degradation was significantly present and needed to be corrected for. A reconstruction technique based on the Optical Transfer Function (OTF) was proposed to correct for the spatial image degradation. First, an outline of previous work on the characterisation of spatial image degradation is given. The discussion proceeds with the formulation of spatial image degradation and the developed model for the OTF. The measurement of the OTF is discussed in detail. Special attention is given to the accuracy of the OTF, particularly the depth-of-field and the temperature dependence. The measured OTF was used to construct a restoration filter which was applied to restore the infrared images.

Literature survey

Several studies reported in literature were already devoted to different measurement techniques to characterise imaging systems with the Optical Transfer Function or its magnitude, the Modulation Transfer Function (MTF). Mullikin et al. [120] discussed the characterisation of different CCD cameras using, among others, the measured spatial frequency responses to a step-edged object in the focal plane of the camera. Marchywka and Socker [100] investigated a modulation transfer function technique which is suitable to characterise large-format, small-pixel detectors for the visible wavelength region. The target image from which they derived the MTF was a sinusoidal holographic test pattern. Daniels et al. [36] measured shift invariant MTFs for scanning imaging systems for application in the visible, $3\text{--}5\mu\text{m}$ and $8\text{--}12\mu\text{m}$ wavelength bands. Their target images were randomly created pixel patterns printed on a transparency and placed in front of a uniform radiation source (black-body). Tzannes and Mooney [168] reported on the characterisation of PtSi infrared cameras. A knife-edge technique was used to obtain edge spread functions, from which the two-dimensional MTF was derived. de Luca and Cardone [92] presented the formulation of a cascade model to predict the MTF of a sampled infrared system.

Bougeard et al. [22] used infrared thermography to evaluate the convective heat transfer from fin-tube heat exchangers. The technique included an image restoration based on Wiener filtering and the Optical Transfer Function. However, the performance of the image restoration was only qualitatively shown and no objective accuracy analysis was provided. Recently, Carlomagno [28] provided a detailed analysis of the various aspects which contribute to the accuracy degradation of convective heat flux measurements. Among others are the inaccuracies involved when using infrared thermography to measure surface temperatures of the test objects. He showed that, in particular for situations with large spatial temperature gradients,

both infrared image and heat flux degradation prevent the accurate determination of heat transfer coefficients. Aliaga et al. [5, 6] used infrared thermography to measure the surface temperature distributions of multiple wall-mounted ribs in channel flow. Lorenz et al. [91] recently reported on results for a similar configuration. Although temperature fields of complex geometries were subject of study, they did not report on the use of image restoration and in situ calibration (the system was calibrated with thermocouples, embedded in the surface).

Formulation of spatial image degradation

The unique relation between the original thermal object $i(x, y)$ and the recorded blurred image $m(x, y)$, expressed in the spatial domain, is given by the 2D Point Spread Function (PSF) $h(x, y)$. It indicates the power distribution in the image plane due to a point source in the object plane [22]:

$$m(x, y) = h(x, y) * i(x, y) + n(x, y), \quad (4.9)$$

where $*$ is the convolution operator, x and y are the spatial coordinates and $n(x, y)$ denotes an additional noise term. The convolution product reduces to an ordinary product after Fourier transformation:

$$M(f_x, f_y) = H(f_x, f_y) I(f_x, f_y) + N(f_x, f_y), \quad (4.10)$$

where $H(f_x, f_y)$ is the Optical Transfer Function (OTF) in spatial frequencies f_x and f_y and M , I and N denote the Fourier transforms of m , i and n , respectively. The modulus of the complex OTF is the Modulation Transfer Function (MTF).

The OTF describes the degradation process, which is determined by the focal-length, optics, lenses, temperature range, etc. Due to the small wavelength bandwidth of the infrared camera (the spectral response is between 3 and 5 microns) and the limited temperature range of the studied objects (about 25 °C), the assumption of a temperature-independent optical transfer function is permitted. This statement was supported by experiments and will be discussed later in this section. Furthermore, the OTF is believed to be in good approximation spatially invariant which is required for applying the measured OTF to the entire image field. Close to the image borders, the imaging system may exhibit invariant behaviour, however, experiments showed randomly distributed noise in the individual horizontal and vertical edge-responses across the entire image and no trend in absolute level was observed. The blurred image can be restored with a restoration filter $W(f_x, f_y)$ yielding the restored image $\hat{I}(f_x, f_y)$ [80]:

$$\hat{I}(f_x, f_y) = W(f_x, f_y) M(f_x, f_y). \quad (4.11)$$

Restoration with the inverse filter, being the reciprocal of the OTF i.e. $W_i(f_x, f_y) = 1/H(f_x, f_y)$, results in unlimited amplification of the response when $H(f_x, f_y)$ approaches zero. This undesired amplification of the response can be avoided by using the Wiener filter $W_w(f_x, f_y)$ [80]:

$$W_w(f_x, f_y) = \frac{H^*(f_x, f_y)}{|H(f_x, f_y)|^2 + (\phi_{noise}/\phi_{image})}, \quad (4.12)$$

where $H^*(f_x, f_y)$ is the complex conjugate of the OTF and ϕ_{noise} and ϕ_{image} are the power spectral densities of the noise and the image, respectively. The additional ratio $\phi_{image}/\phi_{noise}$ is the signal-to-noise-ratio of the image, which attenuates high noise components. The Wiener filter becomes the exact inverse filter for noise-free images, i.e. $\phi_{noise} = 0$.

Model for the OTF

The infrared imaging system is characterised by its PSF or OTF. It is often easier to acquire Line Spread Functions (LSF), from which the OTF can be derived. The camera response to a perfect thermal step-edge, the Edge Spread Function (ESF) $k_x(x)$, is the convolution product of the original one-dimensional step-edge $u_x(x)$ and the Line Spread Function (LSF) $l_x(x)$ (noise is omitted to simplify the derivation):

$$k_x(x) = u_x(x) * l_x(x). \quad (4.13)$$

A similar expression holds for the response in the vertical scan-direction:

$$k_y(y) = u_y(y) * l_y(y), \quad (4.14)$$

where $k_y(y)$ is the ESF, $u_y(y)$ is the step-edge and $l_y(y)$ is the LSF. The relations between the LSF and the PSF are the next integrals:

$$l_x(x) = \int_{-\infty}^{\infty} h(x, y) dy, \quad (4.15)$$

$$l_y(y) = \int_{-\infty}^{\infty} h(x, y) dx. \quad (4.16)$$

Differentiation with respect to the dependent coordinate and using convolution properties result in:

$$l_x(x) = \frac{dk_x(x)}{dx}, \quad (4.17)$$

$$l_y(y) = \frac{dk_y(y)}{dy}. \quad (4.18)$$

Fourier transformation yields the Fourier transformed Line Spread Function $L_x(f_x)$ and $L_y(f_y)$, which are the cross-sections through the centre of the 2D OTF for the horizontal and vertical scan-direction, respectively. A weighted sum of both Fourier transformed LSFs gives an interpolation function for $H(f_x, f_y)$:

$$H(f_x, f_y) = \alpha L_x(f_x) + \beta L_y(f_y), \quad (4.19)$$

where α and β are interpolation coefficients. For a linear dependency on the rotation angle of the edge in the first quadrant θ , α and β can be selected as $\alpha = (90 - \theta)/90$ and $\beta = \theta/90$. This linear approach is verified with measurements of the edge response for certain rotation angles of the thermal step-edge which will be discussed in the next section.

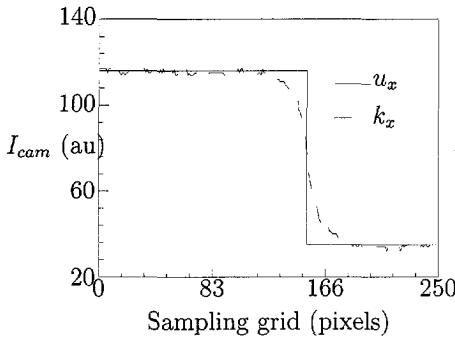


FIGURE 4.17: Cross-sectional slice of the original (u_x) and the measured image (k_x) of a horizontal thermal step-edge for a focal length of $M = 0.19$ m.

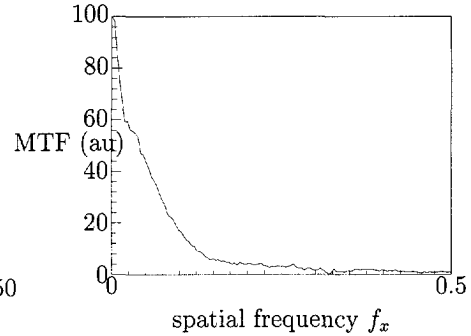


FIGURE 4.18: MTF for a focal length of $M = 0.19$ m characterising the horizontal scan-direction. The MTF was based on the original image size without 2×2 binning.

Measurement of the OTF

Experimental thermal step. Optical systems are commonly characterised by slits or knife-edges, where usually a plate at uniform temperature is covered by a thin strip at a different temperature. The gap between the two materials, since the hot and the cold plane can not be located in the same plane of view, may result in non-ideal behaviour of the thermal step-edge (for instance defocussing). Therefore, a different approach was adapted. An almost perfect thermal step-edge was realized by an abrupt step in the surface emissivity of a heated copper plate. One half of the plate was covered with a very thin coating (< 0.01 mm) with emissivity ϵ_1 and the other half with a thin coating with ϵ_2 . The temperature of the entire surface of the test-plate was uniform (within 0.05°C) because of the high thermal conductivity of copper and the very small layer-thickness of the coating. The difference in surface emissivity of the test-plate caused an abrupt change in the emitted radiation, which resulted in a thermal step-edge, with a hot and a cold part located in the same plane of view.

The setup was equipped with a controlled heater for adjusting the copper plate temperature at the desired temperature levels. Rotation of the set-up permitted measurements of the LSF at different rotation angles (0° , 90° , 180° and 270°) in order to investigate possible anisotropic behaviour. The setup could also be positioned at inclination angles of 30° , 45° and 60° to verify the weighted sum approach for obtaining the 2D OTF. The object-distance could be varied between zero and infinity to examine out-of-focus and object-distance dependencies of the OTF.

Implementation details. A cross-section of the in-focus step-edge response of the infrared camera is shown in Fig. 4.17 for a focal length of $M = 0.19$ m, illustrating the relatively large image degradation. The step function $u_x(x, y)$ was located at the inflection point of the camera response (s-curve). The significant deviation between the original and the measured image would obviously lead to unacceptable errors in

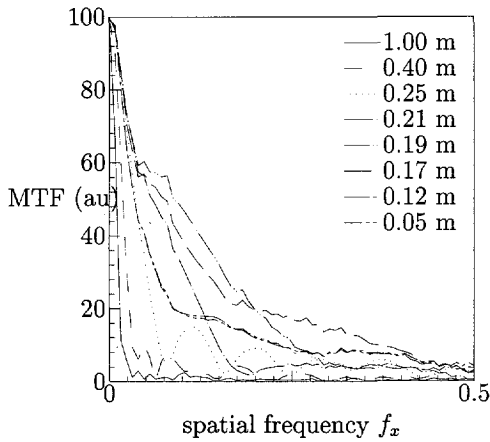


FIGURE 4.19: Out-of-focus behaviour of the MTF for a fixed object distance of $O_b = 0.19$ m, parametric in the focus-length M .

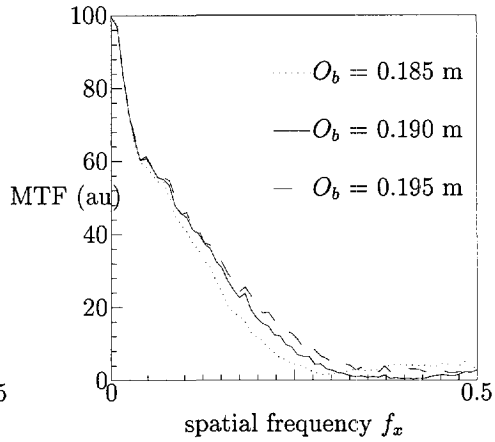


FIGURE 4.20: Out-of-focus behaviour of the MTF for a fixed focal length of $M = 0.19$ m, parametric in the object-distance O_b .

the temperature reading, even after application of an appropriate in situ calibration of the infrared system.

The step-edge was aligned with the sampling grid within 0.1° . For each individual scan-line, the derivative of the ESF was approximated with a sample difference filter:

$$\frac{d}{dx}k_x(i, j) \approx k_x(i, j) - k_x(i - 1, j), \quad (4.20)$$

where the indices i and j denote the discrete pixel location. The derivative of the ESF was averaged over all scan lines (200 and 300 for the horizontal and vertical step edge, respectively), to arrive at the LSF. Misalignment of the edge caused by a slight translation would obviously lead to broadening of this averaged LSF and, therefore, the line response was aligned to the centre of the step-edge at sub-pixel level. The shift was estimated from a least squares fit of a linear function through the phase data of the Fourier transform. The Fourier transform of a shifted function $l(x - x_0)$ will have a phase component proportional to the frequency, i.e. $e^{-jx_0\omega}L(\omega)$. Multiplication of the Fourier transform with the inverse of this phase term resulted in perfect alignment. The noise level of the source image was low since an average of 125 images was used.

The Modulation Transfer Function, characterising the horizontal scan-direction (rotation angle 0°) is shown in Fig. 4.18 for a focus-length of $M = 0.19$ m. The amplitude at zero frequency is scaled at 100. The image size was converted to 256×200 in order to apply the Fast Fourier transformation (FFT). The pixel size was 0.1724 mm and corresponds to a maximum spatial frequency of 5800.5 1/m. The smooth appearance of the MTF is attributed to the noise-limited source image (an average of 200×125 scan-lines). The MTF is peaked in the low frequency region and decays to zero for the larger spatial frequencies. The notable plateau appeared to be an artifact of the camera. The narrow-banded appearance was because of

the oversampling of the camera. Less than half of the entire spatial frequency domain was used and, therefore, the possibility of reducing the bandwidth was examined. With 2×2 binning, the sampling density was reduced with a factor 2 in both scan-directions, yielding a pixel size of 0.3448 mm which corresponds to a maximum spatial frequency of 2900.2 1/m. The incorporated averaging resulted in an improved signal-to-noise-ratio, appearing from the smoother MTF (see Fig. 4.19, $M = 0.19$ m). Despite the reduction in sampling density, the MTF still did not cover the entire spatial frequency range. It is noted that both MTFs for $M = 0.19$ m (shown in Fig. 4.18 and Fig. 4.19) contain basically the same information, except for the noise level. A stronger reduction in resolution will obviously lead to aliasing and, therefore, the 2×2 binning appeared to be the optimum choice for present application.

In the following discussion, all MTFs were determined for the reduced image scale except Fig. 4.25 in which the object distance dependency is shown.

Out-of-focus of object. Out-of-focus of the infrared camera was investigated by considering step-edge responses which were by purpose rendered out-of-focus. Therefore, the focal-length was varied between 0.05 and 1.00 m while the object-distance was fixed at $O_b = 0.19$ m. The corresponding MTFs are shown in Fig. 4.19. When the focal-length of the camera markedly exceeds the object-distance, with infinity as the limit, the step-response approaches a linear profile with a constant derivative. Such a smooth profile corresponds to a narrow, spiked MTF. The same holds for the cases that the focus distance M was smaller than the object distance O_b . However, small deviations are observed for the higher spatial frequencies. All out-of-focus MTFs are enclosed by the in-focus MTF, at least in the low-frequency range. This provides a suitable tool to examine whether an image is in-focus or not which is described in detail in Boddeke et al. [20]. Examination of the behaviour of the MTF shape for small out-of-focus situations provided knowledge of the depth-of-field of the infrared camera. Fig. 4.20 shows the Modulation Transfer Functions for a fixed focus-length of $M = 0.19$ m for differences in object-distance of ± 5 mm. The deviations between the different MTFs were small. To examine the involved temperature effect, a test-image was restored with rotation-symmetric Wiener filters, constructed from these one-dimensional OTFs. The test-image is a two-faces projection of the measurement cube, as given in Fig. 4.21. Horizontal cross sections of the restored test-images are given in Fig. 4.22. The profiles denote intensities but these are directly related to surface temperatures by means of a calibration. It is almost impossible to observe differences between the three profiles which indicates that the discrepancies between the restored images were very small (in the order of magnitude of 0.1°C). This illustrates that the depth-of-field of the infrared camera is sufficiently large to measure the entire surface temperature distribution of the cube.

Wavelength dependency. The surface temperatures of the measurement cubes under operational conditions ranged from 40°C to 65°C . According to Wien's displacement law, the maximum power is emitted at corresponding wavelengths be-

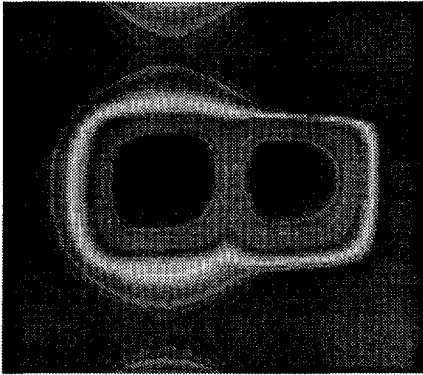


FIGURE 4.21: Test-image: two-faces projection (the top and side face) of the measurement cube.

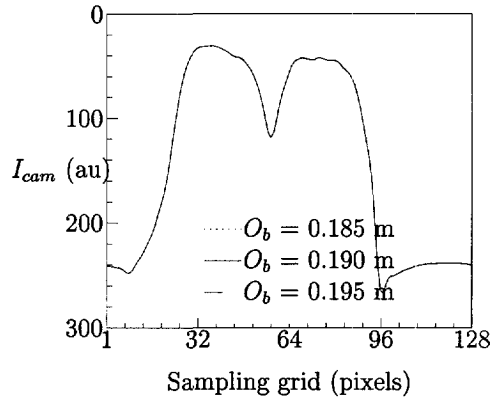


FIGURE 4.22: Horizontal lines through the centre of the test-image after restoration with Wiener filters based on edge responses for slightly out-of-focus thermal step-edges.

tween 8.3 and 8.8 micron [108]. This implies that for the wavelength region of 3 to 5 microns in which the used infrared camera was operated, the Planck curves exhibits strong gradients. A small shift in the surface temperature would result in a strong change in the emitted power. However, since the studied surface temperatures are within a small temperature range of 25 °C, the MTF appeared to be wavelength independent for this small temperature range. This wavelength invariance was verified with a series of measurements for different temperature levels of the thermal step. The MTFs resulting from temperature levels of the step-edge between 35 °C and 75 °C are presented in Fig. 4.23. No significant differences can be distinguished between the different curves. The MTFs for temperature levels close to 55 °C are included to illustrate the reproducibility of the measurements. The thermal step was realized by a pronounced difference in the surface emissivity instead of a difference in the surface temperature. Since the emitted power is proportional to the surface emissivity, the effect of wavelength dependency is negligible for the thermal step itself. A quantitative discussion of the incorporated temperature deviations is possible with restorations of the test-image, shown in Fig. 4.21, with rotation symmetric Wiener filters based on the one-dimensional edge responses. Horizontal lines through the centre of the restored images are given in Fig. 4.24. These profiles are also directly related to surface temperatures. The small deviations in the restored results correspond to temperature differences of about 0.1-0.2 °C. It is important to note that the final OTF, used to restore the real infrared images, was determined at a step-edge temperature of approximately 55 °C. The small temperature differences in the restored results verify the temperature independence of the OTF.

Object-distance dependency. The object-distance dependency of the MTF is illustrated in Fig. 4.25. For four different focal lengths, i.e. $M = 0.15, 0.19, 0.24$

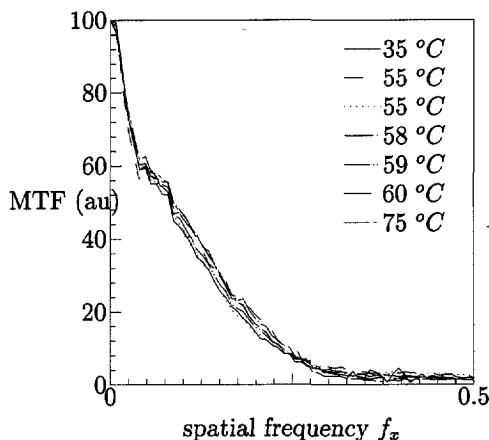


FIGURE 4.23: MTFs for different temperature levels of the experimental thermal step-edge at a focus-length of 0.19 m.

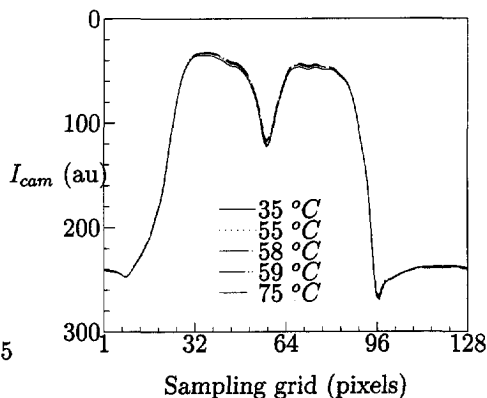


FIGURE 4.24: Horizontal lines through the centre of the test-image after restoration with Wiener filters based on edge responses for different temperature levels of the step-edge.

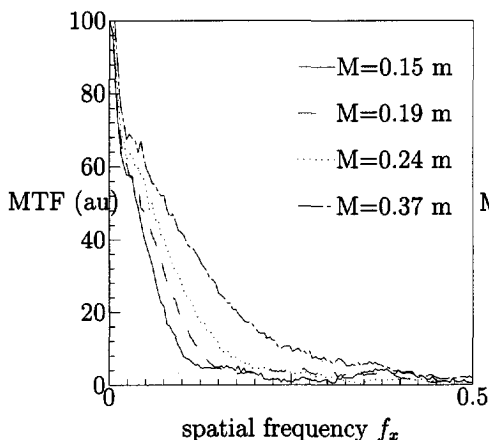


FIGURE 4.25: MTFs parametric in the focus length M .

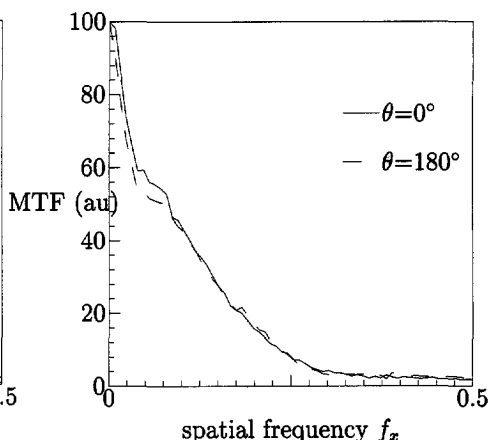


FIGURE 4.26: MTFs for rotation angles of 0° and 180° for $M = 0.19$ m.

and 0.37 m, the MTF was determined for in-focus object images. The bandwidth of the MTF becomes broader with increasing focus length, indicating that a less pronounced optical blurring is accompanied by a decrease in sampling density.

Two-dimensional OTF. Edge responses for various rotation angles of the thermal step-edge for a focus length of $M = 0.19$ m were acquired. The rotation angles 0° and 180° both give an edge response for the horizontal scan-direction. The corresponding MTFs are shown in Fig. 4.26. The small discrepancies between these

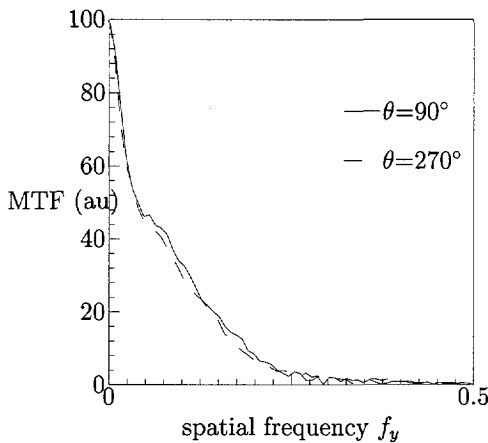


FIGURE 4.27: MTFs for rotation angles of 90° and 270° for $M = 0.19$ m.

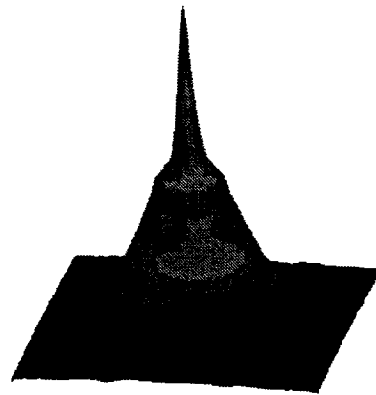


FIGURE 4.28: Two-dimensional Optical Transfer Function $H(f_x, f_y)$.

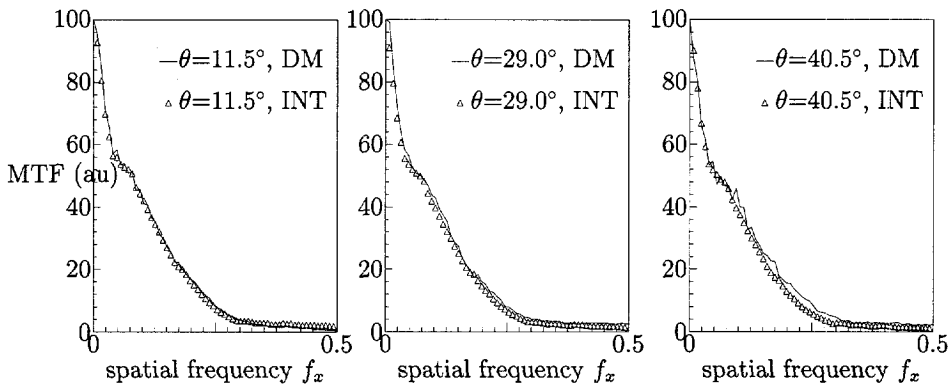


FIGURE 4.29: MTFs for different rotation angles of the thermal step-edge at $M=0.19$ m. Lines denote direct measurements (DM) and markers (Δ) the interpolations (INT): a) 11.5° , b) 29° and c) 40.5° .

edge responses, as can be observed from the figure, are within the experimental uncertainty. The good agreement illustrates that temperature hysteresis, which is a potential problem for high frame-rate cameras, is absent. The edge responses for the rotation angles 90° and 270° , corresponding to the vertical scan-direction, are given in Fig. 4.27. The averaged edge response for the rotation angles of 0° and 180° gives $H(f_x, 0)$ and the averaged edge response for 90° and 270° gives $H(0, f_y)$. The weighted sum of these functions results in the 2D Optical Transfer Function $H(f_x, f_y)$. The magnitude of this OTF $H(f_x, f_y)$ is shown in Fig. 4.28. To verify the interpolation method as suggested, edge responses were measured for rotation angles of 11.5° , 29° and 40.5° . For these angles, the sampling grid of the image is not perpendicular to the step-edge. This problem was addressed by processing

the angled edge responses with a rotation- and bspline interpolation function to allow the analysis presented in this thesis [51, 137]. The corresponding MTFs are shown in Fig. 4.29 as single lines. Cross-sectional slices of the interpolated 2D MTF (Fig. 4.28), were acquired at corresponding rotation angles and presented as markers in Fig. 4.29. The negligible discrepancies between the interpolations and corresponding measurements justify the application of the weighted sum approach.

Restoration filter

With the measured optical transfer function, thoroughly discussed in the previous sections, the Wiener restoration filter can be constructed. The only unknown which remains in this filter is the ratio of the power spectral densities of the noise and the image, ϕ_{noise} and ϕ_{image} , respectively. Although these densities are spatial frequency dependent, it was shown that the variation with spatial frequency was very small. Therefore, it is sufficiently accurate to assume both spectral densities to be constant, at least in the range in which the infrared camera was operated. The ratio $\phi_{noise}/\phi_{image}$, better known as the inverse signal-to-noise-ratio of the image ($1/SNR$), is also constant for this approach, giving the pseudo-Wiener filter:

$$W_w(f_x, f_y) = \frac{H^*(f_x, f_y)}{|H(f_x, f_y)|^2 + 1/SNR} \quad (4.21)$$

This filter was used to restore all infrared images (both from the in situ calibration and real measurements). Based on experimental evidence, a value of $1/SNR = 0.05$ was used for all infrared images.

When the SNR is overestimated, the high-frequency noise contributions are underestimated and, therefore, are amplified by the Wiener filter. This leads to undesirable amplifications of the restored image. But because relatively high frequencies are incorporated, it is possible to recover some of the sharpness of the original image which, as discussed, has to be paid for with additional noise. In contrast, if the SNR is chosen too small, the noise level of the infrared image is overestimated leading to a severe damping of the noise. The higher frequencies are then filtered out which leads to smooth restored images. Due to the absence of the higher spatial frequencies, originally sharp edges are less sharp in the restored image. These effects are illustrated for a copper cube of size 15 mm which was mounted on an insulated base-plate. The copper cube was heated similarly as the copper cubes used for the in situ calibration and controlled to a temperature which was constant within 0.1°C . Because of the large thermal conductivity of copper, the (surface) temperature was uniform within 0.05°C and therefore, this situation resembles a perfect thermal block. The object was scanned perpendicularly at $M = 0.19$ m for which the OTF was determined. A cross-section at the centreline of the unrestored camera response to the thermal block is shown in Fig. 4.30. The profile is smooth, there is hardly no noise but the spatial image degradation is obvious. The other curves refer to image restorations with pseudo-Wiener filters based on different values of the SNR . Especially when the noise contribution is underestimated, for the lower $1/SNR$ values, the restored image is more irregular as appears from the large

fluctuations in the restored profile. However, these restorations are accompanied by a steeper restored step-edge (image). The restored results with the larger values of $1/SNR$ are less irregular but also less sharp at the edges. The somewhat lower level of the unrestored camera response compared to the restored profiles is only a matter of scaling. This test-case is a good example of the compromise between sharp reconstructed gradients and noise-amplification, involved in the Wiener restoration technique.

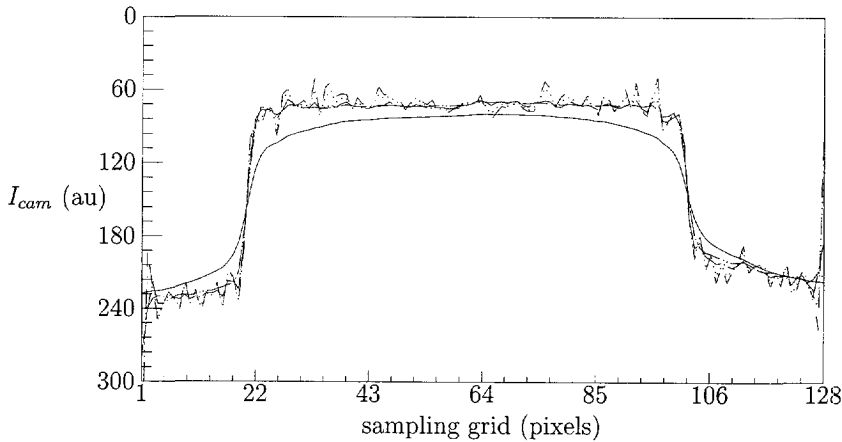


FIGURE 4.30: Cross-sections at the centre-line of camera responses (restored and unrestored) to the heated copper cube: original unrestored image (—) and restored results parametric in $1/SNR$: (---) 0.00001, (···) 0.0001, (- · -) 0.001 and (- · · -) 0.01.

Validation of the accuracy

The improvement of the image restoration technique was confirmed with liquid crystal surface temperature measurements. It was guaranteed that the experimental situation during both thermography measurements were identical to allow a quantitative comparison between both results (an identical experimental situation refers to the power supply, air temperature, air flow, and so on).

Figure 4.31 gives profiles of in situ calibrated infrared surface temperatures along path $ABCD$, which is a cross-section perpendicular to the mounting plate at the symmetry plane ($z/H=0$). The ($\square$) denotes a temperature profile for which both the in situ calibration images and the measurement images were not processed with the restoration filter. The ($\diamond$) indicates the infrared temperatures obtained from the restored calibration and measurement images. It is important to note that the originally recorded infrared images are the same for both temperature profiles; the only difference is the application of the image restoration filter. Despite the use of the in situ calibration, the discrepancies between both temperature profiles (with and without Wiener filtering) are up 3°C for certain areas at the surface of the cube. The previously discussed mixture of five liquid crystals was used to

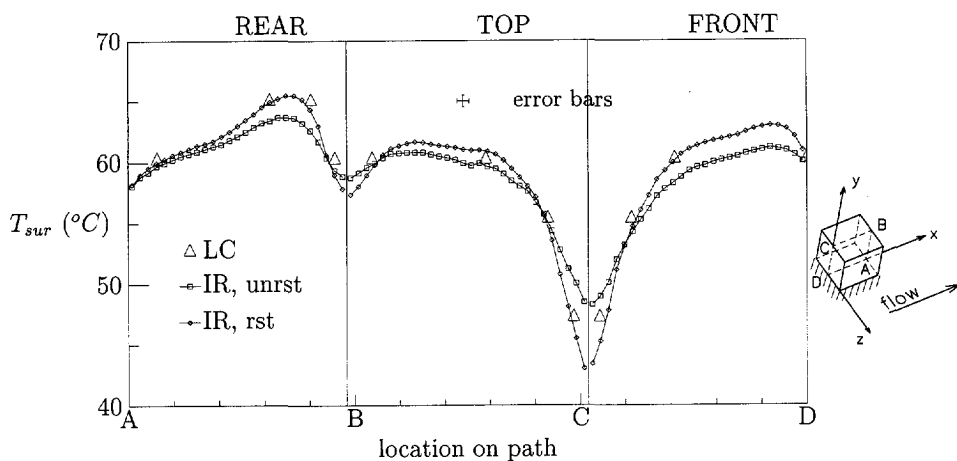


FIGURE 4.31: Comparison between surface temperatures from liquid crystal measurements (LC, Δ) and from infrared thermography without (IR, unrst, $\square$) and with (IR, rst, $\diamond$) application of image restoration. The results are shown along cross-section $ABCD$ as indicated in the accompanying sketch.

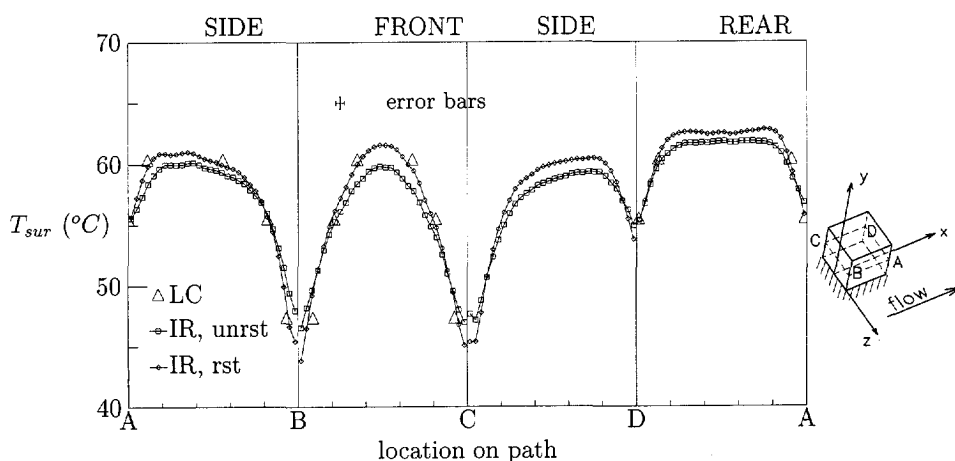


FIGURE 4.32: Comparison between surface temperatures from liquid crystal measurements (LC, Δ) and from infrared thermography without (IR, unrst, $\square$) and with (IR, rst, $\diamond$) application of image restoration. The results are shown along cross-section $ABCD A$ as indicated in the accompanying sketch.

confirm the improvements of image restoration. The calibrated red-green transitions corresponded to 47.5, 55.8, 60.5, 65.2 and 72.8 $^{\circ}\text{C}$.

The temperature and spatial accuracies, evaluated from uncertainty analysis, were estimated to be within 0.4 $^{\circ}\text{C}$ and 0.4 mm, respectively. The spatial reso-

lution is rather poor, but this is of less importance since validation was the only purpose for using these liquid crystals. The LC temperatures are denoted as (Δ) in Fig. 4.31. Error bars for the LC temperatures are also given in the figure. The infrared surface temperatures obtained without the application of image restoration deviate significantly from the LC temperatures. However, the restored infrared results coincide with the LC results within the experimental uncertainty of the LC measurement. Profiles of the surface temperature along path *ABCD*A, which is a cross-section parallel to the mounting plate at half the cube height $y/H = 0.5$, are shown in Fig. 4.32. The (Δ) denotes the LC measurements, the ($\square$) and ($\diamond$) correspond respectively to the unrestored and the restored infrared temperatures. Again, the LC measurements coincide pretty well with the restored infrared measurements.

In conclusion, the coinciding surface temperatures of the two independent techniques illustrate, first, the necessity of image restoration for the considered application. And second, it provides an overall temperature accuracy of 0.4°C for the restored and in situ calibrated infrared measurements. The small discontinuities at the corners are attributed to the somewhat larger inaccuracy in the surface temperature as caused by the mapping of the different faces.

4.3 Laser Doppler Anemometry

Laser Doppler anemometry (LDA), a well established non-intrusive measurement technique, is particularly suited for the investigation of the low Reynolds number flow around small three-dimensional protrusions. However, care must be taken to eliminate numerous causes of error if the LDA measurements are performed in small flow passages and close to solid walls. The LDA technique uses laser beams which intersect in a measuring volume of finite size. The interference of these laser beams causes a fringe pattern. Particles in the flow (the seeding) which pass this measuring volume will shift the frequency of the incident laser light (the Doppler effect). The frequency shift is directly proportional to the velocity of the particles, and with the assumption of no slip between the flow and the particles, the velocity of the flow. Detection of the reflected and frequency-shifted laser light (detection involves also the validation of the signal, meaning that it should satisfy certain quality criteria) allows the measurement of the instantaneous velocity of the particle passing the measuring volume. Consecutive velocity realisations give a time series of the velocity from which, with statistical processing, the first (mean velocity) and higher order moments (Reynolds stresses etc) can be derived. By traversing the measuring volume with an accurate traverse mechanism, profiles of the mean velocity and turbulence fields can be measured in the entire domain.

A detailed discussion of the application and the limitations of Laser Doppler Anemometry in general are given in, for example, Absil [1], Durst et al. [40] and Drain [41]. The discussion in this thesis will be focussed on the specifications and the accuracy of the LDA-system used to acquire velocity data in the flow around the cubical protrusions. Further, the measurement and the involved accuracy of the third velocity component is discussed.

4.3.1 Specifications of the LDA-system

A two-component back-scatter Laser Doppler Anemometer (TSI 9230/9201) was used. The geometrical situation allowed us to select an optical focal length of 120.5 mm which involved a good signal-to-noise ratio and a relatively large beam intersection angle of 0.380 rad. The resulting measuring volume had dimensions of 0.20 mm in length and 0.04 mm in diameter. Care was taken to properly align the optics. It was verified by projection on a wall with a micro-objective that both volumes of the green and the blue pairs of laser beams coincided at least within 95 %.

The measurement of flow reversal was enabled by the selection of a preshift frequency of 5 MHz which was based on the velocity range of interest. A mixture of atomised glycerol/water seeded the flow (the particle sizes were between 0.1 and 1 μm). A TSI IFA-750 was used for data acquisition. Typical data rates ranged between 300-1000 Hz.

Coincidence windows between 50-100 μs were used to resolve the Reynolds shear stresses $\overline{u'v'}$, $\overline{u'w'}$ and $\overline{v'w'}$. This range of coincidence windows was determined from the averaged residence time of a particle in the measuring volume and the micro time-scales in the flow. The residence time was in the order of magnitude of 50 μs , while the micro time scale was estimated from the channel dimension and the averaged bulk velocity. Further, Reynolds shear stresses were measured with various coincidence windows for several locations in the flow indicating that the measurements were invariant for this range of coincidence windows.

4.3.2 Measurement of the third component

The regular orientation of the probe was perpendicular to the channel wall, allowing velocity measurements in the longitudinal (u -component) and spanwise (w -component) direction of the flow. The flow around a cubical obstruction is highly three-dimensional and, therefore, an accurate analysis of the flow structures is hardly possible without knowledge of the velocity components in all three directions. The easiest way is to make the test facility optically accessible from the sides of the windtunnel but this was not a realistic option in the present situation. First, the focal length of the optics needed to be increased with a factor 3-4 for measurements from the sides. This reduces the signal-to-noise ratio of the LDA signals drastically but further, it gives taller measuring volumes. The latter was not desirable due to the large velocity gradients which were present in the flows in question. Second, in the case of multiple-cube configurations with several rows of cubes (for example, the matrix flow) the cubes located at the sides of the cube in question will obstruct the laser beams and thus prevent a LDA measurement. The laser beams, therefore, needed to enter the set-up from the front, the normal direction y . Two different techniques were developed to measure the third velocity component. The first method was based on a direct measurement of the normal velocity with a novel prism cube. In the second method, the normal velocity was reconstructed from different velocity measurements acquired at different inclination angles of the LDA probe.

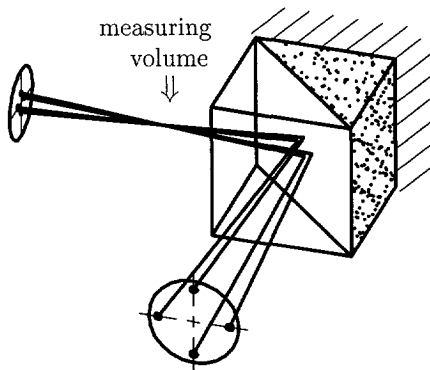


FIGURE 4.33: Schematic of the prism-cube to measure the normal velocity component (v).

Prism-cube

An in-house developed measurement cube was used to measure the normal velocity component directly. The cube consisted of two prisms, one was made of glass and the other was made of bronze with a mirror at the diagonal plane (the hypotenuse). A schematic of this prism cube is given in Fig. 4.33 (the refraction of the laser beams is omitted to simplify the drawing). A photograph of the prism-cube with the laser beams during operation is given in Fig. 4.34. The mirror at the hypotenuse-plane reflects the laser beams at an angle of 90° and creates a measuring volume in the inter-obstacle spacing (in front of the prism cube). For the shown orientation in Fig. 4.33, a traverse of the laser head (LDA probe) in the normal direction results in a traverse of the measuring volume in the streamwise (x) direction. Vertical and horizontal movement of the laser probe equally cause a translation of the measuring volume in the vertical and normal direction, respectively. The reflected measuring volume enables the measurement of the spanwise (w) and the normal velocity (v) components in coincidence mode. A 90° rotation of the whole prism-cube allowed the measurement of the streamwise (u) and the normal velocity components in the region adjacent to the sides of the cube. Measurement of v and w in the wake region was possible with a 180° rotation. The measurement was geometrically constrained by the size of the cube and the intersection angle. Measurements were only possible in a region bound by the side faces of the cube.

Rotation of the laser probe

The second method explored was based on a reconstruction of different velocity measurements for which the optical probe was rotated in the xy -plane. The measured velocity component, oriented perpendicularly to the optical axis of the probe consists then partly of a normal and a streamwise component in the global reference system. A sketch of the measurement situation is given in Fig. 4.35. The inclination angles of α , β and γ refer to the optical axis of the laser probe. Rotation of the probe (or the optical axis) with inclination angle α results in a local reference system in which the velocity component in the local x -direction is measured directly: u_α . The

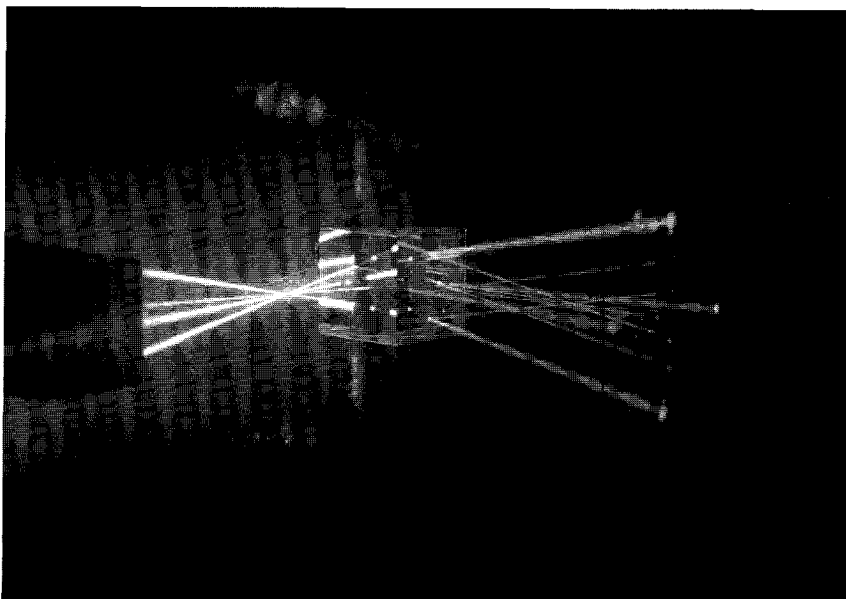


FIGURE 4.34: Photograph of the prism-cube with laser beams during operation. The laser beams are reflected by the mirror at the diagonal plane to create the measuring volume in front of the cube (the left hand side of the picture).

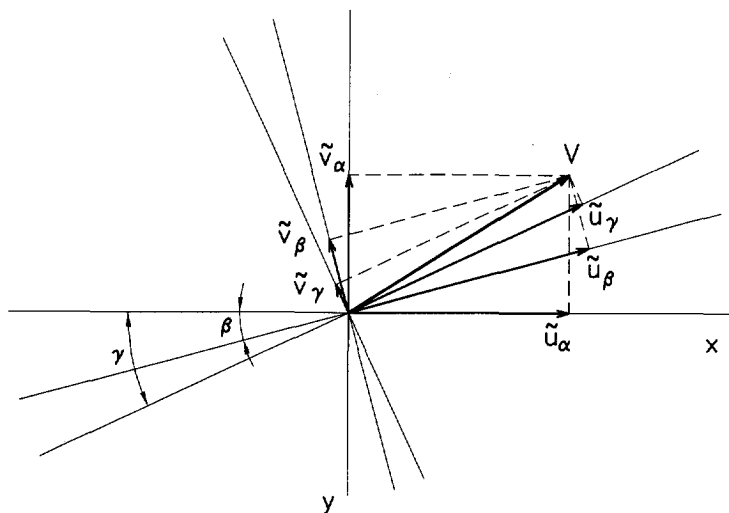


FIGURE 4.35: Schematic of the reconstruction of the third velocity component. α , β and γ refer to the inclination angles of the optical axis of the LDA probe ($\alpha = 0$ in this drawing).

instantaneous velocity u_α can be decomposed according to Reynolds decomposition into a long-term averaged part $\bar{u}_\alpha$ and a fluctuating part u'_α :

$$u_\alpha = \bar{u}_\alpha + u'_\alpha. \quad (4.22)$$

This velocity u_α can then be expressed in terms of the global velocities u and v , which are also Reynolds decomposed:

$$u_\alpha = \bar{u}_\alpha + u'_\alpha = u \cos(\alpha) + v \sin(\alpha) = (\bar{u} + u') \cos(\alpha) + (\bar{v} + v') \sin(\alpha). \quad (4.23)$$

Rotation at a second inclination angle β gives the velocity u_β which is decomposed similarly:

$$u_\beta = \bar{u}_\beta + u'_\beta = u \cos(\beta) + v \sin(\beta) = (\bar{u} + u') \cos(\beta) + (\bar{v} + v') \sin(\beta). \quad (4.24)$$

The long-term averaged velocities $\bar{u}$ and $\bar{v}$, defined in the global coordinate system, are directly resolved from the long-term averaged velocities $\bar{u}_\alpha$ and $\bar{u}_\beta$ according to the next set of coupled equations:

$$\begin{bmatrix} \bar{u}_\alpha \\ \bar{u}_\beta \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ \cos(\beta) & \sin(\beta) \end{bmatrix} \begin{bmatrix} \bar{u} \\ \bar{v} \end{bmatrix}. \quad (4.25)$$

Or, if we write the velocity components $\bar{u}$ and $\bar{v}$ explicitly:

$$\begin{bmatrix} \bar{u} \\ \bar{v} \end{bmatrix} = \frac{1}{\cos(\alpha) \sin(\beta) - \cos(\beta) \sin(\alpha)} \begin{bmatrix} \sin(\beta) & -\sin(\alpha) \\ -\cos(\beta) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} \bar{u}_\alpha \\ \bar{u}_\beta \end{bmatrix}. \quad (4.26)$$

This derivation shows that two measurements at different inclination angles are required to resolve the mean velocities $\bar{u}$ and $\bar{v}$. The situation is different to resolve the fluctuating part in the velocity components. From one single LDA measurement at the inclination angle α , the fluctuation u'_α is expressed as:

$$u'_\alpha = u' \cos(\alpha) + v' \sin(\alpha). \quad (4.27)$$

The square of the fluctuating part, u'^2_α , then becomes:

$$u'^2_\alpha = (u' \cos(\alpha) + v' \sin(\alpha))^2 = u'^2 \cos^2(\alpha) + v'^2 \sin^2(\alpha) + u'v' \sin(\alpha) \cos(\alpha). \quad (4.28)$$

A similar derivation was performed for u'^2_β :

$$u'^2_\beta = (u' \cos(\beta) + v' \sin(\beta))^2 = u'^2 \cos^2(\beta) + v'^2 \sin^2(\beta) + u'v' \sin(\beta) \cos(\beta). \quad (4.29)$$

The measurements for the different inclination angles α and β were performed consecutively. Long-term averaging gives:

$$\overline{u'^2_\alpha} = \overline{u'^2} \cos^2(\alpha) + \overline{v'^2} \sin^2(\alpha) + \overline{u'v'} \sin(\alpha) \cos(\alpha), \quad (4.30)$$

$$\overline{u'^2_\beta} = \overline{u'^2} \cos^2(\beta) + \overline{v'^2} \sin^2(\beta) + \overline{u'v'} \sin(\beta) \cos(\beta). \quad (4.31)$$

As can be seen from these two expressions, a third unknown is introduced by the long-term averaging: the Reynolds shear stress $\overline{u'v'}$. When both velocities u_α and u_β were measured simultaneously (in coincidence), the problem was directly solvable since then $u'v'$ is known for each measurement. However, this is not the case for consecutive measurements. This is a good illustration of the closure problem: the time averaging introduces a new unknown. A third equation, or third measurement at inclination angle γ , is then required to solve the closure problem:

$$\overline{u_\gamma'^2} = \overline{u'^2} \cos^2(\gamma) + \overline{v'^2} \sin^2(\gamma) + \overline{u'v'} \sin(\gamma) \cos(\gamma). \quad (4.32)$$

These three equations were combined to a matrix equation from which the unknowns $\overline{u'^2}$, $\overline{v'^2}$ and $\overline{u'v'}$ were directly resolved:

$$\begin{bmatrix} \overline{u_\alpha'^2} \\ \overline{u_\beta'^2} \\ \overline{u_\gamma'^2} \end{bmatrix} = \begin{bmatrix} \cos^2(\alpha) & \sin^2(\alpha) & \sin(\alpha)\cos(\alpha) \\ \cos^2(\beta) & \sin^2(\beta) & \sin(\beta)\cos(\beta) \\ \cos^2(\gamma) & \sin^2(\gamma) & \sin(\gamma)\cos(\gamma) \end{bmatrix} \begin{bmatrix} \overline{u'^2} \\ \overline{v'^2} \\ \overline{u'v'} \end{bmatrix}, \quad (4.33)$$

$$\begin{bmatrix} \overline{u'^2} \\ \overline{v'^2} \\ \overline{u'v'} \end{bmatrix} = \begin{bmatrix} \cos^2(\alpha) & \sin^2(\alpha) & \sin(\alpha)\cos(\alpha) \\ \cos^2(\beta) & \sin^2(\beta) & \sin(\beta)\cos(\beta) \\ \cos^2(\gamma) & \sin^2(\gamma) & \sin(\gamma)\cos(\gamma) \end{bmatrix}^{-1} \begin{bmatrix} \overline{u_\alpha'^2} \\ \overline{u_\beta'^2} \\ \overline{u_\gamma'^2} \end{bmatrix}. \quad (4.34)$$

Data reduction of the time series of u_α and u_β gave the variances $\overline{u_\alpha'^2}$ and $\overline{u_\beta'^2}$ similarly to the data reduction performed for the LDA measurements with the probe in the regular orientation.

It is noted that the system was operated in coincidence mode during the inclined measurements. Therefore, correlations between the inclined components u_α , etc. and the spanwise velocity w were measured directly in the form of Reynolds shear stresses $\overline{u_\alpha'w'}$, $\overline{u_\beta'w'}$ and $\overline{u_\gamma'w'}$.

It is not obvious to select the inclination angles (α , β and γ) to reconstruct the normal velocity component since the selection is constrained by several aspects. The reconstruction of the velocity is based on measurements which always involve experimental uncertainty. Since only a small part of the normal velocity component is measured for small angles, it is preferable to select a large angle which minimises the uncertainty involved. Another complication is the refraction of the laser beams by the window (a window was needed since the configuration of cubes was placed in a closed wind tunnel, while the Laser probe was located outside the channel). When the probe is oriented perpendicularly to the window, both laser beams are refracted in the same way. This gives only a displacement of the measuring volume which can be seen from the sketch of Absil [1], see Fig. 4.36. The displacement is then expressed as [40]:

$$\Delta z = 1 - \frac{\cos(\theta/2)}{\sqrt{(n_p/n_a)^2 - \sin^2(\theta/2)}}. \quad (4.35)$$

The intersection angle, which determines the fringe spacing, is not changed. In the case that the laser probe is rotated at an angle, the two laser beams in the plane of

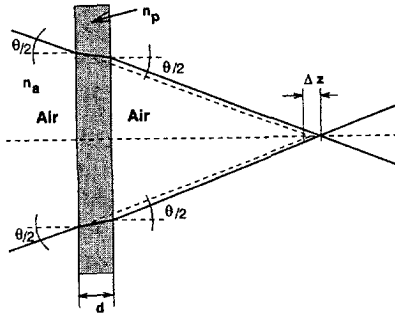


FIGURE 4.36: Schematic of the refraction of two laser beams which pass a window of finite thickness d (from Absil, 1996).

rotation then intersect differently with the glass window which consequently results in differently refracted laser beams. The waists of both laser beams are then shifted differently which implies that the measuring volume is not located at the waist of both laser beams anymore. This causes a fringe gradient in the measuring volume [1, 40]. In other words, a relatively thick window will cause significant misalignment of the optics which has to be paid with the aforementioned fringe gradient and, therefore, artificial turbulence. This argument pleads for a small rotation angle which involves a small difference between the incident angles. A thin sheet has less stiffness than a thick window. Perturbations induced by the turbulent channel flow or by flow periodicities (like the vortex shedding in the wake of the cubes) can cause the window to vibrate. If the amplitude of the induced sheet oscillation is too large, it may also cause misalignment of the optics and, consequently, artificial turbulence.

The optimum solution consisted of a significantly thin window (100 micron) made of a plastic sheet, and rotation angles of -35° , 0° , and 35° . It was estimated that possible sheet oscillations had negligible effects on the accuracy of the measurement for this situation. For the regions in the vicinity of the front-face-channel-floor corner and the rear-face-channel-floor corner, the laser beams were blocked by the cube and, therefore, different sets of inclination angles were selected to overcome these geometrical restrictions. For the front face region, inclination angles of -35° , -20° and 0° were used, for the rear face region, the used angles were 35° , 20° and 0° . A sketch of these angled projections is given in Fig. 4.37.

4.3.3 Experimental accuracy

Absil [1] gave a very detailed and complete analysis of all possible contributions to the uncertainty and errors involved in LDA measurements. In this thesis, we will discuss only the most significant errors involved in the experimental results.

The experimental uncertainty will be determined for the velocity measurements with the LDA probe in the standard orientation, which includes u , w , and also u_α , u_β , etc. In the next section, this analysis is extended to estimate the uncertainty for the normal velocity component (v).

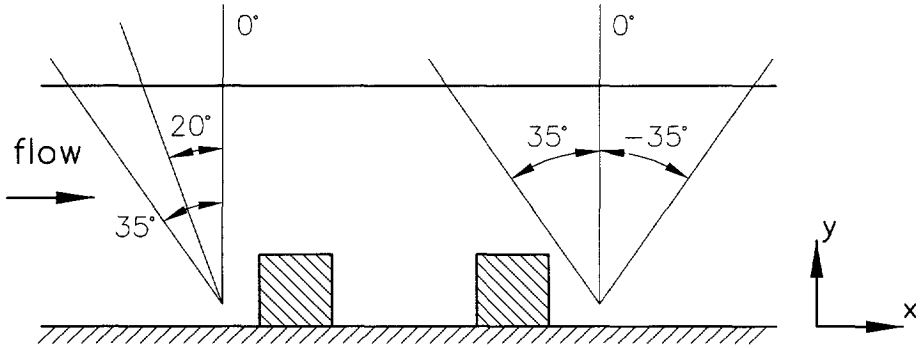


FIGURE 4.37: Sketch of the different inclination angles of the optical axis of the laser probe to measure the third velocity component.

Statistical errors

The first and higher order statistical quantities were mostly determined from 10000 samples which gave a statistical uncertainty of less than 0.02 m/s. This value was determined from an existing dataset of 1,000,000 realisations from which first and higher order moments were calculated with different sample sizes, varying between 100 and 1,000,000. It was concluded that the number of 10,000 samples was sufficiently large to obtain statistically independent first and higher order moments.

Errors related to the system

Several different errors contribute to the total experimental uncertainty during the route from scattered light to the Doppler frequency. An extensive analysis of the inaccuracies involved is reported in, for instance, Absil [1], Durst et al. [40] and Drain [41]. The velocity can be written in terms of the Doppler frequency:

$$|u_x| = \frac{f_D \lambda}{2 \sin(\theta/2)}, \quad (4.36)$$

where f_D is the Doppler frequency, θ is the intersection angle between the laser beams and λ is the wavelength of the laser light. With the standard single uncertainty analysis [109, 110], the experimental uncertainty in $|u_x|$ becomes:

$$\left(\frac{d|u_x|}{|u_x|} \right)^2 = \left(\frac{df_D}{f_D} \right)^2 + \left(\frac{d\lambda}{\lambda} \right)^2 + \left(\frac{d\theta}{4 \tan(\frac{1}{2}\theta)} \right)^2, \quad (4.37)$$

with $d\theta^2$ in rad.

The Doppler frequency is derived from the detector frequency according to $f_{det} = f_D + f_{preshift}$. The uncertainty in the Doppler frequency reads:

$$\left(\frac{df_D}{f_D} \right)^2 = \left(\frac{df_{det}}{f_{det}} \right)^2 + \left(\frac{df_{preshift}}{f_{preshift}} \right)^2. \quad (4.38)$$

Several methods exist to determine the intersection angle. Absil [1] examined four techniques. A distinction between direct and indirect methods is usually made. A direct method is based on the relation between the Doppler frequency and an *a priori* known velocity field, for example, a rotating disk or a well calibrated velocity field [1]. The indirect method is based on the measurement of the wavelength and the intersection angle. The latter is usually measured with a theodolite or the graphic method. Absil [1] and other researchers concluded that the indirect method with use of a theodolite provided accurate and reliable calibration intersection angles. This method was used here. The measurements with a theodolite gave an intersection angle of 0.383 rad which was accurate within 0.001 rad. This accuracy contributes to about 0.3% error in the velocity.

The green and the blue laser beams of the used Argon laser had wavelengths of 514.5 and 488 nm, respectively. The uncertainty in the wavelength was about 0.1 nm which did not contribute significantly to the error in the measured velocity.

The accuracy of the measured Doppler frequency is more complicated. As already pointed out, the error in it consisted of the errors caused by the selected preshift and the detector frequency. The different contributions to the uncertainty in the detector frequency are summarised as follows:

- Finite time resolution: The detector frequency f_{det} is determined from an auto-correlation function in case of the TSI IFA 750 processor. In the TSI-manual, an accuracy of 0.05% is claimed which corresponds to 0.003 MHz in the case of a 6 MHz detector signal.
- The bragg-cell oscillator used in the TSI-system operates at 40 MHz with a deviation of about 0.1%. This uncertainty is carried through the circuit of downshift filters. When a 5 MHz downshift frequency is selected, the additional uncertainty is ± 0.005 Mhz. For a measured signal of 6 MHz, the corresponding Doppler frequency of 1 MHz contains an uncertainty of 0.005 Mhz, which corresponds to an error in the velocity of approximately 0.5%.
- Although an auto-correlation function is used, high frequency noise can still affect the determination of the detector frequency. The good quality of the Doppler burst as encountered in the present study eliminates this error.

Errors caused by misalignment of the optics

- Fringe gradients: out of waist intersections of the laser beams result in a gradually varying fringe spacing, which is known as a fringe gradient [1, 40]. Particles, moving with the same velocity (for example in a laminar flow) which pass the measuring volume at different locations (the measuring volume has a finite size) will experience a different fringe pattern. This leads to different Doppler frequencies and thus to different velocities. Time averaging of these velocity time series will introduce artificial turbulence which is in fact noise. In the case of a turbulent flow, the physical turbulence level will be significantly corrupted by this effect which obviously will lead to misinterpretation of the physics. Proper alignment

minimises this effect. LDA measurements in a laminar flow of uniform velocity showed that the turbulence intensities were smaller than 1%, which is sufficiently low to verify the negligible influence of fringe gradients.

- Fringe chirps: a fringe chirp is also caused by misalignment of the laser beams, when both waists of the laser beams are located in opposite side of the intersection point. The resulting fringes are basically parallel, however with varying spacing. The velocity measurements in the laminar channel flow, see above, verified also the absence of this misalignment.

Errors caused by bias-effects

- Velocity or sampling bias is the most important among the bias effects. It is the effect that for a uniformly seeded turbulent flow, periods of high velocity are benefited to periods of lower velocity due to wrong statistical averaging of the velocity samples. This wrong averaging has the consequence that the calculated velocities are significantly larger than the real time-averaged velocities. For equidistantly sampled processes, the arithmetic-averaged results are not hampered by velocity bias. This is also the case for some other irregular sampling methods as long as the sampling process is independent on the physical process. Arithmetic averages of the quasi Poisson-distributed LDA-realizations are indeed hampered by velocity bias since the detection of a particle is related to the Poisson-distributed particles in the flow which makes the sampling process dependent on the physical process.

Several detection methods are proposed in the literature to indicate the presence of velocity bias in the LDA measurements. The interarrival time between consecutive realisations is a good measure for possible velocity bias. In the case of significant velocity bias effects, fast particles will pass more frequently the measuring volume than slower particles will do. Thus, relatively small interarrival times are more often encountered in the velocity time series. The averaged interarrival time then decreases with increasing velocity. In the case of an unbiased signal, the relation between the averaged interarrival time and the velocity is more or less constant. This detection method was firstly introduced by Adams [2]. An other detection method uses the Poisson distributions of the particles in the flow. The probability density function of the interarrival time between consecutive realisations ($pdf(\Delta t)$) can be described as:

$$pdf(\Delta t) = \nu e^{-\nu \Delta t}, \quad (4.39)$$

where ν is the mean sampling rate, and Δt is the interarrival time. Taking the exponent of relation 4.39 gives a linear relation: $\ln(pdf(\Delta t)) = \ln(\nu) - \nu \Delta t$. Deviations from the linear profile indicate that bias effects were present.

Numerous velocity bias correction methods have been reported in the literature which can be divided into two main categories. The first category covers the sampling techniques, which provide a regular sampling of the times series. Examples include the sample-and-hold method, the saturated processor and the controlled processor which are all discussed in Edwards [43] and Absil [1]. The methods

of other category use weighting factors to obtain a better estimate of the time-averaged quantities. The weighted mean velocity $\bar{u}$ and higher order moments $\overline{u^n}$ are expressed as the next finite summations, illustrated for the streamwise component u :

$$\bar{u} = \frac{\sum_{i=1}^N u_i \omega_i}{\sum_{i=1}^N \omega_i}, \quad \overline{u^n} = \frac{\sum_{i=1}^N (u_i - \bar{u})^n \omega_i}{\sum_{i=1}^N \omega_i}, \quad (4.40)$$

where ω_i is a proper weighting factor, u_i denotes an instantaneous velocity measurement (which is the sum of an average $\bar{u}$ and a fluctuating part u'), N is the number of samples and n denotes the order of the moment. Weighting factors are, among others, the inverse velocity, the residence time and the interarrival time. If a constant weighting factor is selected ($\omega_i = 1$ for all i) the arithmetic average is obtained. The interarrival time and the inverse velocity weighting factors are explained in more detail:

- Interarrival time.

The time-averaged velocity can be approximated by the finite summation over a period T :

$$\bar{u} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T u(t) dt \approx \frac{\sum_{i=1}^N u_i \Delta t_i}{\sum_{i=1}^N \Delta t_i}, \quad (4.41)$$

where Δt_i is the interarrival time weighting factor. Typical velocity time series, acquired with a LDA system operated in the burst detection mode, are not equidistant in time (Poisson distributed). This implies that also relatively large interarrival times occur in the dataset. The above approximation is, therefore, only valid if the interarrival time is much smaller than the micro time scales of the studied flow [43]. This requires relatively large averaged data rates, i.e. $Tu \times \nu > 10$, where Tu is the integral time scale of the flow and ν is the averaged sampling rate [1]. The integral time scale was estimated from the ratio of a characteristic velocity V and a characteristic length L . Typical values for the flow around the cubes are: $V = 1$ m/s and $L = 15$ mm, the cube size. This gives an integral time scale of about 0.015 sec. The integration of the autocorrelation function (see expression 2.28), described in detail in Absil [1], yielded also integral time scales in the order of magnitude of 0.015 sec. To satisfy the constraint $Tu \times \nu > 10$, the averaged sampling rate should be of the order of magnitude of 500. Since data rates were roughly between 400-1000, some assessment of the interarrival time weighting is needed. This is discussed in the next section.

Different choices of the interarrival time are possible. $\Delta t_i = t_{i+1} - t_i$ was proposed by Barnett and Bentley [15] and resembles the sample-and-hold estimator. Two estimators based on other possible time discretisations, $\Delta t_i = 0.5 (t_{i+1} - t_{i-1})$ and $\Delta t_i = t_i - t_{i-1}$, were also examined.

- Inverse velocity.

The averaged number of particles N passing the measuring volume in a time period ΔT can be approximated with $N = |V|A_p\rho_p\Delta T$, in which $|V|$ is the magnitude of the instantaneous mean velocity vector, assumed constant in the measuring volume, A_p is the cross-sectional area normal to the velocity vector and ρ_p is the particle density (number of particles per unit volume in the uniformly seeded flow). For a constant particle density, $N/\Delta T$ is proportional to $|V|A_p$ and, therefore, $1/|V|A_p$ is a proper weighting factor to correct for velocity bias. For an ellipsoidal measuring volume, the cross-sectional area A_p is not constant but varies with the direction of the instantaneous velocity vector. The inverse velocity weighting factor then becomes $1/\sqrt{u_i^2 + w_i^2 + (d_m/d_l)^2 v_i^2}$ in which d_m and d_l are the diameter and the length of the measuring volume, respectively. In the case of a two-dimensional LDA system, the third instantaneous velocity component is unknown. When d_m/d_l is much smaller than 1, which is usually the case for long measuring volumes (due to a small intersection angle), the term $(d_m/d_l)^2 v_i^2$ is negligible compared the other two. In that case, the weighting factor consists of only the velocity components u_i and w_i and is, therefore, called 2D. The measuring volume in our measurements was rather short and thick, the aspect ratio was $d_m/d_l \approx 0.2$. Therefore, the third velocity component was estimated according to Nakayama [121] who suggested to take the average of the variances in the x - and z -direction $\sigma_v^2 = \frac{1}{2}(\sigma_x^2 + \sigma_w^2)$ to approximate the third unknown squared velocity component: $\omega_i = 1/\sqrt{u_i^2 + w_i^2 + (d_m/d_l)^2 \sigma_v^2}$. This estimator is referred to as 2D+.

- Amplitude bias is related to the fact that the signal-to-noise ratio in the LDA signal is dependent on the Doppler frequency: high-frequency Doppler bursts have generally smaller amplitudes than low-frequency Doppler bursts. When the Doppler signals are of the order of magnitude of the noise level of the system, the bursts are not recognised and validated by the processor. This causes that low-frequency bursts are more often validated than high-frequency bursts which consequently leads to biased quantities. For the present study, this type of bias is of minor importance due to the good signal-to-noise ratio of the LDA signals as a result of the small focal length. The effect of amplitude bias is estimated to be less than 0.5% in the velocity measurements.
- Fringe bias can be encountered when a minimum number of fringe crossings is required to be recognised as a valid realisation. Particles which move through the centre might be benefited to particles crossing the edges of the measuring volume. When the preshift frequency is at least twice the dominant Doppler frequency, this effect is negligible [1].
- Velocity gradient bias is caused by the finite dimensions of the measuring volume. When a steep velocity gradient is encountered by the measuring volume, particles with different velocities will add artificial turbulence to the measurements. Durst et al. [40] proposed an estimate of the additional mean velocity and variance,

assuming a Gaussian detection probability function across the measuring volume and approximating the velocity distribution by a Taylor expansion:

$$\bar{u} - \bar{u}_{center} = \frac{d_m^2}{32} \left(\frac{\partial^2 u(y)}{\partial y^2} \right), \quad (4.42)$$

$$\overline{u'^2} - \overline{u'^2}_{center} = \frac{d_m^2}{16} \left(\frac{\partial u(y)}{\partial y} \right)^2, \quad (4.43)$$

where $\bar{u}_{center}$ is the averaged velocity and $\overline{u'^2}_{center}$ is the Reynolds normal stress at the centre of the measuring volume. The small measuring volume used in the present study minimised the effects of velocity gradient bias. The largest occurring velocity gradients were in the order of magnitude of 2000 1/s. The deviations in the mean and variance of the velocity components were estimated with the use of expressions 4.42 and 4.43 and were about 2%.

- The size of the seeding particles is important for accurate LDA measurements. Particles which are significantly larger or smaller than the fringe spacing in the measuring volume will not generate detectable Doppler bursts but generally will contribute only to the background light intensity. Besides this, when the particles are too large to correctly follow the flow (particle slip) the velocity measurements are not accurate. This phenomenon is known as particle lag bias. The size of the particles was Gaussian distributed in the range between 0.1 and 1 micron. This distribution was determined by Opstelten [128].

4.3.4 Evaluation of the experimental uncertainty

The above overview provides already insight in the significance of the most common causes of uncertainty involved in LDA measurements. This paragraph supports that discussion with a quantification of the overall accuracy. The used LDA-system generated signals with a good signal-to-noise-ratio (SNR) although the system was operated in the back-scatter mode. This good signal quality was, among others, caused by the relatively small focal length of the optical system (i.e. 12.05 cm), the spherical seeding particles which reflected the incident laser light pretty well, and the good optical quality and the proper alignment of the system. A good SNR eliminates several sources of inaccuracy related to data acquisition and the processing of the LDA signals.

A quantitative impression of the accuracy of the used LDA system, including signal processing, was obtained from measurements in a laminar uniform flow. A laminar flow was established using smooth channel walls and a sufficiently small Reynolds number. The resulting velocity measurements had small-banded and symmetric probability density functions. The standard deviation was less than 1%. As already mentioned, such a small turbulence intensity denies the existence of numerous misalignment errors.

For turbulent flows, the error analysis is more complicated. The flow around the protruding cubes is characterised by different types of flow: separation, reattachment, recirculations, shear flow. Thorough diagnostics of the LDA-signals is,

therefore, required which had to be performed in all mentioned types of flow. Velocity bias effects are one of the most important contributions to the experimental inaccuracy of LDA measurements in a turbulent flow where large velocity fluctuations are typically present. Therefore, this issue deserves special attention.

Variation of the particle density

The influence of the particle density on the distribution of the interarrival time was examined for the matrix flow at the location $x/H = 1.3$, $y/H = 0.5$ and $z/H = 0.5$. It is assumed that the particles were uniformly distributed in the flow. The seeding density was varied for different experiments. This resulted in variable averaged interarrival times ($\overline{tbd}$) which are plotted in Fig. 4.38. The particle density was largest at run 1, thus giving a small averaged interarrival time (or a high average sampling rate). The opposite applied for run 13, in which the particle density was decreased drastically (low average sampling rate). The corresponding probability density functions of the interarrival time are shown in Fig. 4.39 in log coordinates (the pdf for run 1 was almost identical to the pdf for run 2 and is, therefore, not shown). According to the Poisson distribution of the particles in the flow, the shape of the pdf is linear on log scale. For most of the experiments (runs) shown in Fig. 4.39, this Poisson distribution is obvious. Deviations from this behaviour are observed for the experiment with the largest averaged interarrival time, namely run 13. The tail in the pdf for run 1-4 is probably due to an increased noise level.

The pdf for experiment 13 exhibits more or less a concave profile; this deviation from the linear shape is an indication of velocity bias (see the above discussion). Particles with a relatively high velocity are more often validated than the slower particles. This observation leads to the suggestion that velocity bias is related to the particle density in the flow. Apparently, for a significantly low data rate (as in run 13) the interarrival time of the particles is not longer Poisson distributed. This is in contradiction with the findings from Absil [1]. It was ensured that all LDA experiments shown in this thesis were performed with data rates between 400-1000 which corresponded to the experiments in the intermediate regime.

It is interesting to note that the slope of the linear part of the histogram in log coordinates becomes steeper with increasing particle density (or sampling rate). This is in agreement with expression 4.39 from which the linear relation $\ln(pdf(\Delta t)) = \ln(\nu) - \nu\Delta t$ was deduced: an increase in the sampling rate causes indeed a steeper slope.

Multiple burst validation is the effect that long Doppler bursts are more than once validated. For example, if a burst consists of 100 cycles, and if the validation criterion is based on 30 cycles, then theoretically it is possible that one burst is validated three times. The result is a corrupted velocity time series, which will obviously give wrong statistics, i.e. first and higher order moments. Multiple-burst validation can be detected from the probability density function of the interarrival time. If multiple validation of one burst occurs, the interarrival time between two consecutive velocity measurements is almost zero. If this happens frequently, this will give a lob close to zero interarrival times. For the results presented in this

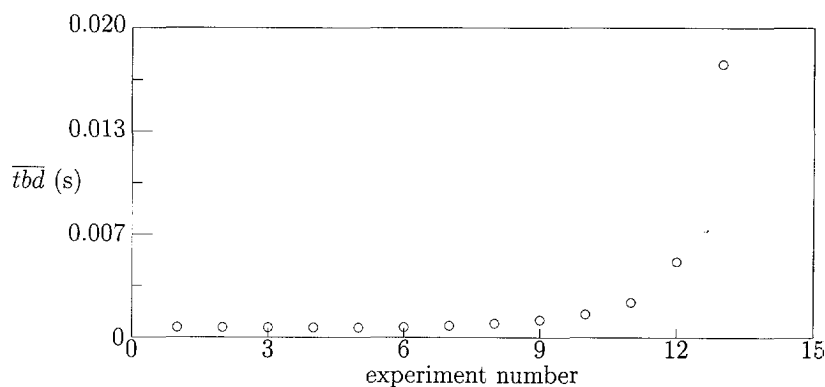


FIGURE 4.38: Averaged interarrival time versus the number of the experiment (the index).

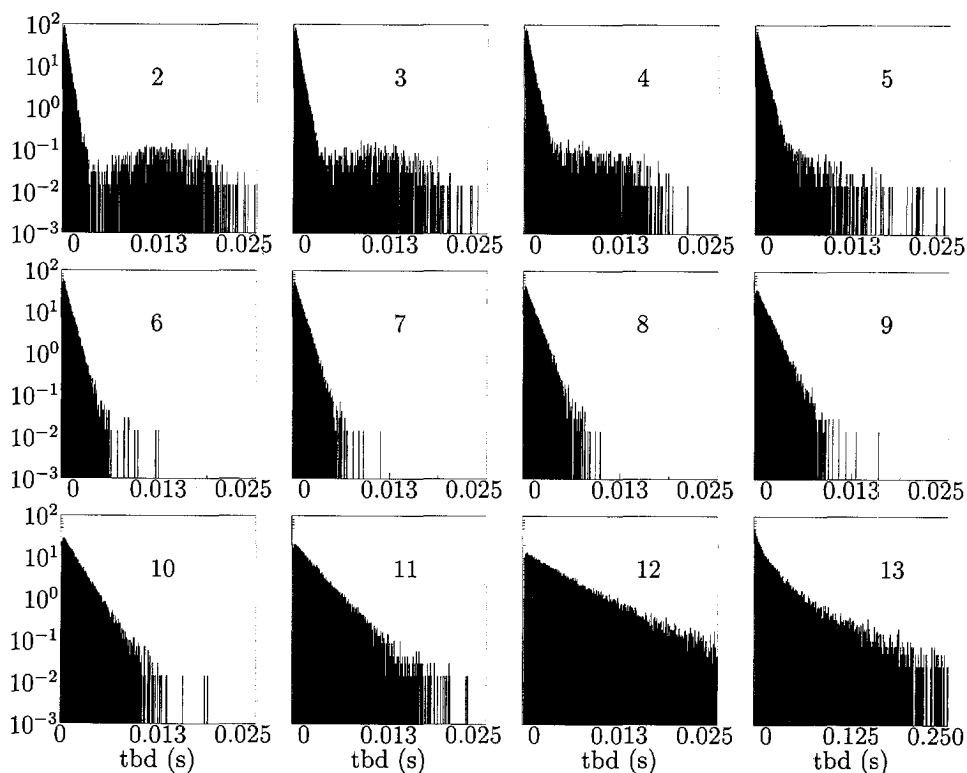


FIGURE 4.39: Probability density functions of the interarrival time corresponding to different averaged interarrival times $\overline{tbd}$: starting from left to right at the upper left pdf, and ending at the lower right pdf, the pdfs refer to experiment number 2 to 13 as indicated in Fig. 4.38 (the pdf of the first experiment is omitted since it was more or less similar to that of the second experiment).

thesis, it was ensured that proper settings of the data acquisition system avoided multiple-burst validation ('one measurement per single burst' mode). Evidence of this is found in the shown pdfs in Fig. 4.39; the pdfs show no significant increase around zero interarrival times.

Velocity bias

The probability density functions of the streamwise velocity component are discussed along a traverse in the normal direction at $x/H = -0.3$ and $z/H = 0$ for the cube matrix. This profile starts at the channel floor, just in front of the cube, passing the turbulent inter-obstacle cavity, the core flow region and ends at the opposing wall. Pdfs of the streamwise velocity are given in Fig. 4.40 for 25 stations along the traverse. The stations y/H are indicated in the figure and correspond to the profile of the averaged velocity given in Fig. 4.41 (the figure shows the comparison between different weighting methods which is discussed later). The pdfs are based on 10000 data points; the velocity in each of the 25 velocity bins is based on approximately 400 data points. The (*)-markers refer to the mean interarrival time, normalised with the maximum averaged value found, as a function of the velocity of the particles passing the measuring volume.

Fig. 4.40 shows that the pdfs in the core region of the flow (Fig. 4.40e-h) are Gaussian shaped. For these stations, the interarrival time of the particles is more or less constant or, in other words, independent of the velocity of the particles. A negative slope in the interarrival time is found for the pdfs at the stations close to the opposing channel wall (Fig. 4.40a-d). This decrease towards the higher velocities indicates that faster particles pass more often the measuring volume than the slower particles do. A similar observation is made for the pdfs in the inter-obstacle cavity (Fig. 4.40m-t) and close to the channel floor (Fig. 4.40u-y).

We notice that the turbulence intensity, in terms of the Reynolds normal stresses $\overline{u'^2}$, is rather small in the core region for the corresponding station at $x/H = -0.3$ and $z/H = 0$ (see Fig. 4.42, this figure also shows the comparison between different weighting methods but the profiles are qualitatively similar), while it increases towards both channel walls with a pronounced maximum at $y/H \approx 1/3$. The rather low turbulence level in the core region explains the constant relation between the mean interarrival time and the velocity in the flow (see Fig. 4.40e-h). The velocity fluctuations are moderate and, thus, bias effects caused by fast alternating velocity differences are less pronounced. The opposite is found for the wall regions. The relatively large velocity fluctuations at these locations in the flow, causing the observed high levels of turbulence, give rise to differences between periods of low and high velocities and thus to the occurrence of velocity bias.

Bias effects become even more pronounced for areas where the averaged velocity approaches zero. The velocity distributions become more or less bimodal since particles hardly pass the measuring volume during periods when the instantaneous velocity approaches zero. This is obvious from the last two pdfs in Fig. 4.40, namely x and y , for which the averaged velocity is close to zero. The mean interarrival time decays towards the higher (absolute) velocities, causing the mountain-type of

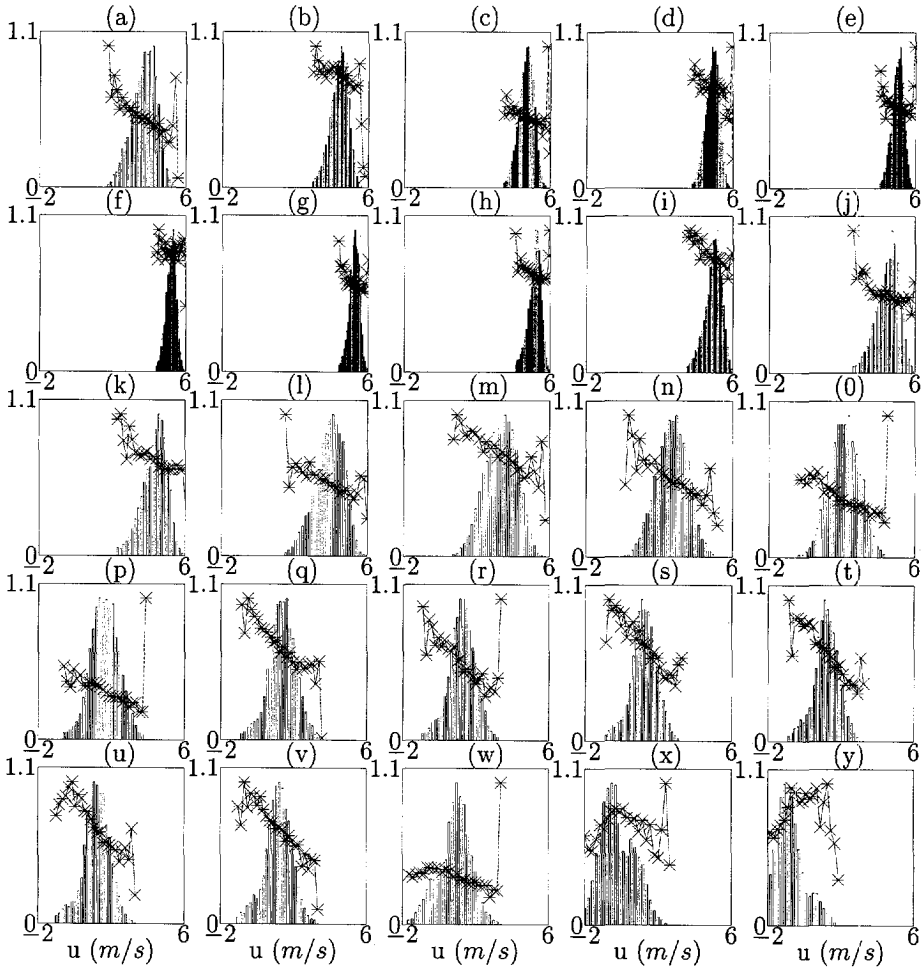


FIGURE 4.40: Probability density functions of the streamwise velocity for different stations along a traverse in the normal direction at $x/H = -0.3$ and $z/H = 0$. The normal coordinates are $y/H = 50/51$ (a); $48/51$ (b); $46/51$ (c); $44/51$ (d); $42/51$ (e); $38/51$ (f); $34/51$ (g); $30/51$ (h); $26/51$ (i); $22/51$ (j); $21/51$ (k); $19/51$ (l); $17/51$ (m); $15/51$ (n); $13/51$ (o); $11/51$ (p); $9/51$ (q); $8/51$ (r); $7/51$ (s); $6/51$ (t); $5/51$ (u); $4/51$ (v); $3/51$ (w); $2/51$ (x); $1.5/51$ (y). The (*)-markers refer to the mean interarrival time per velocity bin.

relation. It is important to note that this bimodal behaviour was not caused by possible coherent structures in the flow, for example the quasi-periodic oscillations in the flow as caused by the vortex shedding from the lateral faces of the upstream cube. Power density spectra, calculated from the velocity time series, did not exhibit an intensification at the predominant frequency of vortex shedding. It is, therefore, concluded that the bimodal behaviour was caused by velocity bias effects.

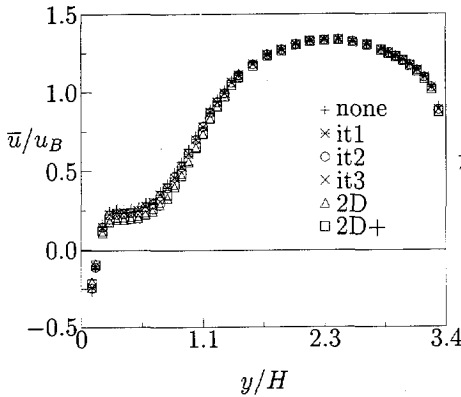


FIGURE 4.41: Profile of the mean velocity at $x/H = -0.3$, $z/H = 0$ for three bias correction methods.

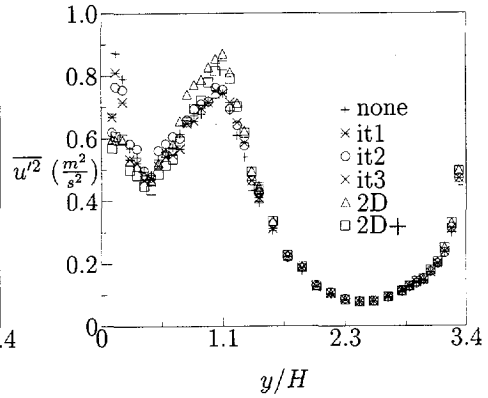


FIGURE 4.42: Profile of the Reynolds normal stresses $\overline{u'^2}$ at $x/H = -0.3$ and $z/H = 0$ for three bias correction methods.

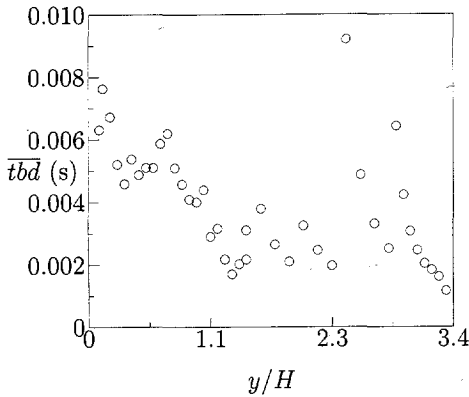


FIGURE 4.43: Mean interarrival time as a function of the normal coordinate y at $x/H = -0.3$ and $z/H = 0$.

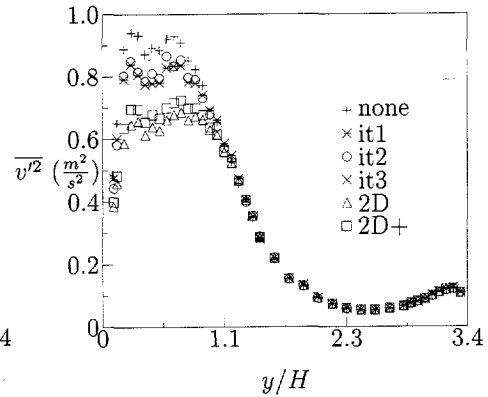


FIGURE 4.44: Profile of the Reynolds normal stresses $\overline{v'^2}$ at $x/H = -0.3$ and $z/H = 0$ for three bias correction methods.

The mean interarrival time of the measurements at the discussed stations, at $x/H = -0.3$ and $z/H = 0$ given in Fig. 4.43, shows an increase towards the channel floor ($y/H = 0$). To explain this shape, we need to consider the velocity fluctuations in the spanwise direction as well. These are represented as the Reynolds normal stresses $\overline{v'^2}$ in Fig. 4.44. While the mean spanwise velocity $\bar{v}$ is zero in the symmetry plane $z/H = 0$, the fluctuations increase significantly in the vicinity of the cube wake. The corresponding high turbulence level causes biased velocity histograms (they become bimodal in the worst case), and a strong reduction in the sampling rate.

Both the inverse velocity and the interarrival time weighting were assessed to obtain a suitable quantification of the error made when no correction is applied. Although this analysis was performed for several locations, the measurements along

the traverse in the normal direction at $x/H = -0.3$ and $z/H = 0$ (matrix flow) are used. The different correction methods are compared for the mean velocity $\bar{u}$ and the Reynolds normal stresses $\overline{u'^2}$ and $\overline{v'^2}$, given in Figs. 4.41, 4.42 and 4.44, respectively. For this profile, the average sampling rate ranged between 500 in the core region and 150-200 in the vicinity of the cubes. The arithmetic average (in which the weighting factor is unity for all realisations!) is indicated as (+). The results obtained with the three considered interarrival time corrections, $\Delta t_i = t_{i+1} - t_i$, $\Delta t_i = 0.5 (t_{i+1} - t_{i-1})$ and $\Delta t_i = t_i - t_{i-1}$, are indicated with (it1, *), (it2, o) and (it3, x), respectively. The results from the three interarrival time weighting schemes differed less than 0.5% from each other. In the case of sufficiently small interarrival times, or large data rates, no differences in discretisation are expected. Therefore, we limit the discussion only to the sample-and-hold estimator ($\Delta t_i = t_{i+1} - t_i$). The inverse velocity corrections, 2D and 2D+, are also indicated in the Figs. 4.41, 4.42 and 4.44.

An examination of the figures learns that the differences between the corrected results (for all the correction methods studied) and those obtained from arithmetic averaging are negligible in the core region. This is expected since in this region no significant bias effects were detected from the probability density functions and the plots of the averaged interarrival time (Fig. 4.40). Differences between the corrected and the uncorrected results become distinguishable in the region close to the opposing channel wall and in the vicinity of the cube, the inter-obstacle wake ($y/H < 2$), where the turbulence level significantly increased. This is also according to the observations made from the velocity histograms. The pronounced differences between the interarrival time correction (either one of the three discretisations shown) and the inverse velocity weighting are most likely due to the somewhat low data rate in these regions to allow the application of the interarrival time correction ($Tu \times \nu > 10$). Because of the relatively small aspect ratio of the measuring volume, the effect of the third velocity component on the particle distribution in the measuring volume is not negligible, and its contribution to the velocity vector should not be underestimated. The 2D+ weighting method differed, therefore, markedly from the 2D approach. This effect becomes more pronounced in the turbulent region, where also large instantaneous normal velocities were found.

From the aforementioned analysis it can be concluded that severe velocity bias effects are most likely to occur at locations where the turbulence intensities are significantly large ($Tu > 5\%$). To support this conclusion, the velocity field was calculated from the LDA measurements with the inverse velocity bias weighting method 2D+ and without weighting (from an arithmetic average). This analysis was performed for the flow over two cubes in the off-axis tandem configuration (the streamwise distance was $S_x/H = 4$, the spanwise distance was $S_z/H = 1$) in a fully developed channel flow at a Reynolds number of $Re_H = 3900$ (see chapter 7). The plane considered is at half the cube height ($y/H = 0.5$) and contains the streamwise (u) and spanwise (w) velocity components. The difference between both averaging methods (2D+ and arithmetic) is expressed in the velocity difference and scaled with the

magnitude of the velocity vector in that plane:

$$\Delta u = (u_{2D+} - u_N) / \sqrt{u_N^2 + w_N^2}, \quad (4.44)$$

$$\Delta w = (w_{2D+} - w_N) / \sqrt{u_N^2 + w_N^2}, \quad (4.45)$$

where the subscripts $2D+$ and N refer to the inverse velocity and arithmetic averaging, respectively. The arithmetic averaged velocity vector is selected for the normalisation (note that this is an arbitrary selection). These velocity deficits are interpreted as a vector from which the magnitude M is calculated:

$$M = \sqrt{\Delta u^2 + \Delta w^2}. \quad (4.46)$$

An iso-value plot of M is given in the upper plot of Fig. 4.45. The corresponding plot of the turbulent kinetic energy k , based on the two measured Reynolds normal stresses ($k = 0.75 (\overline{u'^2} + \overline{w'^2})$) is given in the lower plot of Fig. 4.45. From the comparison between both plots we learn that the largest differences between the weighted and the unweighted velocities (large values of M) occurred at locations where the turbulent fluctuations (turbulence intensities) were large. Note that the scaling factor explodes for zero velocity vectors which is the case in the reattachment region.

Experimental accuracy of the normal velocity component

The two methods to measure the normal velocity component introduce additional experimental error. This is highlighted in the next paragraphs.

Prism-cube. The prism-cube was constructed from two prisms which were glued together at the hypotenuse. This interface, oriented at an angle of 45° , reflected the incident laser beams at an angle of 90° . When the glass prism, through which both laser beams passed, is a perfect perpendicular triangle (i.e. the two side faces of the prism are exactly perpendicular to each other) the paths of both laser beams through the glass are equal. However, a small deviation in this 90° angle will cause that the laser beams have slightly different path lengths while passing through the glass. Since the laser beams are then refracted differently, the waist of the individual laser beams will be shifted differently. The resulting misalignment of the measuring volume may cause fringe gradients which attribute to artificial turbulence [1]. The glass-prism with its mirror at the hypotenuse, a deposited metal sheet, was precisely manufactured. The angle and parallel tolerances were better than 0.01° and 0.01 mm, respectively. Expression 4.35 was used to estimate the displacement difference between both shifted waists. If the path length difference of both laser beams through the glass is estimated to be at maximum 0.01 mm, a shift of less than $3.6 \mu\text{m}$ exists if the angle of incidence was 10° . This error is, therefore, estimated to be less than 0.1%.

Further, the glass prism was carefully attached to the bronze prism to create the prism-cube. Differences in the glue layer thickness may cause that the interface

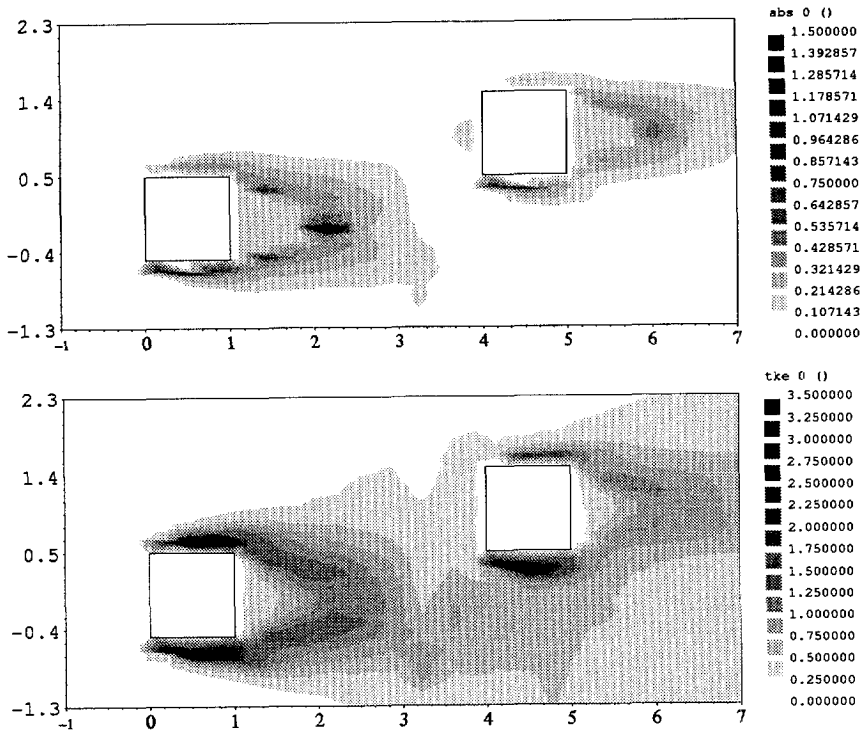


FIGURE 4.45: Upper plot: iso-value plot of the scaled magnitude of the velocity difference vector M for the flow over two cubes in an off-axis arrangement ($S_x/H = 4$ and $S_y/H = 1$) at the plane $y/H = 0.5$ ($Re_H = 3900$). The lower plot shows the corresponding iso-value plot of the turbulent kinetic energy k .

mirror is not exactly oriented at an angle of 45° with respect to the channel floor. Since this causes a translation of the total measuring volume (both laser beams are shifted equally), only a spatial error is introduced. This spatial error is estimated to be less than 0.2 mm.

In order to obtain a quantifiable validation of this estimation, long-term velocity and Reynolds normal stresses data obtained with the prism-cube were compared with measurements obtained with the LDA set-up in the standard orientation. For the prism cube oriented in the second quadrant (a rotation of the cube of 90° with respect to the x -axis, the measuring volume was then projected to the upper side face of the cube), the streamwise and the normal velocity components were measured in coincidence. The streamwise component was compared with direct measurements of the streamwise velocity. Traverses at $x/H = 0.5$ and $y/H = 0.5$ in this quadrant of $\bar{u}$ and u'^2 are compared in Fig. 4.46 for the cube placed in the spatially periodic matrix flow ($S_x/H = S_z/H = 4$). Shown are two sets of data measured with the prism-cube (traverses indicated with $\circ$ and $\times$) and one profile was obtained with

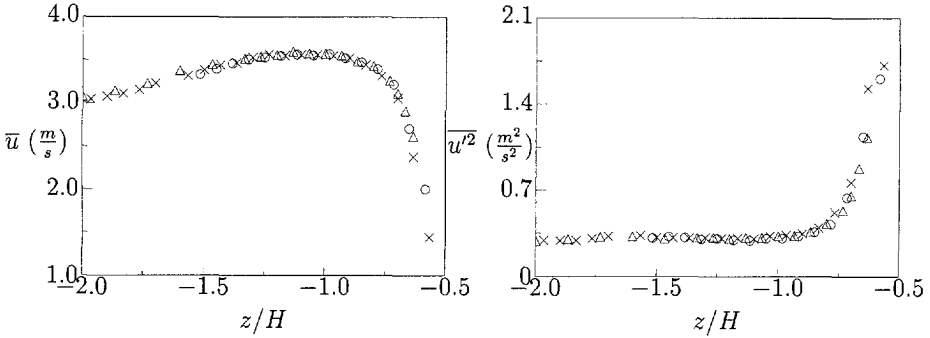


FIGURE 4.46: Comparison between measurements of $\bar{u}$ (left figure) and $\overline{u'^2}$ (right figure) performed with the prism-cube ($\circ$ and $\times$, one profile is a reproduction) and with the cube in the regular orientation (Δ) at $x/H = 0.5$ and $y/H = 0.5$ for the matrix flow ($S_x/H = S_z/H = 4$).

the regular LDA-probe orientation (Δ). The differences between the results of both methods are negligible. This confirms the high accuracy of the prism-cube measurements. The good agreement between the results (even at locations of large velocity gradients) illustrates further the good reproducibility of the measurements.

Reconstruction from the inclined measurements. It was already mentioned that the inclination angles were selected according to certain criteria like the refraction of the laser beams, the optical accessibility of the measurement domain, etc. The method is based upon the principle that a part of the normal velocity is measured when the optical laser probe is used at an inclination angle. For the angles involved (sets of 35° , 0° , -35° and 35° , 20° and 0° and -35° , -20° and 0°) only a small part of this normal velocity can be measured. The reconstruction of the u and v component is based on the subtraction of two velocity components, each scaled with a coefficient which depends on the inclination angles used (see equation 4.26). When the subtracted values are of the same order of magnitude, the difference will be small and, therefore, the reconstruction is very sensitive to experimental error.

The experimental uncertainty in the values of $\bar{u}_\alpha$, $\bar{u}_\beta$, $\overline{u'^2}_\alpha$, $\overline{u'^2}_\beta$ and $\overline{u'^2}_\gamma$ was obtained from the foregoing analysis. Because a very thin sheet was used through which the inclined laser beams entered the wind tunnel section, the measuring volume was still well aligned. Misalignment of the measuring volume, therefore, does not contribute to additional error in the inclined measurements. An attempt was made to evaluate the experimental uncertainty for both the long-term mean velocities and the Reynolds stresses from a single sample uncertainty analysis. But the sinus functions in the coefficients in equations 4.26 and 4.34 gave complex derivatives making this analysis hard to perform. To estimate the sensitivity of the coefficients, a different approach was adapted. The equations were implemented in a spread sheet program and by systematic variation of the coefficients (or, to precise, the angles α , β and γ) the sensitivity of the reconstruction was evaluated.

For the selected angle combinations, an experimental uncertainty of about 10%

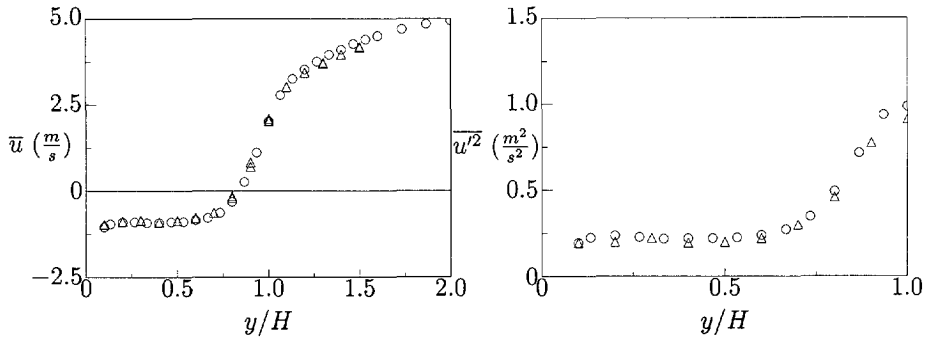


FIGURE 4.47: Comparison between measurements of $\bar{u}$ (left figure) and $\overline{u'^2}$ (right figure) obtained with the reconstruction technique (Δ) and with the cube in the regular orientation ($\circ$) at $z/H = 0$ and $x/H = 1.7$ for the matrix flow.

was found for the mean velocities and about 20% for the Reynolds stresses.

An important contribution to experimental uncertainty is the repeatability of the positioning of the measuring volume. To obtain one data set of two velocity components and three Reynolds stresses, three measurements at the same location were required. In particular, when the measurements are performed in regions with steep velocity gradients, a bad reproducibility of the positioning of the measuring volume will lead to large experimental error and consequently, will lead to the failure of the velocity reconstruction technique. For different inclination angles, the measuring volume was aligned with a reference position using a binocular. This procedure ensured a reproducibility of 0.1 mm which was sufficiently small with respect to the velocity gradients in the flow.

The reconstructed streamwise velocity $\bar{u}$ and the Reynolds normal stress $\overline{u'^2}$ were compared with direct measurements with the laser probe in the regular (perpendicular) orientation to obtain a quantifiable validation of the accuracy. Since a relatively large part of the streamwise component u is measured with the used inclination angles, the agreement between both results should be good. This can be seen in Fig. 4.47 in which profiles in the y -direction of $\bar{u}$ and $\overline{u'^2}$ are compared at $x/H = 1.7$ in the symmetry plane for the spatially periodic matrix flow (see chapter 8). The reconstructed streamwise velocity and Reynolds normal stresses (indicated with (Δ) in Fig. 4.47) are in good agreement with the measurements performed with the laser probe in the regular orientation (denoted as ($\circ$) in Fig. 4.47).

The reconstructed normal velocities $\bar{v}$ and Reynolds normal stresses $\overline{v'^2}$ were compared with velocity measurements acquired with the prism-cube. This comparison is shown in Fig. 4.48 for a traverse in the y -direction at $x/H = -2$ and $z/H = 0$, again for the spatially periodic matrix flow. The reconstructed results are denoted as (Δ), the measurements obtained with the prism-cube are indicated with ($\circ$). The velocities of both techniques compare very well which support the conclusion that the reconstructed normal velocities are accurate within 10%. While the Reynolds normal stresses measured with the prism-cube collapse to a smooth profile, the values of $\overline{v'^2}$ obtained from the reconstruction are scattered around the

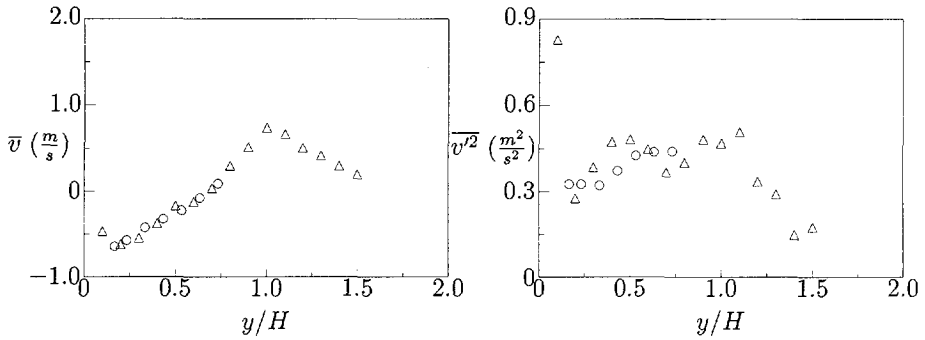


FIGURE 4.48: Comparison between measurements of $\bar{v}$ (left figure) and $\overline{v^2}$ (right figure) obtained with the reconstruction technique (Δ) and with the prism-cube ($\circ$) at $z/H = 0$ and $x/H = -2$ for the matrix flow.

prism-cube measurements. This confirms the estimation of 20% accuracy for the second order moments of the reconstructed results.

4.3.5 Accuracy values for the experimental results

All the LDA measurements presented in this thesis were processed with velocity bias weighting methods to eliminate, or at least to minimise, the error introduced by velocity bias. For the regions in the flow without severe velocity bias effects, for example the core flow in the channel corridor above the cubes, the correction did not effect the results, as was demonstrated in the foregoing paragraphs. In the cases of large turbulence intensities (or velocity bias effects), marked differences between the weighted and unweighted long-term averaged velocities and higher order moments were observed. Most of the velocity data were processed with the inverse velocity bias method $2D+$. In some cases, the interarrival time weighting was used. For these cases, it was demonstrated that no marked differences existed between the inverse velocity and the interarrival time weighting due to the fact that the sampling frequencies were significantly large to satisfy the criterion $Tu \times \nu > 10$ (Tu is the integral time scale and ν is the averaged sampling rate). Based on the evaluation of all possible experimental uncertainties involved, the long-term averaged velocities presented in this thesis are accurate within 5%. The Reynolds stresses were estimated to be accurate within about 7% while a value of 10% pertained to the higher order moments (skewness and kurtosis factors). These values apply also for the normal velocities measured with the prism-cube. The measurements of $\bar{v}$ and $\overline{v^2}$ obtained with the reconstruction technique were accurate within 10% and 20%, respectively.

4.4 Flow visualisations

4.4.1 Oil-film visualisations

The oil-film visualisation technique is a widely used measurement technique to qualitatively visualise the time-averaged flow pattern just above the surface of interest (the near-wall velocity vector). The technique is primarily based on the accumulation and the removal of tracer particles, present at the surface. The transport of these tracer particles and the resulting streak lines on the surface are directly proportional to the surface stress vector, which must be parallel to the near-wall velocity vector. This condition is needed for the successful application of this technique [68]. Particularly for flows over wall-mounted obstacles, this technique is very useful because of the different flow regimes encountered in these kind of flows. Flow reattachment and separation cause significant differences in the surface shear stress and consequently cause clear pigment patterns. This is illustrated from the numerous papers in which oil-film visualisations of the flow over wall-mounted prismatic obstacles are discussed [68, 101, 102, 86]. But also for other configurations, the oil-film technique appeared to provide suitable visualisations of the surface pattern. The flow features around a square cylinder in the vicinity of a channel wall in a turbulent flow was experimentally studied by Wu and Martinuzzi [177, 105]. In addition to LDA measurements, they performed oil-film visualisations to document different types of flow as a function of the distance between the suspended cylinder and the channel wall. Ruderich and Fernholz [143] studied the flow characteristics of a bluff plate with a splitter plate located in the symmetry plane of the bluff plate (the configuration was in fact a T-junction with a long tail). The bluff plate caused the flow to separate where after it reattached on the splitter plate. Oil-film visualisations on both the splitter plate and the side wall were used to derive flow details.

The working principle is as follows. The surface was brushed with a tracer mixture which was composed of jet fuel, low transmission oil and graphite (copier toner). The low transmission oil was added to make the mixture viscous for embedding the graphite tracers. The jet fuel made the mixture more liquid. The mixture composition was chosen in a way, by means of trial and error, that it was viscous enough to prevent surface pattern diffusion due to turbulent fluctuations in the near wall velocity field. Further, it was guaranteed that the mixture was still liquid enough to allow the particles to follow the near-wall velocity field. The typical duration of an oil-film visualisation experiment was about 15-20 minutes. In that period, the jet fuel was sufficiently evaporated in a sense that the graphite patterns were attached to the surface.

Although the oil-film visualisation technique provides information only of the near wall flow features, careful topology analysis allows to extend these results to obtain a qualitative description of the macroscopic flow features away from the wall. Due to the fine detail possible with the technique, it is possible to identify small structures which could be difficult to detect with other methods. In addition to providing quantitative information of the location of separation and reattachment lines, this method could also serve to guide anemometric studies.

4.4.2 Smoke visualisations

Smoke visualisations, obtained from smoke injection and illumination, give instantaneous records of the flow field. The smoke particles were illuminated in a thin sheet of light, created with a coherent laser beam and a rotating mirror (polygon). The smoke was injected in the flow with either a smoke tube or with a hot wire method. For more background information on smoke visualisation methods, we refer to the literature [11, 119]. It was verified for both methods that the smoke injection did not influence the flow field. Images of the smoke traces were recorded on video tape with a CCD-camera. Separate frames were selected from the movie and digitised with a frame grabber. Good contrast in the images was obtained for low Reynolds numbers (1000-2000). Due to turbulent mixing, the smoke traces vanished in the turbulent field for the larger Reynolds numbers and, therefore, no distinguishable flow pattern could be observed from the separate frames. But in that case, the recorded movie revealed sufficient details and information.

Further, a stroboscope was used to freeze the flow structure in time. Predominant frequencies in the flow, for example the vortex shedding from the lateral faces of the cube, could be visualised with this method. In fact, it is a conditional sampling of the velocity field.

4.5 Structure identification with topology principles

The pattern of pigment accumulation or removal in the oil-film visualisation can serve to identify critical points, where the surface stress and the velocity vectors identically vanish, or critical lines along which the surface shear stress component normal to the line is zero. There are two classes of critical points [68, 134]. Nodes, or nodal points, are isolated singularities through which an infinite number of surface-stress trajectories pass either radially (simple nodes) or as high vorticity confocal spirals (foci). Reattachment (stagnation) and separation points correspond to divergent and convergent flow patterns, respectively. Typical examples are shown in Figs. 4.49a and b. Saddle points are singularities through which only two critical lines cross. The flow converges along one line and diverges along the other, see Fig. 4.49c, (the diverging critical line is often rich in pigment). Establishing the physical significance of saddle points depends on a proper interpretation of the local flow.

For a general flow field, the difference between the total sum of nodes, $\sum N$, and saddles, $\sum S$, satisfies:

$$\sum N - \sum S = 2 - 2g, \quad (4.47)$$

where g is the genus or connectivity of the surface. The flow at the channel walls and at the surface of the wall-mounted cubes is topologically equivalent to a sphere ($g=0$) with two additional nodal points (channel inlet and outlet) so that $2 - 2g=0$. Hunt et al. [68] extended the topological principle to the analysis of the velocity field in planes normal to walls. It was shown that, for the symmetry plane $z/H=0$:

$$\left(\sum N + \frac{1}{2} \sum N'\right) - \left(\sum S + \frac{1}{2} \sum S'\right) = 0, \quad (4.48)$$

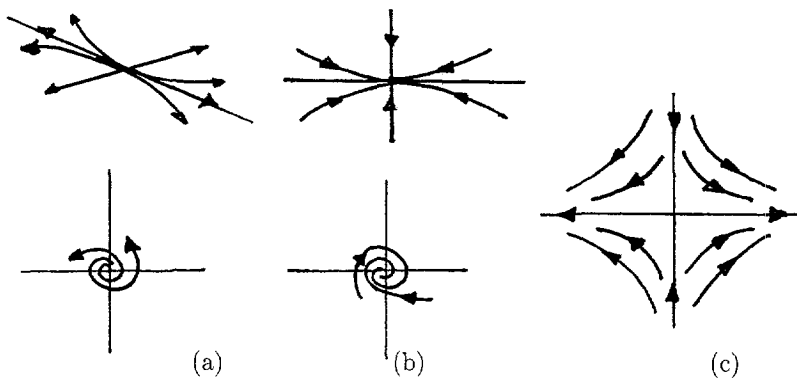


FIGURE 4.49: Types of topological critical points: Nodal points (N): (a) attachment, (b) separation and (c) Saddle point (S).

where N' and S' are half nodes and saddles, respectively, occurring at the solid boundaries.

The three-dimensional mean flow structure was inferred from these critical points and lines at the surface of both the obstacle and the channel wall, together with Laser Doppler Anemometry measurements and smoke visualisations of the flow structure.

Some other structure identification methods are shortly mentioned. Usually velocity vectors and streamlines are examined to study coherent structures in the flow but the substantial computation time is a major drawback of this approach, especially for complex flows. It is therefore beneficial to introduce scalar quantities which represent vortical motion. Robinson [138] introduced the concept of using low-pressure regions in the flow as possible regions with vortex cores. Sandham and Kleiser [144] confirmed that this method represented coherent structures in a channel flow. The modulus of the vorticity is also often used to identify coherent structures in the flow. Joeng and Hussain [81] applied this method successfully to free shear flows. Chong et al. [32, 135] employed a method based on critical points in the flow to derive the flow pattern in the entire flow domain. The flow pattern was defined by the eigenvectors and eigenvalues of the velocity gradient tensor which were evaluated at the critical points. de Souza et al. [156] applied three coherent structure identification techniques to measurements obtained in the near wake of a circular cylinder suspended in a turbulent boundary layer flow. One of the three techniques was based on the principle that the large-scale structures are associated with sudden variations in the velocity. The other two methods were filtering techniques, based on the spectral and on the time-frequency (wavelet) decomposition of the instantaneous velocities. The method of Joeng and Hussain [81] was extended by Michard et al. [107] to detect coherent structures from PIV measurements and DNS data of turbulent rotating flows.

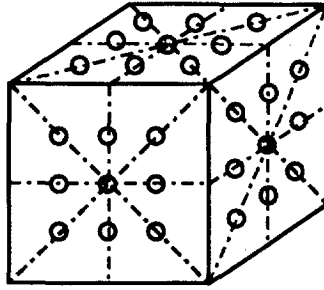


FIGURE 4.50: Sketch of the pressure-cube to measure the local static surface pressure at the five faces of the cube.

4.6 Static pressure measurements

Static pressure measurements were performed at the channel floor and at the five faces of the cube exposed to air flow. In a separate experiment, the channel floor was equipped with small pressure taps. Turbulent fluctuations damped out completely and the technique is, therefore, only suited to measure long-term averages of the static pressure. A special cube was designed to measure the static pressure at the cube faces. It consisted of five separate chambers, one for each face exposed to the flow, with 9 pressure taps at each face. These pressure taps were located at the face in an equidistant arrangement. A schematic of this cube with pressure taps is given in Fig. 4.50. The cube allowed the measurement of the face-averaged static pressure when all the taps are open. By taping 8 of the 9 holes, disconnecting 8 pressure taps from the underlying chamber, a local static pressure measurement was performed. The static pressure was represented in the pressure coefficient, defined as:

$$C_p = \frac{p_s - p_r}{\frac{1}{2}\rho U_\infty^2}, \quad (4.49)$$

where p_s and p_r are the surface static pressure and the reference pressure, U_∞ is the reference velocity and ρ is the fluid density.

Part III

Test cases

CHAPTER 5

Single cube

The single cube is the most basic model considered. The cube was mounted on the channel floor in the closed wind tunnel. The transition from a laminar to a turbulent boundary layer flow was forced by a tripping strip which was placed on the channel floor about 75 cm upstream of the cube. It was experimentally determined that the flow was effectively tripped for bulk velocities $u_B > 2.5$ m/s (which corresponds to $Re_H = 2500$). Therefore, the flow structure around the single cube was studied in the Reynolds number range of $2500 < Re_H < 5000$. The LDA-analysis was conducted at a bulk velocity of 4.47 m/s which corresponds to a Reynolds number of $Re_H = 4440$. Complementary flow information was obtained from visualisation

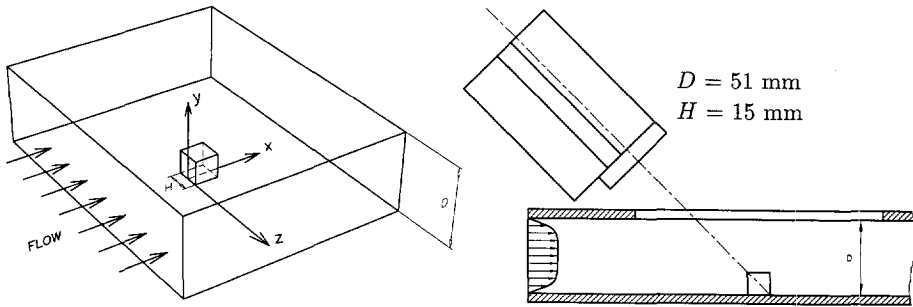


FIGURE 5.1: A side view and three-dimensional view of the wall-mounted cube in the channel flow. experiments (smoke and oil-film visualisations). Differences in the flow field for the different Reynolds numbers appeared to be insignificant. Therefore, the presented flow field analysis applied to the Reynolds number range of about $2500 < Re_H < 5000$.

5.1 Oncoming flow conditions

The tripping of the flow resulted in a developing turbulent boundary layer on the channel floor and a developing laminar boundary layer at the opposing wall. A typical velocity profile at the centreline of the channel at $x/H = 6.7$ is given in Fig. 5.2. Due to the growing boundary layer in the downstream direction, the velocity deficit in the boundary layer resulted in an acceleration of the displaced flow in the core region (due to the incompressibility of the flow). The resulting velocity gradient in the core region was determined to be $dU_\infty/dx = 0.67$ 1/s, see Fig. 5.3.

In order to compare the velocity profiles with literature data from Erm et al. [44] the momentum thickness θ was evaluated. This thickness was used to define

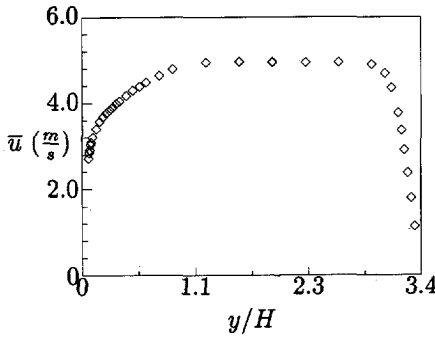


FIGURE 5.2: Profile of the streamwise velocity $\bar{u}$ in the normal direction at $x/H = 6.7$ at the centreline of the channel.

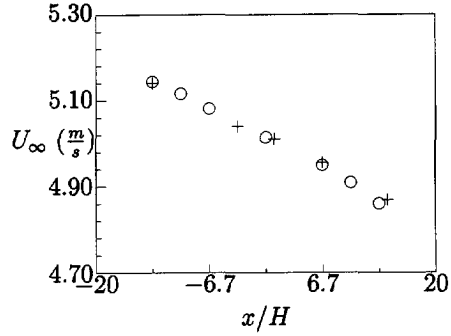


FIGURE 5.3: Variation of the free stream velocity in the channel core as a function of the streamwise coordinate x .

a local Reynolds number Re_θ . In the absence of the cube, this Reynolds number was $Re_\theta = 660$ for $x = 0$ which is the location of the windward face of the cube when the cube is in its position. Profiles of the mean velocity and the second order moments were measured at five stations x/H at the channel centreline, namely three upstream and two downstream of the cube. Clauser plotting was used to determine the shear velocities u_τ . The mean velocity gradients evaluated from the measured velocity profiles close to the wall agreed with Coles method within about 5-10%. These shear velocities were used to non-dimensionalise the velocities and the y -coordinate. A typical value for the shear velocity was $u_\tau = 0.25$ m/s at $x = 0$. The dimensionless mean velocity $\bar{u}/u_\tau$ and turbulent intensities $\sqrt{u'^2}/u_\tau$ were scaled with inner variables. The resulting profiles are plotted on semi-log scale in Fig. 5.4 and are compared with results of Erm et al. [44] for $Re_\theta = 617$. Although these

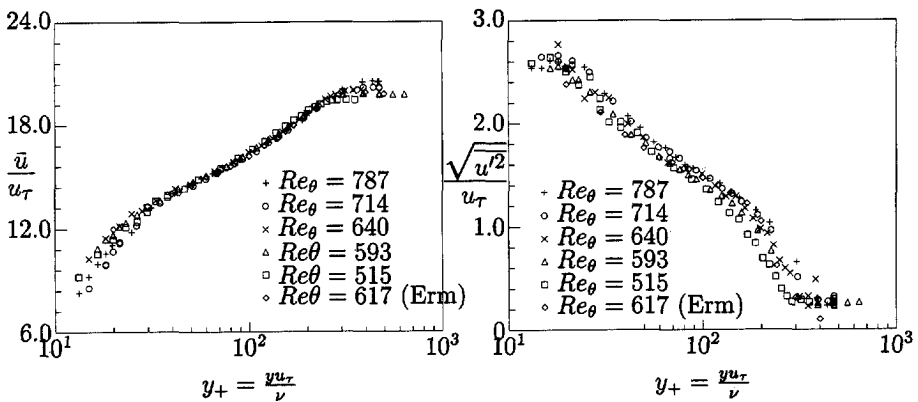


FIGURE 5.4: Profiles of $\bar{u}/u_\tau$ and $\sqrt{u'^2}/u_\tau$ for different Reynolds numbers: comparison between present data and literature (Erm et al., 1987).

literature data refer to a turbulent developing boundary layer flow in the absence of a pressure gradient, the results are in good agreement with the present data.

A consistency check was conducted with conservation of mass at the five subsequent stations. The axial mass flow rate was determined from:

$$\dot{m}_x = \rho dz \int_0^D \bar{u} dy, \quad (5.1)$$

where D is the channel height and ρ is the fluid density. The mass flow was obtained from a numerical integration of the individual profiles which yielded an averaged mass flow rate of 0.262 kg/s per unit area (dz). The averaged bulk velocity, derived from the ratio of the averaged mass flow rate and the integration area, was 4.47 m/s. The discrepancies between the calculated bulk velocities for the five profiles were within 1%. This result confirms conservation of mass and illustrates the good quality of the LDA measurements.

5.2 Vortex shedding past the cube

The vortex shedding from the side faces of the cubes caused the wake flow to fluctuate periodically. The coherent structures were detectable from all three velocity components. But because of the strong spanwise coupling of the shed vortices, the velocity amplitude of the periodic motion of the spanwise velocity w was the largest. Maximum amplification of the periodic motion was observed for locations around $x/H = 2$, $y/H = 0.5$ and $z/H = 0.5$, the oscillations were attenuated for larger streamwise locations. The power density spectrum was then calculated from the measured velocity time series. Long-term averaged spectra were obtained from time durations of 500,000 data points which corresponded to about 20,000-40,000 vortex shedding cycles. This number of realizations was required to ensure a sufficiently large signal-to-noise ratio. A slotting, sample-and-hold and re-sampling method were compared to confirm the accuracy of the calculated power density spectra and of the obtained wake frequencies. Details of these methods can be found in Absil [1] and Tummers and Passchier [167]. The wake frequencies obtained from the calculated spectra were within 0.5 Hz. This value pertains also to the uncertainty of estimating the frequencies from the spectra and to the experimental uncertainty involved.

A typical example of a spectral power density function is given in Fig. 5.5 for a station at $x/H = 2.7$, $y/H = 0.5$ and $z/H = 0.5$. The peak in the spectrum around $f = 33$ Hz is associated with the passage of shed vortices. The coherent structures in the flow carry a substantial part of the turbulent kinetic energy which appears from the observed peak. The dashed line, also indicated in Fig. 5.5, is related to the incoherent structures (stochastic motion) in the flow. It is observed that the spectral energy level of the periodic component is approximately three times larger than that of the incoherent structures. This separation between organised and disorganised structures in the flow can be elucidated by writing the periodically fluctuating velocities, analogously to Reynolds decomposition, as a superposition of a long-term average $\bar{u}$, a turbulent or random fluctuating part u'_t and the periodic mean component u'_p (triple decomposition):

$$u = \bar{u} + u'_t + u'_p. \quad (5.2)$$

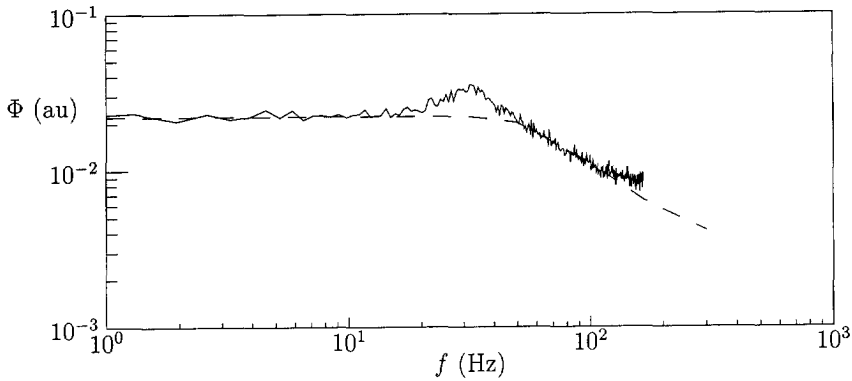


FIGURE 5.5: Spectral power density function of the w -velocity measured at the location $x/H = 2.7$, $y/H = 0.5$ and $z/H = 0.5$ ($Re_H = 5305$).

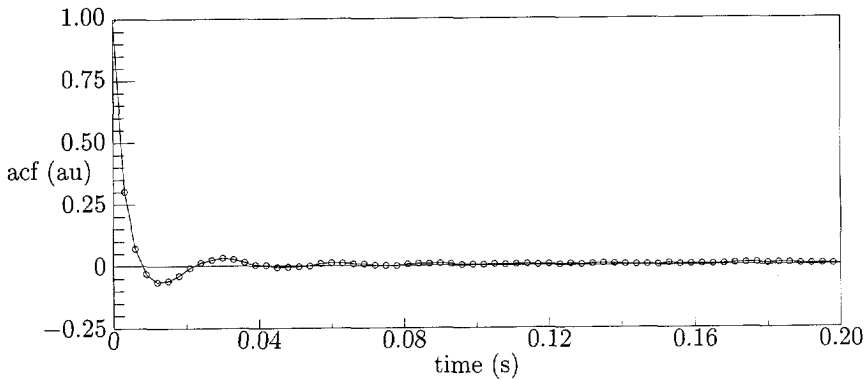


FIGURE 5.6: Autocorrelation function of the w -velocity measured at the location $x/H = 2.7$, $y/H = 0.5$ and $z/H = 0.5$ ($Re_H = 5305$).

The coherent component can then be considered as the sum of the long-term average and the periodic mean component. It can also be considered as a phase average. This decomposition is commonly used to separate the coherent contribution from the incoherent part, for example, by Balachandar et al. [14].

Since a highly turbulent flow was considered, vortex shedding was not activated at one distinguishable frequency but it occurred at a broader range of frequencies. The dominant characteristic frequency was derived from the frequency in the spectrum which corresponded to the maximum energy level. The peak in the spectrum given in Fig. 5.5 is related to a wake frequency of 33.2 Hz. The corresponding Strouhal number $St = fH/u_B$, based on the bulk velocity and the cube height, was $St = 0.093$. The periodical behaviour of the spanwise velocity can also be observed from the corresponding autocorrelation function, given in Fig. 5.6, which is based on the same dataset of 500,000 points. The autocorrelation function consists of an

incoherent or random part and a coherent part. The random part with the value 1 for time delays of 0 sec, the origin, and with a gradual decay to 0 for large time delays, is indicative for turbulent flows. The coherent part caused the sinusoidal shape of the autocorrelation function around the x -axis. Integral time scales T_u were derived from the time integration of the autocorrelation function (expression 2.28). Typical absolute values were of the order of magnitude of 12-13 msec; values from a true integration, with nullifying positive and negative contributions, were about 7 msec. The ratio of the characteristic length (the cube height $H = 15$ mm) and the characteristic velocity (about 1 m/s) gave also a value of about 15 msec for the integral time scale.

From Taylor expansion, the discretized second time derivative at $t=0$ was obtained to estimate the Taylor micro scale λ_T :

$$\lambda_T^2 = -\frac{\Delta t^2}{2(\rho(0) - \rho(1))}, \quad (5.3)$$

where $\rho(0)$ and $\rho(1)$ are the values of the autocorrelation at respectively $t = 0$ and $t = \Delta t$ sec. For the isolated wall-mounted cube, this value was in the order of magnitude of 2.5 msec.

Wake frequencies were measured for a range of mass flow rates. The results are plotted versus the bulk velocity in Fig. 5.7. All shown frequencies were determined from 500,000 data points. The relationship is clearly linear which indicate that the results are independent of the Reynolds number. This was expected since the considered range of Reynolds numbers covered less than one decade. The somewhat larger discrepancies from the linear regression for the lower bulk velocities are probably due to the somewhat larger experimental uncertainty. The Strouhal number of $St = 0.095$ was determined from the slope of the linear regression and was accurate within 5%.

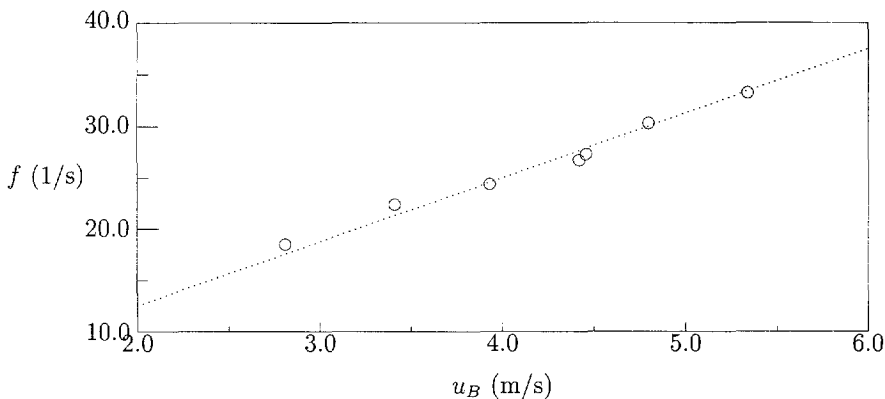


FIGURE 5.7: Wake frequencies as a function of the bulk velocity. The slope determines the Strouhal number, namely $f/u_B = St/H$.

Wake frequencies for a single cube were measured for different flows and different oncoming flow conditions. The results include a cube with height $H = 25$ and $H =$

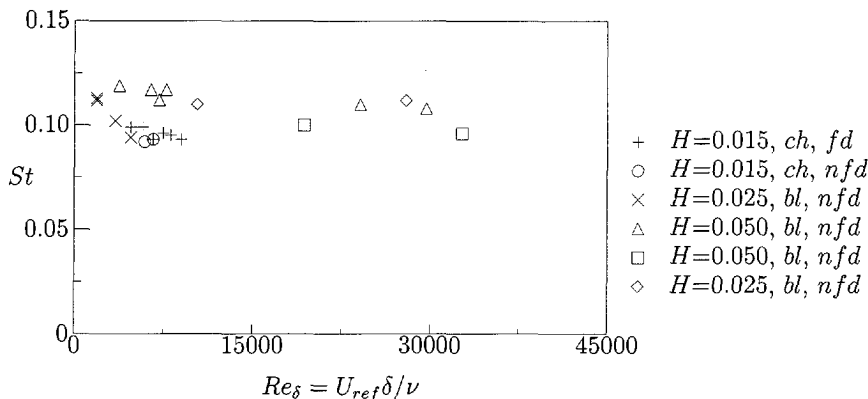


FIGURE 5.8: Wake frequencies in terms of the Strouhal number for a single cube as a function of $Re_\delta = \delta U_{ref} / \nu$. *bl* refers to boundary layer, *ch* refers to channel, *fd* refers to fully developed and *nfd* refers to not fully developed.

50 mm which was mounted on a flat plate in a free turbulent boundary layer flow. Only for these cases, the wake frequencies were determined from hot-wire velocity measurements. Further, a single cube in a fully developed and a developing channel flow were considered. The Strouhal numbers for these cases are plotted in Fig. 5.8 as a function of the Reynolds number $Re_\delta = \delta U_{ref} / \nu$, based on the boundary layer thickness (δ) and a reference velocity (U_{ref}). It is seen that the values are scattered around a mean value of about $St = 0.10$. The deviations are approximately 15 % and the Strouhal numbers do not fully collapse on a single curve. Further, these values differ significantly from $St = 0.145$ as measured by Martinuzzi [101] in the wake flow of a single cube in a fully developed channel flow at $Re_H = 40,000$ and a channel-height-to-cube-height ratio of 2. It is believed that the observed differences between the present results as well as compared to the value of $St = 0.145$ were caused by different levels of the free stream turbulence which was significantly higher in the fully developed high Reynolds number case of Martinuzzi. Another reason may be the difference in aspect ratio which impose different blockage effects (the channel-height-to-cube-height ratios are different).

5.3 Time-averaged flow field

The sketch given in Fig. 2.1 in chapter 2 is representative for the flow over the isolated wall-mounted cube in the developing turbulent channel flow. In the following discussion of the relevant flow details, reference is also made to that sketch.

5.3.1 Front of the obstacle

The flow in the front region of the cube is primarily dominated by the horseshoe vortex. The trace of this vortex in the upstream stagnation region along the centre plane ($z/H = 0$) is readily identified in the smoke visualisation patterns, shown in

Fig. 5.9. The figure gives an instantaneous image for $Re_H = 1000$, flow is from right to left. The smoke, injected upstream, is convected smoothly in the vortex core of the primary vortex. Smoke visualisation experiments, performed at higher Reynolds numbers, revealed similar observations but the smoke traces were more distorted. Although the smoke traces were not distinguishable anymore from single shots, the recorded movie clearly showed the paths of the smoke particles. Since no significant differences were observed between the experiments for low and high Reynolds numbers, it is concluded that the flow visualisations given in this thesis are representative for the Reynolds number range of $2500 < Re_H < 5000$.

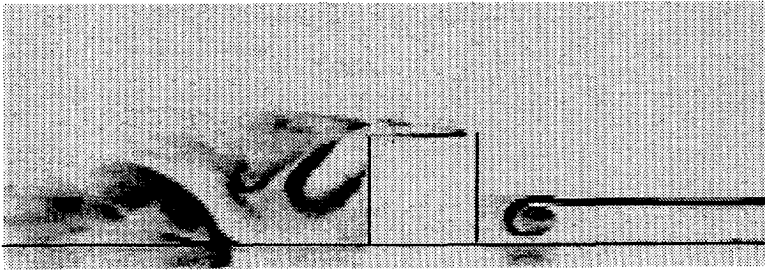


FIGURE 5.9: Instantaneous smoke visualisation of the flow pattern in the symmetry plane $z/H = 0$. Flow is from right to left.

The surface imprint of the horseshoe vortex is observed from the oil-film visualisation given in Fig. 5.10. This flow visualisation was obtained for $Re_H = 8000$. For the oil-film visualisations also, it is noted that experiments at lower and higher Reynolds numbers revealed similar oil pattern. It is, therefore, concluded that the shown oil-film results are representative for the Reynolds number range of interest ($2500 < Re_H < 5000$). The primary horseshoe vortex is observed from the pigment pattern marked **A** in Fig. 5.10. It caused a flow separation which resulted in the separation saddle, denoted as S_a in Fig. 5.10. A system of induced vortices is located upstream of the primary horseshoe vortex. It extends up to the primary separation saddle point S_b , seen as the accumulation of pigment in Fig. 5.10 which is located approximately $1.4H$ upstream of the leading face. The flow separation line originates at this saddle point and is delimited by the pigment line which is extended laterally and then further downstream from this point.

The structure of this vortex system appears clearly from the smoke visualisation given in Fig. 5.11. This image shows a cross-section of the three-dimensional flow structure illuminated by a laser sheet at the symmetry plane $z/H = 0$. The smoke visualisation observations showed the cyclical growth and collapse of the horseshoe vortex structure. Similar observations were made by Martinuzzi et al. [86, 101] for a single cube in a fully developed channel flow. They postulated a hypothesis that two different flow modes counterbalance each other. The alternation between these two states caused bimodal velocity distributions in the horseshoe vortex region. Such bimodal behaviour was also detected from LDA measurements in the horseshoe

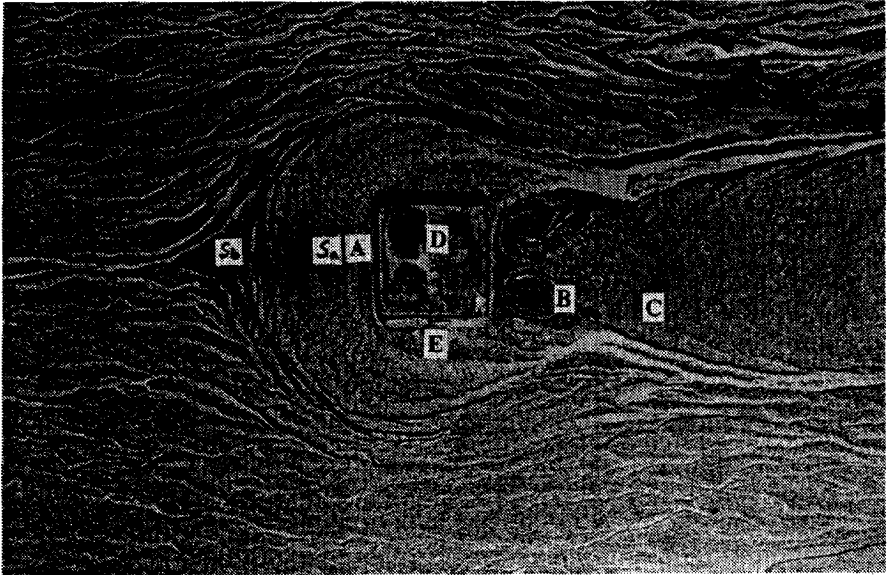


FIGURE 5.10: Oil-film visualisation of the surface flow pattern for the single cube in the turbulent channel flow ($Re_H = 8000$).

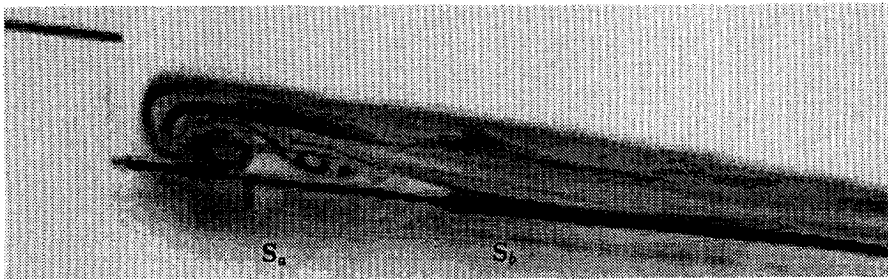


FIGURE 5.11: Smoke visualisation of the front face of the leading cube in the array of cubes (flow is from right to left). S_a and S_b correspond to the notation in the oil-film results.

vortex region for the single cube case considered in this thesis. A typical probability density function of the streamwise velocity u measured at the station $x/H = -0.5$, $y/H = 0.1$ and $z/H = 0$ is given in Fig 5.12. Three distinguishable peaks are observed in this histogram. Two peaks are concentrated around zero velocity which is expected due to velocity bias. Particles with low velocities passed the measuring volume less frequently than particles with higher velocities did. This led to the marked minimum around zero velocity (see the discussion of velocity bias in chapter 4). The peak in the interarrival time around zero velocity, which is

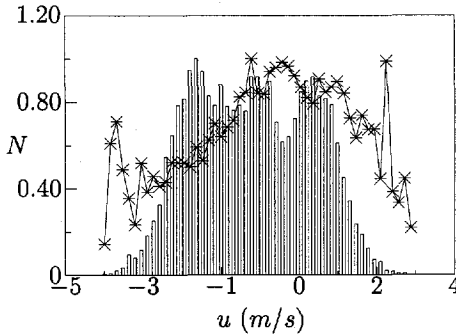


FIGURE 5.12: Histogram of the streamwise velocity u at $y/H = 0.1$, $z/H = 0$ and $x/H = -0.5$ ($Re_H = 4440$).

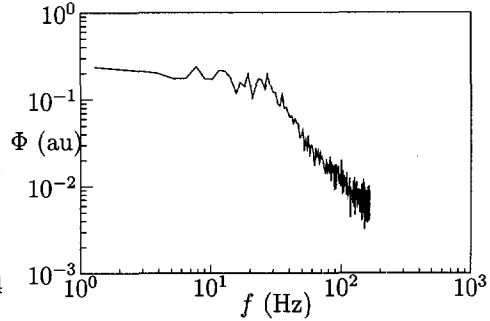


FIGURE 5.13: Power spectral density of the streamwise velocity u at $y/H = 0.1$, $z/H = 0$ and $x/H = -0.5$ ($Re_H = 4440$).

given as a function of the corresponding velocity (see (*) in Fig 5.12), supports this conclusion. If the velocity probability density function was weighted, for example with the reciprocal of the instantaneous velocity, this minimum around zero would diminish. A third peak is observed around $u \approx -1.5$ m/s. With the knowledge that velocity bias corrupted the histogram around zero velocity, we clearly observed a bimodal behaviour which supports the hypothesis of the two preferred flow states. Devenport and Simpson [38] measured also bimodal pdfs in the horseshoe vortex region where always one peak was found at zero velocity. Power density spectra of the velocity datasets were calculated in the area where the probability density functions displayed bimodal (not due to velocity bias) behaviour to examine whether or not the dynamic behaviour involved a characteristic frequency. The spectrum shown in Fig. 5.13, which corresponds to the dataset shown in Fig 5.12, only decays with a $-5/3$ slope when plotted in log-log coordinates. No increased activity was observed around a specific characteristic frequency which suggests that the dynamics occurred randomly. Therefore, it is not clear if this dynamic process is coupled with the vortex shedding from the side faces of the cube. With the employed experimental techniques, it was not possible to measure velocities in the wake and in front of the cube simultaneously and, therefore, it was not possible to detect a coupling between the dynamics in the horseshoe vortex and the vortex shedding from the sides of the cube. Martinuzzi [101] calculated also power spectral densities from the velocity time series in the horseshoe vortex region and he also did not observe predominant frequencies.

The stagnation flow in front of the cube, with the vortex roll-up, was also observed from pressure measurements at the channel floor. The local static pressure coefficient C_p was measured at the centre-line upstream of the cube from $x/H = 0$ to approximately $x/H = -2.7$. The atmospheric pressure and the bulk velocity were used to define the pressure coefficient. The results are shown in Fig. 5.14 and correspond to four different Reynolds numbers. Although some minor differences are observed, the profiles are qualitatively similar for the different Reynolds numbers.

Far upstream, C_p asymptotes to a constant value of about $C_p \approx 0.2$. In the direction towards the cube, the pressure increases almost monotonically and becomes maximum close the windward face of the cube. The results differ somewhat from the experiments performed by Martinuzzi et al. [101, 102]. They detected a local maximum in the static pressure at the time averaged stagnation point, at about $x/H \approx -0.5$. During the period when the four-vortex structure exists, the stagnation point is the location where the surface pressure becomes maximum. They argued that due to the larger relaxation time of the pressure field compared to that of the velocity field, the pressure imprint of the flow onto the surface is long enough preserved to be measured as a time-averaged increase. This maximum was not very

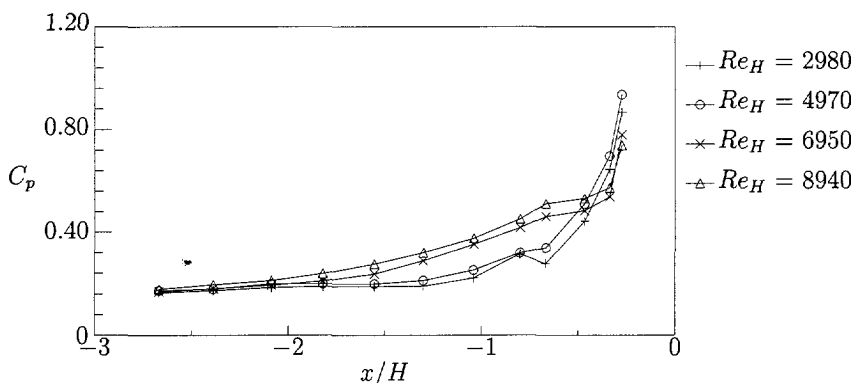


FIGURE 5.14: Profiles of the pressure coefficient C_p for four different Reynolds numbers: (+) $Re_H = 2980$, (o) $Re_H = 4970$, (x) $Re_H = 6950$ and (Δ) $Re_H = 8940$.

clear from the present measurements but a somewhat increased pressure coefficient was observed around $x/H = -0.7$. Another difference with the results of Martinuzzi et al. [101, 102] is that the values of the pressure coefficient are in general somewhat lower. This is explained by the different channel-height-to-cube-height ratio D/H . While the results from Martinuzzi were obtained for $D/H = 2$, in our study this ratio was $D/H = 3.4$. In the smaller aspect ratio case, the flow is more constrained by the channel roof and, therefore, causing a larger build up of the pressure. When the aspect ratio is larger, which implies that the oncoming flow can be trapped easier over the top face, the stagnation pressure is lower.

LDA data of the dimensionless mean streamwise velocity $\bar{u}/u_B$ and Reynolds normal stresses $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ are plotted in Fig. 5.15 for five different heights at the front centreline of the cube. The location of the horseshoe vortex can be identified as the region of negative streamwise velocities and the vortex extends up to approximately half of the cube height. The peaks in the Reynolds normal stresses $\overline{u'^2}$ are found in the proximity of the horseshoe vortex region and coincide with areas of maximum turbulence production due to large velocity gradients. Further, the instable behaviour of the horseshoe vortex contributes also to the increased turbulence production.

Measurements of the local static pressure at the faces of the cube were performed

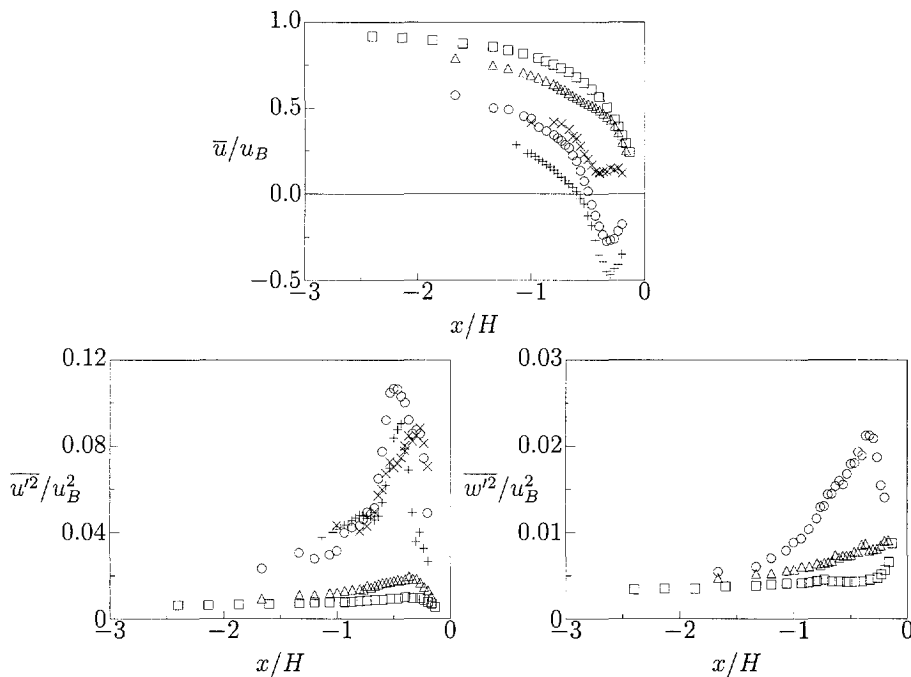


FIGURE 5.15: Profiles of the scaled mean velocity $\bar{u}/u_B$ and Reynolds normal stresses $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ upstream of the cube along the symmetry-plane $z/H = 0$: (+) $y/H = 0.07$, (o) $y/H = 0.1$, (x) $y/H = 0.13$, (Δ) $y/H = 0.3$ and ($\square$) $y/H = 0.5$.

with the pressure-cube, discussed in the chapter on measurement techniques. The reference pressure p_r was measured at the location of the cube while the wind tunnel was running but without the cube in position. This reference pressure was taken instead of the atmospheric pressure to enable the comparison with data from Castro and Robins [30]. The results are shown along cross-section $ABCD$, which denotes a plane parallel to the mounting base at $y/H = 0.5$ and cross-section $ABCD$, which denotes a plane perpendicular to the mounting base at the symmetry plane $z/H = 0$. Both cross-sections are indicated in the figure. The shown pressure measurements were performed for $Re_H = 3900$. For these measurements also, it was shown that the scaled static pressures applied to the whole Reynolds number range of interest ($2500 < Re_H < 5000$).

C_p becomes maximum ($C_p \approx 0.8$) at the centreline of the front face ($B - C$) at approximately $y/H = 2/3$ which corresponds to the location where the flow stagnation point was found. Flow stagnation involves zero velocities and is, consequently, accompanied by a maximum in the static pressure. These values were compared with the static surface pressures, around the stagnation point at the windward face of a cube mounted on a plate in a boundary layer flow, as studied by Castro and Robins [30]. The boundary layer thickness in their study was about ten times the cube height, which is significantly different from the situation studied in this thesis.

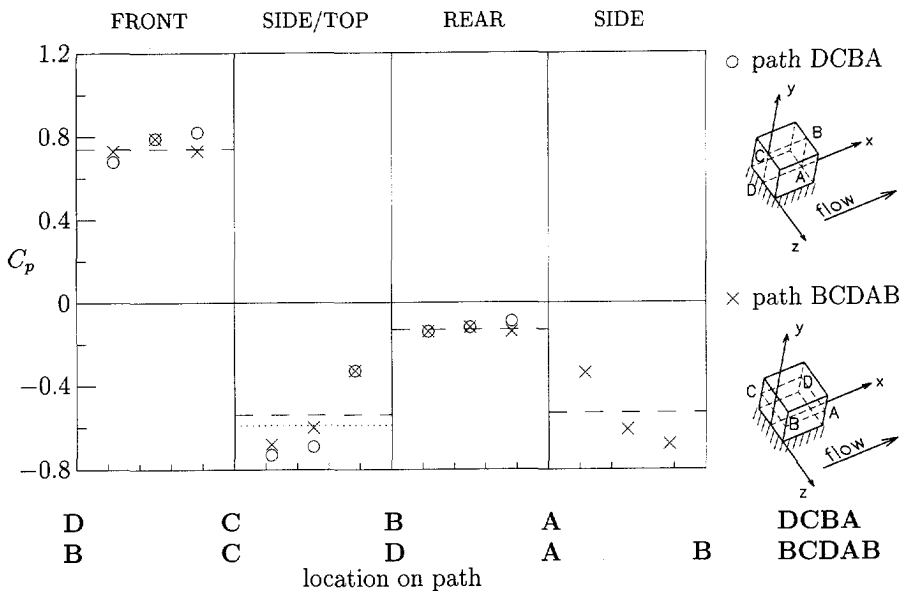


FIGURE 5.16: Profiles of the pressure coefficient C_p for $Re_H = 3900$ along path $ABCD$ (o) and path $ABCD A$ (x). The lines refer to the face-averaged static pressure $\bar{C}_p$: (---) front, rear and side faces, (···) top face.

Further, their Reynolds number was about one order of magnitude larger than the Reynolds number in this study, i.e. $Re_H = 3.0 \cdot 10^4$. Despite these differences, the shape of the profiles and the value of the maximum pressure found in the stagnation point were both in good agreement with the present findings. It is further observed from the profile at $y/H = 0.5$ (x) that the pressure distribution is symmetric. The face-averaged static pressure coefficient $\bar{C}_p$ is indicated as the dashed lines (---) for the front, the rear and the side faces. The dotted line (···) gives the averaged static pressure for the top face. These measurements were performed with the pressure-cube when all 9 pressure taps were connected to the underlying pressure chamber and, therefore, enabling the measurement of the face-averaged pressure. Due to the flow impingement at the front face, the averaged static pressure is positive compared to the other faces.

The skewness S_u ($S_u = \overline{u^3}/\sigma_u^3$) is given in Fig. 5.17 for some profiles in the symmetry plane $z/H = 0$. The profiles which cross the primary horseshoe vortex at $y/H = 0.07-0.1$, around $x/H = -0.5$, have a large positive peak. This deviation from a Gaussian velocity distribution (the skewness is 0 for a Gaussian pdf) illustrates that the time series of the streamwise velocity contained dominant positive spikes. These large velocities indicate that high momentum fluid is integrated in the primary horseshoe vortex. The pronounced minimum around $x/H \approx -0.8$ indicates that strong negative velocity spikes dominate the velocity statistics. The appearance of the strong positive and negative spikes were caused by the flow dynamics observed in the horseshoe vortex region. Despite significant variation in S_u , the skewness for

the spanwise velocity S_w was more or less 0 (with some experimental uncertainty) which is expected for the symmetry plane in which the time-averaged spanwise velocity $\bar{w}$ is zero. The kurtosis K_u ($K_u = \overline{u'^4}/\sigma_u^4$), also shown in Fig. 5.17, deviates significantly from 3 ($K_u = 3$ is denoted as the solid line). Close to the front face of the cube, K_u exceeds this nominal value. In the horseshoe vortex region ($y/H = 0.07-0.1$, $x/H = -0.5$) K_u becomes significantly small (about 2) which indicates that the distribution becomes broader. This is also the location where the turbulence intensities (seen as the peaks in $\overline{u'^2}$ and $\overline{w'^2}$, Fig. 5.15) becomes maximum. The kurtosis of the pdf of the streamwise velocity, K_w , was more or less 3 according to that of a Gaussian distribution.

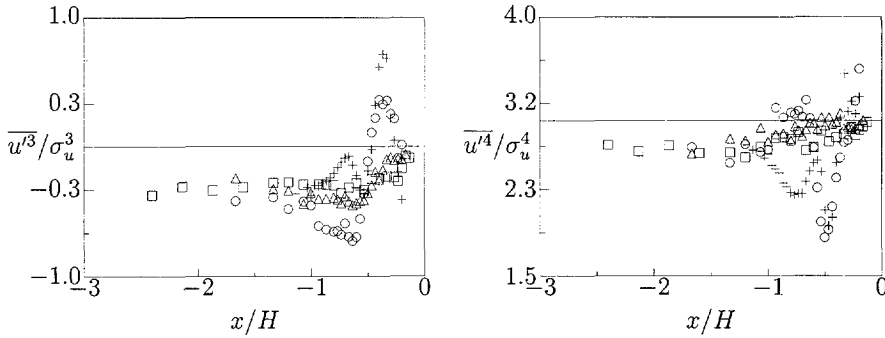


FIGURE 5.17: Profiles of the skewness S_u ($S_u = \overline{u'^3}/\sigma_u^3$) and the kurtosis K_u ($K_u = \overline{u'^4}/\sigma_u^4$) for the streamwise velocity upstream of the cube along the symmetry-plane $z/H = 0$: (+) $y/H = 0.07$, (o) $y/H = 0.1$, (Δ) $y/H = 0.3$ and ($\square$) $y/H = 0.5$.

A close up of an oil-film visualisation at the front face of the cube is given in the left image in Fig. 5.18. The pattern at the front face is dominated by a smooth pigment distribution centred around the stagnation node which is located at the symmetry plane $z/H = 0$ at approximately $2/3$ of the cube height from the base. The removal of pigment close to the cube base (the white area) is caused by the secondary recirculation induced by the main horseshoe vortex.

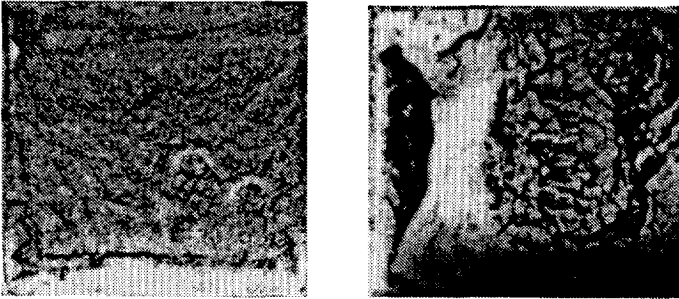


FIGURE 5.18: Close-ups of the oil-film pattern at the front face (left image) and the side face (right image, the flow is from left to right, see section 5.3.4) of the cube.

forced. The wake vortex is clearly present and extends, for this location $y/H = 0.5$, to approximately one cube size behind the cube. The plot clearly shows a saddle at approximately one cube distance (at $x/H = 2$) in the wake of the cube.

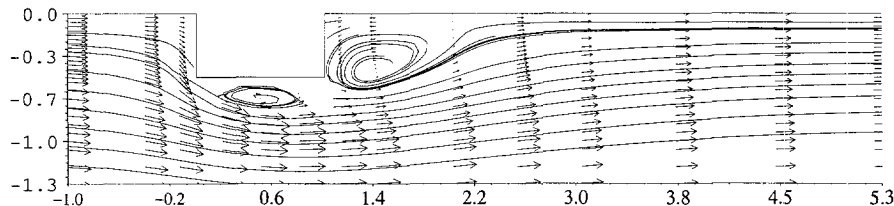


FIGURE 5.20: Long-term averaged vector plot of the flow around the cube at $y/H = 0.5$ for one half of the symmetric flow domain.

Profiles of $\bar{u}/u_B$ at five y/H -locations in the symmetry plane $z/H = 0$ are given in Fig. 5.21. The dashed line at $x/H = 1$ indicates the location of the rear face of the cube. The arch-shaped vortex is apparent from the negative velocities. The line where the velocity changes sign is referred to as the separation line. The reattachment point at the channel floor was derived from an extrapolation of the separation line to the channel floor. The reattachment point found in this way occurred at $x/H = 2.4$ - 2.5 , which is in good agreement with the location of the reattachment point observed from the oil-film observations, see Fig. 5.10. As would be expected, the magnitude of the Reynolds normal stresses $\bar{u}^2$ and $\bar{w}^2$, given in Fig. 5.21 for five stations in the symmetry plane $z/H = 0$, are relatively high in the shear layer (the profiles at $y/H = 0.7$ and 0.9). The strong velocity gradients caused high shearing and thus a significant production of turbulence. Further, the shear layer which separated from the leading top edge reattached at the top face more downstream. Due to Kelvin-Helmholtz instabilities, the free shear layer in the wake will also carry a quasi-periodic component. These fluctuations contribute also to the high turbulence level. It is noted that this periodic motion is significantly weaker in strength than the vortex shedding from the side faces since it is suppressed by the channel roof and wall. The skewness, given in Fig. 5.22 along profiles at locations which are similar to the stations shown in Fig. 5.21, does not show large variations. It seems that the skewness becomes maximum at a location of maximum turbulence intensity (see Fig. 5.21). The same holds for the kurtosis, also given in Fig. 5.22, where the values correspond to more or less Gaussian-shaped velocity distributions. Again it is noted that the skewness and the kurtosis of the spanwise velocity were also more or less equivalent to those for a Gaussian-shaped pdf ($S_w = 0$ and $K_w = 3$, respectively) because of the flow symmetry.

Profiles of $\bar{u}^2/u_B^2$, $\bar{w}^2/u_B^2$ and $\overline{u'w'}/u_B^2$ are given in Fig. 5.23 for three stations in the wake of the cube at $y/H = 0.5$. The profiles show large Reynolds normal stresses, in particular close to the side edge at $z/H = 0.4$. The large values are again explained by the high shearing at locations of large averaged velocity gradients.

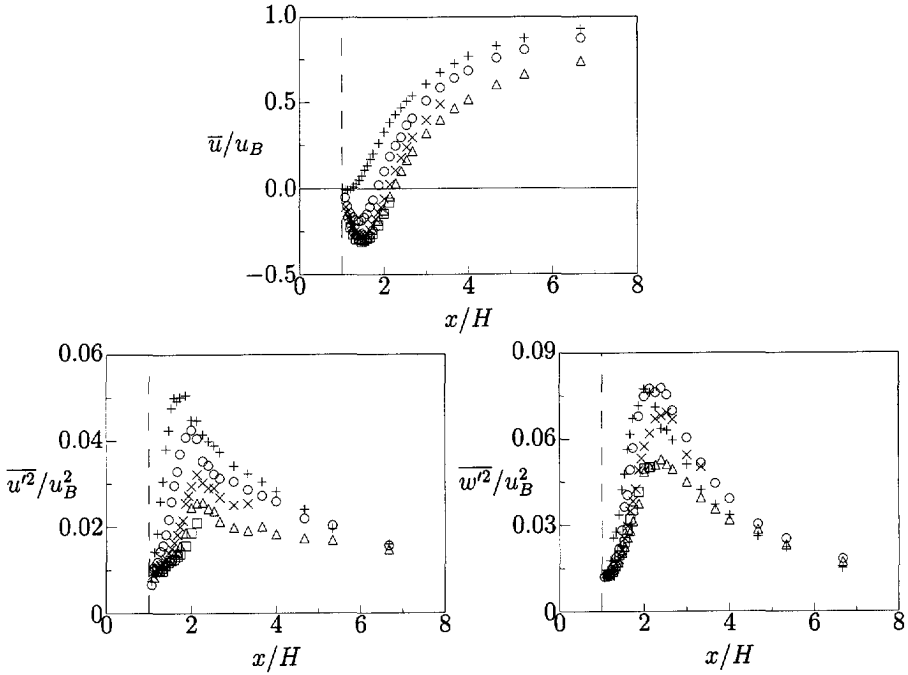


FIGURE 5.21: Profiles of $\bar{u}/u_B$, $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ downstream of the obstacle in the symmetry-plane $z/H = 0$: (+) $y/H = 0.9$, ($\circ$) $y/H = 0.7$, ($\times$) $y/H = 0.5$, (Δ) $y/H = 0.3$ and ($\square$) $y/H = 0.1$.

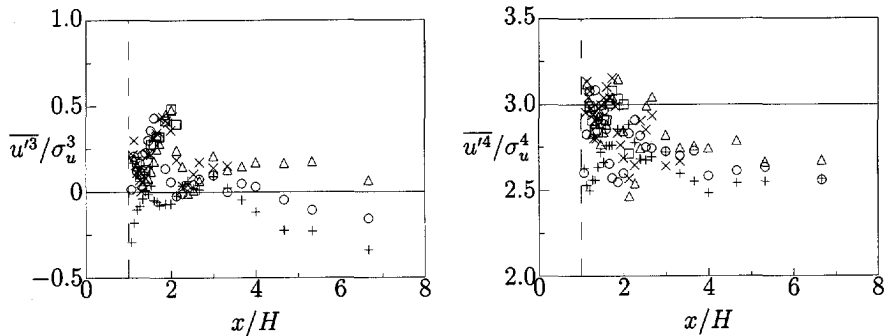


FIGURE 5.22: Profiles of the skewness and the kurtosis for the streamwise velocity in the wake of the cube in the symmetry-plane $z/H = 0$: (+) $y/H = 0.9$, ($\circ$) $y/H = 0.7$, ($\times$) $y/H = 0.5$, (Δ) $y/H = 0.3$ and ($\square$) $y/H = 0.1$.

Further, the vortex shedding at the lateral faces caused the wake flow to fluctuate periodically. It was already argued that the periodic component in the fluctuating velocity carried a substantial part of the total turbulent kinetic energy. Since the vortex shedding occurred at a rather broad frequency range (see for example the

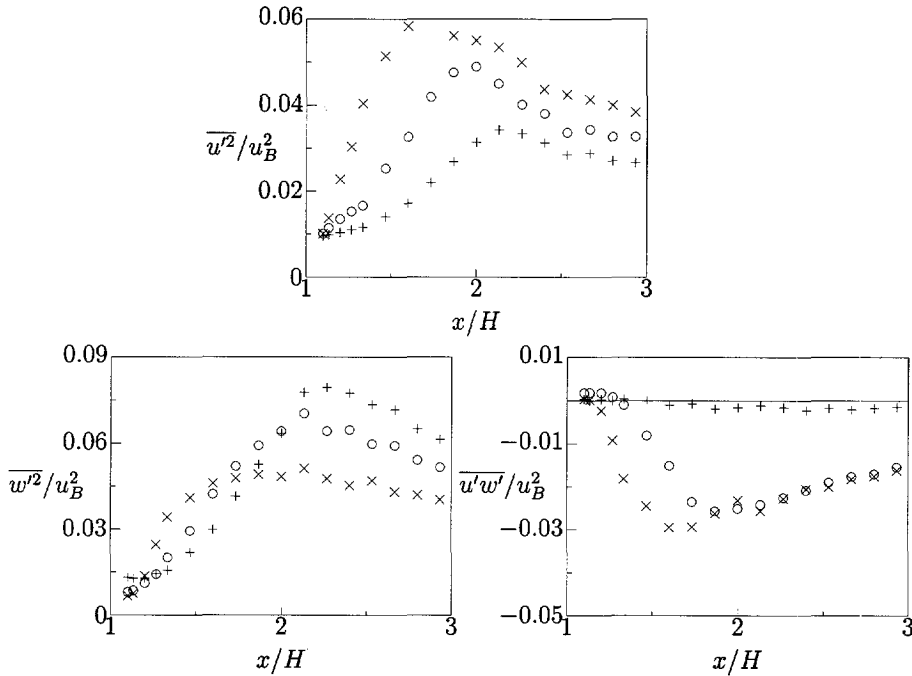


FIGURE 5.23: Profiles of $\overline{u'^2}/u_B^2$, $\overline{w'^2}/u_B^2$ and $\overline{u'w'}/u_B^2$ in the wake of the cube at $y/H = 0.5$: (+) $z/H = 0$, (o) $z/H = 0.2$, and (x) $z/H = 0.4$.

power density spectrum in Fig. 5.5), it was difficult to resample the datasets (with the characteristic frequency) to obtain a separation between the turbulent and the periodic fluctuations. The increased turbulence in the shear layer was shown from the turbulent kinetic energy k . Since the third component, $\overline{v'^2}$, was not measured for all configurations and stations, this component was estimated from: $\overline{v'^2} = 0.5(\overline{u'^2} + \overline{w'^2})$. Substitution then leads to the next expression for k :

$$k = 0.75(\overline{u'^2} + \overline{w'^2}). \quad (5.4)$$

Although the Reynolds normal stresses are anisotropic (this can readily be seen from the profiles of $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ given in Fig. 5.23), this approach leads to reasonably accurate values for k (within 10%). This was verified for the matrix flow, to be discussed in chapter 8, with additional measurements of the normal velocity v with the prism-cube. An iso-value plot of the turbulent kinetic energy in the plane $y/H = 0.5$, given in Fig. 5.24, shows the increased activity in the free shear layer behind the side face of the cube. Downstream of the reattachment point ($x/H = 2.5$), the shear layer has broadened significantly due to diffusion. The turbulent kinetic energy is rather low in the centre of the wake vortex which indicates that the perturbations induced by vortex shedding are damped out towards the centre. The recovery of the flow further downstream is also observed from Fig. 5.24.

The static pressure measurements, the data along the cross-sections $ABCD$ and

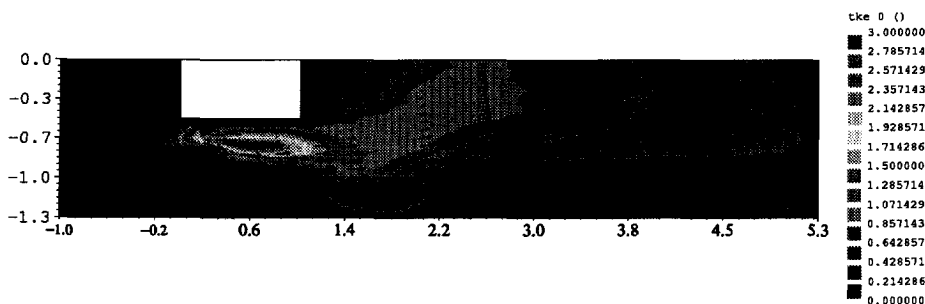


FIGURE 5.24: Iso-value plot of the turbulent kinetic energy k with velocity vectors in the plane $y/H = 0.5$.

$ABCD$ A shown in Fig. 5.16, are rather constant for the leeward face. This further illustrates that the wake is a 'dead-water' zone. Fig. 5.16 further shows the averaged static pressure for the rear face, indicated with the dashed line (---), which is smaller than that for the front face but larger than those for the top and side faces.

5.3.3 Obstacle top face

The top face is characterised by a bound vortex in the vicinity of the leading edge. The oil-surface image of Fig. 5.10 shows two vorticity concentration nodes marked **D**, which are attributed to the footprints of this vortex. Fig. 5.25 gives vector plots of the velocity field in two parallel planes at a distance of 1 mm ($z = 16$ mm, left plot) and 1.5 mm ($z = 16.5$ mm, right plot), just above the top face. The velocity field in the lower plane ($z = 16.0$ mm) is in agreement with the features derived from the pigment pattern at the top face. The smaller pitch between the counter-rotating centre nodes for $z = 16.5$ mm is consistent with the bent shape of the top vortex. An extrapolation of the separation line with $\bar{u} = 0$ suggests that the reattachment of the shear layer at the top face occurred almost at the trailing edge. This is consistent with the oil-film plot in Fig. 5.10 which shows a smeared pigment pattern in the proximity of the trailing edge. Traverses of $\bar{u}/u_B$, $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ at the top centreline of the cube are plotted in Fig. 5.26 for different values of x/H . A strong recirculation zone progresses in the downstream direction where the largest negative velocities are observed for $x/H = 0.5$. The velocity gradients are particularly large in this recirculation area and are of the order of magnitude of $d\bar{u}/dx \approx 5000$ 1/s. It was already discussed in chapter 4 that these strong velocity gradients may possibly lead to velocity gradient bias due to the finite size of the measuring volume. But because of the relatively small measuring volume encountered in the LDA optics used, this error is much smaller than the experimental uncertainty. The profiles of $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ show maxima at approximately $0.13H$ above the top wall of the cube which coincides with areas of large local production of turbulence. The from the leading top edge separated shear layer reattached intermittently at the top face close to the trailing edge which also contributed to the increased turbulence

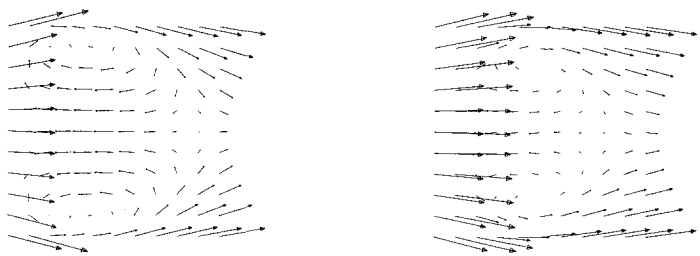


FIGURE 5.25: Vector plots of the velocity field in planes at 1 mm ($y = 16$ mm, left plot) and 1.5 mm ($y = 16.5$ mm, right plot) above the top face.

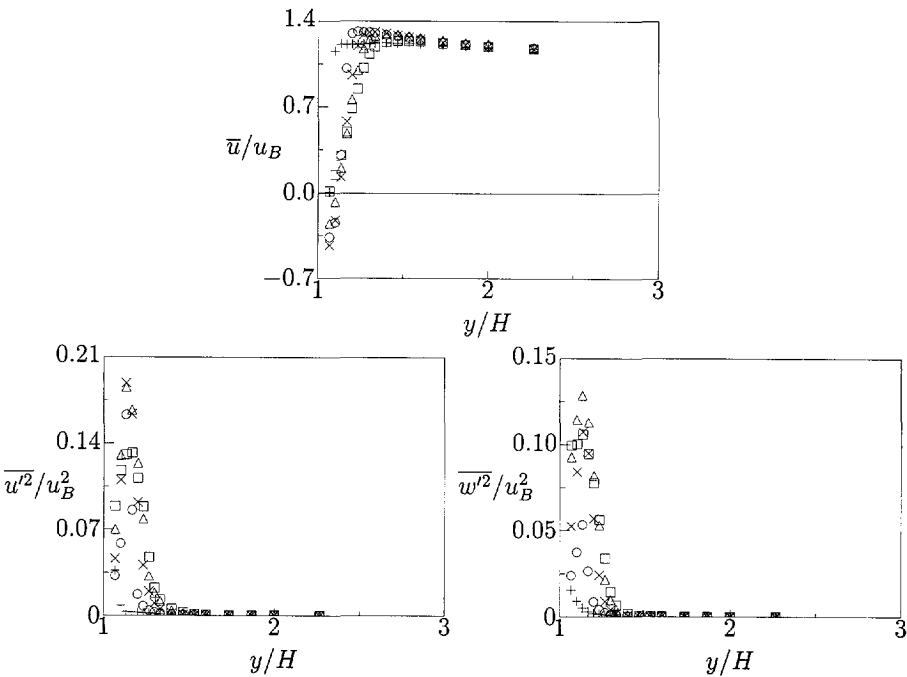


FIGURE 5.26: Profiles of $\bar{u}/u_B$, $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ at the top of the obstacle along the plane $z/H = 0$: (+) $x/H = 0.1$, (o) $x/H = 0.3$, (x) $x/H = 0.5$, (Δ) $x/H = 0.7$ and (□) $x/H = 0.9$.

intensities.

The skewness and kurtosis factors varied significantly in the proximity of the top vortex. While the skewness is positive close to the solid top face, it decayed steeply to -0.5 at the centre of the recirculation (see Fig. 5.27). Apparently, while the streamwise velocity had strong positive spikes close to the solid wall, the spikes were directed in opposite direction in the vortex core. The kurtosis K_u , given in Fig. 5.27, is significantly smaller than 3 at the locations close to the cube face

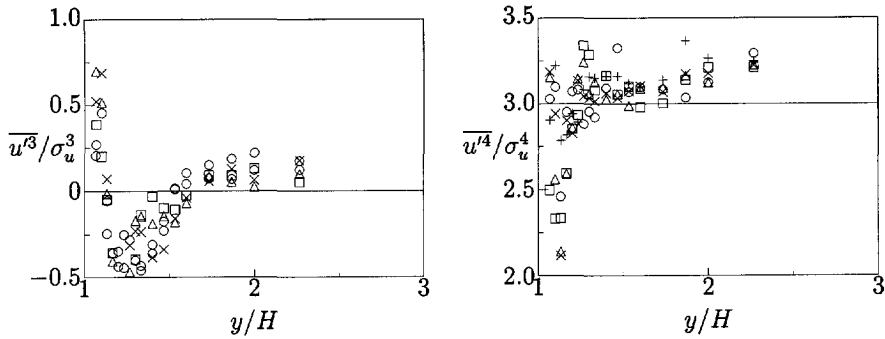


FIGURE 5.27: Profiles of the skewness S_u and kurtosis K_u at top of the cube in the symmetry-plane $z/H = 0$: (+) $x/H = 0.1$, ($\circ$) $x/H = 0.3$, ($\times$) $x/H = 0.5$, ($\triangle$) $x/H = 0.7$ and ($\square$) $x/H = 0.9$.

which indicated that the fluctuations were distributed more uniformly. The kurtosis gradually becomes larger than 3 just outside the vortex core.

The top vortex caused a pronounced suction peak in the surface static pressure, see Fig. 5.16. A pressure peak was found at the reattachment point, close to the trailing edge.

5.3.4 Obstacle side face

The flow at the two sides of the cube was dominated by a side vortex close to the leading edge, confined by the main stream and the cube. These side vortices had origins at the nodes close to the leading edge corner on the channel floor which are marked with **E** in Fig. 5.10. The imprints of the vortex on the side faces can be seen in the oil-film close-up given in the right image in Fig. 5.18. The removal of visualisation pigment close to leading edge, the white band, is the result of the rotation of the vortex. The pigment structure close to the trailing edge is caused by the flow reattachment. This side vortex was also demonstrated from LDA-

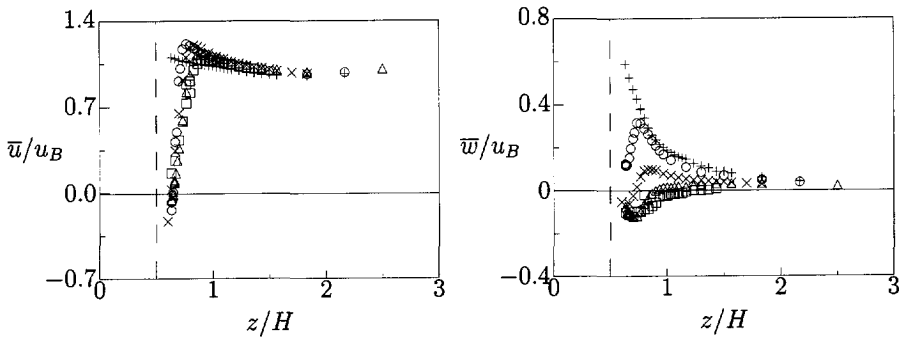


FIGURE 5.28: Profiles in the spanwise direction of the scaled mean velocities $\overline{u}/u_B$ and $\overline{w}/u_B$ at the lateral face at $y/H = 0.5$: (+) $x/H = 0.1$, ($\circ$) $x/H = 0.3$, ($\times$) $x/H = 0.5$, ($\triangle$) $x/H = 0.7$ and ($\square$) $x/H = 0.9$.

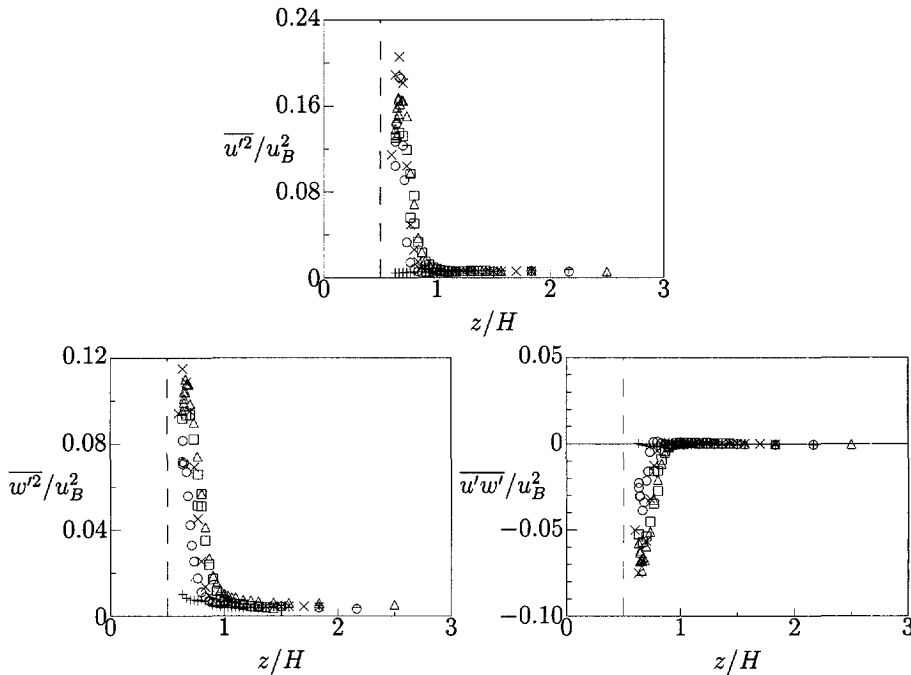


FIGURE 5.29: Traverses in the spanwise direction of the Reynolds stresses $\overline{u'^2}/u_B^2$, $\overline{w'^2}/u_B^2$ and $\overline{u'w'}/u_B^2$ at the lateral face at $y/H = 0.5$: (+) $x/H = 0.1$, (o) $x/H = 0.3$, (x) $x/H = 0.5$, (Δ) $x/H = 0.7$ and ($\square$) $x/H = 0.9$.

measurements of the mean velocities, given in Fig. 5.28 for five spanwise traverses at $y/H = 0.5$. The negative velocities extend to approximately $x/H = 0.75$. The side vortex can also be observed from the vector plot in Fig. 5.20. The scaled Reynolds stresses $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$, given in Fig. 5.29 at corresponding stations, are maximum in the vicinity of the cube wall, at approximately $y/H = 1.1$ - 1.2 , again due to local turbulence production in the shear layer. The intermittent reattachment of the separated shear layer contributed also to the increased turbulence production. Far away from the cube surface, the profiles asymptote to the boundary layer values. The dimensionless Reynolds shear stresses $\overline{u'w'}/u_B^2$ in the xz -plane, also shown in Fig. 5.29, are almost zero at $x/H = 0.1$ and become maximum for $x/H = 0.5$ - 0.7 . As we would expect, this location coincides with the location where the turbulent kinetic energy is maximum and where the mean velocity gradients are steep. Since the correlation between the streamwise and spanwise velocity vanishes for large distances away from the cube, the Reynolds shear stresses decrease to zero. An iso-value plot of $\overline{u'w'}$ is shown in Fig. 5.30 for $y/H = 0.5$. The plot shows large values of $\overline{u'w'}$ in the vicinity of the side vortices and the free shear layers just behind the cube.

The skewness factor S_u , see Fig. 5.31, displays a marked peak close to the side wall at approximately 1-2 mm away from the wall, similarly as for the top face.

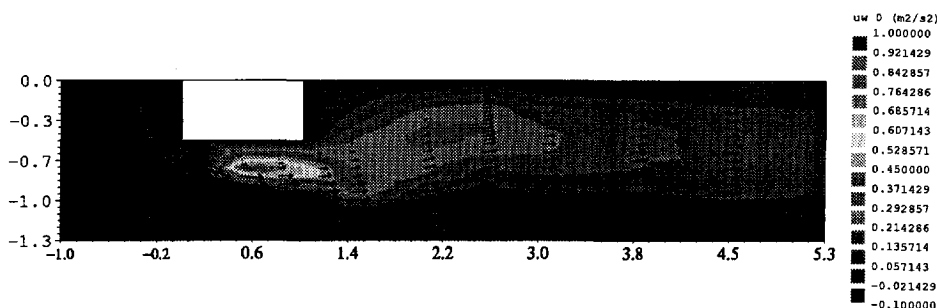


FIGURE 5.30: Iso-colour plot of the Reynolds shear stresses $\overline{u'w'}$ in the plane $y/H = 0.5$.

The location of this peak corresponds to the location where the negative streamwise velocity is maximum in the side vortex. The peak indicates that positive velocity spikes occurred frequently in the velocity histograms. It is noted that all five profiles coincide remarkably well. The skewness of the spanwise velocity distribution, S_w , also given in Fig. 5.31, had strong positive peaks approximately in the core of the side vortex. This illustrates that the intermittent reattachment was associated with strong wallward jets.

The kurtosis K_u is significantly smaller than 3 close to the side wall for all five locations shown in Fig. 5.32. It arises the value for Gaussian distributions (i.e. 3) and become maximum at about the location where the steepest gradients in $\bar{u}$ were found. The K_w -values, also given in Fig. 5.32, are more or less scattered around 3.

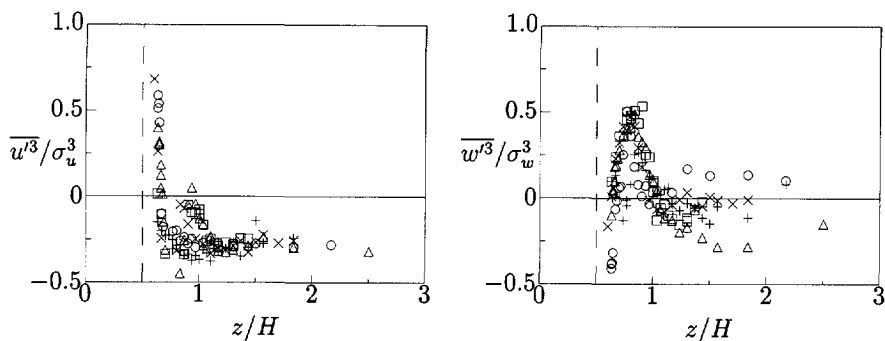


FIGURE 5.31: Profiles of the skewness S_u and S_w at the side face of the cube at $y/H = 0.5$: (+) $x/H = 0.1$, (o) $x/H = 0.3$, (x) $x/H = 0.5$, (Δ) $x/H = 0.7$ and (□) $x/H = 0.9$.

The sides of the cubes exhibit the largest normal vorticities ω_y ($\omega_y = (\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z})$). This is illustrated in the iso-colour plot of ω_y in the xz -plane at $y/H = 0.5$ (again, only half of the domain is shown). The largest values of the vorticity are centred around the front of the side vortex due to the strong curvature of the flow.

Similarly as for the top vortex, a suction peak in the static pressure was found below the side vortices, see Fig. 5.16. The local static pressure became maximum at the back half of the side face where the shear layer reattached.

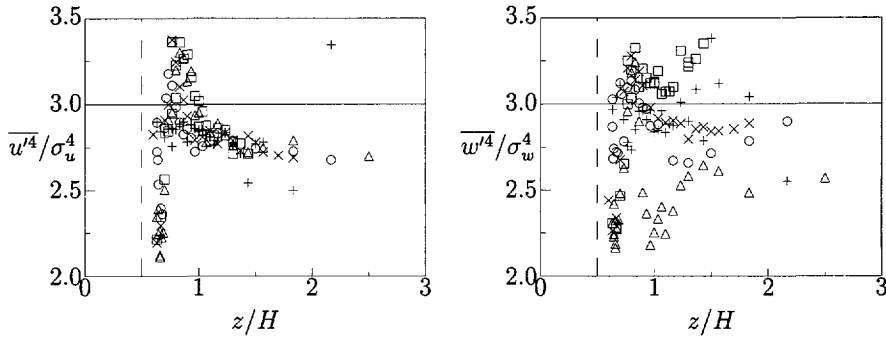


FIGURE 5.32: Profiles of the kurtosis K_u and K_w at the side face of the cube at $y/H = 0.5$: (+) $x/H = 0.1$, (o) $x/H = 0.3$, (x) $x/H = 0.5$, (Δ) $x/H = 0.7$ and ($\square$) $x/H = 0.9$.

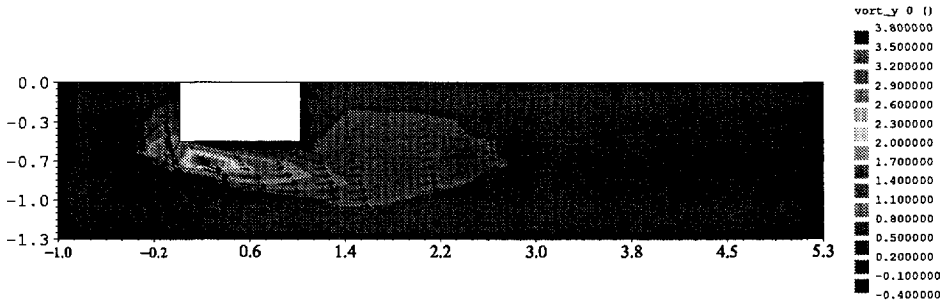


FIGURE 5.33: The vorticity $\omega_y = (\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z})$ in the xz -plane at $y/H = 0.5$.

5.3.5 Comparison with literature

The single cube case discussed in this thesis differed in several aspects from the case of Martinuzzi et al. [101]. The isolated cube as discussed here was placed in a developing turbulent channel flow at $Re_H = 4440$, while Martinuzzi et al. considered a fully developed channel flow for a Reynolds number which was almost one order of magnitude higher (i.e. $Re_H = 40,000$). Further, the channel-height-to-cube-size ratio was different ($D/H = 3.4$ in the present case versus $D/H = 2$ in Martinuzzi's case).

Although, the flow fields were in general more or less qualitatively similar, we found shear layer reattachment at the top and side faces of the cube while this was absent in the fully developed case. Further, the flow reattachment at the channel floor downstream of the cube occurred at a larger distance away from the leeward face than in the fully developed channel flow. The flow dynamics was also different as illustrated by the measurements of the vortex shedding frequency. Although the present flow is superficially comparable to the fully developed channel flow, these noticeable differences in the flow will have a significant effect on the local convective heat transfer from the cube.

Profiles of the scaled Reynolds normal stresses $\overline{u^2}/\overline{u^2}_{max}$ and $\overline{w^2}/\overline{w^2}_{max}$ and mean velocity $\overline{u}/\overline{u}_{max}$ are compared in Fig. 5.34 to results from Martinuzzi et al.

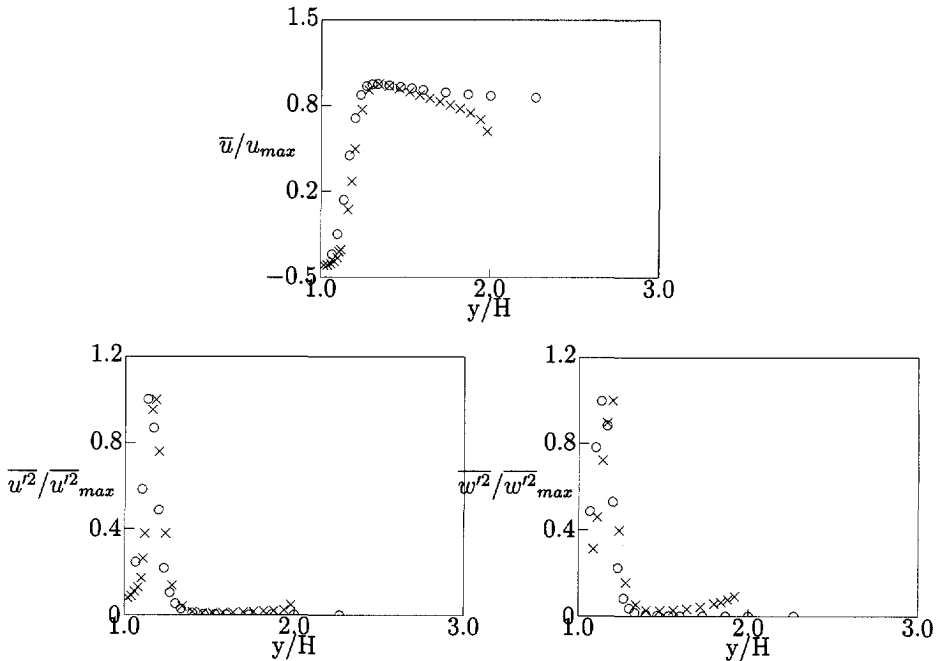


FIGURE 5.34: Comparison between the present data and literature data: traverses of $\bar{u}/\bar{u}_{max}$, $\overline{u'^2}/\overline{u'^2}_{max}$ and $\overline{w'^2}/\overline{w'^2}_{max}$ in the normal direction at $z/H = 0$ and $x/H = 0.5$: (o) current data and (x) literature (from Martinuzzi and co-workers).

[101]. To allow the comparison, the values were scaled with the maximum value in the profile. The coordinate was scaled with the cube height. The agreement in the proximity of the cube is superficially good, the differences were caused by the aforementioned flow differences. The deviations observed around $y/H = 2$ are due to the presence of the channel roof for the literature data (the aspect ratio was $D/H = 2$ in that case).

5.4 Convective heat transfer

Heat transfer measurements were performed for bulk velocities of 2.8, 3.2, 4.0, 4.5 and 5.0 m/s which correspond to Reynolds numbers of respectively $Re_H = 2750$, 3200, 4000, 4440 and 4970. For these Reynolds numbers, the channel flow was turbulent. While the flow field was documented with LDA for $Re_H = 4440$, flow visualisation experiments and measurements of vortex shedding frequencies in the wake of the cube illustrated that the derived flow field applied to the entire range of Reynolds numbers in which the heat transfer measurements were performed (i.e. $2500 < Re_H < 5000$).

Additional measurements were performed for laminar developing channel flows. Such a situation was achieved by running the wind tunnel at a sufficiently low speed

such that the tripping wire was not effectively tripping the flow to turbulence. The Reynolds numbers for these situations were $Re_H = 533, 888, 1332, 1776$ and 2220 . Since no LDA data were collected for these cases, only some flow visualisations, the interpretation of the results is limited to a qualitative one.

The temperature of the internal copper core was adjusted to $75.0\text{ }^\circ\text{C}$, which was regulated to the supplied power through a feedback loop. For one Reynolds number, $Re_H = 4440$, the copper core temperature was varied between $60\text{ }^\circ\text{C}$ and $80\text{ }^\circ\text{C}$ to investigate the influence of the surface temperature boundary condition on the heat transfer coefficient. Further, these experiments illustrated the accuracy of the surface temperature measurements. The inlet temperature of the air was kept at room temperature ($20\text{--}21\text{ }^\circ\text{C}$). The real ambient temperatures during the different experiments were measured and used to calculate the radiative heat fluxes and the heat transfer coefficients.

5.4.1 Distributions of the surface temperature

Verification of the accuracy of the infrared thermography was enabled by comparing the infrared surface temperature distributions with liquid crystal temperature measurements. The improvement of infrared image restoration was already confirmed in the discussion of the infrared image restoration technique (see chapter 4). The results shown here refer to different Reynolds numbers.

Liquid crystal results are shown in Fig. 5.35 along two paths at the cube surface (cross-sections $ABCD$ and $ABCD_A$) for three Reynolds numbers, $Re_H = 2220, 3200$ and 4440 . The lowest Reynolds number corresponds to the laminar developing channel flow. The results for this Reynolds number were added to demonstrate the agreement between infrared and liquid crystal measurements in the entire range of Reynolds numbers considered. The upper figure corresponds to cross-section $ABCD_A$ which denotes a plane parallel to the mounting base at $y/H = 0.5$ and cross-section $ABCD$ denotes a plane perpendicular to the mounting base at the symmetry plane $z/H = 0$. The LC measurements, only available at discrete locations at the cross-sections of isotherms, are indicated with $\triangle$, $\square$ and $\diamond$ for $Re_H = 2220, 3200$ and 4440 , respectively. Notice that liquid crystal data are not available for all the cube faces for $Re_H = 2200$. The infrared measurements, obtained under similar experimental conditions, are also plotted in the figures (lines with $+$, $\circ$ and $\times$ for $Re_H = 2220, 3200$ and 4440 , respectively). The surface temperatures from the two different thermography techniques agree very well. This confirms the accuracy of $0.4\text{ }^\circ\text{C}$ which is claimed for the IR temperature measurements.

Isotherm maps of the five faces exposed to air flow for the five turbulent Reynolds number cases, shown in Fig. 5.36, are very similar. The five faces are folded out and mapped into one plane. The flow is from left to right. Together with the oil-film results it is reasonable to assume that the surface heat transfer and flow field are qualitatively similar throughout the turbulent Re_H -range studied. The topology of the surface temperature plots show a marked similarity with the pigment structure at the top face of the cube (see Fig. 5.10). An accumulation of pigment, the imprint of the bound top vortex, is observed at the location of the hot spot. Since the cubes

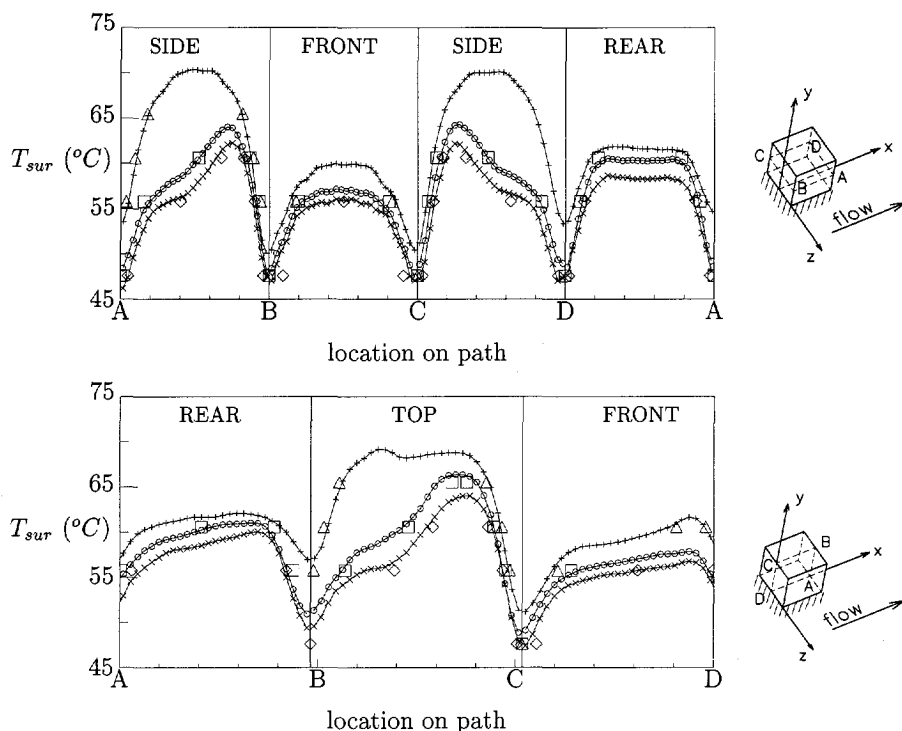


FIGURE 5.35: Surface temperatures along path $ABCD$ A (upper plot) and along path $ABCD$ (lower plot) for $Re_H = 2220$, $Re_H = 3200$ and $Re_H = 4440$ respectively: comparison between LC (Δ , $\square$ and $\diamond$) and infrared thermography (lines with +, $\circ$ and $\times$).

were mounted vertically, it is important to note that the isotherm distributions are symmetric. This symmetry indicates that buoyancy effects are negligible, which was to be expected since the ratio of the Grashof number to Reynolds number squared (Gr/Re^2) is significantly smaller than one ($Gr/Re^2 = Hg\Delta T/TU^2 \approx 0.001$, where g is gravity, $H = 0.015$ m is the cube size, $\Delta T = 40$ $^{\circ}\text{C}$ is the temperature difference between the surface and the ambient, $T = 333$ $^{\circ}\text{C}$ is the absolute temperature and $U = 4$ m/s is the characteristic velocity).

Variation of the copper temperature leads to different levels of the surface temperature distributions. For five copper temperatures, i.e. $T_{co} = 60, 65, 70, 75$ and 80 $^{\circ}\text{C}$, the surface temperature distribution was measured with the infrared system. The surface temperatures along cross-sections $ABCD$ A and $ABCD$ are given in Fig. 5.37. It is noted that the temperature profiles are more or less congruent, this illustrates again the accuracy of the measurements. Further, the heat transfer coefficients calculated from these surface temperature boundary conditions differed about 5%. These differences are within the experimental uncertainty of the method and measurement technique. Taking this experimental uncertainty into account, it can be concluded that the convective heat transfer did not depend on the boundary condition imposed by the different surface temperatures.

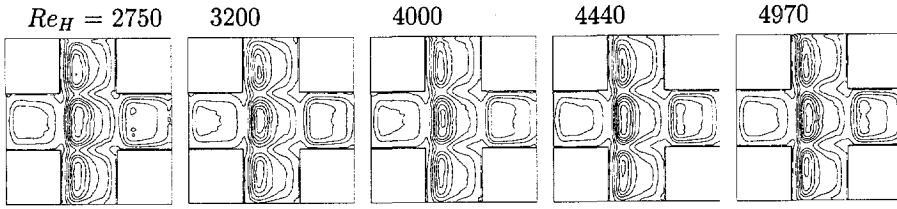


FIGURE 5.36: Infrared surface temperature maps for Reynolds numbers of $Re_H = 2750$, 3200 , 4000 , 4440 and 4970 . Flow is from left to right.

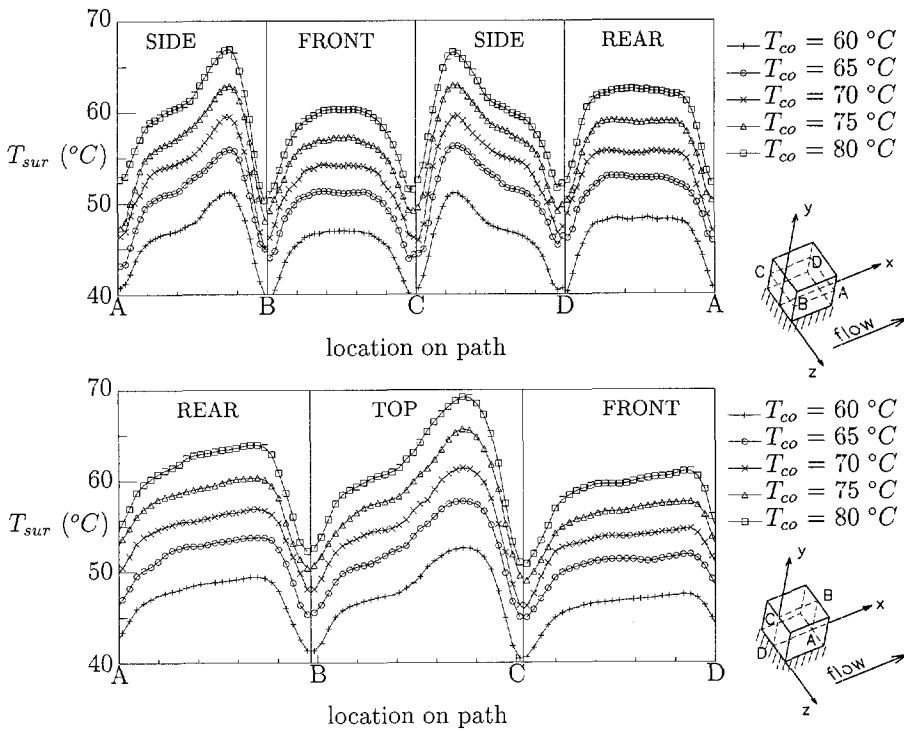


FIGURE 5.37: Profiles of the surface temperature along path ABCDA (upper plot) and along path ABCD (lower plot) for $Re_H = 4440$ for different internal copper temperatures.

5.4.2 Distributions of the local heat transfer coefficient

Profiles of the surface temperature and heat transfer coefficient are given in Figs. 5.38 and 5.39 along paths ABCDA and ABCD, respectively. The local heat transfer coefficients were made dimensionless with the cube-averaged heat transfer coefficient $\bar{h}$. The coinciding profiles of the heat transfer coefficients again illustrate that similarity was verified throughout the range of Reynolds numbers considered for

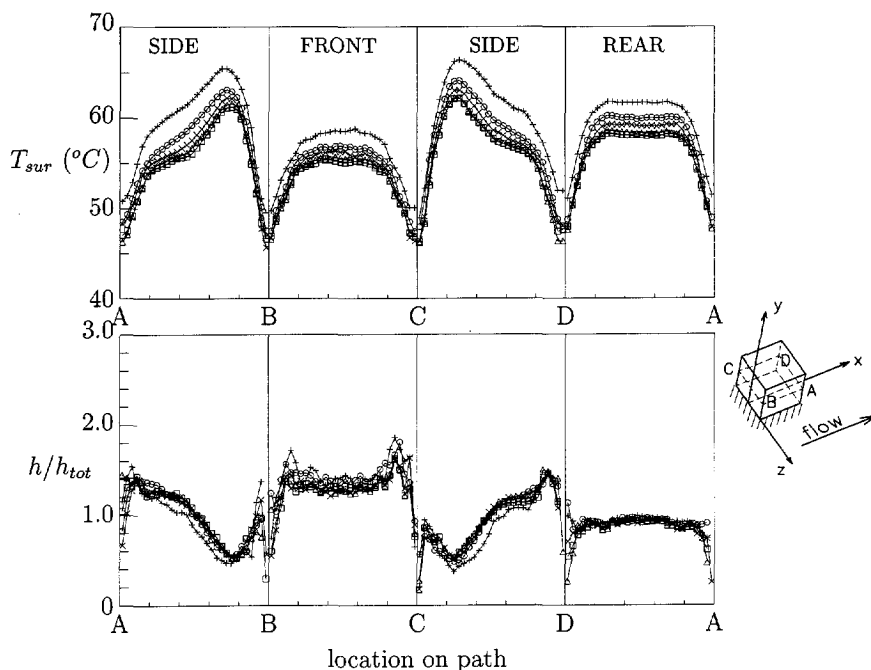


FIGURE 5.38: Distributions of the surface temperature and heat transfer coefficient along path $ABCD A$ parametric in the Reynolds number: (+) $Re_H = 2750$, (o) $Re_H = 3200$, (x) $Re_H = 4000$, (Δ) $Re_H = 4440$ and ($\square$) $Re_H = 4970$.

the turbulent heat transfer measurements ($2750 < Re_H < 4970$) and that the flow field determined for $Re_H = 4440$ applied for this range. The temperature profiles are added to support the discussion of the marked variations in the local convective heat transfer. Further, these are the measurements from which the convective heat fluxes were derived.

Front face

From the flow study it was found that the oncoming flow impinged at the front face of the cube leading to a flow stagnation point at about $y/H \approx 2/3$ at the symmetry line $z/H = 0$ (see Fig. 2.1). Flow impingement generally causes a large and uniform convective heat transfer in the proximity of the stagnation point. This is clearly seen in the profiles along path $ABCD A$ and $ABCD$ in Figs. 5.38 and 5.39, partition BC and CD , respectively. The values of h in the proximity of the stagnation point compared well with data found in the literature.

The small minimum in the heat transfer coefficient, found at the foot of the front face close to the mounting base (see Fig. 5.39, partition CD), corresponds to the centre of the horseshoe vortex while h slightly increases towards the channel floor, point D , which is in opposite direction to the local fluid velocity in the horseshoe vortex.

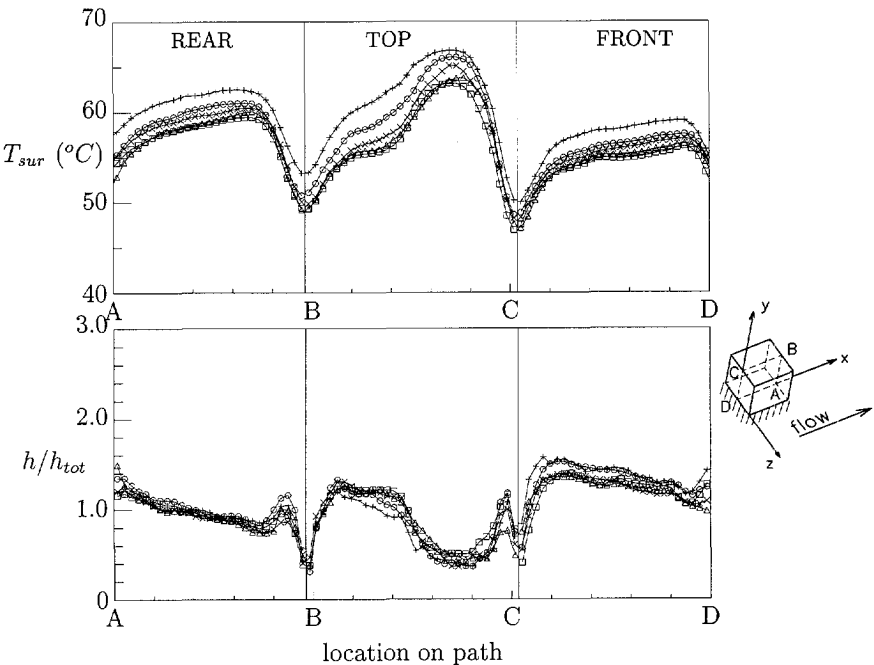


FIGURE 5.39: Distributions of the surface temperature and heat transfer coefficient along path ABCD parametric in the Reynolds number: (+) $Re_H = 2750$, (o) $Re_H = 3200$, (x) $Re_H = 4000$, ($\triangle$) $Re_H = 4440$ and ($\square$) $Re_H = 4970$.

The study of the flow field revealed that the Reynolds normal stresses (or turbulent kinetic energy) showed large peaks in the proximity of the horseshoe vortex. The peaks in the Reynolds normal stresses coincided with maximum turbulence production areas due to large velocity gradients. Furthermore, the unstable behaviour of the horseshoe vortex contributed to the increased turbulence production. Since this region was identified as an area of minimum convective heat transfer, it is suggested that a dissimilarity existed between the heat and momentum transfer: although the level of turbulence generally intensifies the heat transport leading to higher h values, this does not apply for the horseshoe vortex region. An explanation for this departure was sought in the local fluid temperature. The rotational behaviour of the horseshoe vortex caused the residence time of the fluid in the vortex to be high, allowing the local fluid temperature to increase and, in turn, reduced the local convective heat transfer. Further, heat leakage via conduction from the cube through the mounting base might cause a slight increase in temperature of the base plate upstream of the cube. The rotating horseshoe vortex was, therefore, also heated by this secondary effect, although the effect was estimated to be smaller than 5% (this was estimated from a global energy balance). An opposite observation was made around the stagnation point. Although the heat transfer coefficients are high and rather uniform, the turbulent kinetic energy was moderate compared to those

measured in the horseshoe vortex region. But for this area, the high-velocity and low-enthalpy impinging flow caused a beneficial convective cooling.

A similar mechanism was observed by Suzuki et al. [82, 160, 161, 162] for the surface heat transfer in a boundary layer flow disturbed by an inserted cylinder normal to the flow direction. They found a significant enhancement in the local surface heat transfer at the location where the friction factor became minimum. The observed dissimilarity was assumed to be directly related to the vortex shedding from the cylinder. The so-called washing action caused entrainment of cold air at the location where the vortex was directed downward to the surface. An ejection of hot fluid existed at the location where the flow separated from the surface. They supported their hypothesis with time-dependent simulations [77] showing that this dissimilarity also existed for the instantaneous friction and surface heat transfer.

More or less similar observations were made by Natarajan and Chyu [123, 33] for distributions of the mass transfer coefficient of a cube in a thin boundary layer flow at Reynolds numbers between 3.1×10^4 and 1.1×10^5 . The horse vortex region was also indicative for lower mass transfer coefficients while the remainder of the front face exhibited a more or less constant mass transfer rate.

Rear face

The convective heat transfer from the leeward face of the cube was more or less uniform, except for the local maximum at the cube upper edge (see partition *DA* and *AB* in Fig. 5.38 and Fig. 5.39, respectively). The flow on this face is dominated by the arch-shaped vortex. Although the turbulence intensity was rather low in the core of the vortex, the outer region of the vortex was rather turbulent because of the vortex shedding and the turbulence production due to large averaged velocity gradients. This is seen in the plot of the turbulent kinetic energy k in the plane $y/H = 0.5$, given in Fig. 5.24. While the velocity fluctuations were moderate in the vortex core, an increased activity was found in the free shear layer behind the side faces of the cube. Further downstream, the flow recovers to an equilibrium state, as is seen from the decreasing turbulence intensity. Partially due to the mechanism of vortex shedding, the fluid recirculation in the wake caused an efficient mixing which led to higher and more uniformly distributed local fluid temperatures. This in turn caused uniform distributions of the local heat transfer coefficients. The local maximum at the cube upper edge of the leeward face is explained by entrainment of cold air from the shear layer over the top face which locally benefits the convective heat transfer. The heat transfer coefficient increases somewhat in the direction of the channel floor. This is explained by the rotation of the wake vortex and the impingement of low-temperature fluid which entered the wake recirculation from the bottom.

Natarajan and Chyu [123, 33] reported different local mass transfer coefficients at the rear face of the cube. While in the present study the distributions of h were more or less uniform except in the vicinity of the upper top and side edges, where a local maximum and sharp minima were respectively found, the distributions of the mass transfer coefficient showed uniform coefficients across the entire face with

exception of a maximum towards the channel floor. They argued that close to the channel floor, high-rotational fluid from the horseshoe vortex entrains in the wake region and caused the mass transfer to become locally maximum. The maximum close to the upper edge was not present in their case. The shear layer reattachment at the top face of the cube, as found to be an important mechanism at the top face in the present study, caused the flow to be directed from the top wall and, therefore, such a pronounced entrainment was not detected by Natarajan and Chyu.

Side faces

The profiles of the heat transfer coefficient at the side faces are shown between partitions *AB* and *CD* of path *ABCD*A in Fig. 5.38. A marked minimum was found at the front half of the side face, a marked maximum existed at the back half of the side face, close to the trailing side edge. The variation in h along the face is up to 100 %. The flow analysis showed that the flow separated at the leading side edge and reattached downstream at the side face at approximately $x/H = 0.9$. The flow separation resulted in a side vortex, which is oriented nearly in parallel to the leading side edge. The 'artist's impression' shown in Fig. 2.1 is a long-term averaged illustration of the flow structure. Flow instabilities will cause somewhat a-symmetric flow patterns in time: in one cycle, a vortex is shed from the left side face while it is shed from the right face in the next cycle. This alternating vortex shedding resulted in quasi-periodic separation and reattachment. The side vortex, therefore, vibrates somewhat around an equilibrium state. But despite of this vibration, the side vortex was dominantly present. From the h -profiles in Fig. 5.38 it is seen that the location of minimum heat transfer coefficients corresponded to that of the vortex core. This centre of the vortex is associated with a region of high wall shear stress and low static surface pressure. According to Reynolds analogy, regions of high wall shear stress should imply large heat transfer coefficients and this observation thus suggests that Reynolds' analogy locally fails. An explanation is again found in the local fluid temperature in the vortex. Heat is convected in the recirculation region, leading to an increased fluid temperature and the vortex is, therefore, acting as a kind of insulation layer. This buffer prevents beneficial convective heat transfer. On the contrary, the intermittent flow reattachment convects fresh and relatively cold fluid to the surface. And together with strong velocity fluctuations normal to the wall, this mechanism caused the convective heat transfer intensification observed from the local maximum of the heat transfer coefficient in the proximity of the reattachment point ($x/H \approx 0.9$). Areas of flow reattachment are usually characterised by low wall shear stresses, and contrary to Reynolds analogy, it is the large velocity fluctuations normal to the wall as well as the low-enthalpy fluid which intensify the local removal of heat.

The mass transfer coefficients of Natarajan and Chyu [123, 33] showed a slight increase in the streamwise direction for the side face. This was caused by the flow recirculation which enveloped the entire side face. Due to the oncoming flow conditions, the shear layers which separated from the leading side edges did not reattach on the side faces. Therefore, the gradients in the local mass transfer coefficients re-

mained moderate in contrast to the present results which were primarily dominated by flow separation and subsequent reattachment.

Top face

Profiles of the heat transfer coefficients for the top face are shown in Fig. 5.39, partition *BC*. These profiles show also large spatial streamwise gradients: the heat transfer coefficient drops to a minimum where after is recovers downstream to a plateau of maximum convective heat transfer. Similarly to the less extensive side vortices at the side faces, the top vortex dictates the local convective heat transfer at the top face. It was concluded from LDA measurements that no detectable vortex shedding occurred from the top trailing edge. Instabilities in the flow over the top face did not lead to vortex shedding since the necessary second shear layer was absent. Further, it is believed that due to the vicinity of the opposing channel wall (roof) and the channel floor possible flow oscillations were suppressed. Despite that, the separated top shear layer reattached intermittently at approximately $x/H = 0.9$ at the top face. This caused local heat transfer intensification and led to the local maximum in the convective heat transfer towards the trailing edge. Again, a local heat transfer minimum is observed in an area of high wall shear stresses (on the wall directly below the bound vortex core) and a maximum heat transfer in an area of low wall shear stresses (the reattachment region).

Natarajan and Chyu [123, 33] observed monotonically increasing mass transfer coefficients in the streamwise direction at the top face. They attributed this to the recirculation of the top vortex which enveloped the entire top face. Again, the significant differences between the local heat and mass transfer distributions were caused by the different oncoming conditions as encountered in both studies.

Correlations

Due to the coupling between velocity and pressure field and between velocity and convective heat transfer, a correlation between the local static surface pressure and the surface heat transfer was sought. This attempt is shown in Fig. 5.42 where the local heat transfer coefficients are plotted versus the local static pressure coefficient C_p , as given in Fig. 5.16. Only the results for one typical Reynolds number are shown since it was proven that for the different Reynolds numbers studied the scaled heat transfer coefficients coincided with each other. The results shown were measured along the cross-sections *ABCD*A and *ABCD*, at locations for which the local surface static pressure was measured (see Fig 4.50 for the locations in question). The results for the top ($\times$) and side ($\circ$) faces show marked similarities. The most negative pressure coefficients were found underneath the centre of the top and side vortices. Areas of low static pressure are usually associated with high velocities. These locations corresponded to areas of minimum convective heat transfer. Further downstream, both the pressure and the local heat transfer coefficient increase monotonically. Flow reattachment is subsequently characterised by maxima in both the heat transfer coefficient and surface static pressure. The strong gradient dh/dC_p supports the hypothesis that the strong differences in the heat transfer coefficients

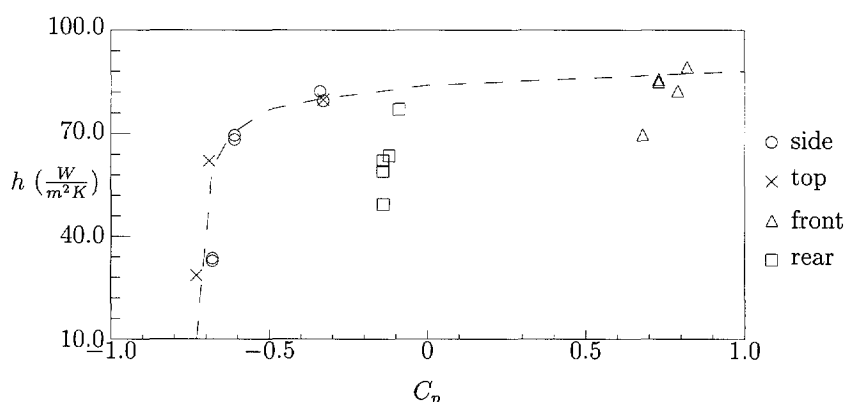


FIGURE 5.40: Local heat transfer coefficients versus the local static pressure coefficient C_p . The markers refer to measurements at different faces of the cube.

are partly caused by temperature effects which are associated with the rotating vortices. The C_p -values for the rear face deviate somewhat from the general trends. Although a marked variation was found in the heat transfer coefficient distribution at the rear face, the pressure distribution is rather flat. Apparently, the static pressure in the wake region is flattened by the wake vortex ('dead-water' zone). The differences in the convective heat transfer are then again explained by temperature effects. Almost flat distributions of both C_p and h were found at the front face. The stagnant flow caused the static pressure to be maximum while the impingement of the cold flow caused large values of the heat transfer coefficient.

If the values for the rear face are omitted, the data can be fitted with a function, asymptoting the h -values found at the front face. This is indicated with the dashed line in Fig. 5.42.

5.4.3 Time-dependent heat transfer

As discussed earlier, the oncoming main flow separated at the sharp leading side edges of the cube, and intermittently reattached further downstream at the side faces of the cube. To investigate a possible coherence between the dynamics of reattachment of the side shear layers and the time-dependent convective heat flux, complementary time-dependent heat flux measurements were performed with the active heat flux sensor as described in chapter 3. The sensor was placed in the back half of the side face of the cube as illustrated in Fig. 3.13. If periodicities were present in the flow, it was expected that these also show up in the heat flux signal due to the direct coupling. From power density spectra of the time-dependent heat flux measurements no increased activity was detected at the frequency of vortex shedding. The fluctuations in the reattached flow were probably too small to have a measurable impact on the heat transfer from the side faces. Further, a kind of spatial averaging was caused by the rather large effective sensor area (4×4 mm) which damped the heat flux fluctuations a bit.

5.4.4 Surface-averaged heat transfer coefficients

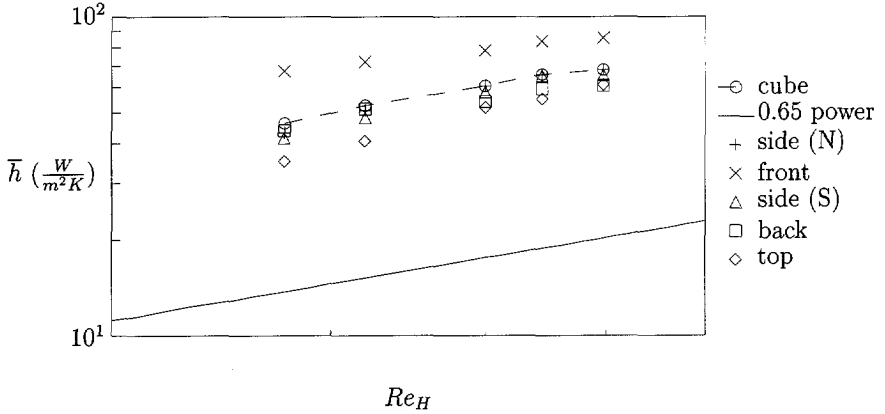


FIGURE 5.41: Cube-averaged and face-averaged heat transfer coefficients as a function of the Reynolds number Re_H : (○) cube average, (+) north, (×) east, (△) south, (□) west and (◇) top. The solid line denotes the relation $\bar{h} = A(Re_H)^{0.65}$.

Surface-averaged heat transfer coefficients were derived from the ratio of the surface-integrated convective heat flux and the difference between the averaged surface temperature and a reference temperature:

$$\bar{h} = \frac{1}{A} \int_A \frac{q_{conv}}{T_{sur} - T_{ref}} dA, \quad (5.5)$$

where A is the total surface of the cube exposed to air flow, q_{conv} and T are the distributions of respectively the local convective heat flux and the surface temperature, and T_{ref} is the reference temperature. This integral can be approximated by the next finite summation:

$$\bar{h} = \frac{\frac{1}{M} \sum_{i,j} q_{conv}}{\frac{1}{M} \sum_{i,j} T_{sur} - T_{ref}}, \quad (5.6)$$

where i and j are the indices denoting the location at the surface A and M is the total number of surface cells per face. Cube-averaged heat transfer coefficients were derived in a similar way, but the integration was then performed over all five faces of the cube exposed to air flow. Note that a different way of defining the averaged heat transfer coefficient is given by averaging the local heat transfer coefficient:

$$\bar{h} = \frac{1}{M} \sum_{i,j} h = \frac{1}{M} \sum_{i,j} \frac{q_{conv}}{T_{sur} - T_{ref}}. \quad (5.7)$$

Differences between both averages appeared to be only a couple of percent (1-3%) for all heat transfer results given in this thesis. Since the former definition (eq. 5.6) is

more related to literature work in which the cube-averaged heat transfer coefficients are usually directly determined from the averaged convective heat flux and the isothermal cube temperature, we will only present these averages.

The cube-averaged heat transfer coefficients $\bar{h}$ are given on log-log scale in Fig. 5.41 as a function of Re_H . On logarithmic scale, $\bar{h}$ is linearly dependent on the Reynolds number. A linear regression suggests a fit of the type $\bar{h} = A(Re_H)^B$, where A and B denote constants. These kind of expressions are commonly found for forced convective heat transfer situations. A least squares fit through the data suggests an exponent of $B = 0.65$. From the naphthalene sublimation experiments of Natarajan and Chyu [123, 33] a coefficient of $B = 0.63$ was found. Although the Reynolds number range considered in their study differed approximately one decade from the present Reynolds number range, the exponents are in good agreement.

The face-averaged heat transfer coefficients, also given in Fig. 5.41 for the different faces, are also linearly dependent on Re_H . The high heat transfer coefficient at the front face (indicated with $\times$) is indicative for the impinging flow at the front face. The low-temperature oncoming flow caused high rates of convective heat transfer. The averaged heat transfer at the remaining faces is of comparable order of magnitude but markedly smaller (about 50 %) than that at the front face. These faces suffer all from strong flow recirculations which have apparently a reducing effect on the convective heat transfer. Among these faces, the top face has the lowest averaged heat transfer coefficient (indicated with $\diamond$ in Fig. 5.41). Besides the strong top vortex found above this face, it is assumed that a blockage effect was present. The channel roof imposes a significant constraint since the channel-height-to cube-height ratio was $D/H = 3.4$. Although flow was trapped over the top face of the cube, the corresponding mass flow rate is somewhat smaller than that along the side faces. A further effect which intensified the convective heat transfer at the side faces was vortex shedding from the trailing side edges. Natarajan and Chyu [123, 33] found exponents between $B = 0.54$ (front face) and $B = 0.67$ (side face) for the different faces of the cubical obstacle, which are in good agreement with the present results.

The face-averaged heat transfer coefficients are plotted versus the face-averaged static pressure $\bar{C}_p$, as given in Fig. 5.16, for five different values of the Reynolds number. It was already mentioned that the pressure measurements corresponded to the whole range of Reynolds numbers of $2500 < Re_H < 5000$. As expected, the windward face ($\bar{C}_p = 0.74$) had the largest convective heat transfer due to flow impingement. Further, this windward face had the largest face-averaged static surface pressure coefficient as the result of the pressure build-up of the stagnant flow. The correlation for the other faces is not ambiguous. The convective heat transfer for the leeward face ($\bar{C}_p = -0.13$) is significantly lower than that at the front face. But the heat transfer for the other faces (top face, $\bar{C}_p = -0.59$, and side face, $\bar{C}_p = -0.13$) tends to increase for even more negative pressure coefficients.

5.4.5 Influence of the oncoming flow conditions

If no tripping strip is used to accelerate the transition to a turbulent flow, the flow is identified as a developing laminar boundary layer flow. Both the flow field and the

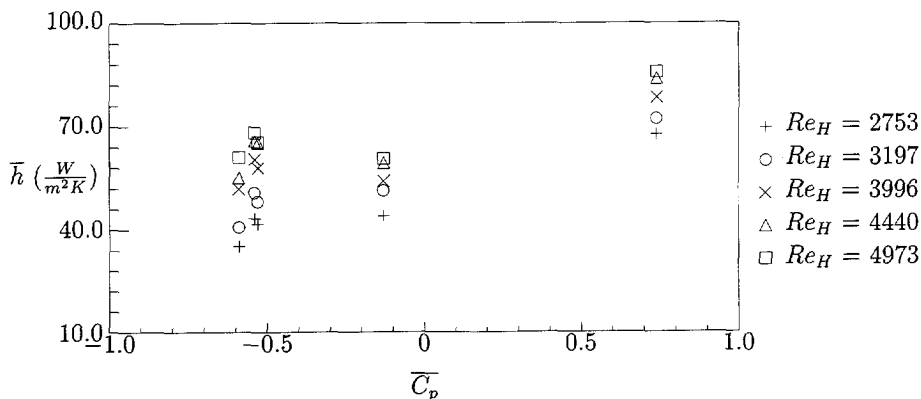


FIGURE 5.42: Face-averaged heat transfer coefficients as a function of the averaged static pressure coefficient $\overline{C_p}$. The markers refer to 5 different Reynolds numbers. The five series of values refer to the five faces: top face ($\overline{C_p} = -0.59$), side faces ($\overline{C_p} = -0.53, -0.54$), rear face ($\overline{C_p} = -0.13$) and front face ($\overline{C_p} = 0.74$).

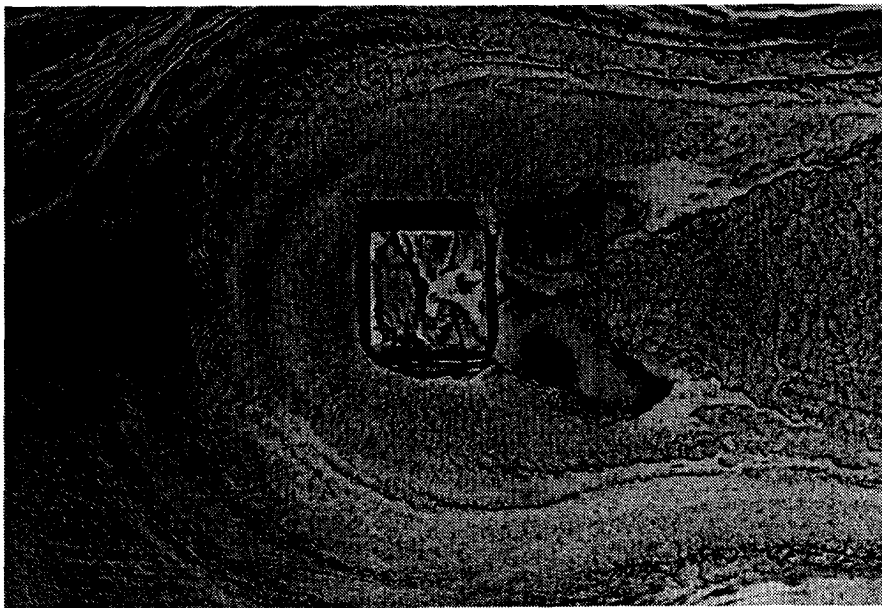


FIGURE 5.43: Oil-film visualisation of the surface flow pattern of a single cube in a developing laminar channel flow at $Re_H = 5000$.

distributions of the local convective heat transfer differ significantly from the turbulent case. A laminar developing boundary layer was also encountered for the cases that the bulk velocity (or Reynolds number) was below the critical value needed to

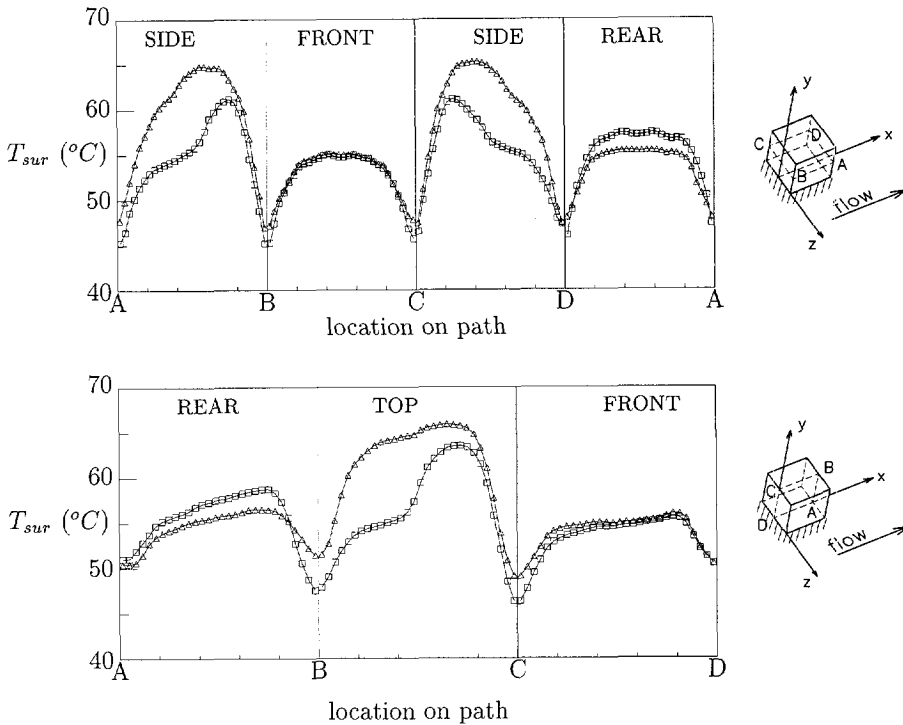


FIGURE 5.44: Comparison of surface temperature measurements for the single cube in a developing turbulent ($\square$) and laminar ($\triangle$) channel flow. The temperatures are shown along the paths $ABCD$ A and $ABCD$ as indicated in the sketches ($Re_H = 4440$).

effectively trip the flow (i.e. $Re_H = 2500$). It is too ambitious to thoroughly investigate the flow structure with the LDA-system in this case. However, an attempt was made to analyse the flow field with the visualisation techniques. An example of an oil-film visualisation of the flow around a cube in a laminar developing channel flow is given in Fig. 5.43. The Reynolds number for this case was $Re_H = 5000$. The flow features observed are at least qualitatively similar to those found in the flow over the cube in the developing turbulent channel flow, discussed in the preceding sections. From a detailed analysis of Fig. 5.43, it appeared that the recirculation regions are generally larger in the non-tripped case. The larger size of the horseshoe vortex affects the slope of the shear layer as it separates from the leading edge. For the laminar case, the top and the side vortices covered these cube faces entirely (see the imprints in Fig. 5.43) and shear layer reattachment did not occur at these faces. This is in contrast with the turbulent case for which the top and side shear layers reattached at the back half of the top and side faces. This flow reattachment caused that the vortices were much smaller and that they were located close to the leading edges of the cube.

The effect of boundary layer tripping and, therefore, of the oncoming flow condi-

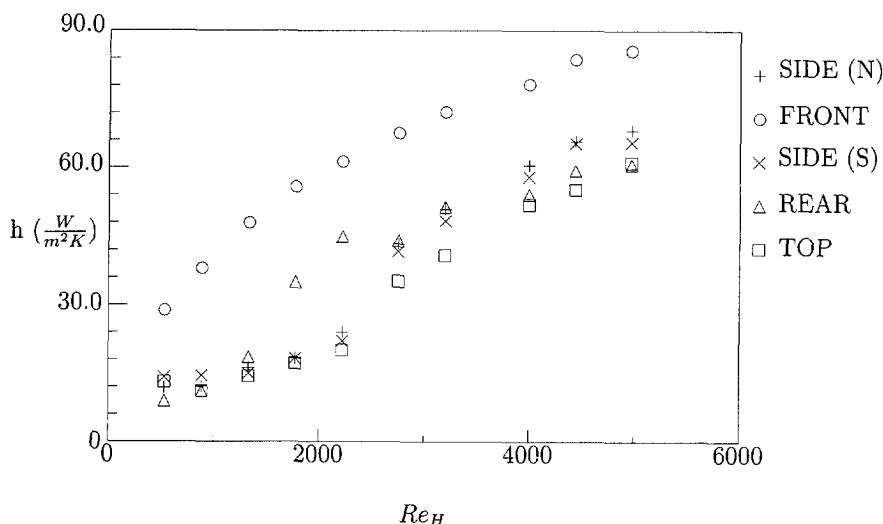


FIGURE 5.45: Face-averaged heat transfer coefficients for the single cube as a function of the Reynolds number Re_H .

tions, is directly reflected in the convective heat transfer. Infrared surface temperature measurements are given in Fig. 5.44 along cross-sections $ABCD$ and $ABCD$. The Reynolds number was $Re_H = 4440$ for both cases. Significant differences in the surface temperature are observed, in particular at the top and side faces which were covered by the top and side vortices. Because these vortices were much smaller in the turbulent case compared to the laminar case, the corresponding temperature peaks are, therefore, much thinner. Further, flow reattachment downstream on these faces in the turbulent case intensified the convective heat transfer which is reflected in a decrease in the surface temperature. The high surface temperatures for the laminar untripped case are explained by the absence of flow reattachment. The differences in the surface temperatures and, therefore, in the surface heat transfer, are very small for the rear face. Differences diminish even within the experimental uncertainty for the front face of the cube. Apparently, the heat transfer in the front stagnation region is hardly effected by the oncoming flow conditions.

Face-averaged heat transfer coefficients are given in Fig. 5.45 for the Reynolds number range $533 < Re_H < 4973$. It is noted that the results for the five largest Reynolds numbers are reproduced from Fig. 5.41. Some interesting features appear from the figure. A clear jump in the heat transfer coefficient for the top and side faces occurred at a critical Reynolds number of about $Re_H \approx 2500$. It is obvious that this jump in the heat transfer is caused by the tripping strip which becomes affective for a certain Reynolds number. A laminar boundary layer thickness scales with $Re_x^{-1/5}$ while a turbulent boundary layer varies with $Re_x^{-1/2}$ (Re_x denotes a local Reynolds number, i.e. $Re_x = Ux/\nu$). When the boundary layer is too thick compared to the tripping strip, which is the case for the lower values of Re_H , the strip is not effectively tripping the flow and, therefore, no transition to a turbulent

boundary layer flow will occur. Only the convective heat transfer from the top and side faces of the cube were significantly affected by the transition to a turbulent channel flow. The averaged heat transfer coefficient at the front face increased gradually without a significant jump. This was already concluded from the surface temperatures shown in Fig. 5.44.

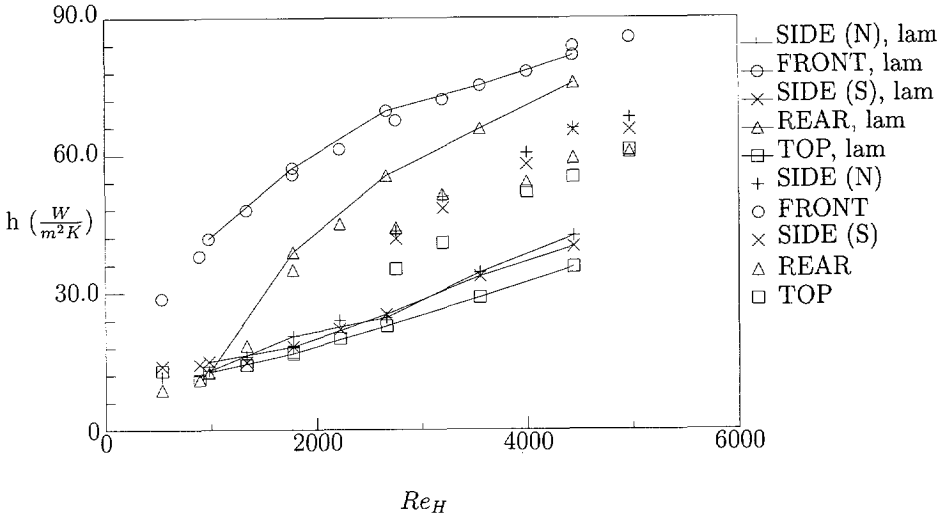


FIGURE 5.46: Face-averaged heat transfer coefficients for a single cube in a tripped and an untripped channel flow as a function of the Reynolds number.

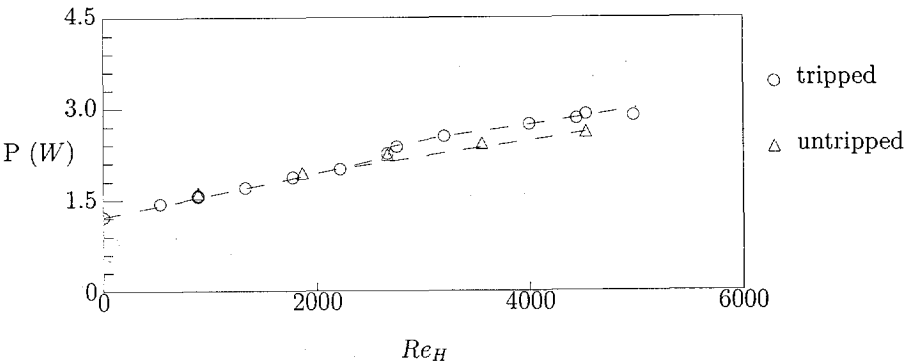


FIGURE 5.47: Power supply as a function of the Reynolds number. Shown are the results for the tripped and untripped channel flow.

Additional heat transfer measurements were performed for the isolated cube in the channel flow without the tripping strip (a developing laminar channel flow resulted) for the Reynolds number range of $2000 < Re_H < 5000$. From these measurements, face-averaged heat transfer coefficients were determined. These are given in Fig. 5.46 together with the previous results. The results for Reynolds numbers

smaller than the critical number ($Re_H \approx 2500$) collapse pretty well. This illustrates the good reproducibility of the experiments. While the profiles of the averaged heat transfer coefficient of the front and the rear faces collapsed more or less for the tripped and untripped cases for Reynolds numbers larger than $Re_H \approx 2500$, significant differences are observed between those of the top and side faces. The monotonically increasing laminar results for the top and the side faces illustrate that the top and side vortices enveloped the entire faces and that flow reattachment was not present.

The jump in the convective heat transfer at $Re_H \approx 2500$ can be traced back to an increased power supply P . The measurement assemblies were heated with an electrical resistance heat source in the core of the cube. The power supply was determined from the measured voltage drop V and the electrical current I . The supplied power for the heat transfer measurements in the developing turbulent (tripped) and the laminar (not tripped) developing channel flow are given in Fig. 5.47. The transition around $Re_H = 2500$ is clear from the results for the tripped case. In case that the flow is not tripped, the supplied power increased gradually.

CHAPTER 6

Array of 9 cubes

The in-line array of nine cubical protrusions, size $H = 15$ mm, was placed at half the channel width on the channel floor. The centre-to-centre spacing between the cubes was $S_x/H = 2$. A schematic of the configuration is given in Fig. 6.1. A boundary layer flow developed along the test-section walls. This resulted in a boundary layer thickness of approximately half the cube height at the location of the row. The first six cubes in the in-line row were used to determine local heat transfer coefficients. Two additional cubes enabled the in situ calibration (cube 7) and the location identification of the obstacles (cube 9). Cube 8 was used to establish an appropriate environment for the in situ calibration. This nomenclature is used in the sketch in Fig. 6.1 (right plot).

Although the general flow direction in this thesis is from left to right, in some exceptions flow results (images) are presented in the opposite direction, from right to left. It will be explicitly mentioned if this is the case.

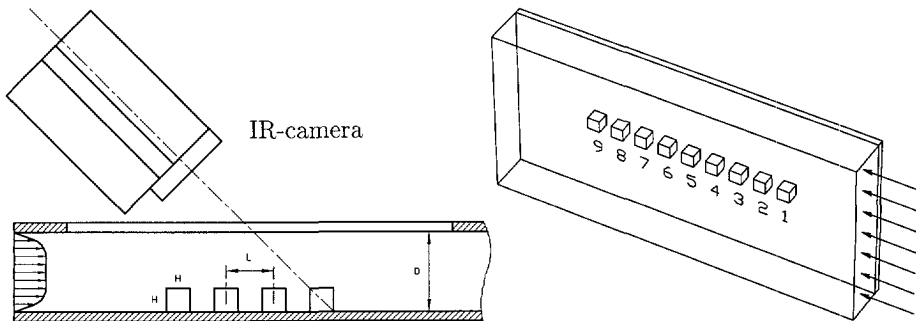


FIGURE 6.1: Schematics of the in-line row of cubes. The left sketch is a side view of the test-section with the infrared camera in position. The right plot shows a three-dimensional view of the set-up.

6.1 Flow field

The flow field around the cubes was derived from LDA-measurements, smoke and oil-film visualisations. The 'artist's impression' given in Fig. 6.2 shows the flow field around the first three cubes in the in-line row. Cross-sections at the symmetry plane $z/H = 0$ and a plane parallel to the channel wall at $y/H = 0.5$ are shown in Fig. 6.3, the upper and lower plot, respectively. The LDA-measurements were performed at a bulk velocity of $u_B = 4.47$ m/s which corresponds to a Reynolds number of $Re_H = 4440$. Both the smoke and oil-film visualisations were carried out for Reynolds number in the range $1000 < Re_H < 8000$. It is noted that no significant

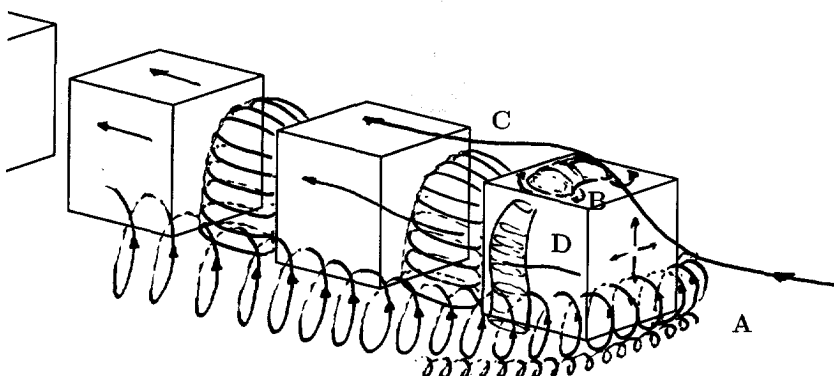


FIGURE 6.2: Three-dimensional 'artist's impression' of the mean flow field around the first three cubes in the in-line row of cubes. Flow is from right to left.

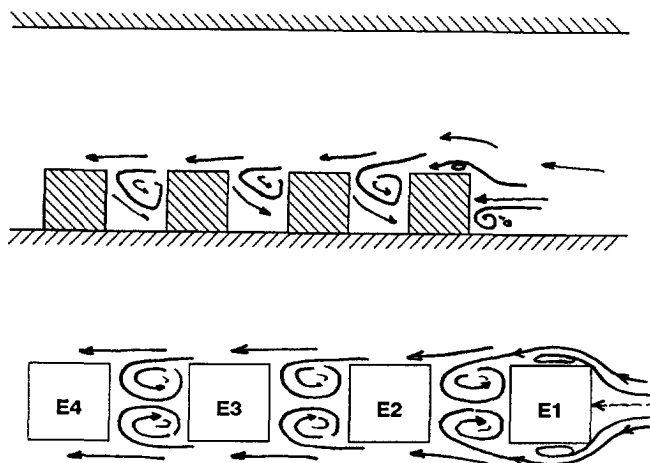


FIGURE 6.3: Schematics of the flow pattern at the symmetry plane $z/H = 0$ (upper plot) and at a plane parallel to the channel floor at $y/H = 0.5$ (lower plot). Flow is from right to left.

flow differences were detected in the range $2000 < Re_H < 5000$. Therefore, the derived flow field was applicable for this range which corresponded to the three largest Reynolds numbers studied. For the lowest Reynolds number studied, i.e. $Re_H = 795$, thermally induced buoyant flow directed perpendicularly with respect to the horizontally oriented row of cubes, caused a significant deviation from the symmetric flow pattern as obtained for the higher Reynolds numbers.

An overview of the oil-film visualisation given in Fig. 6.4 reveals that the pigment

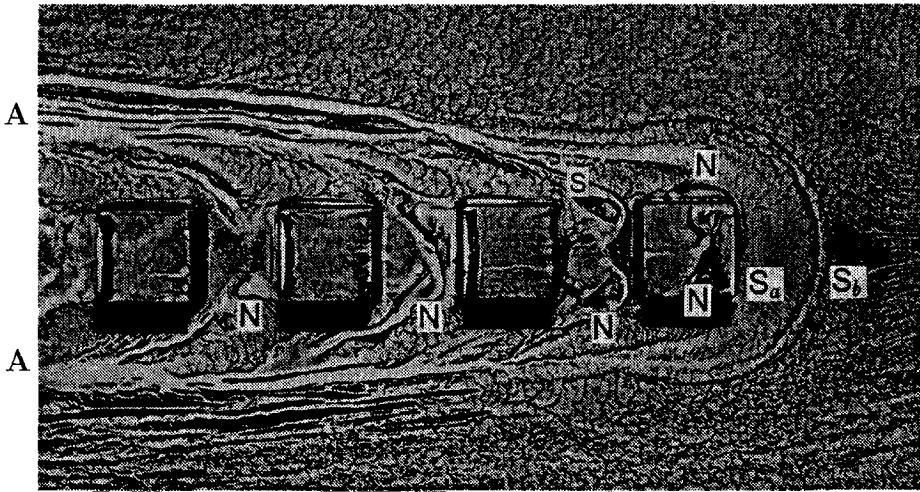


FIGURE 6.4: Oil-film visualisation of the surface flow pattern of an in-line row of cubes in a developing turbulent channel flow. Flow is from right to left.

patterns on and around all but the first obstacle are qualitatively similar. However, the flow patterns are not fully developed (i.e. no spatial periodicity in the streamwise direction) even after six cubes as can be seen from the nearly linear growth of the region bound by the pigment line **A** (see Fig. 6.4). The rather complex surface pigment patterns directly upstream of the leading cube are due to the horseshoe vortex (this vortex is marked with **A** in Fig. 6.2). The cross-section of this vortex structure upstream of the leading cube at $z/H = 0$ can clearly be observed in the instantaneous smoke visualisations given in Fig. 6.5, obtained with a laser light sheet technique. The smoke experiments presented here were performed at a Reynolds number of $Re_H = 1000$. For the experiments at larger Reynolds numbers, the flow became more turbulent and the smoke traces diffused in the turbulent flow field. The flow structure appeared then only from the recorded movie and not from the single shots as presented here. But as already stated, no severe differences were observed between the lower and the higher Reynolds number experiments. It is explicitly mentioned that the flow visualisations were performed in an isothermal situation without powering of the cubes which excluded buoyancy effects. While in the upper two images the smoke traces only elucidate the horseshoe vortex in front of the leading cube, the lower two images also elucidate the flow structure in the inter-obstacle spacings. The horseshoe vortex structure found upstream of the leading cube is similar to the one found for the single cube case (chapter 5). The flow separation caused by the primary horseshoe vortex resulted in two separation saddles, denoted as S_a and S_b in Fig. 6.4. The critical lines through S_a and S_b mark the location of the horseshoe vortex system as it is deflected downstream along the lateral sides of the array. These join approximately $1.5H$ downstream of the leading obstacle trailing edge indicating that the horseshoe vortex structure is consolidated in the primary vortex. From the instantaneous smoke visualisations it was observed

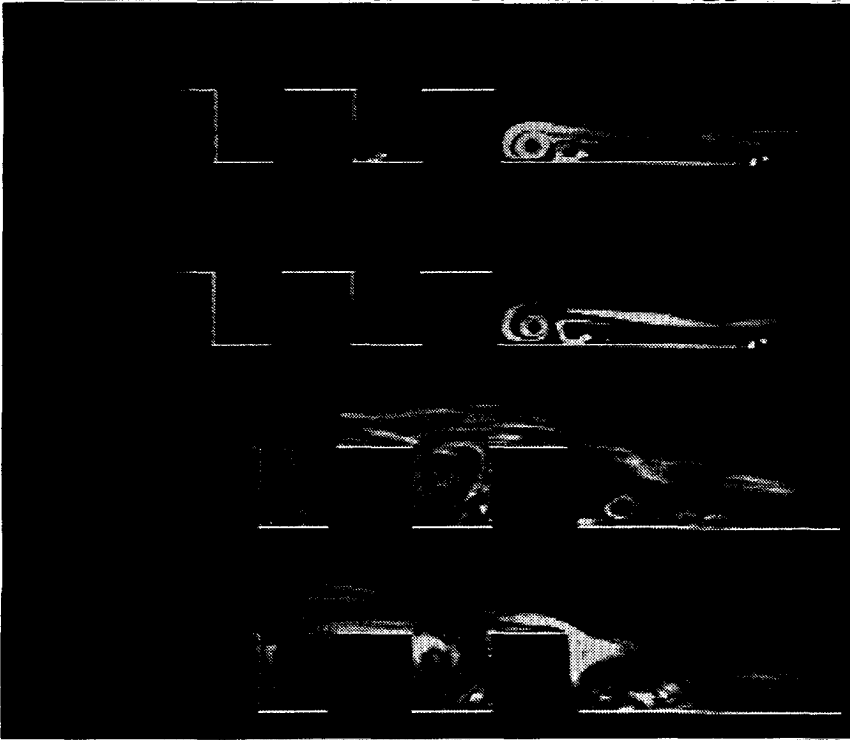


FIGURE 6.5: Four smoke visualisation images of the instantaneous flow pattern around the in-line row of cubical protrusions at the symmetry plane $z/H = 0$. Flow is from right to left.

that the horseshoe vortex was unstable. Such a behaviour was also observed for the single cube in the developing turbulent channel flow.

The appearance of two counter-rotating foci on top of the leading cube (marked with N in Fig. 6.4), the imprint of the bound vortex at the top face, indicates flow reversal. This top vortex is indicated with B in Fig. 6.2. The vortex was confirmed by LDA-measurements of the streamwise velocity component at the centreline of the cube, shown in Fig. 6.6. The profiles of the mean velocity and the Reynolds normal stresses at the centreline are plotted for five different locations in the downstream direction: $x/H = 0.1, 0.3, 0.5, 0.7$ and 0.9 . The dashed line, at $y = 15$ mm, refers to the solid top face of the leading cube. Strong peaks in the Reynolds normal stresses were found in the vicinity of the recirculation region, at approximately $y = 17$ mm, which is at 2 mm distance from the top wall of cube. This is a result of strong shearing at locations of large velocity gradients. Similarly as for the single cube case, flow instabilities caused the flow to intermittently reattach at the back half of the side faces which also contributed to the large values of u'^2 .

This flow pattern differs markedly from those on the top faces of the downstream cubes, where the lack of pigment at the leading edge suggests attached flow. The profiles of $\bar{u}/u_B$ and u'^2/u_B^2 for five locations at the top face of the second cube,

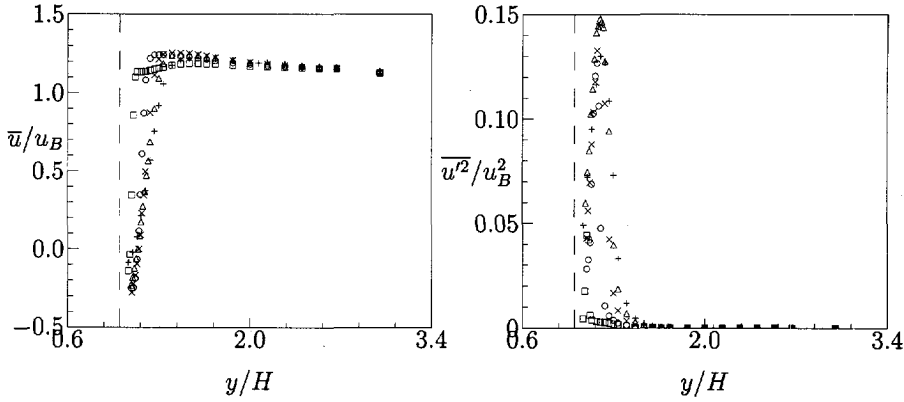


FIGURE 6.6: Profiles in the normal direction of $\bar{u}/u_B$ and $\overline{u'^2}/u_B^2$ at the centreline of the top face ($z/H = 0$) of the leading cube: ($\square$) $x/H = 0.1$, ($\circ$) $x/H = 0.3$, ($\times$) $x/H = 0.5$, ($\triangle$) $x/H = 0.7$ and ($+$) $x/H = 0.9$.

shown in Fig. 6.7, indicate simple shear flow, further supporting this interpretation. Hence, the shear layer separating from the upstream obstacle must reattach on the front face of the subsequent obstacle. An inflection point appeared at approximately 5 mm above the top face, as a result of the interaction between the boundary layer shear flow at this face and the accelerating main stream. The peak in the Reynolds normal stresses $\overline{u'^2}$, shown in Fig. 6.7, occurs in the same region and is again explained by the strong shearing at locations of large velocity gradients. The maximum in $\overline{u'^2}$ shifts towards the wall with upstream location. It is observed that the peaks in $\overline{u'^2}$ are much smaller than those at corresponding locations for the leading cube.

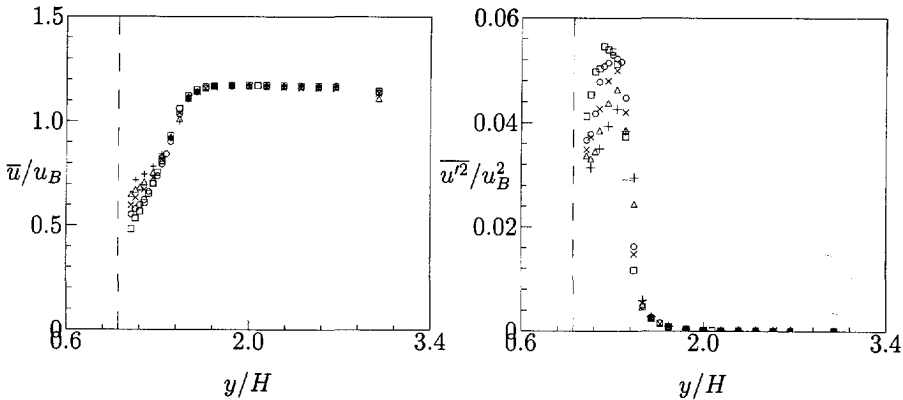


FIGURE 6.7: Profiles in normal direction of $\bar{u}/u_B$ and $\overline{u'^2}/u_B^2$ at the centreline of the top face ($z/H = 0$) of the second cube: ($\square$) $x/H = 0.1$, ($\circ$) $x/H = 0.3$, ($\times$) $x/H = 0.5$, ($\triangle$) $x/H = 0.7$ and ($+$) $x/H = 0.9$.

Flow development along the array of cubes is illustrated in Fig. 6.8, where profiles

of $\bar{u}/u_B$ and $\overline{u'^2}/u_B^2$ are given at $z/H = 0$ and the centre of the top faces for the first four subsequent cubes. In the proximity of the solid wall, the mean velocity profiles of the third and fourth cube are enclosed by those of the first and second cube. The interaction between the channel core flow and the boundary layer flow for the second cube, indicated by the inflection point, vanished in the downstream direction. Further away from the walls, in the core region of the channel, the four profiles coincided more or less except for a small increase. Due to the developing channel flow, the core flow accelerated which subsequently caused the small increase in the mean velocity. The maximum in the Reynolds normal stresses, as observed for the first and second cube, vanishes for the third and fourth cube. This is explained by the diminishing velocity gradients during the recovery of the flow. Further, flow instabilities are damped effectively downstream of the leading cube. Again, the profiles asymptote more or less far away from the solid wall.

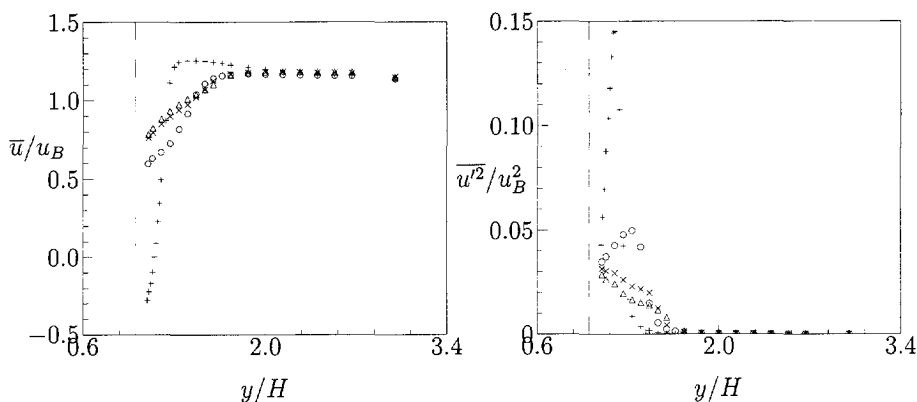


FIGURE 6.8: Profiles of $\bar{u}/u_B$ and $\overline{u'^2}/u_B^2$ at $z/H = 0$ and the centre of the top face: (+) E1, (o) E2, (x) E3 and (Δ) E4.

The significance of the pigment accumulation at the trailing edge of the obstacle top face is not clear. It may be a result of either surface adhesion (evacuation of the pigment being blocked by the return flow up the leeward face) or due to a local flow reversal. Fortunately, since both structures are topologically closed (i.e. removable mathematical singularities), their interpretation will not affect the deduced flow pattern for the remaining field (see Figs. 6.13 and 6.14).

In the wake of the leading and the subsequent cubes, two vorticity concentrations are denoted by the counter-rotating foci (indicated with N in Fig. 6.4). These are the imprints of an arc-shaped vortex, confined in the inter-cube spacing. This arc vortex is indicated with C in Fig. 6.2. An instantaneous smoke visualisation along the plane $y/H \approx 0.3$, given in Fig. 6.9, shows the horseshoe vortex structure in front of the leading cube and the cross-section of the arc-shaped vortex in the gap between the leading and the second cube. Two saddles existed in the inter-cube area (S in Fig. 6.4) suggesting that fluid is exhausted from the wake region between these points and the downstream windward face. Hence, the gap area does not form a closed recirculation bubble. This is dictated by conservation of mass since fluid is



FIGURE 6.9: Smoke visualisation of the flow pattern around the single array of cubes in a plane parallel to the base plate at $y/H = 0.3$. Flow is from right to left. The plane was illuminated with the laser sheet from above which caused the shadow below the cubes.

entrained by the top and the side shear layers.

The existence of a vortex tube extending up the cube side faces was inferred from its footprint on the channel floor (marked with **N** in Fig. 6.4) directly downstream of the leading side edge. The side vortex is marked with **D** in Fig. 6.2.

Vortex shedding was not detected in the wake of the leading and the subsequent cubes. This is in agreement with the observations made for the in-line tandem arrangement of cubes, where the onset to vortex shedding was detected at a minimal streamwise distance of $1.4 H$ between the cubes (see chapter 7).

Measurements of the mean velocities and the Reynolds normal stresses in the xz -plane are given in Fig. 6.10 and 6.11 for the u and w component, respectively. Shown are seven traverses in the spanwise direction at the mid-height of the cube ($y/H = 0.5$) and for x -stations of respectively $x/H = -0.1, 0.1, 0.5, 0.7, 1.1, 1.9$ and 2.5 . The solid side cube wall is indicated with the dashed line. The side vortex appeared from the negative velocities found at the side face of the first cube. Further, the arc-vortex in the inter-obstacle spacing was demonstrated from the negative streamwise velocities.

The Reynolds normal stresses $\overline{u'^2}$ and $\overline{w'^2}$ become maximum at the location $x/H = 0.5-0.7$ for the leading cube where the separated shear layer reattached, similarly as for the top face. The maximum occurred at approximately 2-3 mm from the solid walls. Again these maxima were attributed to the strong shearing at locations of large velocity gradients and to the intermittent flow reattachment.

Far away from the cube surface, all lines asymptote to the undisturbed boundary layer values. The Reynolds shear stresses $\overline{u'w'}$ in the xz -plane are shown in Fig. 6.12 for the same traverses as discussed above. Marked maxima in $\overline{u'w'}$ are found at locations with relatively large velocity gradients, as appears from the maxima at roughly $x/H = 0.5-0.7$. The correlation $\overline{u'w'}$ remains relatively large in the free

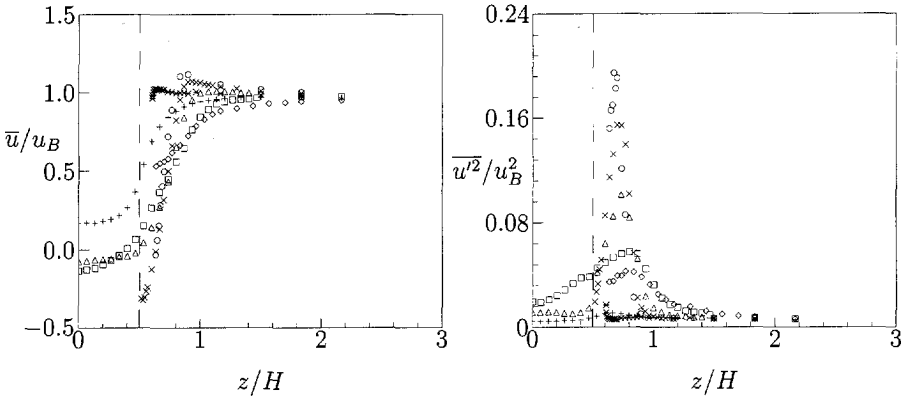


FIGURE 6.10: Profiles of the scaled mean velocity $\bar{u}/u_B$ and Reynolds normal stress $\overline{u'^2}/u_B^2$ in the spanwise direction for 8 different locations at $y/H = 0.5$: (+) $x/H = -0.1$, (x) $x/H = 0.1$, (o) $x/H = 0.5$, (x) $x/H = 0.7$, (Δ) $x/H = 1.1$, ($\square$) $x/H = 1.9$ and ($\diamond$) $x/H = 2.5$.

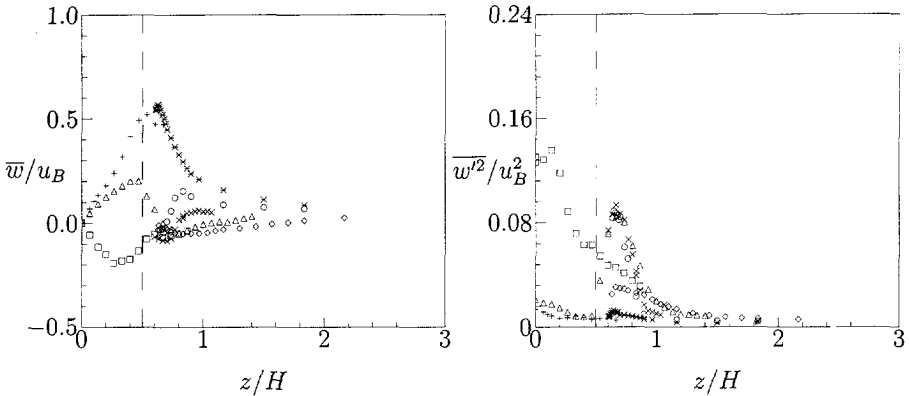


FIGURE 6.11: Profiles of the scaled mean velocity $\bar{w}/u_B$ and Reynolds normal stress $\overline{w'^2}/u_B^2$ in the spanwise direction. For symbols see Fig. 6.10.

mixing layer behind the leading obstacle ($x/H = 1.1$ and 1.9). In the proximity of the leading side edge, at $x/H = -0.1$ and 0.1 , $\overline{u'w'}$ approaches zero as would be expected. Far away from the cubes, at approximately $z/H = 4/3$, the values collapse to zero.

The location of the aforementioned critical points on the channel surface $y/H = 0$ are schematically summarised in Fig. 6.13. The flow field was then derived from a topology analysis. Since the flow pattern is topologically similar around each obstacle, except for the leading and the trailing obstacle, an abridged array representation is shown. Note that the flow, however, is not fully developed (spatially periodic) since the pigment line A, indicated in Fig. 6.13, is still diverging. Also note that the removal of the obstacle surfaces is equivalent to increasing the surface connectivity. Hence, with four obstacles removed, the topological constraint of Hunt et al. [68]

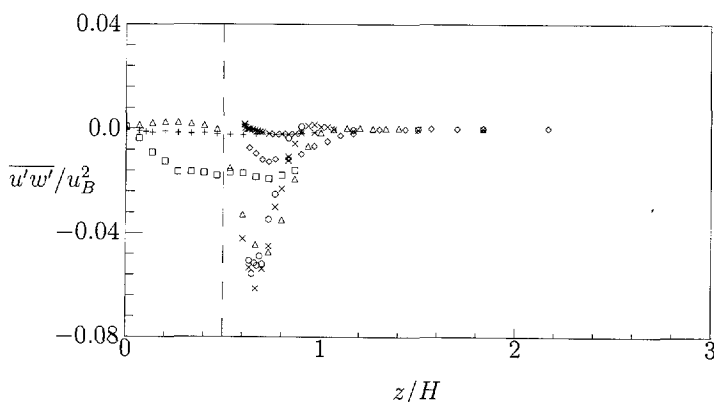


FIGURE 6.12: Profiles of the scaled mean Reynolds shear stress $\overline{u'w'}/u_B^2$ in the spanwise direction. For symbols see Fig. 6.10.

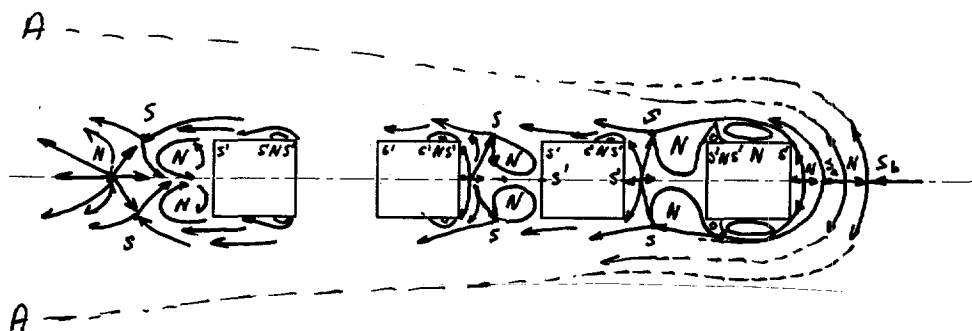


FIGURE 6.13: Sketch of the critical points and streamlines for the plane $y/H = 0$ (bottom of channel). Flow is from right to left.

becomes: $(\sum N + \frac{1}{2} \sum N') - (\sum S + \frac{1}{2} \sum S') = -4$. The total number of saddles, half saddles and nodes were $\sum S = 12$, $\sum S' = 32$ and $\sum N = 24$, respectively which satisfies the constraint: $(\sum N) - (\sum S + \frac{1}{2} \sum S') = -4$.

The critical points on $y/H = 0$ were transposed on the plane of symmetry $z/H = 0$ in Fig. 6.14. For example, while a reattachment point appeared as a simple node on $y/H = 0$, it appeared as a half saddle in the plane $z/H = 0$. The existence and the location of free saddle and nodal points in the plane of symmetry were then deduced to satisfy the topological constraints. The sums of the critical points were: $\sum N = 17$, $\sum S = 2$ and $\sum S' = 30$ and satisfy the topological constraint $(\sum N) - (\sum S + \frac{1}{2} \sum S') = 0$.

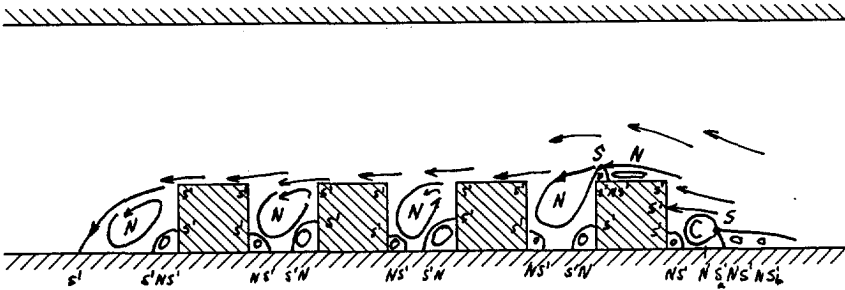


FIGURE 6.14: Sketch of the critical points and the stream lines for the symmetry plane $z/H = 0$. Flow is from right to left.

6.2 Convective heat transfer

The heat transfer measurements were performed for Reynolds numbers of $Re_H = 795, 2086, 3278$ and 5066 . The aforementioned flow field applied to the three largest Reynolds numbers studied, while the flow deviated significantly from that for the lowest Re-number studied ($Re_H = 795$).

6.2.1 Surface temperature measurements

A comparison between liquid crystal and infrared temperature measurements is given in Fig. 6.15 for the fifth cube along paths $ABCD$ and $ABCD A$. Path $ABCD A$ denotes the cross-section of a plane at half the cube height, parallel to the mounting plate; path $ABCD$ is the cross-section of the plane at the centreline of the cube oriented perpendicularly to the base plate. The liquid crystal measurements (isotherms at the surface and, therefore, only available at certain temperature levels) are indicated by the symbols ($\diamond$) and ($\triangle$) for $Re_H = 2086$ and $Re_H = 5066$, respectively. Infrared temperature measurements, based on images which were processed with the restoration filter and the in situ calibration, are indicated as lines with symbols ($\square$) and ($\circ$) for the corresponding Reynolds numbers. The temperatures of both methods coincide within the experimental uncertainty of the LC measurements. The good agreement provides an accuracy estimate of 0.4°C for the infrared temperatures. The small deviations between the results of each of the techniques are attributed to experimental uncertainty in both the spatial coordinate and the temperatures.

Isotherm maps of the five faces exposed to air flow for the first 5 subsequent cubes in the packed array are given in Figs. 6.16 and 6.17 for Reynolds numbers of $Re_H = 5066$ and $Re_H = 795$, respectively. The cube faces are folded out and projected in one plane. It is important to note that the isotherm maps for $Re_H = 5066$ are all symmetric. Asymmetric effects caused by buoyancy induced fluid motion are not observed from these plots. Even the inter-obstacle regions which have flow recirculations, places where the local velocity drops to zero and where

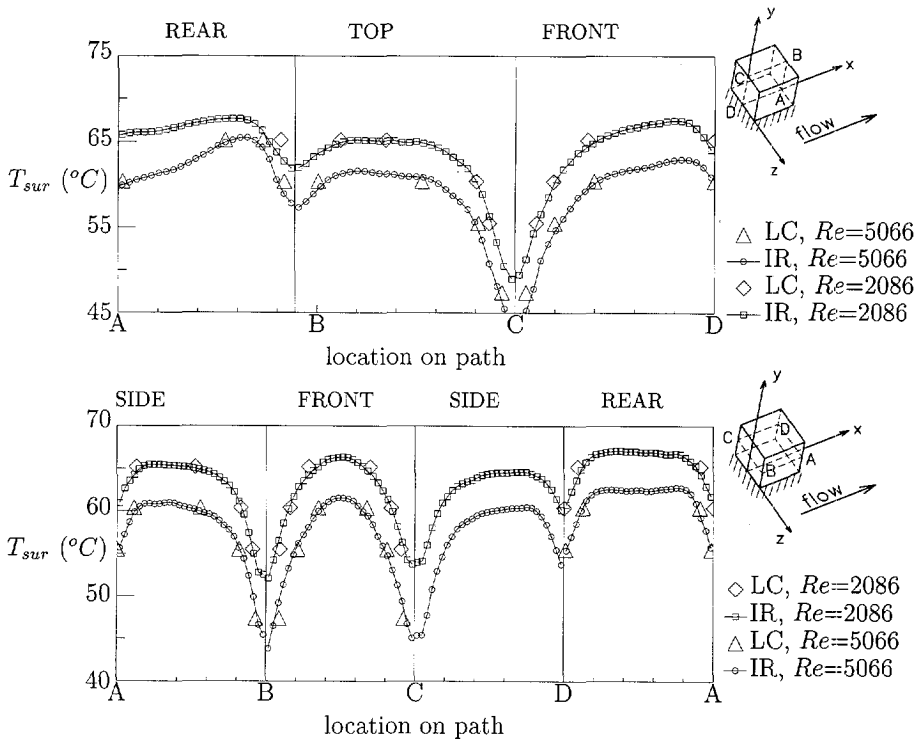


FIGURE 6.15: Surface temperatures along path $ABCD A$ (upper plot) and path $ABCD$ (lower plot) for the fifth cube for $Re_H = 2086$ and $Re_H = 5066$: comparison between liquid crystal measurements (LC, $\diamond$ and $\triangle$) and infrared thermography measurements (IR, $\square$ and $\circ$).

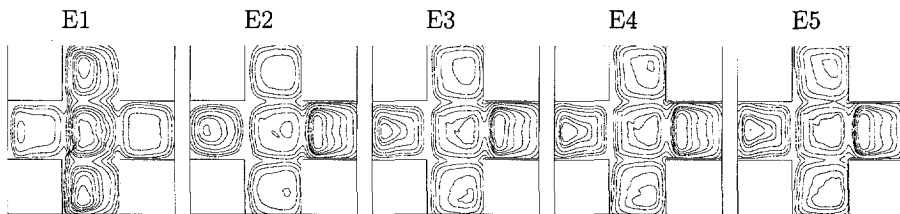


FIGURE 6.16: Infrared surface temperature maps of cubes 1-5 for $Re_H = 5066$.

buoyancy can locally become important, do not show asymmetric behaviour. The surface temperature distribution of the first cube for $Re_H = 795$ is also symmetric. However, pronounced temperature gradients in the vertical (spanwise) direction are observed for the downstream cubes, especially for their front and rear faces. Since all cubes were powered during the experiments, strong thermally induced fluid motion occurred, in particular in the corridors between the obstacles. The convective heat transfer was, therefore, mostly affected at the rear and the front face. The large

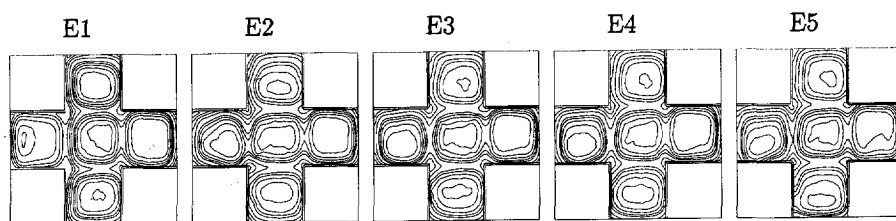


FIGURE 6.17: Infrared surface temperature maps of cubes 1-5 for $Re_H = 795$.

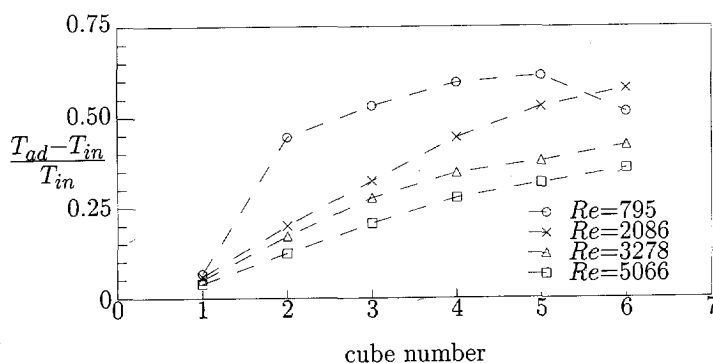


FIGURE 6.18: Normalised adiabatic cube temperatures for the six cube locations and parametric in the Reynolds number: ○ $Re_H = 795$, × $Re_H = 2086$, △ $Re_H = 3278$ and □ $Re_H = 5066$.

spatial gradients in the surface temperature at the individual cubes, as well as the differences between subsequent cubes, already indicate that the heat transfer coefficients will vary significantly.

6.2.2 Local convective heat transfer

The normalised adiabatic cube temperatures ($(T_{ad} - T_{in})/T_{in}$, with T_{ad} the adiabatic temperature and T_{in} the inlet temperature) are depicted in Fig. 6.18 for the first six cubes in the array as a function of the Reynolds number. This adiabatic temperature was measured in a different experiment. The increased air temperature, as a result of the heat generation in the upstream cubes, caused a monotonic increase in the adiabatic temperature. Larger Reynolds numbers caused larger rates of cooling and convective transport and thus, caused lower adiabatic temperatures. The remarkable shape of the curve for $Re_H = 795$ was caused by mixed convection. The lower adiabatic temperature for cube 6 compared to that for cube 5 is because of the vertical motion of the fluid directing cold fluid to the cube. The normalised adiabatic cube temperature for natural convection ($Re_H = 0$) was 0.23 for all cubes in the row.

The heat transfer coefficient distribution along path $ABCD$ A is given in Fig. 6.19 as a function of the six cube positions and parametric in the Reynolds number. Local

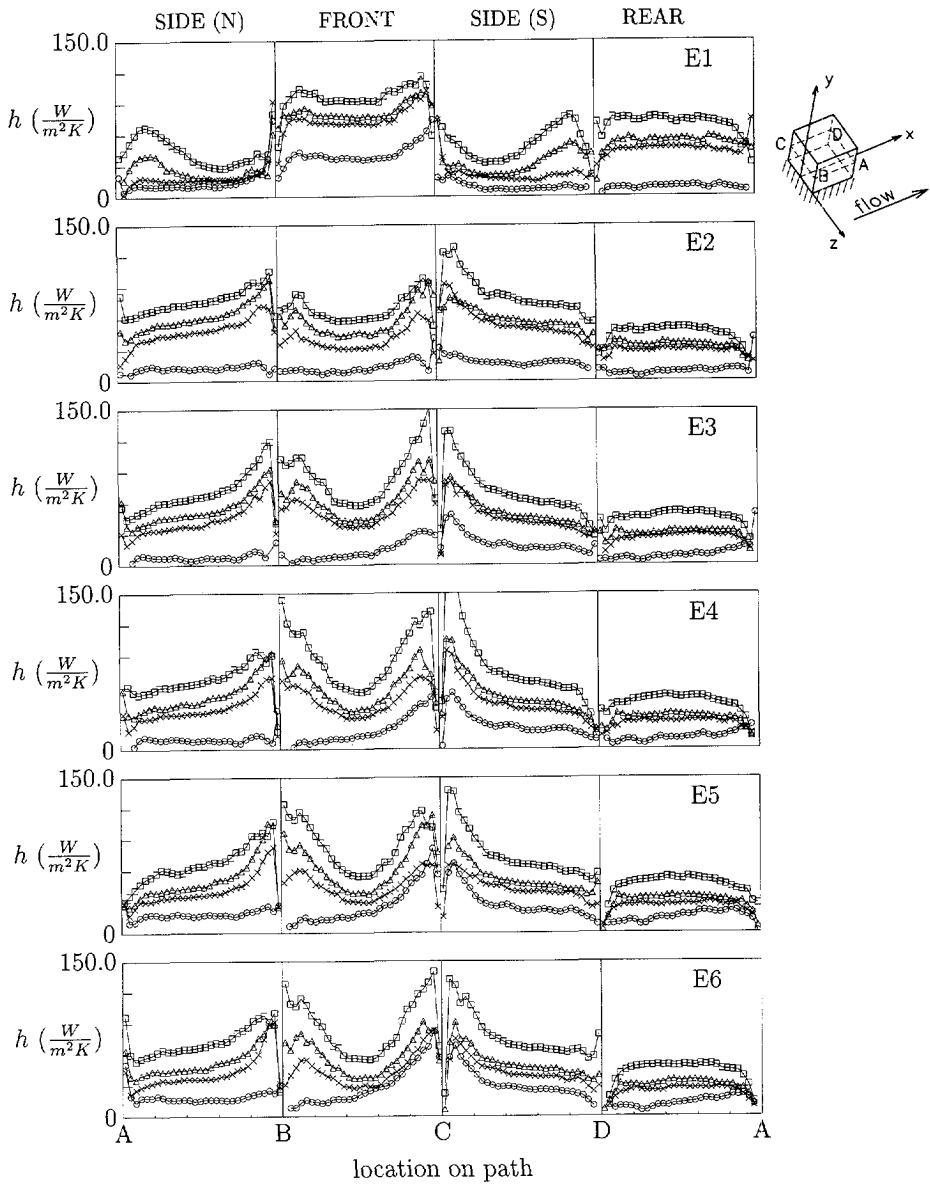


FIGURE 6.19: Heat transfer coefficients along cross-section $ABCD$ for cubes 1-6 parametric in the Reynolds number: $\circ Re_H = 795$, $\times Re_H = 2086$, $\triangle Re_H = 3278$ and $\square Re_H = 5066$.

heat transfer coefficients along path $ABCD$ are given in Fig. 6.20. The uppermost plot corresponds to the leading cube, the lowest plot to the sixth cube. Buoyancy effects, induced by the temperature difference between the heated cubes and the surrounding air, are of negligible importance for the three largest Reynolds numbers studied. For these situations, symmetric profiles were found on the faces of all cubes.

The local heat transfer phenomena at the top and the side faces of the leading cube are dominated by the top and the side vortices. These features are most prevalent for the largest Reynolds number. These vortices were caused by the flow separation at the leading top and side edges. Similarly as for the single cube case, a minimum was found at the front half of the top and side faces underneath the vortices in question. Flow reattachment at the back half of these faces close to the trailing edges caused the heat transfer enhancement. The low-temperature oncoming flow impinged at the front face (partition BC) of the leading cube, consequently leading to a high level of heat transfer. Close to the channel wall, the local minimum at this front face is explained by the rotating horseshoe vortex, see Fig. 6.20, similarly as for the single cube case.

The convective heat transfer from the top and side faces of the second cube, governed by the attached boundary layer flow, is characterised by a strong decay in the direction of the rear edge. The relatively high heat transfer coefficients are caused by the reattaching horseshoe vortex and the main flow. The concave profiles in the heat transfer coefficients at the east (front) face are explained by the counter-rotating extensions of the arch-shaped vortex which envelopes the gap between the leading and the second cube. The heat transfer at the rear face is uniform at the cross-section at $z/H = 0.5$, however, a local minimum, accompanied by a small local maximum, is found at the upper edge. These local extrema are caused by the rotating arch-shaped vortex. The convective heat transfer at the lateral side faces for the subsequent cubes is dictated by boundary layer flow. Along the surface, the heat transfer decays in the downstream direction. Aliaga et al. [5, 6] observed similar behaviour at the top face of multiple wall-mounted ribs in a fully developed channel flow. The heat transfer at the front face is still characterised by concave profiles, caused by the arch vortex. The local extrema at the rear face also remain. No significant differences are observed downstream of the third protrusion.

Significant buoyancy effects appear for $Re_H = 795$, classifying the situation as mixed convection. The temperature-driven buoyancy flow is relatively strong in the corridors between the subsequent cubes. The arch-shaped vortex, which was found for the forced convection dominated flow situations (the three largest Reynolds numbers studied), broke up because of the induced flow motion directed perpendicular to the main flow. The resulting flow pattern caused an increase in the heat transfer at the south-front-edge, decaying in the downstream direction along $C-D$ (south face) and in the opposite direction of gravity along $C-B$ (back face), see Fig. 6.19. The opposite behaviour occurs at the opposing face, although the effects are weaker. This can also be seen in Fig. 6.19. These buoyancy effects become more pronounced for the downstream protrusions. The induced buoyancy flow appears also from the isotherm plots, shown in Fig. 6.17. Differences in temperature gradients indicate the direction of the flow.

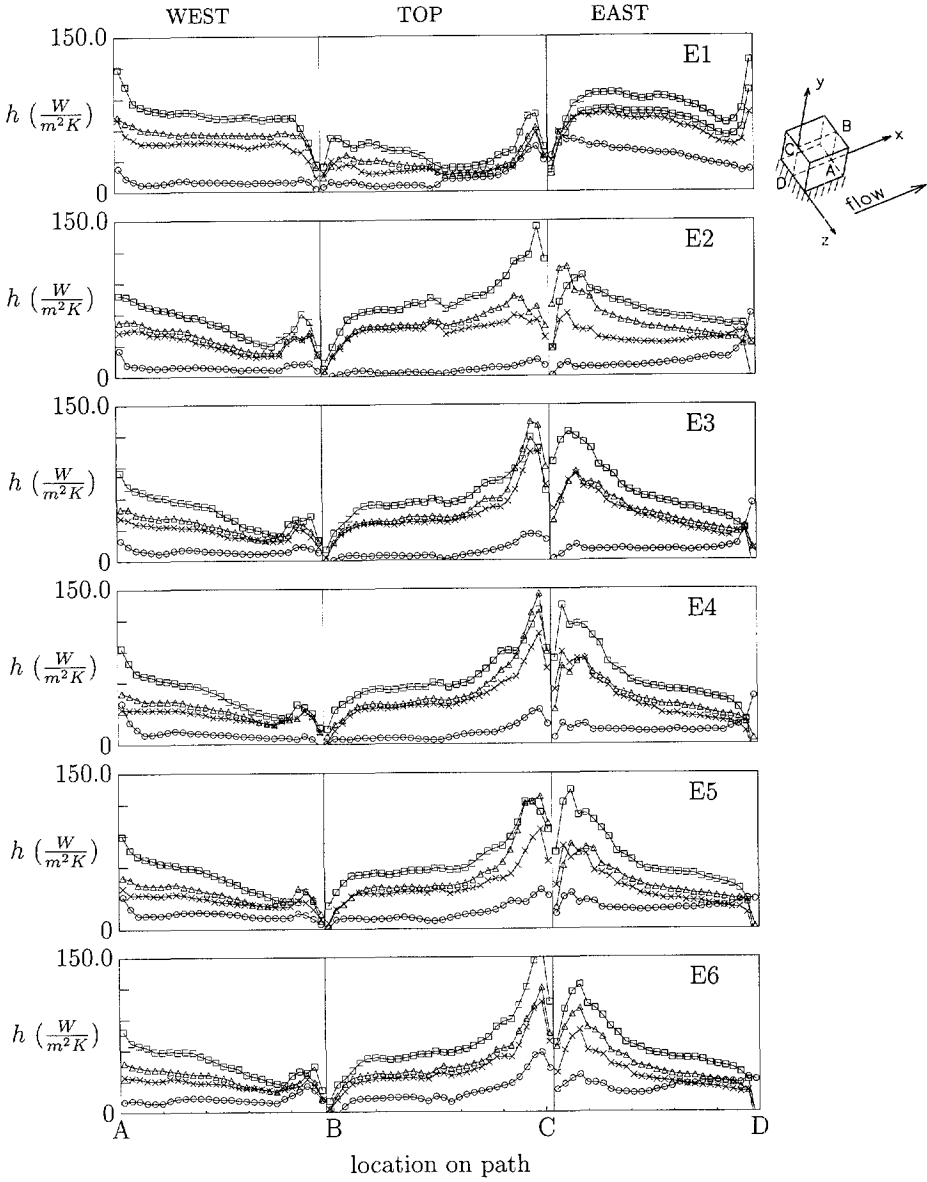


FIGURE 6.20: Heat transfer coefficients along cross-section $ABCD$ for cubes 1-6 parametric in the Reynolds number: $\circ Re_H = 795$, $\times Re_H = 2086$, $\triangle Re_H = 3278$ and $\square Re_H = 5066$.

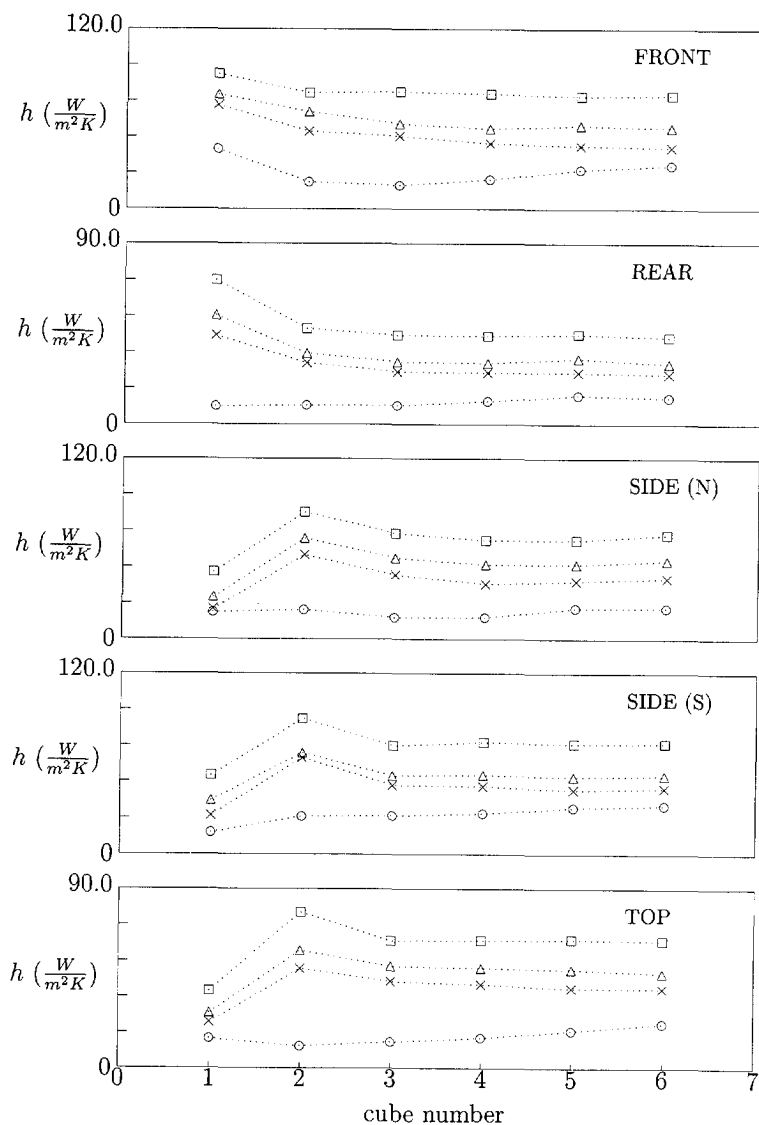


FIGURE 6.21: Face-averaged heat transfer coefficients, parametric in the cube location: $\circ$ $Re_H = 795$, $\times$ $Re_H = 2086$, $\triangle$ $Re_H = 3278$ and $\square$ $Re_H = 5066$ for the east, west, north, south and top face.

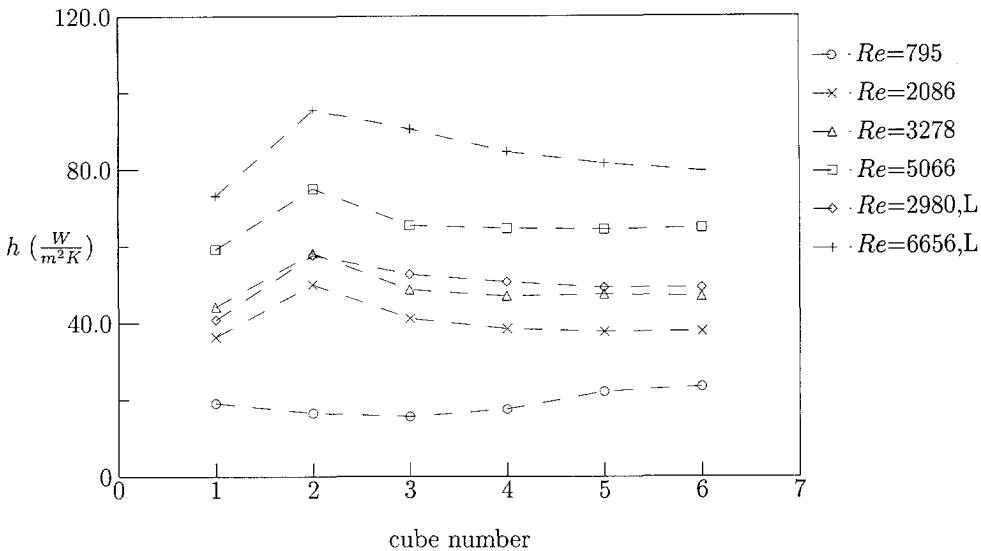


FIGURE 6.22: Cube-averaged heat transfer coefficients, parametric in cube location: $\circ$ $Re_H = 795$, $\times$ $Re_H = 2086$, $\triangle$ $Re_H = 3278$ and $\square$ $Re_H = 5066$. Literature data (L): $\diamond$ $Re_H = 2980$ and $+$ $Re_H = 6656$ (Anderson and Moffat, 1990).

6.2.3 Averaged convective heat transfer

The surface-averaged heat transfer coefficients, derived similarly as for the single cube experiments, for the five separate faces are shown in Fig. 7.22 as a function of the cube location and parametric in the Reynolds number. As is usually the case, the largest Reynolds number caused the largest average convective heat transfer coefficients and vice versa. The oncoming flow impinges at the front face of the leading cube and, hence, results in a large heat transfer coefficient. The front of the second cube is exposed to the arch-shaped vortex. This vortex is continuously supplied with cold air from the horseshoe vortex and the main stream with the consequence of relatively high cooling rates. This also applies to the subsequent cube downstream in the array. The convective heat transfer at the rear face of the first cube is relatively high because of the prevalent arch-shaped vortex between the first two cubes. The heat transfer at the rear faces of the second and third protrusion is much smaller, which is explained by the decreasing impact of the bounded vortices further downstream. The strong flow reversals at the lateral faces of the first cube cause a notable minimum in the convective heat transfer. In contrast, the second cube is exposed to the reattached flow from the horseshoe vortex and main stream and, therefore, its convective heat transfer rate is significantly higher. This feature appears at the three highest Reynolds numbers. The impact of the horseshoe vortex diminishes for the downstream cubes, which results from the lower heat transfer

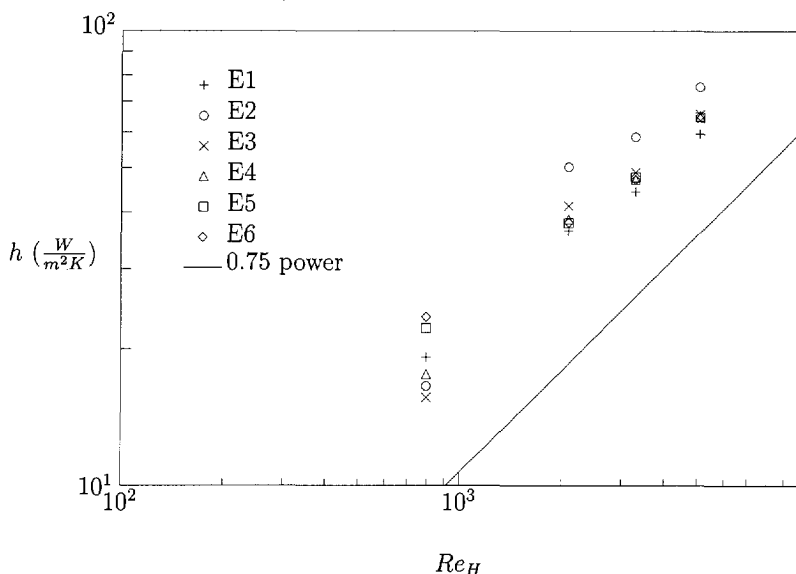


FIGURE 6.23: Cube-averaged heat transfer coefficients as a function of the Reynolds number based on the channel width: + E1, o E2, x E3, Δ E4, □ E5 and ◇ E6.

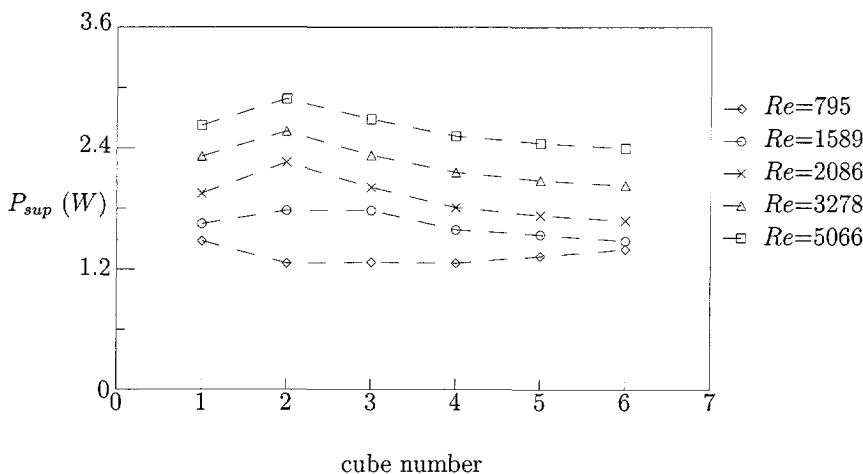


FIGURE 6.24: Supplied power as a function of the cube position: ◇ $Re_H = 795$, o $Re_H = 1589$, x $Re_H = 2086$, Δ $Re_H = 3278$ and □ $Re_H = 5066$.

rate for the third cube. A bounded vortex system envelopes the top face of the leading cube, as a consequence of the separating main stream, thus resulting in a significantly lower convective heat flux. The flow attaches at the top of the second cube and causes a high convective heat transfer rate at this face.

Substantial variation in the averaged heat transfer coefficients is found for the

first three protrusions in the array. After the third cube, the flow becomes gradually fully developed and no noticeable variation in the convective heat transfer is observed. This is the consequence of using the concept of the adiabatic heat transfer coefficient. The small deviations for the remaining cubes, as appear from the plots, are attributed to experimental uncertainty.

The average convective heat transfer for the mixed convection case, $Re_H = 795$, differs somewhat from the others. In particular, for the cubes further downstream in the array, the heat transfer benefits from the buoyancy-induced vertical fluid motion, which results in an enhancement of the heat transfer coefficient.

The heat transfer coefficients averaged over all five cube faces are given in Fig. 6.22 as a function of the cube location. Even at cube level, the recirculative flow-field around the leading cube dictates a pronounced minimum. The second cube has a maximum cube-averaged heat transfer coefficient because of the interaction with the horseshoe vortex and the main flow. Anderson and Moffat [8] performed cube-average heat transfer experiments for a regular matrix of wall-mounted aluminium cubes in a channel flow. The spacing between the cubes was 2 cube widths, i.e., $S_x/H = 3$. The results of their study are also plotted in Fig. 6.22 for $Re_H = 2980$ and $Re_H = 6656$ at a channel-height-to-cube-height ratio of $D/H = 4.6$. Although a somewhat different geometry was studied, their results are in good agreement with the present findings. They also found that the convective heat transfer is minimum for the leading cube and maximum for the second cube.

Fig. 6.23 gives the Reynolds number dependence on the heat transfer coefficient in log coordinates. The cube-averaged heat transfer coefficients versus the Reynolds number, based on the cube height and bulk velocity, are shown parametric in the cube location. The correlation $Nu \propto Re^{0.75}$, which usually represents forced convective heat transfer problems, seems to be applicable only to the two largest Reynolds numbers studied.

An integral analysis was conducted to estimate the conductive heat losses through the interface between the channel wall and cube and through the lead wires. The supplied power, determined from the measured voltage and current, is depicted in Fig. 6.24. The power supply to the second cube is maximum for the three largest cooling rates, which confirms the observations made above. Since the main stream is preheated by upstream cubes, a continuous decrease in the supplied power is encountered in the downstream direction. The power supply for $Re_H = 795$ differs from that of the other cases. It is maximum for the leading cube and, after a decrease in the downstream direction, a slight increase is observed. The power supply for $Re_H = 1589$ is added to illustrate the transition regime between $Re_H = 795$ and $Re_H = 2086$. Also for this cooling rate, the power supply was slightly larger for the second cube, but the transition is obvious. The heat losses in terms of percentage of the power supply range between approximately 25% for $Re_H = 795$ and 10% for $Re_H = 5066$.

CHAPTER 7

Tandem of two cubes

The tandem configuration of two cubes was placed in a fully developed channel flow for most of the results discussed here. The centre-to-centre distance between

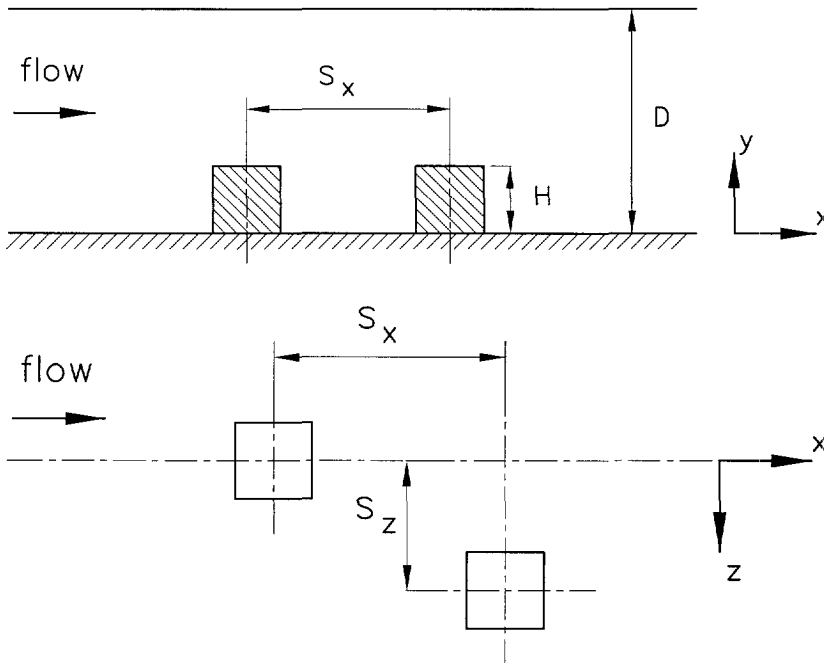


FIGURE 7.1: Sketches of the side view (upper plot) and the plan view (lower plot) of the tandem-arrangement of wall-mounted cubes in the channel.

the cubes in the streamwise direction, S_x/H , was varied between $1 < S_x/H < \infty$. Further, experiments for off-axis or staggered arrangements of the tandem were performed by varying the centre-to-centre spanwise distance S_z/H . Sketches of the side and plan view of the test-configuration are given in Fig. 7.1. The coordinate system is also indicated in the figure where x , y and z denote the streamwise, the normal and the spanwise direction, respectively. The cube size was $H = 15$ mm for most of the heat transfer and flow field measurements. The channel height D was 51 mm, thus giving a channel-height-to-cube-height ratio of $D/H = 3.4$. Some of the frequency measurements were performed in a developing turbulent boundary layer flow with cubes of size $H = 25$ and 50 mm.

The fully developed channel flow was achieved by tripping the flow at 2 m upstream of the cube. Fully development of the flow was demonstrated from the velocity profiles across the channel height at different streamwise locations. Profiles of the mean velocity $\bar{u}$ and the Reynolds normal stresses $\overline{u'^2}$ are given in Fig. 7.2 for different bulk velocities and different streamwise locations in the channel. The cubes were temporarily removed during these measurements. The four velocity profiles at subsequent streamwise locations ($x/H = -13.3, -6.7$ 0 and 13.3) but for a constant Reynolds number show that the channel flow was fully developed. The other two profiles refer to a lower and a higher bulk velocity. The negligible dis-

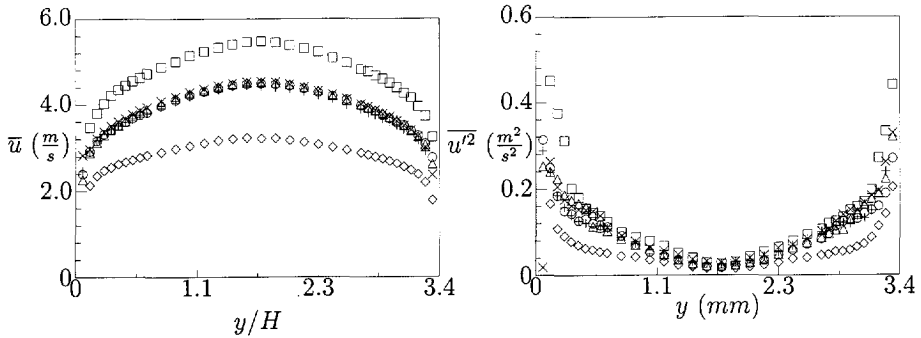


FIGURE 7.2: Profiles of the mean velocity and the Reynolds normal stresses. for a bulk velocity of $u_B = 3.95$ m/s: (+) $x/H = 0$, (o) $x/H = -13.3$, ($\times$) $x/H = 13.3$, (Δ) $x/H = -6.7$. Profile for $u_B = 4.77$ m/s: ($\square$) $x/H = -6.7$. Profile for $u_B = 2.83$ m/s: (o) $x/H = -6.7$.

crepancies between the velocity data for the different locations confirmed the fully developed character of the flow. The integration of the four coinciding velocity profiles provided the average flow rate from which the bulk velocity of $u_B = 3.9$ m/s was derived. The corresponding Reynolds number was $Re_H = 3900$.

All the flow measurements with the tandem in the fully developed channel flow were performed at $Re_H = 3900$. Some additional frequency measurements were performed for a tandem configuration on a flat plate in a developing turbulent boundary layer flow. The Reynolds number for these measurements was based on the free stream velocity, U_∞ , and had a value of $Re_H = 21,500$. To avoid misinterpretation, we will explicitly indicate in the text when these measurements are discussed.

The local structure of the mean flow and the turbulence fields will have a significant effect on the distributions of the local convective heat transfer. The aerodynamic interference between the two cubes gives rise to quasi-periodic phenomena which may lead to an augmentation or degradation of the local and the averaged convective heat transfer. The vortex shedding at the lateral faces of the leading cube caused flow periodicities in its wake that may either be attenuated or amplified by the interaction with the flow structure around the downstream cube. The distance between the cubes is an important parameter for this time-dependent flow behaviour. Flow interference may occur over a range of distances. When the spac-

ing approaches zero, the mutual flow interaction vanishes and the resulting obstacle is considered to be an extended cube of size $2H$. In the opposite limit, when the distance approaches infinity, flow interference also vanishes and the cubes behave like two aerodynamically independent cubes. The convective heat transfer from both cubes may either be augmented or degraded by the flow field interaction. The oscillatory behaviour of the reattaching flow may locally augment the convective heat transfer. On the other hand, flow interference may cause heat to be convected slowly away in areas where the time-mean velocities are low.

7.1 In-line tandem

7.1.1 Mean flow structure

Oil-film visualisations were performed for various streamwise distances to qualitatively examine the influence of the distance on the flow structure around the cubes. The results of these visualisations showed that the three distances $S_x/H = 2, 4$ and 8 incorporated the most significant aerodynamic aspects. It was decided to document only these three cases with the LDA system.

$$S_x/H = 2$$

An oil-film visualisation of the surface flow pattern is given in Fig. 7.3 for $S_x/H = 2$. The Reynolds number was $Re_H = 7000$, based on the cube height and the bulk velocity. It was confirmed that this Reynolds number was representative for the Reynolds number range of interest.

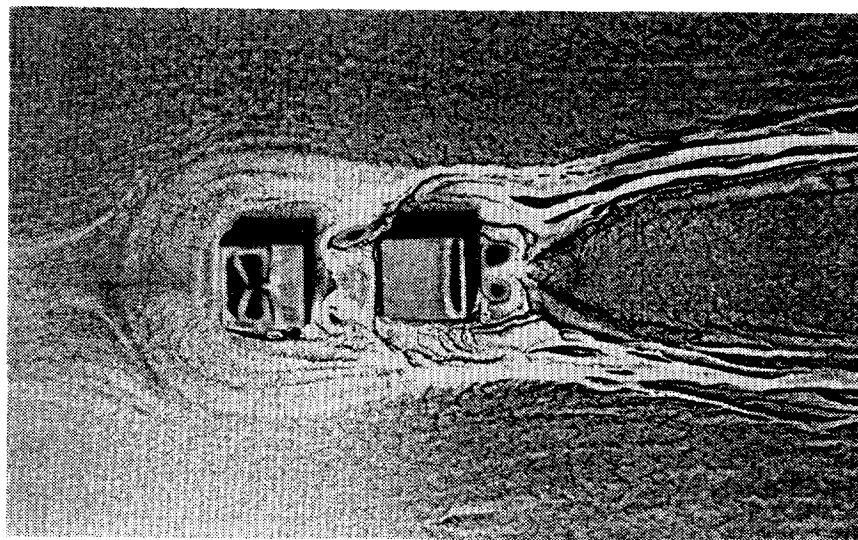


FIGURE 7.3: Oil-film visualisation of the surface flow pattern of an in-line tandem arrangement of cubes for $S_x/H = 2$ ($Re_H = 7000$).

A horseshoe vortex formed upstream of the front face of the leading cube and was deflected downstream along the sides. The shear layer separated at the top leading edge of the upstream cube and reattached at the top face of the downstream cube. The pigment pattern at the top face of the leading cube clearly shows the imprint of the resulting bound vortex. The inter-obstacle wake is characterised by an arc-shaped vortex, confined by the side flow and the separating shear layer. Its imprint appeared as the nodal pigment concentrations in the inter-obstacle spacing. Such a wake vortex was also observed for the downstream cube. The oncoming flow impinged on the front face of the leading cube, separated at the side leading edges and caused vortex tubes to exist along the side edges of the leading cube. Their imprints appear as the nodal pigment concentrations at the channel floor, close to the leading edge. The corresponding LDA measurements are shown in Fig. 7.4

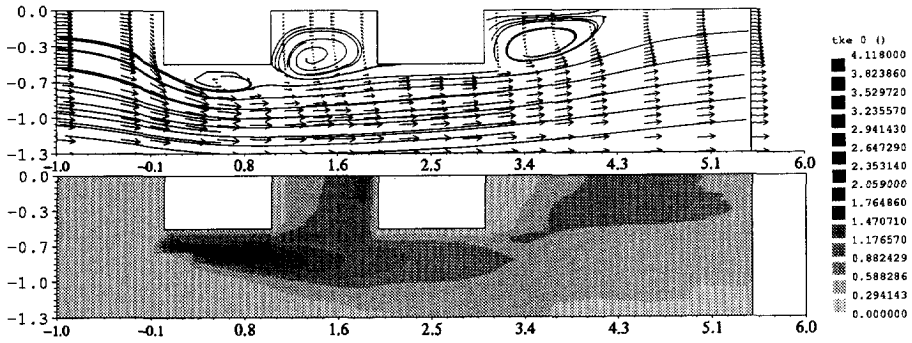


FIGURE 7.4: Vector plot with streamlines and an iso-value plot of the turbulent kinetic energy in a plane parallel to the channel floor at $y/H = 0.5$ for an in-line tandem of cubes with $S_x/H = 2$ ($Re_H = 3900$).

for $Re_H = 3900$. The vector plot with streamlines shows the flow field in a plane parallel to the channel floor at half a cube height $y/H = 0.5$. Only one half of the flow domain is shown since flow symmetry was demonstrated from measurements across the entire domain. The flow clearly separates at the leading side edges and subsequently creating the side vortices. The shear layer reattached at the front side edges of the downstream cube where it separated in the boundary layer flow at the sides and the recirculation in the cavity. The turbulent kinetic energy, $k = 0.75 (\overline{u'^2} + \overline{w'^2})$, becomes maximum at the interface between the side vortices and the shear layer flow around it. An intensification of the turbulence level also occurred in the downstream shear layer. Strong shearing and the quasi-periodic velocity fluctuations both contributed to the high turbulence levels.

$$S_x/H = 4$$

With increasing S_x/H , the shear layer separated from the side and the top edges, penetrated the inter-obstacle spacing and reattached on the channel floor. For $S_x/H = 4$, shear layer reattachment was observed at approximately 1.5 cube distances from

the leeward face. This can be clearly seen from the oil-film visualisations, given in Fig. 7.5, in which the smeared pigment pattern in the reattachment region indicates the highly unsteady nature of the reattaching flow. The imprint of the horseshoe vortex is also clear from the oil-film visualisation. Velocity measurements in the plane of symmetry $z/H = 0$, given in vector representation in Fig. 7.6, indicate the location of the horseshoe vortex in that plane. The normal velocities were

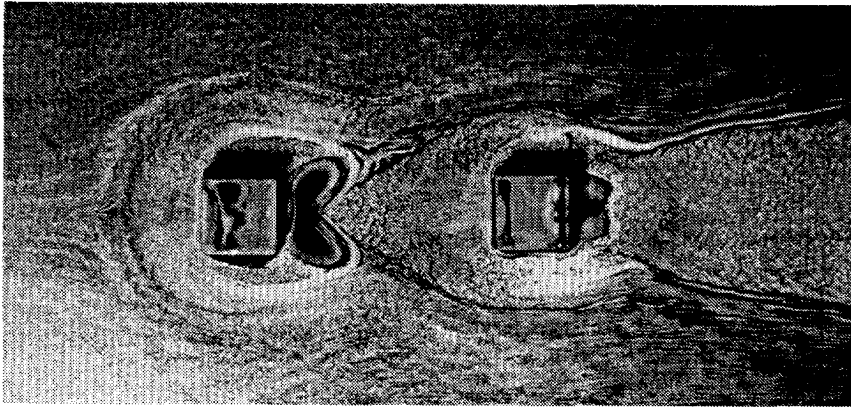


FIGURE 7.5: Oil-film visualisation of the surface flow pattern of an in-line tandem arrangement of cubes for $S_x/H = 4$ ($Re_H = 7000$).

obtained from the reconstruction technique as described in chapter 4. The presence of a second vortex upstream of the main horseshoe vortex can be concluded from the somewhat distorted flow upstream of the primary vortex. The corresponding increase in the turbulent kinetic energy, given in the lower plot in Fig. 7.6, is also indicative for the second vortex. The flow reattachment caused a two-cell structure to exist in the inter-obstacle spacing. The arc-shaped vortex in the wake of the leading cube is identified from the confocal pigment concentrations in Fig. 7.5. The time-averaged vector plot in Fig. 7.6 also elucidates this vortex. The shear layer reattachment further downstream caused that a second horseshoe vortex was formed in front of the downstream cube. The imprint of this vortex is observed from the oil-film pattern, see Fig. 7.5.

Due to the flow impingement at the windward face of the second cube, the flow separated at the top and the side edges causing small vortices to exist near the leading edges. The vector plot of the velocity field at $y/H = 0.5$ (Fig. 7.7) supports the observations made from the visualisation experiments. The flow separation at the windward side of the leading cube is clear from the plot. The resulting flow recirculation at the sides is well resolved with the LDA system. The flow separation at the leading side edge of the downstream cube can be deduced from the curvature of the streamlines following the flow outward the cavity. Arc-vortices are found in the wake of both cubes and their locations are in accordance with the oil-film observations. The turbulent kinetic energy, given in Fig. 7.7, is again

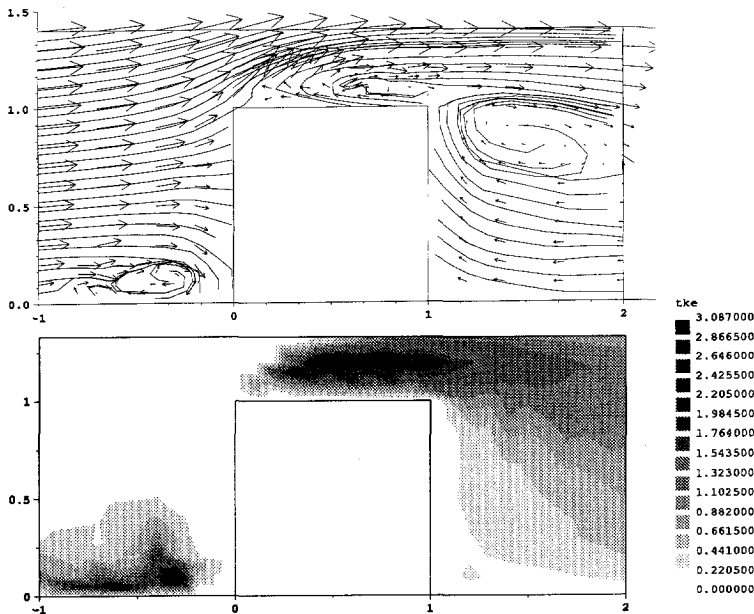


FIGURE 7.6: Vector plot with streamlines and an iso-value plot of the turbulent kinetic energy in the symmetry plane $z/H = 0$ for the leading cube in the in-line tandem of cubes with $S_x/H = 4$ ($Re_H = 3900$).

maximum at the interface of the side vortex and the shear layers. The intermittent reattachment of the shear layer in the inter-obstacle spacing contributed significantly to the production of turbulence, as appears from the relatively large levels of k in this region.

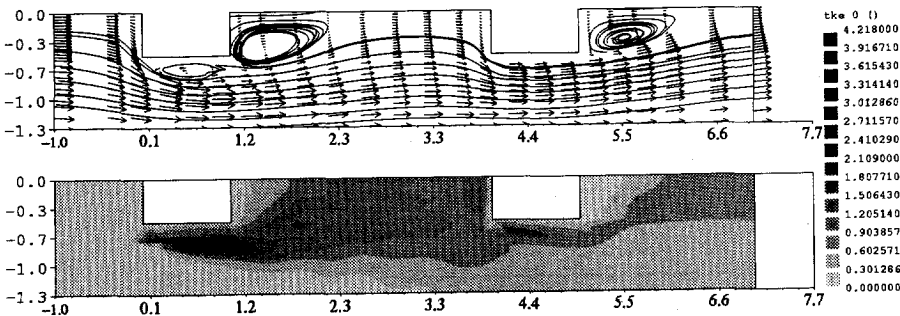


FIGURE 7.7: Vector plot with streamlines and an iso-value plot of the turbulent kinetic energy in a plane parallel to the channel floor at $y/H = 0.5$ for $S_x/H = 4$ ($Re_H = 3900$).

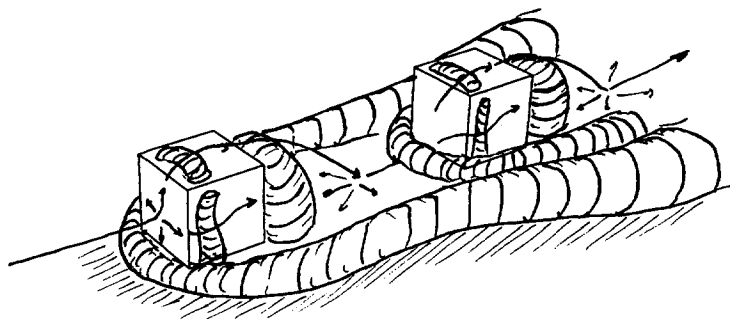


FIGURE 7.8: 'Artist's impression' of the flow structure around a tandem of two cubes in a turbulent channel flow ($S_x/H = 4$).

The flow features are summarised in the 'artist's impression' given in Fig. 7.8. The side, top, horseshoe and wake vortices are indicated in the figure.

$S_x/H = 8$

The LDA results for $S_x/H = 8$ are given in Fig. 7.9 and Fig. 7.10 for respectively the flow around the first and second cube in the tandem arrangement. The shear layer reattached at the channel floor in the inter-obstacle spacing at approximately $x/H = 2.5$. The flow around the leading cube is again characterised by the vortex tubes at the side, the bound vortex at the top (observed from the pigment concentrations at the top face in Fig. 7.11) and the wake vortex. The relative large distance between the cubes allowed the flow to recover in a time-averaged sense as is observed from the mean flow field after the flow reattachment at the channel floor. However, the upstream protrusion disturbed the flow which led to a rather high level of the turbulence intensity as experienced by the downstream obstacle. The mean flow and turbulence fields around the downstream cube exhibit a large similarity with those around the upstream cube, but the recirculations were generally smaller. This is in agreement with earlier observations made in chapter 5 for the single cube test case.

7.1.2 Flow dynamics: vortex shedding

The aerodynamic interference between the two cubes in a tandem arrangement diminishes for a large value of S_x/H , asymptoting to a single cube situation in the case of an infinite distance. These measurements have already been discussed.

An example of a smoke visualisation for the in-line tandem with $S_x/H = 4.5$ is given in Fig. 7.12. The visualisation was performed with three-dimensional light illumination from a stroboscope. The pulsed illumination freezes in fact the time-dependent flow pattern. Illumination with a stroboscope can be regarded as a kind of phase-averaging to detect coherent structures in the flow. In particular in the wake of the leading cube, the flow appeared to be very instable. The technique

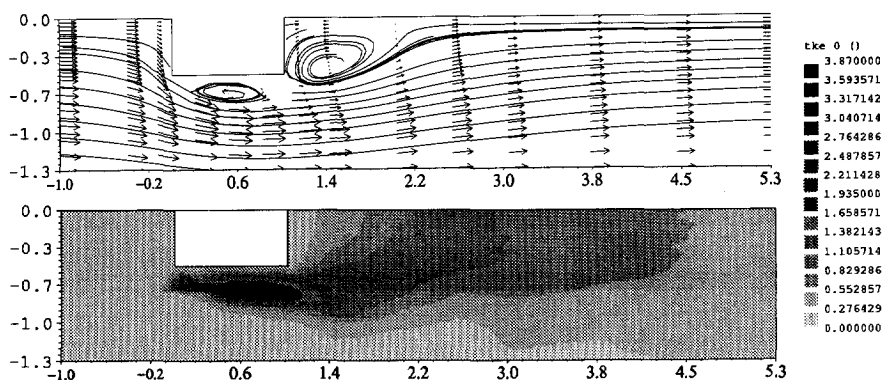


FIGURE 7.9: Vector plot with streamlines and an iso-value plot of the turbulent kinetic energy in a plane at $y/H = 0.5$ for the first cube in the in-line tandem with $S_x/H = 8$ ($Re_H = 3900$).

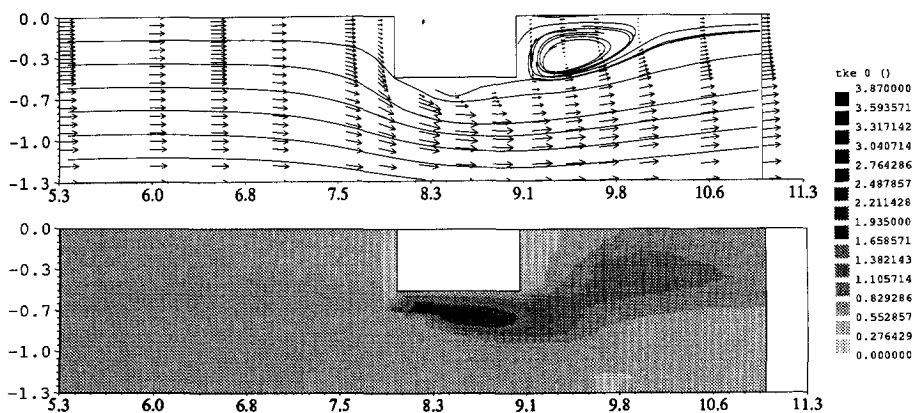


FIGURE 7.10: Vector plot with streamlines and an iso-value plot of the turbulent kinetic energy in a plane at $y/H = 0.5$ for the second cube in the in-line tandem with $S_x/H = 8$ ($Re_H = 3900$).

was not accurate enough to derive values of the shedding frequencies from it. But nevertheless, it illustrated the flow dynamics in the wake region.

LDA measurements at stations in the wake of the leading cube were performed to determine the shedding frequencies. Typical stations were at $x/H = 1.2-2$, $y/H = 0.5$ and $z/H = 0.5$. Dimensionless wake frequencies in terms of the Strouhal number St ($St = fH/u_B$, where f is the frequency, H is the cube size and u_B is the bulk velocity) for the leading cube in the tandem arrangement are shown in Fig. 7.13 as a function of S_x/H . The results for the fully developed turbulent channel flow, indicated with ($\circ$), show a minimum in the dominant frequency at a gap size of

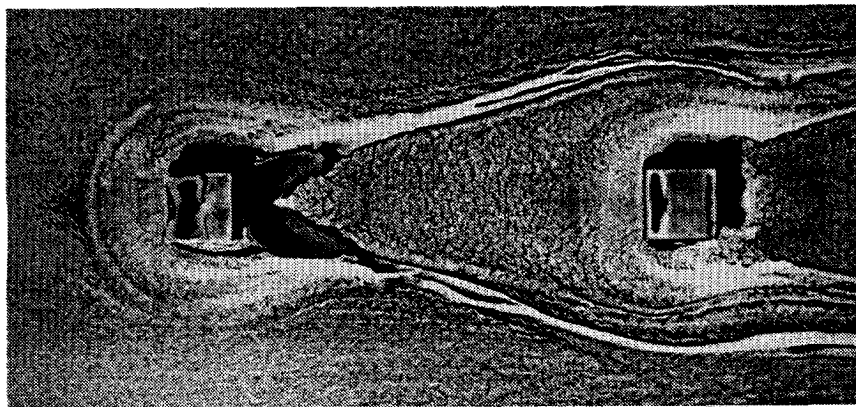


FIGURE 7.11: Oil-film visualisation of the surface flow pattern of an in-line tandem arrangement of cubes for $S_x/H = 8$ ($Re_H = 7000$).

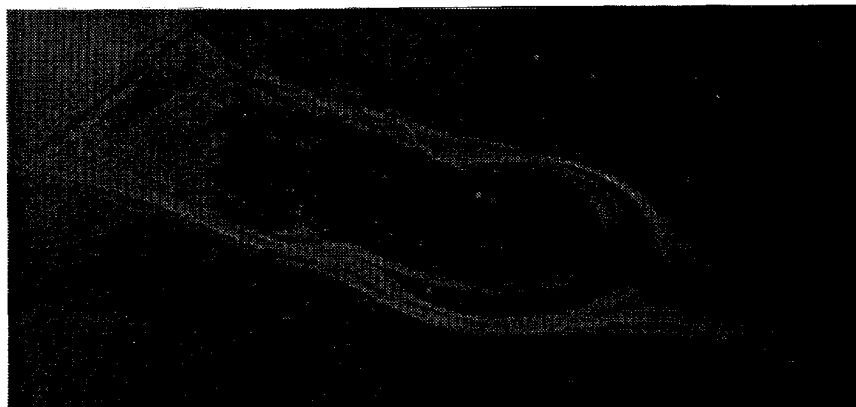


FIGURE 7.12: Smoke visualisation of the flow around an in-line tandem of cubes with $S_x/H = 4.5$ ($Re_H = 1000$). Flow is from right to left.

$S_x/H = 2.9$. The onset of vortex shedding activity was detected at $S_x/H = 2.4$. With increasing S_x/H , the shedding frequencies gradually converged to the value found for the infinite distance, a situation which corresponds to a single cube. The variation of S_x/H appeared to have a significant effect on the curvature of the shear layer separated from the leading edge of the upstream cube. For $S_x/H = 2$, it was found that the shear layer reattached at the leading edges of the downstream cube while flow reattachment at the channel floor at $x/H = 2.5$ was found for $S_x/H = 4$. From the flow visualisation observations made for $S_x/H = 3$, it was concluded that the shear layer reattaches at the lower edge of the windward face. Differences in the curvature caused the adverse pressure gradient in the cavity to be different,

which consequently resulted in the observed minimum in the shedding frequency. The Strouhal numbers for the leading cube in a tandem arrangement placed in a

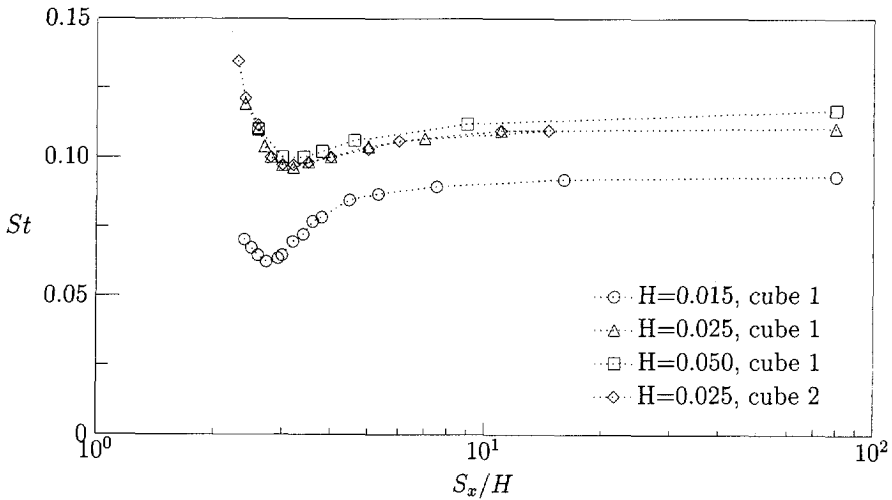


FIGURE 7.13: Dominant wake frequencies of the leading cube as a function of S_x/H for two cubes in the fully developed turbulent channel flow ($H = 1.5$ cm, $\circ$) and in a thin turbulent boundary layer ($H = 2.5$ cm, $\triangle$ and $H = 5.0$ cm, $\square$). The markers $\diamond$ indicate the wake frequencies for the second cube.

thin turbulent boundary layer at $Re_H = 21,500$ are indicated with $\triangle$ and $\square$, for a cube size of respectively $H = 2.5$ and 5.0 cm. Although these measurements also show a minimum frequency at a gap size of $S_x/H = 3.0-3.2$, the Strouhal numbers are significantly higher than those for the fully developed channel flow. The small shift in the critical distance, for which the shedding frequency becomes minimum, is most likely due to flow structure differences around the cube, for example the curvature of the separated shear layer. A minimum in frequency was also observed by Takeuchi and Matsumoto [163, 164, 165] who studied the interference between two square cylinders suspended in a uniform free stream. They found a critical gap space of $S_x/H = 3$. It is interesting to note that despite totally different oncoming flow conditions were encountered, the agreement is remarkably good, suggesting that the mechanism of vortex shedding is dominated by inviscid flow effects.

Non-dimensionalised shedding frequencies measured in the wake of the second cube (with $H = 2.5$ cm) are indicated with ($\diamond$) in Fig. 7.13. It is observed that the shedding frequencies in the wake of the first and the second cube are the same, at least within the experimental uncertainty in determining the shedding frequency. We note that the frequencies behind the two cubes were not measured simultaneously. Observed differences in the frequencies as seen in Fig. 7.13 are most likely due to experimental artifacts, for example duplicating S_x , between two separate experiments. These small differences are amplified in the vicinity of the critical region because of the strong sensitivity of the frequency to spacing. The similarity never-

theless indicates that the vortex shedding of both cubes is frequency locked. At the moment it is not clear if a phase shift exists between the two shedding mechanisms.

7.1.3 Convective heat transfer

The heat transfer measurements were performed at $Re_H = 3900$. The streamwise distance S_x/H was varied between $S_x/H = 0.5$ and ∞ while the spanwise distance was kept at zero ($S_z/H = 0$). Both cubes were separately powered to enable the use of the inlet flow temperature as the reference temperature (if both cubes were powered, the effect of upstream heating needed to be taken into account which involved a different experiment to determine the actual adiabatic temperature of the cube in question).

Distributions of the local convective heat transfer

Distributions of the heat transfer coefficient for the leading cube in the in-line tandem are given in Fig. 7.14 for $S_x/H = 2, 4$ and 8 . The distributions along two paths are shown: path $ABCD$ denotes a plane parallel to the channel floor at half a cube height $y/H = 0.5$; path $ABCD$ denotes a plane perpendicular to the channel floor at $z/H = 0.5$. It is noteworthy that the h -profiles are symmetric around the plane of symmetry, indicated with the dotted lines in the upper plot in Fig. 7.14.

Although a large spatial variation in the local heat transfer coefficient is observed, the differences between the results for the different distances ($S_x/H = 2, 4$ and 8) are too small to attribute to physical phenomena. The profiles of the heat transfer coefficient at the top and side faces show a marked minimum and maximum, located at the front half and the back half of these faces, respectively. For all three values of S_x/H , a vortex tube was observed to exist at the side faces and a bound vortex on the top of the cube. The heat removed from the hot cube faces recirculated in the vortex and caused the local fluid temperature to increase. This caused the convective heat transfer to decrease which led to the observed minimum in h . The intermittent reattachment of the cold shear layer close to the trailing edges benefited the convective heat transfer.

From the previous discussion of the mean flow and turbulence fields it can be concluded that only minor differences existed in the flow structure in the vicinity of the leading cube. The distributions of the heat transfer coefficients for $S_x/H = 2, 4$ and 8 support this conclusion. Since these three cases are assumed to cover the different types of flow, it is unlikely that distributions of the heat transfer coefficient for other spacings will differ greatly from those presented here for $S_x/H = 2, 4$ and 8 .

Distributions of h along path $ABCD$ for the second cube in the in-line tandem arrangement are given Fig. 7.15 as a function of S_x/H . The distributions are separated in two plots to increase the clarity of the results. It is seen from the figures that the convective heat transfer at the back (west) face is hardly affected by a variation in S_x/H . This is expected since the wake vortex of the second cube was more or less similar for all the considered in-line tandems. Further, the shape of these heat transfer coefficient distributions is also very similar to the those of

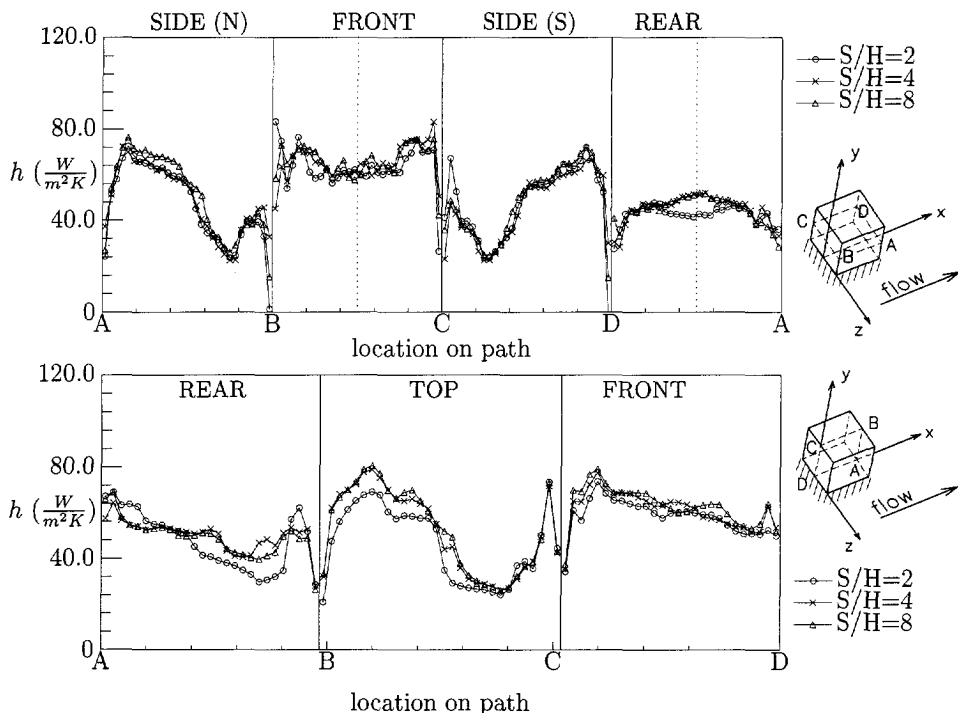


FIGURE 7.14: Heat transfer coefficient distributions along path $ABCD$ A (upper plot) and along path $ABCD$ (lower plot) for the leading cube in the tandem as a function of S_x/H .

other studied configurations, like the isolated cube and the in-line row of cubes. The h -profiles at the front face of the downstream cube have a more or less concave shape, in particular for the smaller streamwise spacings. Such a shape in the local heat transfer coefficient was also found for the front face of the downstream cubes in the in-line row (see chapter 6). It was explained that the concave shape was caused by the counter-rotating extensions of the arch-shaped vortex in the inter-obstacle spacing.

The distributions of the heat transfer coefficient of both the side faces show large differences for the different streamwise spacings. For the smallest spacing considered, $S_x/H = 2$, h decays monotonically in the downstream direction. The separated shear layer from the leading cube reattached at the front edges without causing the flow to recirculate. The resulting boundary layer flow explains the observed behaviour. The heat convected away from the hot side surfaces caused an increase of the local fluid temperature, which subsequently caused the gradual drop in h . For $S_x/H = 4$, a small vortex tube was found at the side faces close to the leading edge (see Fig. 7.7). The influence of this rather small vortex tube is reflected in the minimum in h , close to the leading edge, followed by a slight increase in h . As argued before, the latter is caused by the reattachment of the separated shear

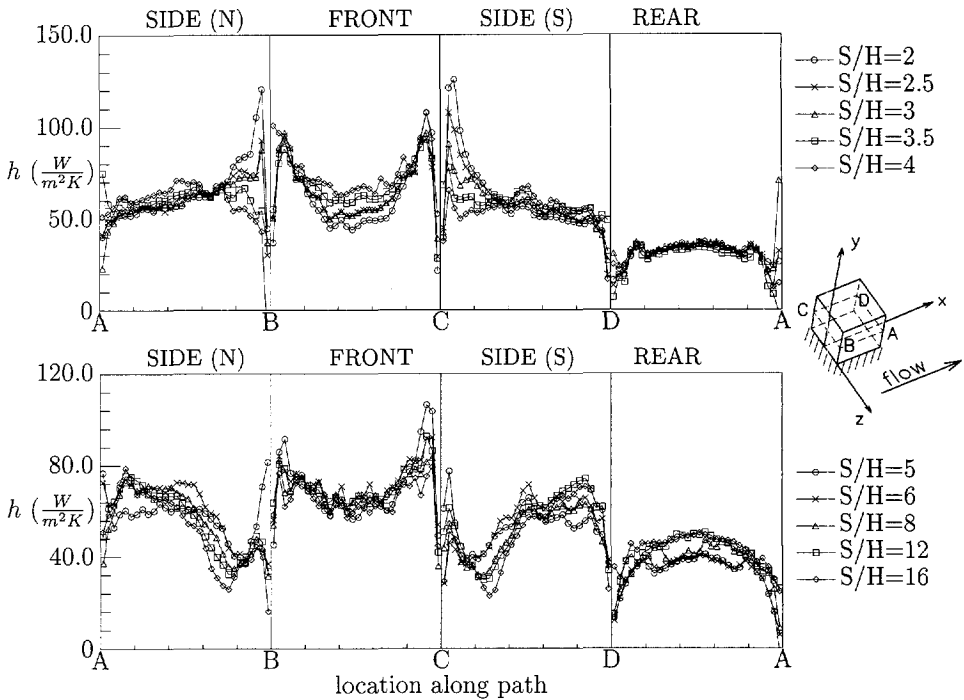


FIGURE 7.15: Heat transfer coefficient distributions along path $ABCD$ A for the second cube in the tandem as a function of S_x/H . The upper figure shows the results for $S_x/H = 2, 2.5, 3, 3.5$ and 4. The lower figure refers to $S_x/H = 5, 6, 8, 12$ and 16.

layer. An examination of the convective heat transfer for the intermediate distances shows the gradual transition from h -distributions caused by boundary layer flow to h -distributions caused by the flow recirculations.

For larger distances, the vortex increased in size and strength. This is reflected in the more pronounced heat transfer minima and maxima. It was shown from the mean flow and turbulence fields that the flow structure around the upstream and downstream cube was similar for $S_x/H = 8$. Therefore, the distributions of h are qualitatively similar to those of the leading cube, given in Fig. 7.14.

The distributions of h along a cross-section perpendicular to the channel floor, $ABCD$, are shown in Fig. 7.16. Again the heat transfer characteristics at the back face (west) hardly depend on S_x/H . The transition from convective heat transfer caused by boundary layer flow, for $S_x/H = 2$, to a situation in which the heat transfer is dominated by the bound vortex on top of the obstacle, for $S_x/H > 4$, is obvious from the distributions. An interesting feature, the marked minimum at the lower edge of the front (east) face, is caused by the flow recirculation in the inter-obstacle space. For $S_x/H = 2$, a recirculation in the cavity caused a gradual decrease towards the lower edge (towards D). The development of a horseshoe vortex

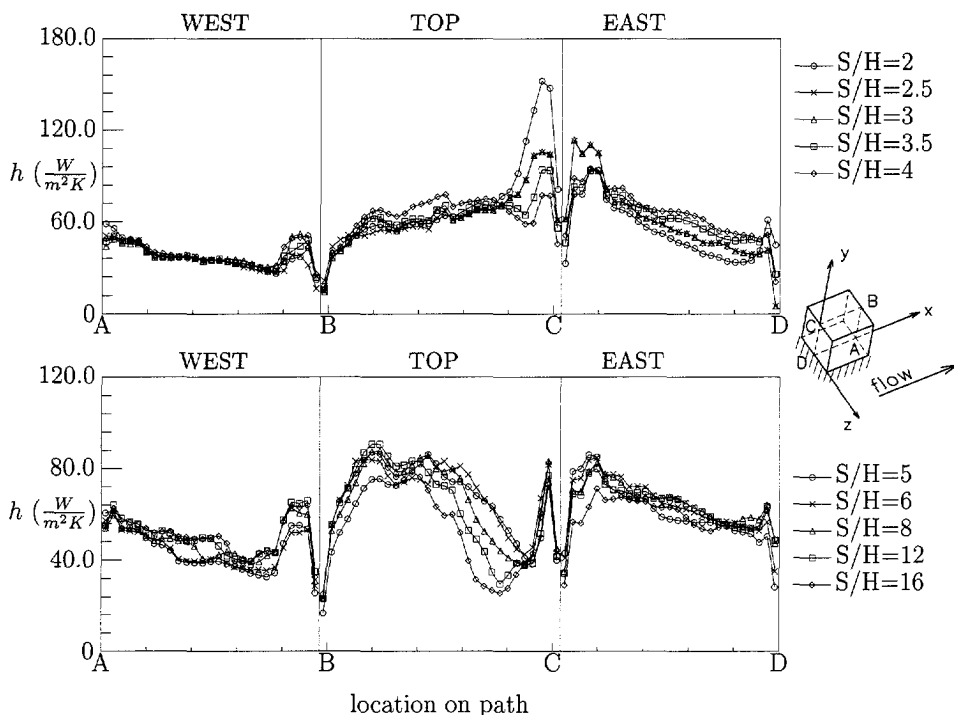


FIGURE 7.16: Heat transfer coefficient distributions along path $ABCD$ for the second cube in the tandem as a function of S_x/H . The upper figure shows the results for $S_x/H = 2, 2.5, 3, 3.5$ and 4 . The lower figure refers to $S_x/H = 5, 6, 8, 12$ and 16 .

for the intermediate spacings $S_x/H = 4$ caused the pronounced minimum close to the channel wall. The heat convected from the wall, recirculated in the vortex and caused the local fluid temperature to be elevated.

The aforementioned differences in the local convective heat transfer for the various distances S_x/H studied can be observed from the folded-out isotherm maps of the surface temperature distributions for the five faces of the cube, as given in Fig. 7.17. The plots show a qualitatively similar temperature pattern for the east and west face for all values of S_x/H . For these faces, it is noted that for the larger distances fewer isotherms are shown due to the high temperature levels at the other faces. Substantial differences between the isotherm patterns of the small and large distances S_x/H are observed at the side and the top faces.

Time-dependent convective heat transfer

From the flow study, it appeared that variation of S_x/H had a significant effect on the flow structure around the downstream cube while the flow pattern around the leading cube was only a little affected. For $S_x/H = 2$, the separated shear layer

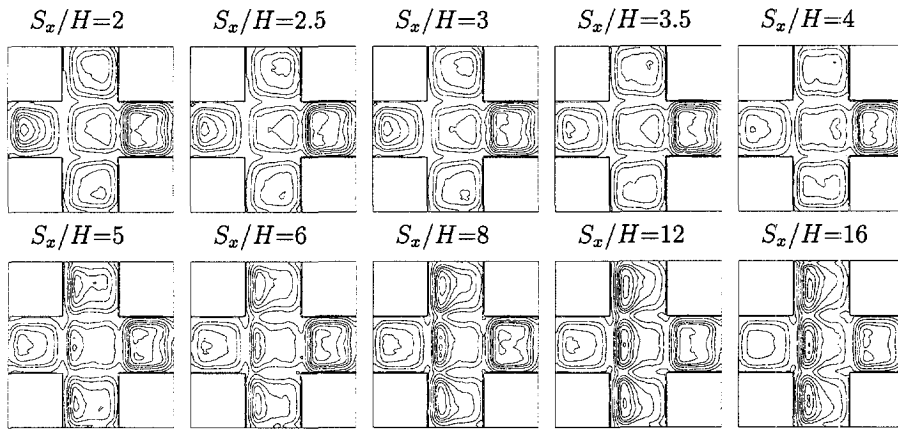


FIGURE 7.17: Isotherm maps of the surface temperature of the second cube in the tandem arrangement for various values of S_x/H . Flow is from left to right.

reattached more or less at the side edge of the downstream cube; it reattached in the inter-obstacle spacing at $x/H = 2.5$ for $S_x/H = 4$ and larger. The flow structure around the upstream and downstream cube were more or less similar for $S_x/H > 8$. The wake flow of the upstream cube showed oscillatory behaviour due to the shed vortices behind its side faces. The frequency of vortex shedding was modulated by the aerodynamic interaction with the downstream cube resulting in a dependency on the streamwise distance S_x/H . The vortex shedding frequency was minimum for $S_x/H = 2.9$. Dependent on S_x/H , the shed vortices reattach more or less periodically at the side faces of the downstream cube, presumably causing the time-dependent heat flux to fluctuate also periodically. This coupling was demonstrated for geometries other than wall-mounted cubes [146, 147, 148, 155, 99, 142] and, therefore, expected in this case to be present as well.

Measurements of the time-resolved convective heat flux were performed with the active heat flux sensor (see chapter 3). The two dynamic calibrations described in chapter 3 ensured a sufficiently high frequency-response in the order of magnitude of at least 200 Hz. The absolute value of the heat flux signal was not calibrated because of the time-dependent heat losses which were significantly large but difficult to estimate correctly. Therefore, we will only focus on the frequency information contained by the time-dependent convective heat flux. The sensor was flush mounted in the side face of a cube of formaldehyde. Formaldehyde was chosen since it was easy to machine and it had a low thermal conductivity of $\lambda = 0.33 \text{ W/mK}$ (see appendix A). The data acquisition and the sensor controlling were similar to that of the frequency-response calibration, see chapter 3.

Simultaneously, a hot-wire anemometer was used to measure the spanwise velocity in the wake of the upstream cube. The vortex shedding frequency was then obtained from the corresponding power density spectra. A typical example is shown in Fig. 7.18 for $S_x/H = 3$ and $u_B = 2 \text{ m/s}$. The spectrum was based on 500,000 data

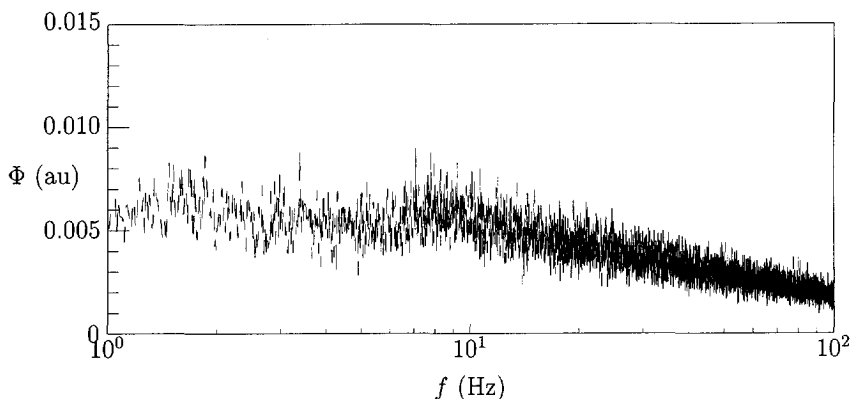


FIGURE 7.18: Power density spectrum of the spanwise velocity measured with a hot-wire anemometer in the wake of the upstream cube for $S_x/H = 3$ and $Re_H = 2000$.

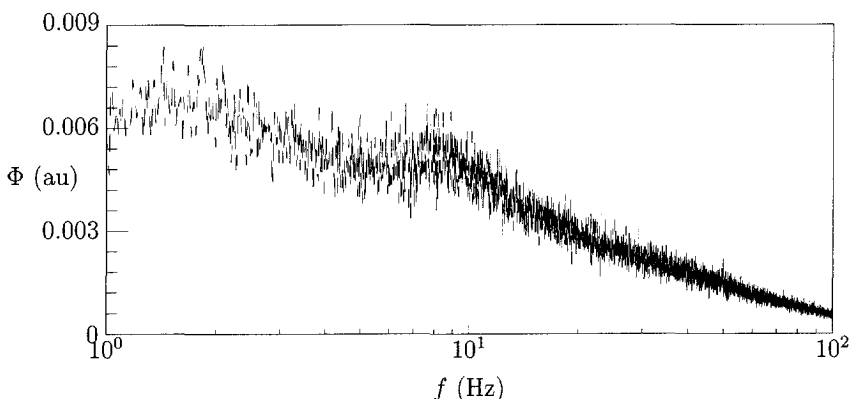


FIGURE 7.19: Power density spectrum of the time-dependent heat flux at the side face of the downstream cube in the in-line tandem for $S_x/H = 3$ and $Re_H = 2000$.

points. Standard recipes with windowing were used to calculate the spectra. The spectrum in Fig. 7.18 shows a peak at $f = 8 \pm 1$ Hz which corresponds to a Strouhal number of $St = fH/u_B = 0.06$. A spectrum of the corresponding time-dependent heat flux from the downstream cube is shown in Fig. 7.19 for the same situation. A marked peak is found at the vortex shedding frequency: $f = 8 \pm 1$ Hz. Apparently, the periodic flow reattachment at the sides of the downstream cube causes the time-dependent heat flux to fluctuate periodically as well. The peak is even more pronounced when the cross spectrum of both the heat flux and the velocity signal is calculated. This spectrum is shown in Fig. 7.20. A strong correlation was found at $f = 8 \pm 1$ Hz.

The Strouhal number of $St = 0.06$ is in good agreement with the results given in Fig. 7.13 for the tandem of two cubes in the fully developed turbulent channel flow. These results are reproduced in Fig. 7.21. The Strouhal number obtained with the

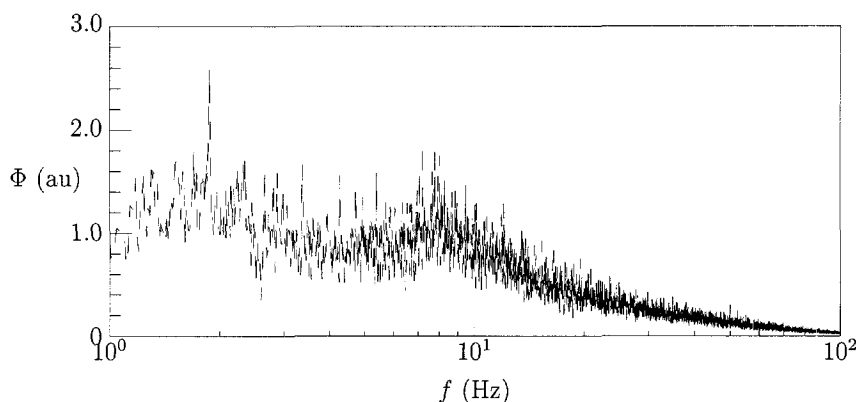


FIGURE 7.20: Spectrum of the cross-correlation between the time-dependent heat flux and the velocity signal at the side face of the downstream cube in the in-line tandem for $S_x/H = 3$ and $Re_H = 2000$.

heat flux sensor is indicated with ($\triangle$) for $S_x/H = 3$.

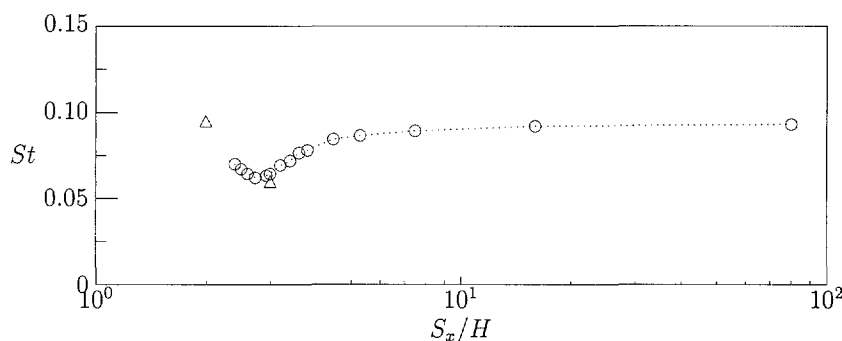


FIGURE 7.21: Dominant wake frequencies of the leading cube as a function of S_x/H for two cubes in the fully developed turbulent channel flow. LDA-measurements of the vortex shedding are indicated with ($\circ$), the dominant frequencies in the heat flux signal are indicated with ($\triangle$).

A periodic heat flux is only present when the sensor is subject to oscillatory flow which is only the case for some combinations of S_x/H and the bulk velocity. A distinguishable peak was also observed for $S_x/H = 2$ and $u_B = 1.9$ m/s. The Strouhal number for this case was $St = 0.095$. This value is also in good agreement with shedding frequency results, as shown in Fig. 7.21.

The noise level in the hot-wire signals was rather high due to the limited number of samples used to calculate the spectrum. Further, the hot-wire probe was positioned in an area where the amplitude of the oscillating flow was small. Since the vortex shedding frequency was clear from the spectrum, we did no attempt to improve the signal-to-noise ratio. However, the noise level in the heat flux signals was large due to the small signal. The periodic variation in the heat flux signal

was only millivolts while the dc-component was in the order of magnitude of volts. Further the sensor area was not infinitely small but $4 \times 4 \text{ mm}^2$ which caused a spatial averaging of the event.

7.1.4 Averaged heat transfer coefficients

In an attempt to correlate the average heat transfer coefficients with the observed minimum in the wake shedding frequency of the leading cube, the face-averaged heat transfer coefficients of the downstream cube, determined from the face-averaged convective heat flux and surface temperature, are plotted versus S_x/H in Fig. 7.22. The average heat transfer coefficient of the side and top faces exhibits a small minimum around $S_x/H = 3$. This minimum occurred at the critical distance for which the shedding frequency was minimum and it strongly suggests that this minimum is correlated to the shedding frequency and/or mean flow structure. At the moment, it is not clear if the minimum in h is directly caused by the frequency of vortex shedding or that both minima (in the frequency and the heat transfer coefficient) are caused by the local flow structure. A plausible argument for the former possibility, a direct correlation, is that the quasi-periodic flow to and from the walls augments the convective transport of heat, as for example occurs in intermittent flow reattachment. It is noted that the causality between the minimum in the shedding frequency and the convective heat transfer is very weak as appears from the minor differences. Further, the cube-averaged heat transfer coefficient, which was determined from the cube-averaged heat flux and surface temperature, shows also a small minimum around $S_x/H = 3$. The face-averaged heat transfer at the east and west face is more or less constant around $S_x/H = 3$ and apparently, the side and top faces contribute markedly to the cube-averaged heat transfer. The dip in $\bar{h}$ at $S_x/H = 5$ is explained by experimental uncertainty and a possible variation in the averaged flow rate. The latter explanation was made plausible by plotting the supplied power, the product of the voltage and current, versus S_x/H . The results indicated with ($\circ$), given in Fig. 7.23, clearly show a dip in the supplied power at $S_x/H = 5$. The second measurement series of the supplied power, obtained from an independent measurement and indicated with ($\triangle$) in Fig. 7.23, shows a monotonic increase up to $S_x/H = 6$, indicating that the local minimum around $S_x/H = 5$ is an experimental artifact. The figure further shows a minimum in the supplied power around $S_x/H = 3$, which supports the previous conclusion. In general, the variations in the face-averaged and cube-averaged heat transfer coefficient as a function of S_x/H are rather small compared to the variations in the local heat transfer coefficient. Besides the minimum around $S_x/H = 3$, the only marked variation is found for the leeward face (west) for which $\bar{h}$ slightly increases with increasing distances. The averaged heat transfer coefficient for the other faces decreases slightly.

7.2 Off-axis tandem arrangement

The off-axis arrangements of two cubes were also placed in the fully developed channel flow. While the long-term averaged flow field and the convective heat

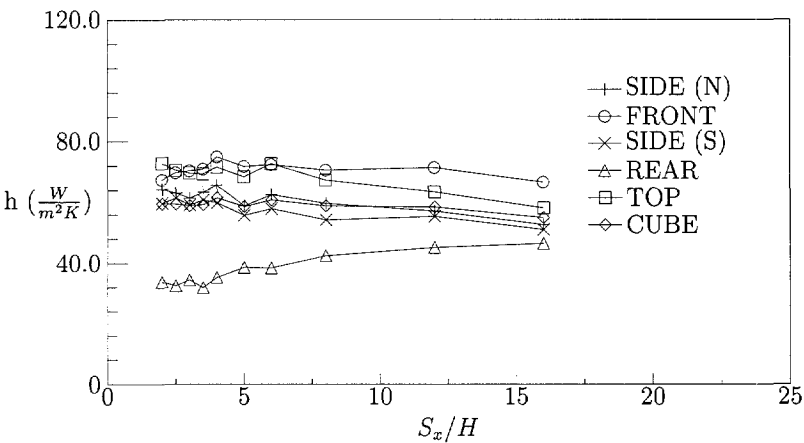


FIGURE 7.22: Spatially-averaged heat transfer coefficients as a function of S_x/H for the second cube in the in-line tandem: (+) side (north), (○) front, (×) side (south), (△) rear, (□) top and (◇) cube average.

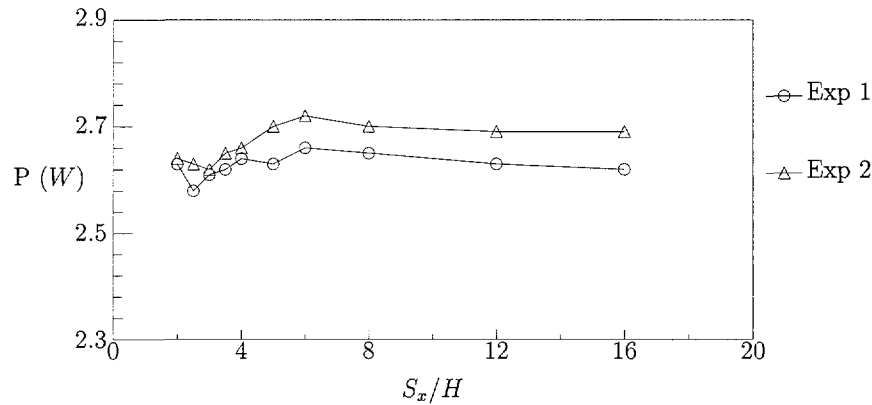


FIGURE 7.23: Total supplied power as a function of S_x/H . The markers refer to experiment 1 (○) and experiment 2 (△).

transfer distributions were symmetric in the in-line tandem cases, the results for the staggered arrangements were significantly asymmetric for some combinations of the streamwise and the spanwise distance. For either a large streamwise or spanwise distance, no significant asymmetry of the long-term averaged flow field is expected and, therefore, these arrangements represent more or less the flow around a single cube. But for relatively small streamwise and spanwise distances, the flow will be significantly directed by the downstream staggered cube which results in asymmetry of both the averaged flow pattern and the convective heat transfer. Oil-film

and smoke visualisation experiments were performed for a large range of streamwise and spanwise distances (S_x/H and S_z/H) to qualitatively study the effect of off-axis arrangements on the averaged flow structure. From these observations, three cases were selected to be documented quantitatively with the LDA-system, namely: $S_x/H = 2$ and $S_z/H = 1$; $S_x/H = 4$ and $S_z/H = 1$; $S_x/H = 4$ and $S_z/H = 2$. The results showed that the flow structures around both cubes generally resembled much of the features found for the in-line tandem arrangements of cubes. However, for some configurations the flow asymmetry caused significant departures from these. Heat transfer measurements were performed for a larger series of streamwise and spanwise distances, i.e. $S_x/H = 2$ and $S_z/H = 0.5, 1, 1.5, 2$ and 3 ; $S_x/H = 4$ and $S_z/H = 0.5, 1, 1.5, 2$ and 3 ; $S_x/H = 8$ and $S_z/H = 0.5, 1, 1.5, 2$ and 3 . Both the convective heat transfer and the flow field measurements were performed at the Reynolds number of $Re_H = 3900$, which corresponded to the bulk velocity of $u_B = 3.9$ m/s.

For the interpretation of the off-axis results, we will distinguish between the inner and outer side face of the leading and the downstream cube. The outer side face is the face pointing outwards the tandem, both inner side faces (of the upstream and downstream cube) point to each other. See the sketch of the plan view of the off-axis tandem arrangement given in Fig. 7.1.

7.2.1 Mean flow structure

The first off-axis configuration discussed had a streamwise and spanwise distance of $S_x/H = 2$ and $S_z/H = 1$, respectively. An oil-film visualisation of the surface streaklines is given in Fig. 7.24. The overall pigment pattern indicate the flow asymmetry for this slightly staggered configuration. The horseshoe vortex in front of the leading cube can clearly be seen from the pigment pattern. The symmetric pigment pattern in this upstream region suggests that the flow impinges perpendicularly on the front face. This more or less undistorted oncoming flow can also be seen from the long-term averaged vector plot of the velocity field in a plane at half the cube height ($y/H = 0.5$), given in Fig. 7.25. The downstream off-axis obstacle caused an extra blockage effect which imposed additional flow resistance. While the separated side shear layer reattached at the outer side wall of the leading cube, and thus causing a small side vortex to exist close to the leading edge, the inner side face is covered entirely by a side vortex without flow reattachment at the back half of the face. This is obviously caused by the direction of the flow as oppressed by the downstream cube. The extra downstream blockage caused the oncoming main flow to be more directed towards the inner side face of the first cube. This caused a large recirculation and, therefore, the absence of flow reattachment at the back half of the side face. The flow accelerated through the corridor between the cubes. Since the flow was about 45° inclined with respect to the system centre-line, it separated at the inner side face of the downstream cube and caused the rather large side vortex at this side face (see Fig. 7.25). Further, a big flow recirculation in the wake of the upstream cube resulted from the inclined oncoming flow. The arch-shaped vortex, as found for the symmetric flow cases, was then replaced by a one-centre

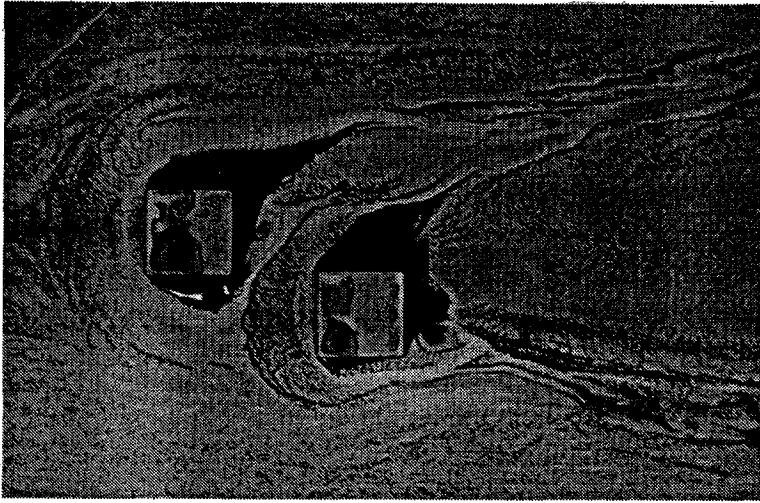


FIGURE 7.24: Oil-film visualisation of the surface flow pattern of the staggered tandem with $S_x/H = 2$ and $S_z/H = 1$ ($Re_H = 7000$).

wake vortex. The wake of the downstream cube exposed a weird recirculation area. Here also, due to the asymmetry of the flow, the arch-shaped vortex was replaced by a one-centre recirculation.

From the oil-film visualisation it is seen that the extension of the horseshoe vortex, induced in front of the first cube, interacts with the vortex in front of the second cube. This caused a strong intensification of the turbulence production which subsequently will cause heat transfer enhancement.

Strong peaks in the turbulence intensity were found at the interfaces between the side vortices and the shear layers for both cubes. Both the intermittent flow reattachment due to flow instabilities (at the outer and inner side face of the upstream and the downstream cube) and strong shearing are mentioned as causes for the large values of the turbulent kinetic energy. This is observed from the iso-value plot of k , given in Fig. 7.26 for a plane at half a cube height $y/H = 0.5$. Further, the free mixing layers behind the lateral faces have relatively large values of the turbulent kinetic energy. This is also due to the strong shearing and the vortex shedding from the side faces. The vortices in the wake of both cubes are rather stable, as is seen from the low values of the turbulence intensity. The region in front of the cubes have the lowest level of k . Since the plane considered ($y/H = 0.5$) does not intersect with the horseshoe vortices found upstream of both cubes, an increased turbulence level as usually found in the horseshoe vortex is not visible from the plot.

The influence of the off-axis placed downstream cube on the time-averaged flow field diminishes for a larger streamwise distance. This can clearly be seen from the oil-film visualisation given in Fig. 7.27 and from the vector plot of the velocity field in the plane at $y/H = 0.5$, shown in Fig. 7.28 for the tandem with a larger streamwise distance ($S_x/H = 4$), but the same spanwise distance ($S_z/H = 1$). The flow around

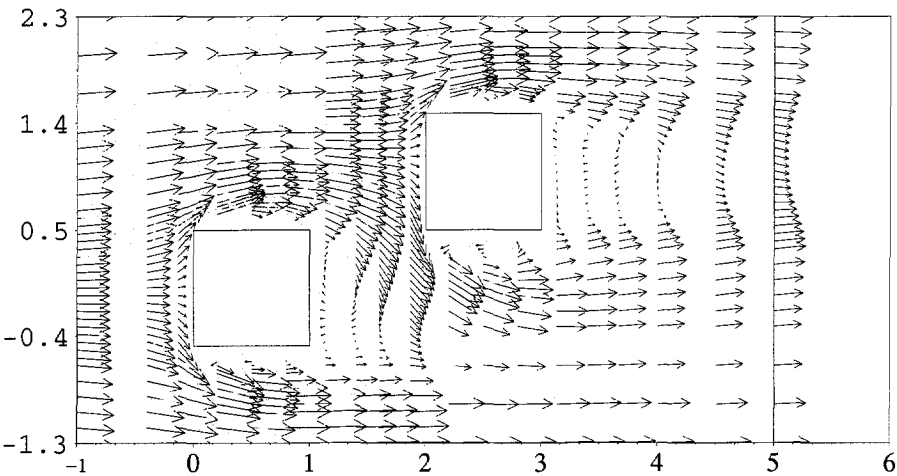


FIGURE 7.25: Long-term averaged vector plot of the velocity field with streamlines in a plane parallel to the channel floor at $y/H = 0.5$ for the off-axis tandem with $S_x/H = 2$ and $S_z/H = 1$ ($Re_H = 3900$).

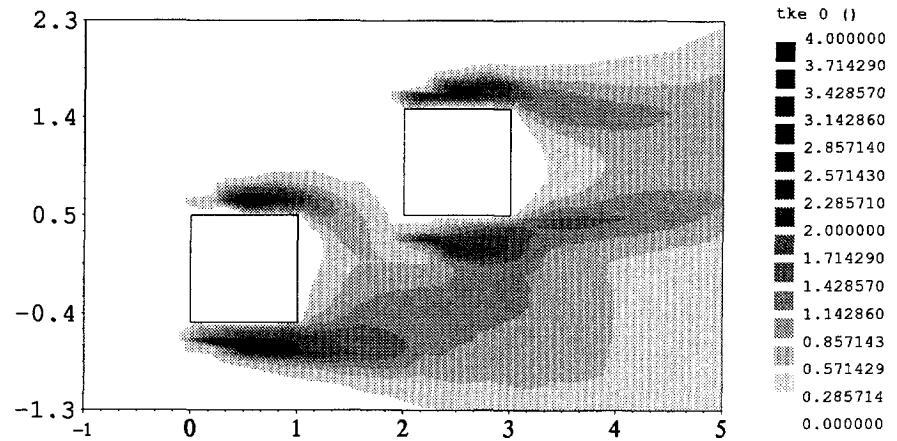


FIGURE 7.26: Iso-colour plot of the turbulent kinetic energy in a plane parallel to the channel floor at $y/H = 0.5$ for the off-axis tandem with $S_x/H = 2$ and $S_z/H = 1$ ($Re_H = 3900$).

the leading cube is almost unaffected by the downstream cube. The oncoming flow impinged perpendicularly on the front face of the leading cube where after it separated symmetrically at both side faces of the cube. It resembles more or less the flow field as found around the leading cube in the in-line tandem of cubes. The flow acceleration in the corridor between the cube was less strong but still observable.

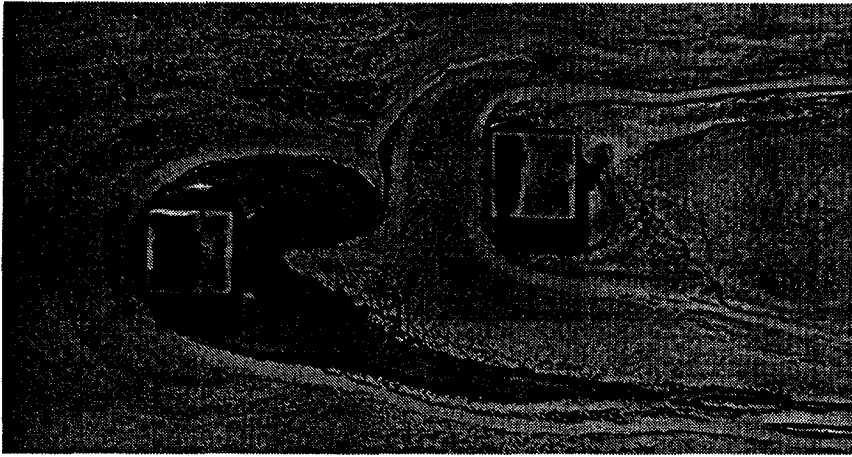


FIGURE 7.27: Oil-film visualisation of the surface flow pattern of the staggered tandem with $S_x/H = 4$ and $S_z/H = 1$ ($Re_H = 7000$).

The curvature associated with this corridor flow caused flow reattachment at the inner side face of the downstream cube. From the vector plot it is concluded that no flow reattachment occurred at the inner side face of the leading cube and outer side face of the downstream cube, but instead, the flow recirculations enveloped the entire face. Further, the protruding shape of the leading cube and the induced flow instabilities caused the wake flow to be highly turbulent which caused smaller flow recirculations around the downstream cube, as is seen in Fig. 7.28.

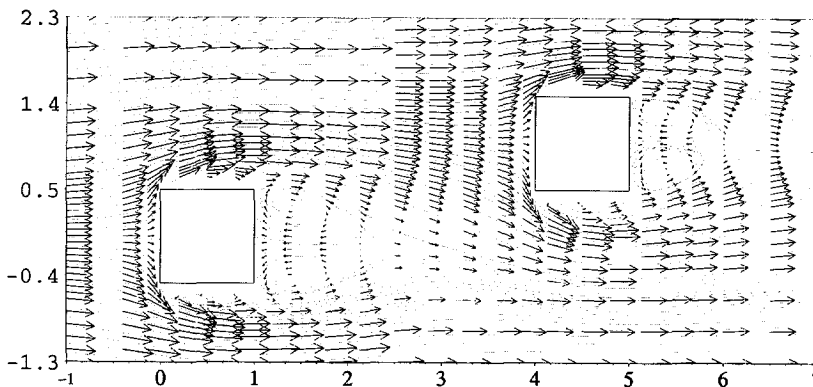


FIGURE 7.28: Long-term averaged vector plot of the velocity field with streamlines in a plane parallel to the channel floor at $y/H = 0.5$ for the off-axis tandem with $S_x/H = 4$ and $S_z/H = 1$ ($Re_H = 3900$).

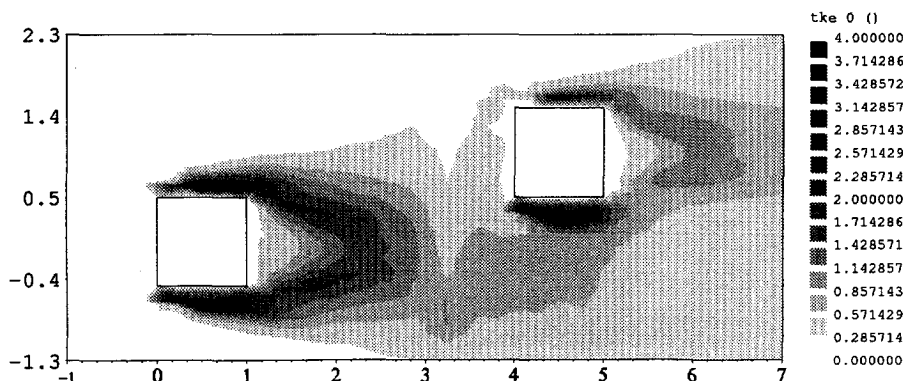


FIGURE 7.29: Iso-colour plot of the turbulent kinetic energy in a plane parallel to the channel floor at $y/H = 0.5$ for the off-axis tandem with $S_x/H = 4$ and $S_z/H = 1$ ($Re_H = 3900$).

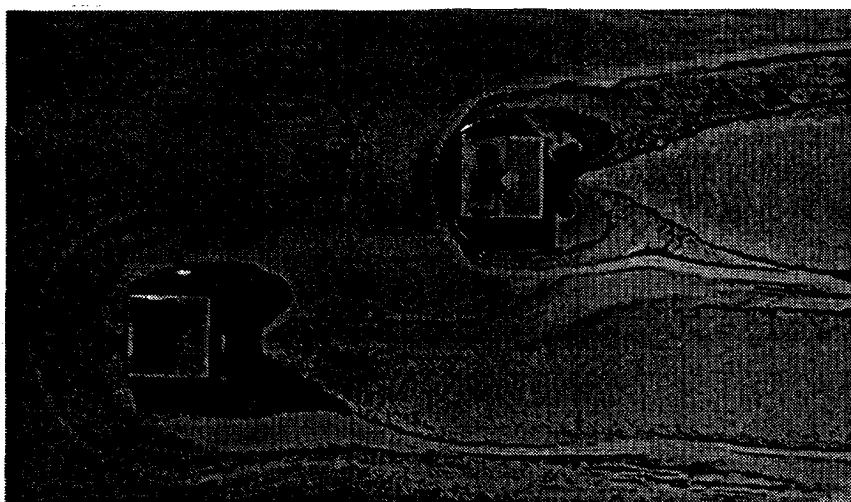


FIGURE 7.30: Oil-film visualisation of the surface flow pattern of the staggered tandem with $S_x/H = 4$ and $S_z/H = 2$ ($Re_H = 7000$).

While the plot of the turbulent kinetic energy is more or less symmetric for the upstream cube (see the iso-value plot of k in Fig. 7.29), a slightly increased turbulent intensity is observed at the inner side of the second cube. This was caused by the interaction with the extension of the upstream horseshoe vortex.

When the spanwise distance is increased to $S_z/H = 2$, with an unchanged streamwise distance $S_x/H = 4$, the situation becomes, as expected, more symmetric. This is observed from the oil-film visualisation and the LDA-results given in Figs. 7.30, 7.31 and 7.32.

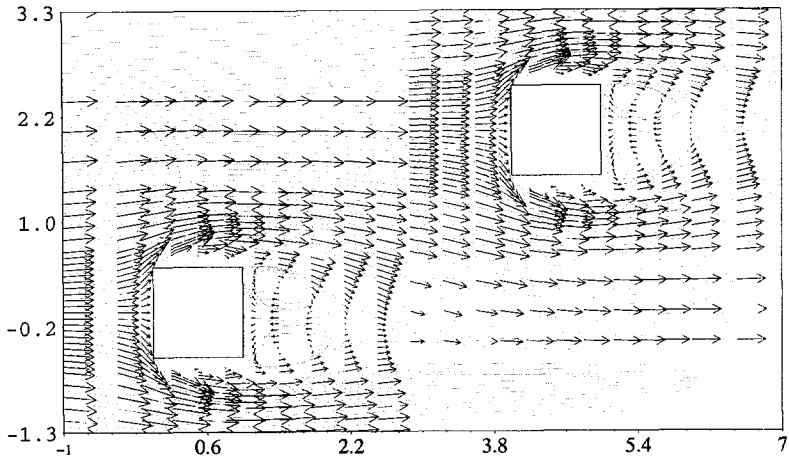


FIGURE 7.31: Long-term averaged vector plot of the velocity field with streamlines in a plane parallel to the channel floor at $y/H = 0.5$ for the off-axis tandem with $S_x/H = 4$ and $S_z/H = 2$ ($Re_H = 3900$).

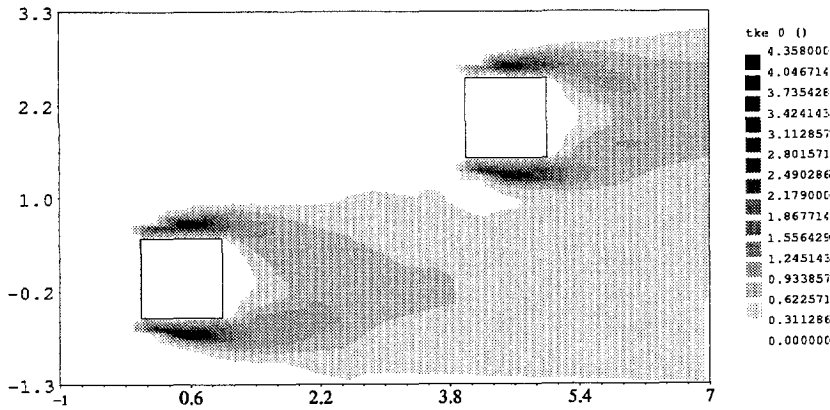


FIGURE 7.32: Iso-colour plot of the turbulent kinetic energy in a plane parallel to the channel floor at $y/H = 0.5$ for the off-axis tandem with $S_x/H = 2$ and $S_z/H = 1$ ($Re_H = 3900$).

7.2.2 Distributions of the local heat transfer coefficient

Because the local heat transfer characteristics and their associated flow features were already discussed in detail for the in-line tandem, the discussion in this section

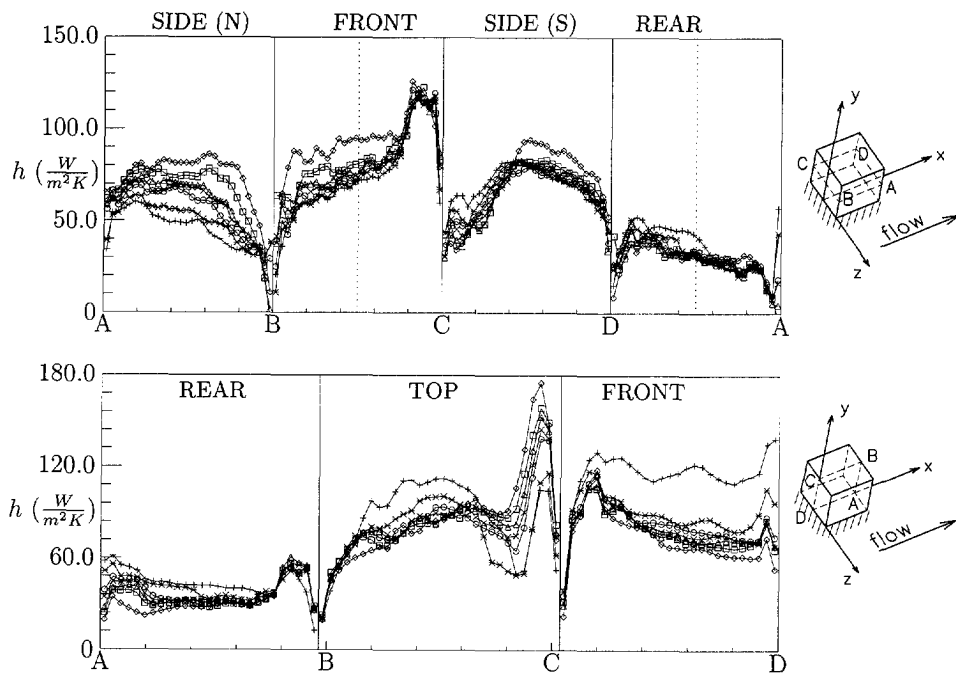


FIGURE 7.33: Heat transfer coefficient distributions along path $ABCD$ (upper plot) and along path $ABCD A$ (lower plot) for the downstream cube in the tandem for $S_x/H = 2$ and $S_z/H = 0.5$. The different symbols refer to different locations of the cross-sections.

will primarily be on the asymmetry of the local convective heat transfer. Significant asymmetry effects were detected in the distributions of the heat transfer coefficients for both the upstream and downstream cube for small values of S_x/H and S_z/H . Profiles of the local heat transfer coefficient are given in Fig. 7.33 along paths $ABCD A$ and $ABCD$ for the downstream cube in a typical off-axis arrangements with severe asymmetry effects, i.e. $S_x/H = 2$ and $S_z/H = 0.5$. The dotted lines refer to the centreline of the cube and denote the eventual symmetry lines ($x/H = 0.5$). It is seen that the profiles are significantly asymmetric. The flow acceleration through the corridor between the two cubes caused the distinct peak in h at the front face. This flow acceleration was shown for $S_x/H = 2$ and $S_z/H = 1$ (see Fig. 7.25), but is more pronounced for smaller spanwise distances. The inner side face of the downstream cube, indicated with BC in the upper plot in Fig. 7.33, shows a marked minimum close to the leading edge, caused by the side vortex. The maximum located at the middle of this side face is due to flow reattachment. Although these flow features were observed from the flow field analysis for $S_x/H = 2$ and $S_z/H = 1$, similar flow features apply for this case. The heat transfer coefficient profiles at the outer side is caused by the flow recirculation which enveloped more or less

the entire face. The convective heat transfer at the back face of the downstream cube decays monotonically towards the outer side face of the cube. This direction is opposite to the local air speed of the rotating wake vortex. The heat removal from the surface caused an increase in the fluid temperature and, therefore, reduced the convective heat transfer. The top face (shown in the lower plot in Fig. 7.33) exhibits more or less similar heat transfer features as for the single cube. The heat transfer coefficient decays strongly up to approximately $x/H = 1/4$, where a local minimum is found. This is the location of the top vortex. The subsequent flow reattachment caused the heat transfer to become maximum at about $x/H = 1/2$.

7.2.3 Average heat transfer

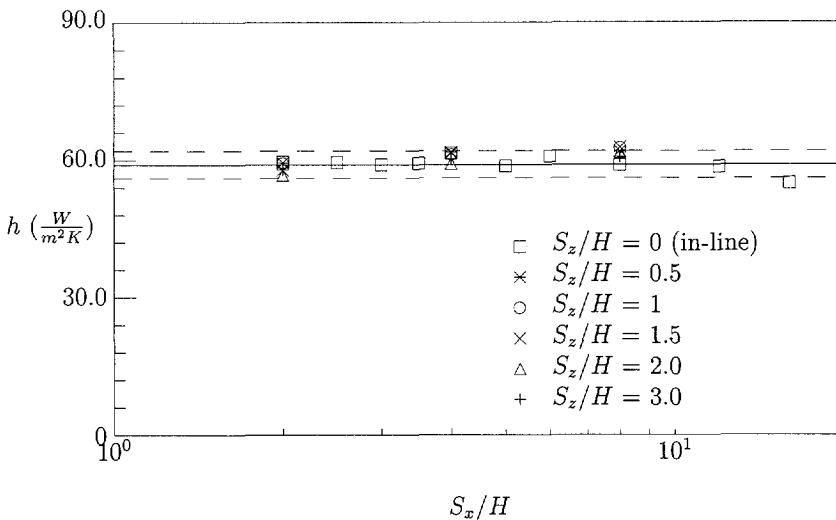


FIGURE 7.34: Cube-averaged heat transfer coefficient for different spanwise distances S_z/H as a function of the streamwise distance S_x/H . The solid line indicates the averaged value of $\bar{h} = 59 \text{ W/m}^2\text{K}$, the dashed lines indicate a bandwidth of 5%.

The local convective heat transfer appeared to be strongly correlated to the local flow structure in the proximity of the cubes. Even face-averaged heat transfer coefficients showed significant differences for the different off-axis situations considered (this is not further discussed here). But if we plot all the measured cube-averaged heat transfer coefficients for the downstream cube in the off-axis tandem as a function of the streamwise distance S_x/H , the results coincide pretty well. This can be seen in Fig. 7.34 in which the different markers refer to different spanwise distances S_z/H . The results for $S_z/H = 0$, the in-line tandem arrangements, are also included in the figure. All results were obtained for $Re_H = 3900$. The solid line refers to a cube-averaged heat transfer coefficient of $h = 59 \text{ W/m}^2\text{K}$. The dashed lines indicate a bandwidth of 5% experimental uncertainty. The scatter is remarkably

small which suggests that the overall cooling of the cube is dominated by the bulk flow. Although severe differences were observed in the distributions of the local heat transfer coefficient, the cube-averaged heat transfer coefficient apparently is not a function of the location of this cube with respect to its upstream neighbour. A similar observation was already made for the leading cube.

CHAPTER 8

Matrix of cubes

The experimental work discussed in this chapter served as reference dataset for the validation of numerical simulations at ERCOFTAC workshops held at the Delft University of Technology, at June 6-7, 1997 [62] and at UMIST, May 28-29, 1998, Manchester [88]. The usefulness of this test case is directly related to the well defined periodic boundary conditions involved which can easily be implemented in numerical simulations. Periodic boundary conditions make simulations much more reliable, reproducible and comparable. For accurate experimental results and for a fair validation of the results, it is of crucial importance that the experimental flow boundary conditions are periodic. In order to achieve this periodicity, we needed at least 15 rows of cubes to make the flow fully periodic.

An equidistantly-spaced matrix of wall-mounted cubes was placed on the channel floor of the rectangular windtunnel section of 51 mm height ($D = 51$ mm, $D/H = 3.4$) and 600 mm width. A schematic of the configuration is given in Fig. 8.1. The matrix consisted of a total of 25×10 cubes in respectively the streamwise and the spanwise direction. The inflow conditions were controlled by tripping the boundary layer approximately 1 m upstream of the first column of cubes. The cube size was $H = 15$ mm. The local origin of the cube system is defined as shown in Fig. 8.1. The x , y and z direction are in the streamwise, normal and spanwise direction, respectively. The streamwise and spanwise distances between the cubes were 45 mm, giving centre-to-centre distances of $S_x/H = S_z/H = S/H = 4$.

The flow and heat transfer measurements were carried out around the 18th row from the inlet, at mid-height of the channel. This location ensured a spatially periodic flow in both the streamwise and the spanwise direction independent of the inflow conditions, as was confirmed by velocity measurements at several locations across the entire matrix (see below). The velocity and Reynolds stress data were duplicated for several stations to verify the reproducibility of the measurements. The discrepancies between the different measurements were less than 1%.

The mounting base was machined from formaldehyde which had a low thermal conductivity of 0.33 W/mK. This low value minimised the conductive heat losses through the base plate. The flow field was measured with the LDA system for a Reynolds number of $Re_H = 3854$, based on the cube height and the bulk velocity. Smoke and oil-film visualisations were performed in a broader range of Reynolds numbers (1000-8000) to complement the LDA measurements. The convective heat transfer measurements were performed for the range: $2380 < Re_H < 5280$.

8.1 Flow field

8.1.1 Spatial periodicity

A typical example which demonstrates the spatial periodicity in the streamwise direction is given in Fig. 8.2. Shown are profiles of the mean streamwise velocity

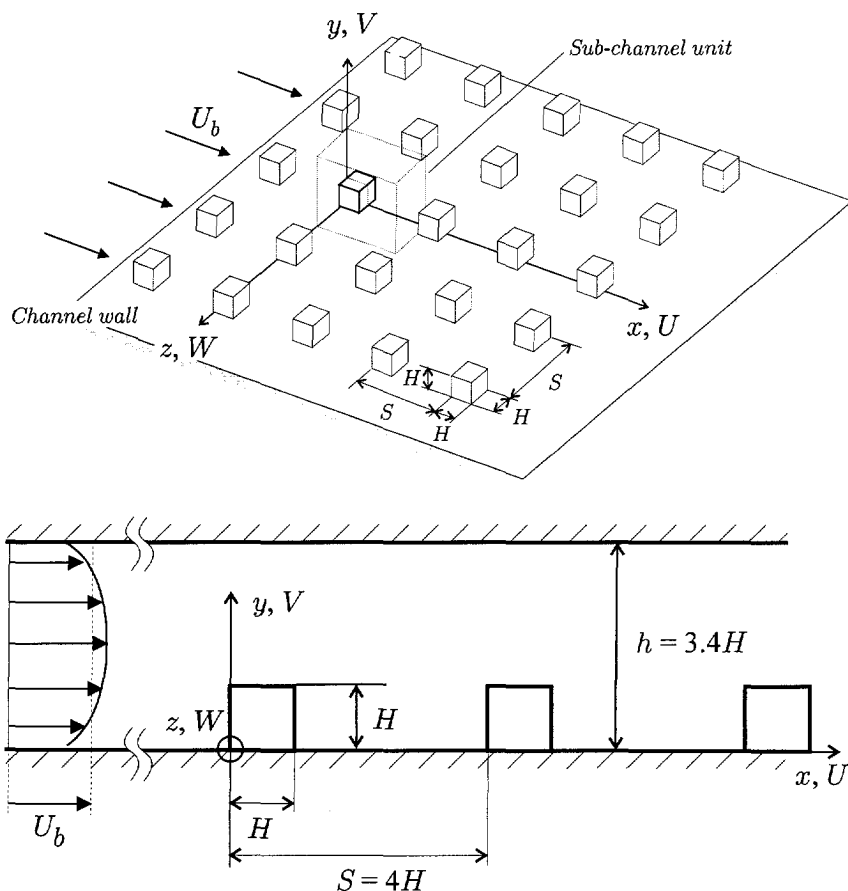


FIGURE 8.1: Three-dimensional sketch of the matrix configuration of cubes on the channel wall (upper plot) and the side view of the configuration (lower plot).

$\bar{u}/u_B$ and the Reynolds normal stresses $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$, scaled with the bulk velocity in the channel u_B , through the centre of the streamwise corridor at $z/H = 2$ and at height $y/H = 2$. It is clearly observed that both the mean velocity and Reynolds stresses asymptote to a maximum value. After roughly 15 cube rows, the flow appeared to be spatially periodic which is now referred to as fully developed. Similar experiments were performed to confirm the spatial periodicity in the spanwise direction. Profiles of $\bar{u}/u_B$, $\bar{w}/u_B$, $\overline{w'^2}/u_B^2$ and $\overline{u'w'}/u_B^2$ are given in Fig. 8.3 for a traverse in the z -direction at $x/H = 2$ and $y/H = 0.5$. The flow symmetry is apparent from the profiles.

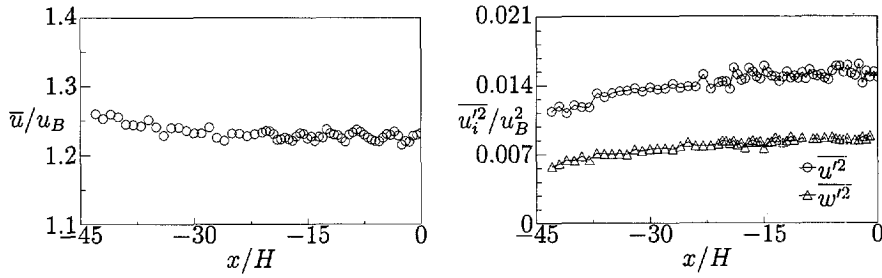


FIGURE 8.2: Profiles in the streamwise direction of $\bar{u}/u_B$ (left) and $\bar{u'^2}/u_B^2$ and $\bar{w'^2}/u_B^2$ (right) at $z/H = 2$ and $y/H = 2$.

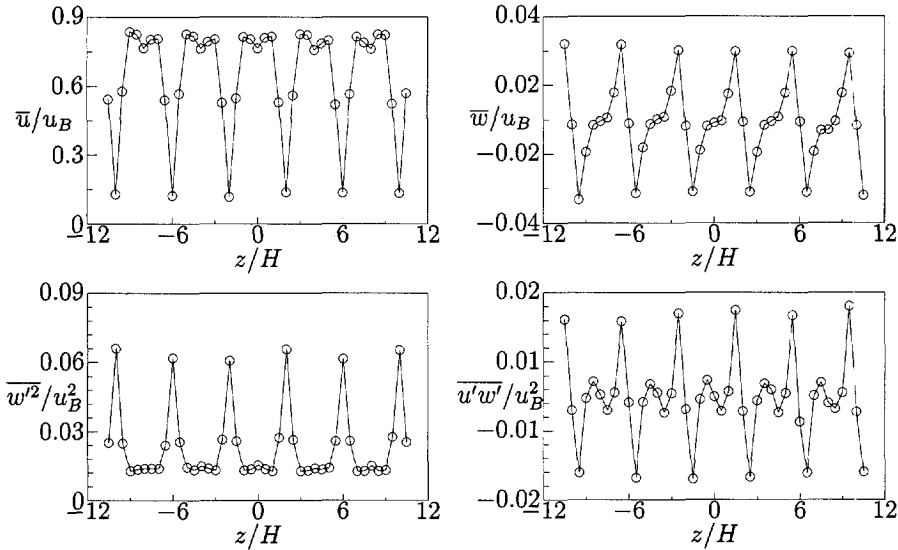


FIGURE 8.3: Profiles in the streamwise direction of $\bar{u}/u_B$ (upper left), $\bar{w}/u_B$ (upper right), $\bar{w'^2}/u_B^2$ (lower left) and $\bar{u'w'}/u_B^2$ (lower right) at $z/H = 2$ and $y/H = 2$.

8.1.2 Consistency validation

A consistency validation was performed from mass conservation through four different sub-planes at periodic up- and downstream locations of the cube. These planes ($x/H = -1.5$ and 2.5 , $z/H \geq 0$ and $x/H = -1.5$ and 2.5 , $z/H \leq 0$) are centred around the local symmetry plane $z/H = 0$ at the middle of the cube in question. The spatially periodic and symmetric situation requires a necessarily equal mass flow rate through the four sub-planes. For each sub-section, seven velocity profiles in normal direction were measured at equidistant locations. The profiles were measured at the following locations: $x/H = -1.5$ and 2.5 , $z/H = 0, \pm 0.33, \pm 0.66, \pm 1, \pm 1.33, \pm 1.66$ and ± 2 . The axial mass flow rate was determined from a numerical integration. The distance between the individual stations (5 mm) was sufficiently small

to ensure a reasonably accurate estimate of the velocity gradients in the spanwise direction. The integrations of the velocity profiles yielded an averaged mass flow rate of $6.85 \cdot 10^{-3}$ with deviations less than 1%. These small discrepancies showed the good quality of the spatial periodicity and the symmetry of the matrix flow. The averaged velocity, derived from the ratio of the mass flow rate and the integration area, yielded a value of $u_B = 3.86$ m/s. The Reynolds number based on the bulk velocity and the cube size was $Re_H = 3854$.

8.1.3 Comparison with numerical simulations

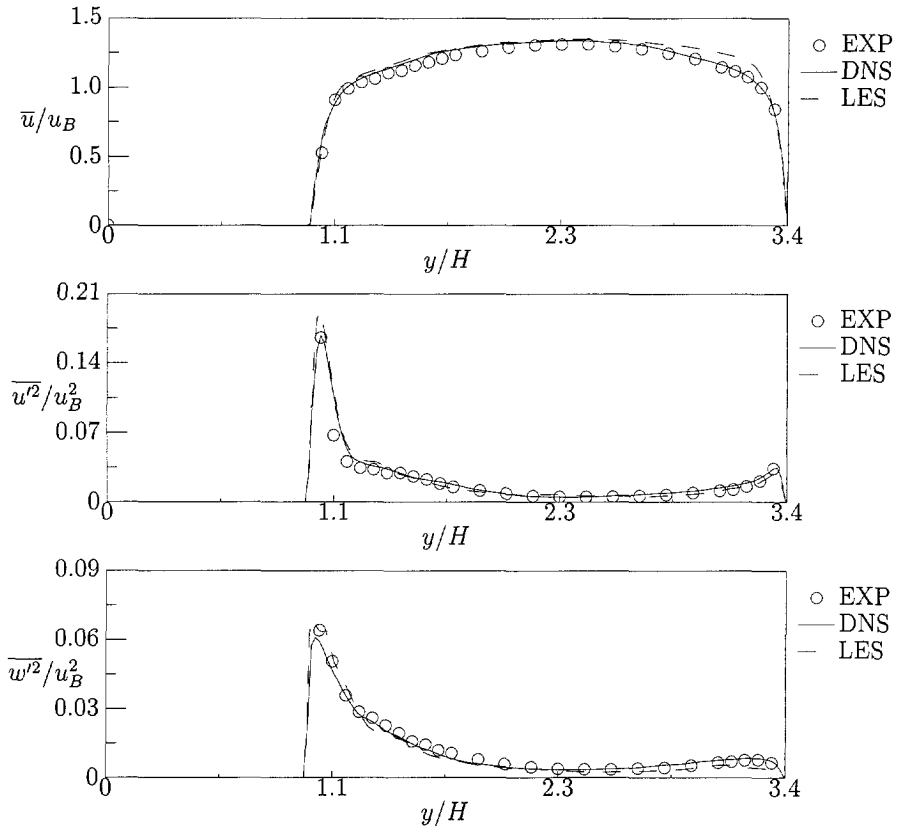


FIGURE 8.4: Profiles of $\bar{u}/u_B$ (first), $\bar{u'^2}/u_B^2$ (second) and $\bar{w'^2}/u_B^2$ (third) in the normal direction at $z/H = 0$ and $x/H = 0.3$. The markers ($\circ$) refer to the experiments (EXP), the solid and dashed lines refer to the Direct Numerical Simulations (DNS) of Verstappen and Veldman [170,171] and the Large Eddy Simulations (LES) of Mathey et al. [106], respectively.

Before starting the actual discussion of the flow and heat transfer results, measured velocity data will be compared to numerical results submitted to the ERCOF-TAC workshops on refined flow modelling (held at Delft University of Technology,

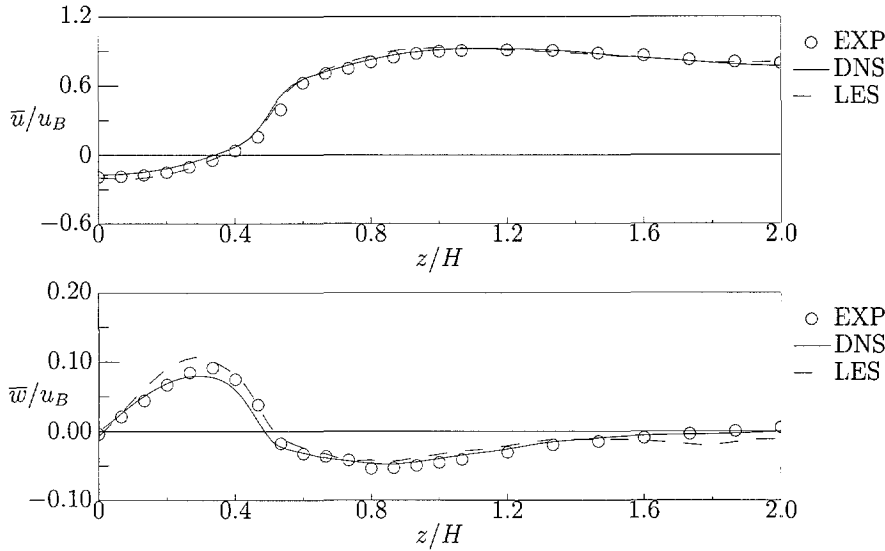


FIGURE 8.5: Profiles of $\bar{u}/u_B$ (first) and $\bar{w}/u_B$ (second) in the spanwise direction at $x/H = 1.3$ and $y/H = 0.5$. The markers ($\circ$) refer to the experiments (EXP), the solid and dashed lines refer to the DNS [170,171] and LES [106] data, respectively.

June 6-7, 1997 [62] and at UMIST, May 28-29, 1998, Manchester [88]) for two different stations. The periodic flow boundary conditions and the different flow types involved (flow recirculations, separation and reattachment) made this test case very suitable for an objective evaluation of different turbulence models and computational methods. Verstappen and Veldman [170, 171] performed a kind of Direct Numerical Simulation (DNS) which was based on a high-order, spectro-consistent discretisation method. The results of Mathey et al. [106] are large eddy simulations (LES). These methods were allowed due to the relatively low Reynolds number of the flow. Fig. 8.4 shows profiles of $\bar{u}/u_B$, $\overline{u'^2}/u_B^2$ and $\overline{w'^2}/u_B^2$ in the normal direction at $z/H = 0$ and $x/H = 0.3$. Profiles of $\bar{u}/u_B$, $\bar{w}/u_B$, $\overline{u'^2}/u_B^2$, $\overline{w'^2}/u_B^2$ and $\overline{u'w'}/u_B^2$ are given in Figs. 8.5 and 8.6 for a traverse in the spanwise direction at $x/H = 1.3$ and $y/H = 0.5$. It is observed that the numerical results, the averaged velocities as well as the Reynolds stresses, are in good agreement with the experimental results. Even the strong peaks in the Reynolds stresses are well predicted by the numerical simulations. Besides that this good agreement is a complimenting result for the computers, it also indicates that the LDA measurements were accurately performed. The numerical results show that LES and DNS are promising methods to accurately resolve this kind of bluff-body flows.

8.1.4 General flow structure

A three-dimensional impression of the flow field around the cube was derived from the LDA-measurements and flow visualisations (see Fig. 8.7). Distinct vortex struc-

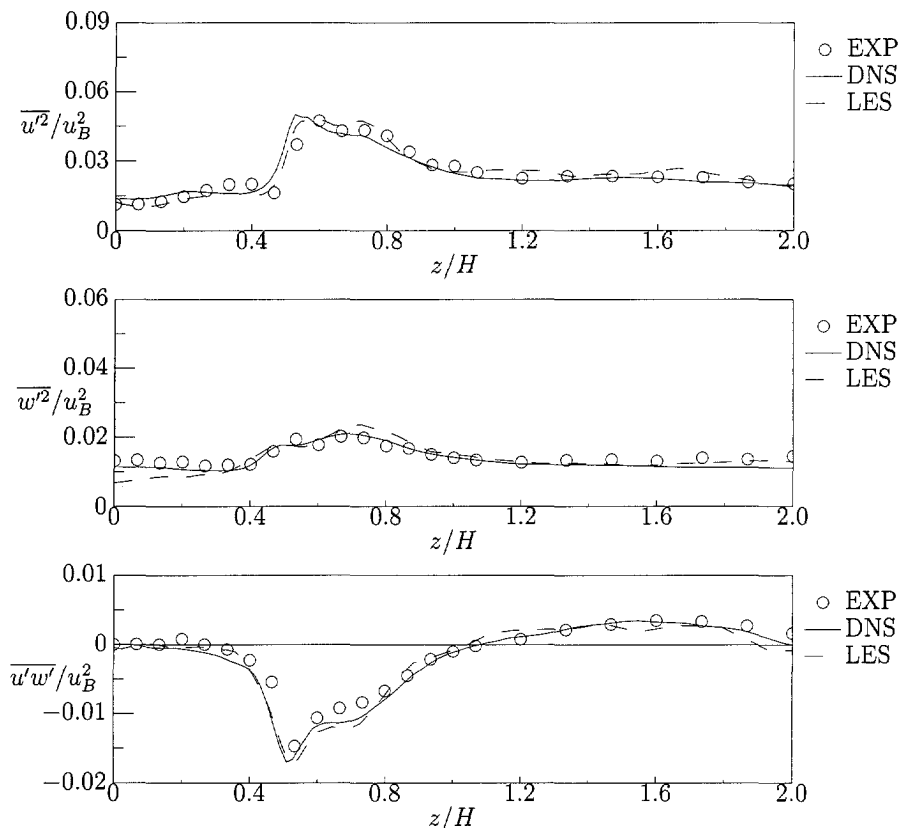


FIGURE 8.6: Profiles of $\overline{w'^2}/u_B^2$ (first), $\overline{w'^2}/u_B^2$ (second) and $\overline{u'w'}/u_B^2$ (third) in the spanwise direction at $x/H = 1.3$ and $y/H = 0.5$. The markers ($\circ$) refer to the experiments (EXP), the solid and dashed lines refer to the DNS [170,171] and LES [106] data, respectively.

tures were only observed in the immediate proximity of the obstacle, while the core flow in the region above the cubes and in the streamwise corridors between them remained almost undistorted. The flow in the inter-obstacle space is characterised by a two-cell structure arising from the intermittent reattachment of the shear layer separated at the leading top edge of the upstream cube. The long-term averaged flow reattachment occurred at a distance of approximately 1.5 cube heights ($x/H = 2.5$) from the leeward face. This shear layer bounds the arc-shaped vortex in the wake of the upstream cube. A horseshoe type of vortex was formed at the windward face of the cube and was deflected downstream along both sides of the cube, weakening further downstream through the interaction with the counter-rotating vortex along the neighbouring cubes.

A vector plot of the velocity field in the symmetry plane $z/H = 0$ is given in Fig. 8.8. The plot is a collection of results obtained from two different measurement techniques. The vectors, located in the two triangular areas upstream and

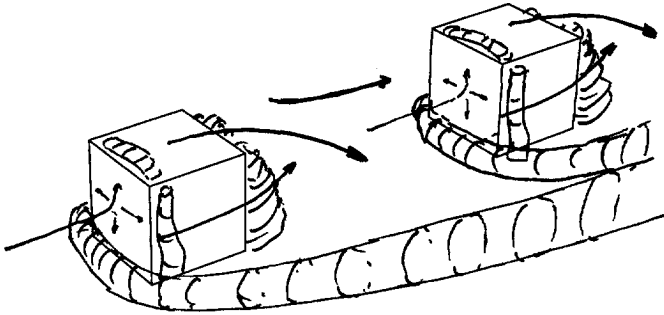


FIGURE 8.7: Three-dimensional schematic of the flow pattern around cubes in the spatially developed matrix flow.

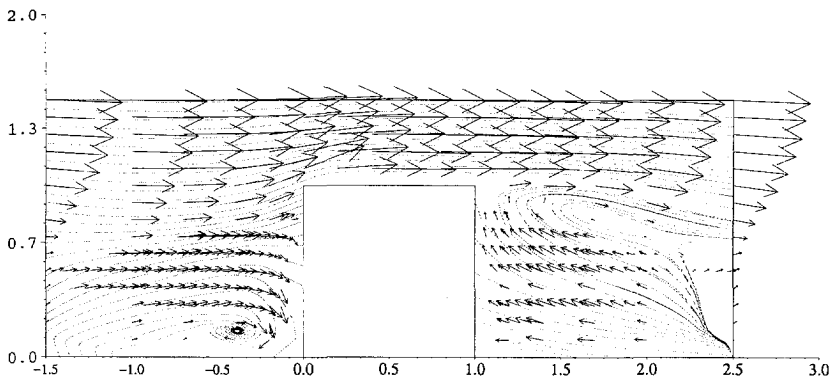


FIGURE 8.8: Long-term averaged vector plot with streamlines in the xy -plane for one spatial period in the matrix at $z/H = 0$.

downstream of the cube are composed out of the normal velocities measured with the prism cube and the streamwise velocities obtained from the direct measurement with the laser probe in the standard orientation. The figure shows that due to the geometrical constraints imposed by the prism-cube, the normal velocities could only be measured in the limited triangular regions. The other vectors were obtained with the reconstruction technique from two separate velocity measurements acquired at different inclination angles of the laser probe. Although the results obtained with the reconstruction technique were less accurate than those acquired with the prism-cube, the agreement between the results in the overlapping region provides sufficient evidence of the reliability of the reconstruction technique. The vortex structures can be readily identified from the vector plot in Fig. 8.8. A downwash of the flow, originating from the horseshoe vortex, is characterised by negative velocities in the

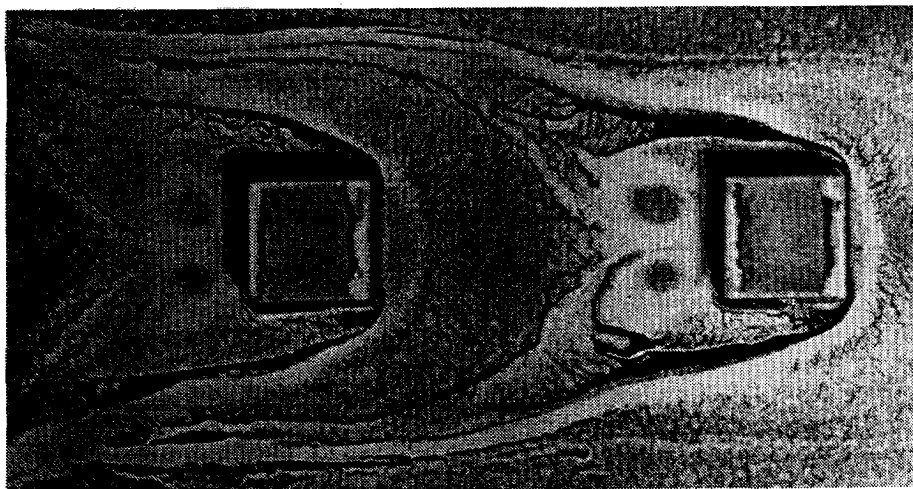


FIGURE 8.9: Oil-film visualisation of the surface streaklines of the flow in the matrix flow. It looks like that the oil-film pattern is not fully developed but this was caused by the lack of pigment upstream of the cube when the experiment was performed.

proximity of the windward cube wall. The wake flow is dominated by the arch vortex, causing a strong upwash close to the leeward face. The velocity vectors close to the top leading edge, Fig. 8.8, indicate that the flow separated and caused a recirculation bubble to exist in the vicinity of the leading top edge. Although it was not possible to measure this with the used LDA set-up, the flow reversal was apparent from the surface streak lines of the oil-film visualisations given in Fig. 8.9. The top face shows pigment removal close to the leading edge which originated from the recirculating flow. Flow reattachment was found at approximately 4 mm downstream of the leading edge, as appears from the more or less smeared pigment pattern.

A further analysis of the visualisation in Fig. 8.9 reveals the imprint of the horseshoe-type vortex in front of the obstacle as the white line, bending around the cube along the separation line. This vortex joined the horseshoe vortex generated upstream in the corridor region, see Fig. 8.9. The influence of the extensions of other horseshoe vortices, induced in front of the cubes more upstream, has diminished as is seen from the oil-film visualisation. At $x/H = 0.5$, both co-rotating horseshoe vortices interacted with each other but did not fully merge. If they are of comparable strength, the interaction at this location leads to negative normal velocities close to the obstacle side face, positive values at the symmetry line $z/H = 2$ and zero values at the interface. This hypothesis is supported by the v -measurements given in Fig. 8.10 in the yz -plane at $x/H = 0.5$. Note that the traverses shown were measured with the prism cube. Also for this orientation, the measurement was geometrically constrained. Negative velocities were found around $z/H = 0.5$ to 1 due to the first horseshoe vortex, which decreased gradually with an increase in height y/H . Positive velocities are observed around $z/H = 2$ due to the extensions of the upstream horseshoe vortex with zero velocities around $z/H = 1.4$ due to the

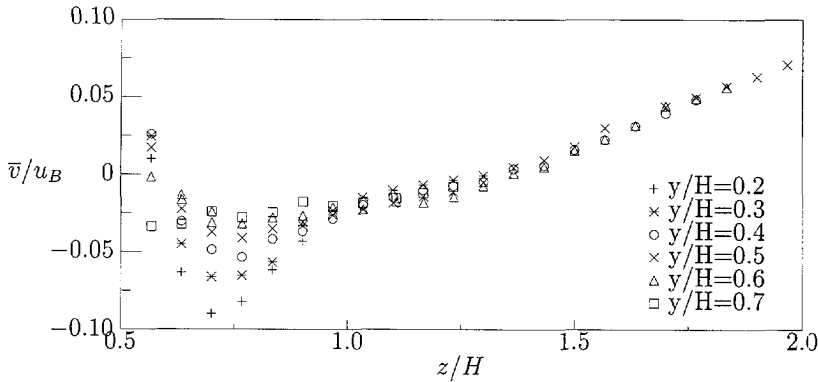


FIGURE 8.10: Profiles of the normal velocity $\bar{v}$ at $x/H = 0.5$.

superposition of the two co-rotating vortices. Because of the flow symmetry, the effects of the two extensions of the upstream horseshoe vortex are superimposed, resulting in an intensification of the effect.

Vector plots of the streamwise and spanwise velocity components u and w with superimposed trajectories are given in Fig. 8.11 for one half of a spatial period in the matrix for $y/H = 0.25, 0.50$ and 0.75 . The flow symmetry was verified with velocity measurements across the entire xz -plane at different stations showing negligible differences. This symmetry allowed us to concentrate on only one symmetric half of the spatially periodic domain. An almost undistorted corridor region between the cube arrays is obvious from the plots. The oncoming flow is bended towards the cube sides to separate at the leading side face, creating a small recirculation bubble close to the leading side edge. As a consequence, the separated shear layer reattaches at approximately 5 mm downstream from the side edge. This was demonstrated from measurements of the streamwise component at 0.5 mm distance from the wall ($z/H = 8/7.5$). These measurements show a negative velocity up to 5 mm distance ($x/H = 1/3$) from the leading edge, Fig. 8.12. The side vortices are aligned along the leading side edges with their origins at the channel floor. Their imprints are seen from the oil-film picture as the nodal pigment accumulation close to the leading side edge (see Fig. 8.9). The oil-film pattern at the side faces (not shown in the thesis) showed a white feather-type imprint of the side recirculation close to the leading edge. This structure is qualitatively similar to that found for a single cube in a developing turbulent channel flow (see Fig. 5.18).

The wake flow shows the recirculation of the arc-type of vortex. The results at the three different levels y/H show a gradual growth of the recirculation area towards the channel wall. An extrapolation of the separation line, where the velocity changes sign, gives a reattachment point at $x/H = 2.5$. The oil-film visualisation in the wake of the cube indicates two pigment concentrations which refer to two counter-rotating foci. These vorticity concentrations are the imprints of the wake

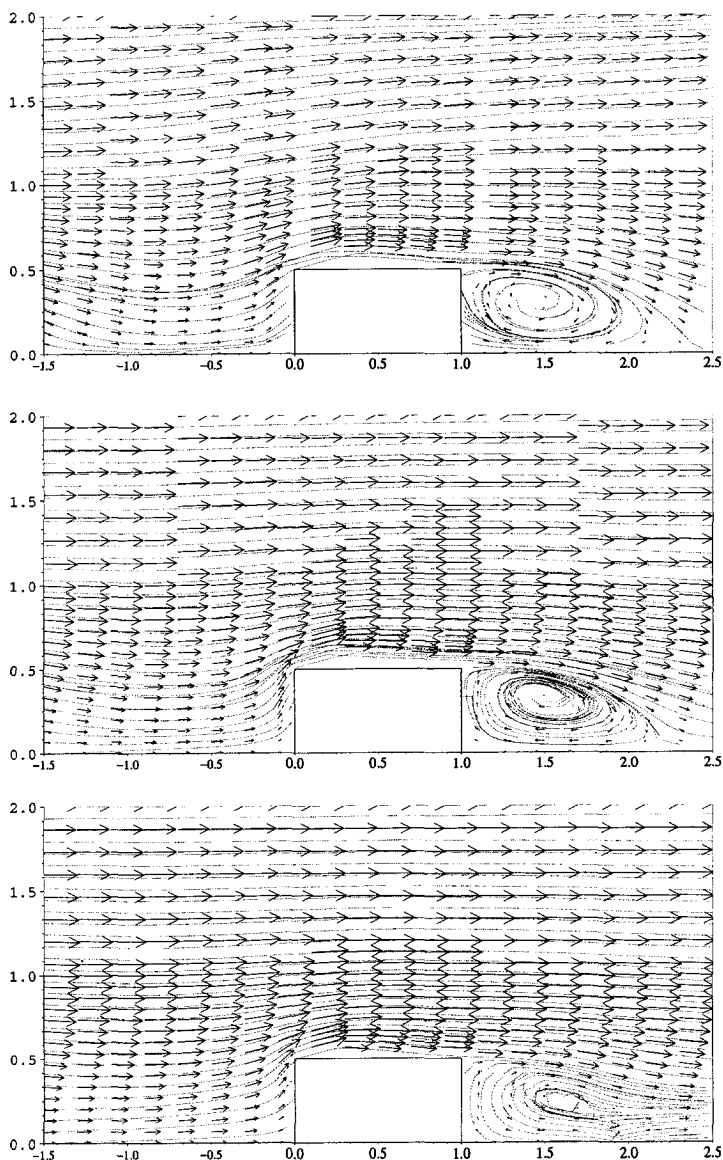


FIGURE 8.11: Vector plots with streamlines in the xz -plane for one half of a spatial period in the matrix for $y/H = 0.25$ (first), $y/H = 0.50$ (second) and $y/H = 0.75$ (third).

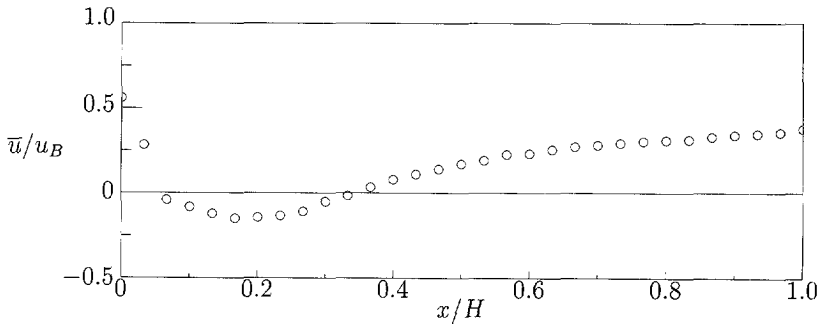


FIGURE 8.12: Profile of the mean velocity $\bar{u}$ at $y/H = 0.5$ and $z/H = 8/7.5$.

vortex and its location agrees reasonably well with the LDA measurements at $y/H = 0.25$, extrapolated towards the channel wall. The shear layer reattached in the inter-obstacle spacing at $x/H = 2.5$, as observed from the pigment accumulation. This location is in good agreement with the LDA observations. The three vector plots show that the centres of the two recirculations are at about the same location. If the wake vortex was closed, the centres of the recirculations would shift towards the symmetry line for cross-sections at increasing height y/H . This shift is not observed which suggests that the wake vortex is not closed at its top. The results indicate that the wake vortex consists of two separate spirally-winded vortices with origins at the channel floor and that their endings are diffused in the downstream free shear layer.

The vector plots further show that the streamlines in the inter-obstacle space upstream of the cube are more or less parallel and undistorted for $y/H = 0.75$, with exception of the wake vortex region and close to the windward face, whilst the streamlines are significantly curved at the plane $y/H = 0.25$. This curvature indicates that much air is advected away along the side whilst at higher locations, much more air flows over the obstacle top.

The co-rotating horseshoe vortices can hardly be seen from the u -component at $y/H = 0.25$ since no local acceleration in the streamwise direction is generated by this vortex structure. This means that the vortex is advected away with the velocity of the bypassing flow.

8.1.5 Dynamic behaviour: vortex shedding

Flow instabilities which are initiated at the side faces caused the flow to fluctuate in a periodic manner. These coherent structures in the wake flow were detected from power density spectra of the time series of the spanwise velocity component. Since the flow was highly turbulent, vortex shedding occurred at a broader range of frequencies. The dominant characteristic frequency was derived from the location in the spectrum which corresponded to the maximum energy. Long-time averaged spectra were obtained from time durations of 500,000 data points which corresponded to about 20,000-40,000 vortex shedding cycles. This number of realizations was required to ensure a sufficiently large signal-to-noise ratio. A slotting,

sample-and-hold and re-sampling method were compared to justify the accuracy of the calculated power density spectra, and thus the accuracy of the wake frequencies [1]. The three methods reproduced the wake frequencies within 0.5 Hz. This value also pertains to the uncertainty of estimating the frequencies from the spectra and the experimental uncertainty involved. A typical example of a power density spectrum is given in Fig. 8.13. The coherent structure in the flow (vortex shedding) is apparent from the increased activity around $f = 27$ Hz. The dashed line, the turbulent power density spectrum, indicates the contributions from the incoherent structures in the flow. The corresponding autocorrelation function calculated

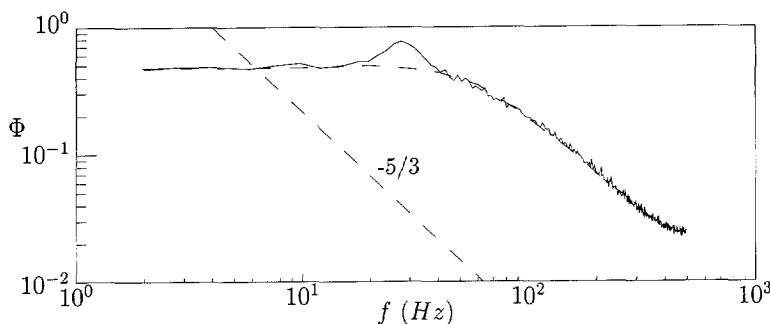


FIGURE 8.13: Power density spectrum of the time series of the v -component at the station $x/H = 2$, $y/H = 0.5$, $z/H = 0.5$ for $u_B = 3.9$ m/s.

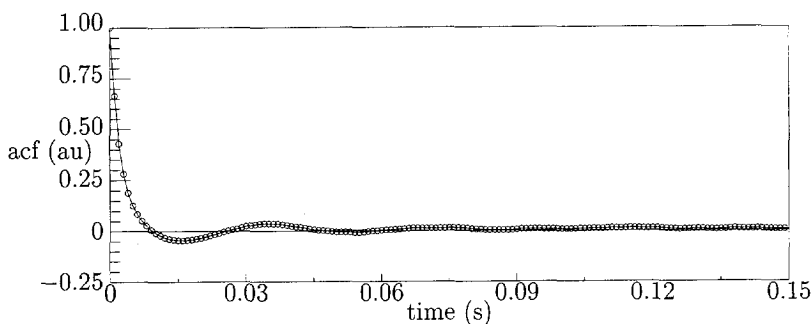


FIGURE 8.14: Autocorrelation function of the time series of the v -component at the station $x/H = 2$, $y/H = 0.5$, $z/H = 0.5$ for $u_B = 3.9$ m/s.

from the dataset in question is given in Fig. 8.14. The function is a superposition of a coherent part (a sinusoidal profile), related to the large-scale structures and of an incoherent part (the decaying profile from 1 for a zero time delay to 0 for large time delays). Typical values of the integral time scale Tu , obtained from the integration of the autocorrelation function, were of the order of 15 msec. This value was also derived from the characteristic length (the cube $H = 15$ mm) and the characteristic velocity (about 1 m/s). The second time derivative at $t=0$ gives the value for the

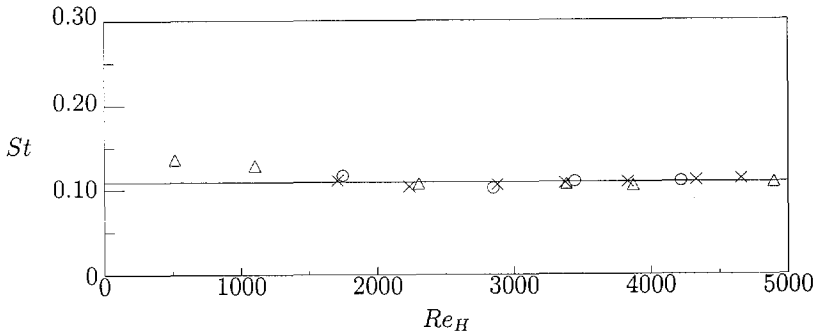


FIGURE 8.15: Strouhal numbers ($St = Hf/u_B$) versus the Reynolds number Re_H measured in the wake of a cube in the spatially periodic matrix flow.

Taylor micro scale which was in the order of magnitude of 1 msec. These values were more or less similar to the ones found for the isolated cube in a channel flow (chapter 5).

Wake frequencies were measured for a range of mass flow rates. These frequencies were made dimensionless with the bulk velocity and the cube size to yield the Strouhal number, $St = Hf/u_B$. These Strouhal numbers are plotted versus the Reynolds number in Fig. 8.15. All shown results were determined from 500,000 data points. When an experimental uncertainty of 5% is taken into account, a constant Strouhal number of $St = 0.109$ is found for the Reynolds number range of $2000 < Re_H < 6000$. The deviations for $Re_H < 2000$ are explained by larger experimental uncertainty or possible Reynolds number effects.

The value of $St = 0.109$ agrees well with the Strouhal number measured in the wake flow of the leading cube in the tandem configuration of cubes placed in a turbulent boundary layer flow for $S_x/H = 4$ (see Fig. 7.13). The tandem of cubes placed in the fully developed turbulent channel at the much lower Reynolds number of $Re_H = 3900$, subject of study in chapter 7, was characterised by a significantly lower Strouhal number, i.e. about $St = 0.08$ for $S_x/H = 4$. The large turbulence intensity of the oncoming flow in the matrix configuration of cubes is a possible explanation for this good agreement with the Strouhal number for the turbulent boundary layer flow. This observation is supported by the conclusion drawn in chapter 5 that, among others, the increased turbulence level of the oncoming flow caused a higher shedding frequency for the single cube placed in the high-Reynolds number fully developed channel flow: compare $St = 0.145$ for $Re_H = 40000$ [101] and $St = 0.095$ for $Re_H = 3900$, chapter 7.

It is noted that the power density spectrum in Fig. 8.13 decays with a power of $-5/3$ for the inertial subrange which is usually found in turbulence spectra for high Reynolds numbers away from the wall. Also note that the noise level becomes important around the truncation of the spectrum, $\Phi \approx 0.03$.

8.1.6 Second order moments

Grey-value representations of the Reynolds normal stresses $\overline{u'^2}$ and $\overline{w'^2}$ in the plane of symmetry, given in Figs. 8.16 and 8.17, directly show the anisotropy of the turbulent stress fields. Note that both plots are presented at the same scale to allow a

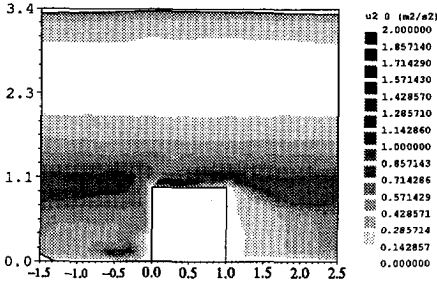


FIGURE 8.16: Grey-values plot of the Reynolds normal stresses $\overline{u'^2}$ for one spatial period at the symmetry plane $z/H = 0$.

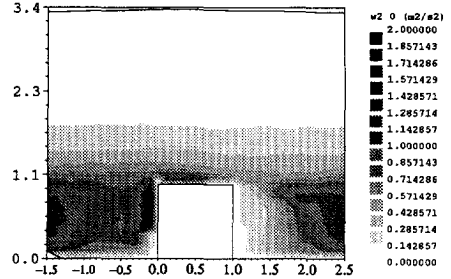


FIGURE 8.17: Grey-values plot of the Reynolds normal stresses $\overline{w'^2}$ for one spatial period at the symmetry plane $z/H = 0$.

quantitative comparison. While the streamwise fluctuations $\overline{u'^2}$ become largest in the shear layer and horseshoe vortex region, the fluctuations in the spanwise velocity $\overline{w'^2}$ exhibit maxima where the side shear layers interfere, which is approximately around the reattachment point $x/H = 2.5$. The vortex shedding from the lateral faces caused significant fluctuations in the z -direction. However, the fluctuations $\overline{w'^2}$ in the wake vortex region are rather small. Apparently, the unstable shedding behaviour is only prevalent in the shear layer enveloping the arc vortex and is almost diminished at the centreline $z/H = 0$. A qualitatively similar maximum in the spanwise velocity fluctuations around the reattachment point was also observed for a single wall-mounted cube in a fully developed turbulent channel flow [101]. For this single cube configuration, the mechanism of vortex shedding contributed also to the production of turbulence.

The maxima in $\overline{u'^2}$ in the vicinity of the top face, where the shear layer separates and subsequently reattaches, are accompanied by a strong streamwise velocity gradient $\frac{\partial \overline{u}}{\partial x}$, see Fig. 8.8, leading to large values of the streamwise normal stress production term $\overline{u'^2} \frac{\partial \overline{u}}{\partial x}$. Although the velocity gradient diminishes further downstream in the shear layer, $\overline{u'^2}$ contributes still significantly to the streamwise normal stress production.

Measurement of the Reynolds normal stresses $\overline{v'^2}$ with the prism-cube provided the third component for the turbulent kinetic energy $k = 0.5 (\overline{u'^2} + \overline{v'^2} + \overline{w'^2})$ in the triangular region. It is noted that the Reynolds normal stresses $\overline{v'^2}$ obtained with the reconstruction technique were not accurate enough and are therefore, not considered in this thesis. Values of k are given in Fig. 8.18 in the two triangles in front and in the wake of the obstacle. In the remaining part of the spatial period, $\overline{v'^2}$ is assumed as $\overline{v'^2} = 0.5 (\overline{u'^2} + \overline{w'^2})$ leading to $k_{2D} = 0.75 (\overline{u'^2} + \overline{w'^2})$ where $\overline{u'^2}$ and $\overline{w'^2}$ were

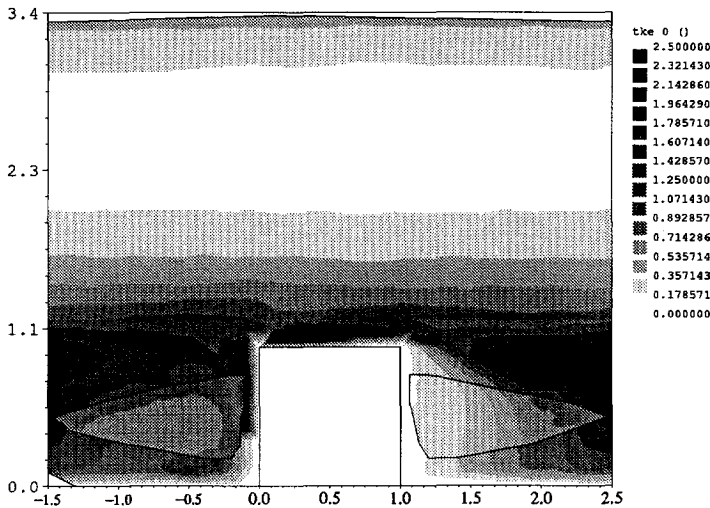


FIGURE 8.18: Turbulent kinetic energy k for one spatial period at the symmetry plane $z/H = 0$.

obtained from the LDA measurements with the probe in the standard orientation. These values of k_{2D} are also given in Fig. 8.18. Although the above assumption is not fully accurate, particularly in the wake and reattachment region (see Figs. 8.16 and 8.17) the agreement between k and k_{2D} is sufficiently good (differences are at maximum 10 %) to allow at least a qualitative interpretation of the turbulent kinetic energy field with k_{2D} .

The turbulent kinetic energy above the cubes (the core region of the channel) is rather small and increases gradually towards the opposing channel wall. The large Reynolds normal stresses found in the shear layer region and around the reattachment point lead to local maxima in the turbulent kinetic energy. A maximum in k was found downstream in the shear layer at approximately one cube distance behind the leeward face where the shear layer is broadening and the turbulence level is decaying due to turbulent advection. The large values of k in the horseshoe vortex region at the windward face of the obstacle indicates that this vortex structure is unstable and highly turbulent. Unstable behaviour of the horseshoe vortex was also observed for a single cube in a fully developed channel flow [101, 86]. For this single cube, bimodal but non-periodic behaviour of the velocity components was observed which was attributed to the flipping between two flow modes. In the case of a spatially periodic matrix flow, no bimodal behaviour was detected from the measured probability density functions in the horseshoe region.

The turbulent kinetic energy, $k_{2D} = 0.75 (\overline{u'^2} + \overline{w'^2})$ in the xz -plane is given in Fig. 8.19 for one half of a spatial period in the cube matrix for $y/H = 0.25$, 0.50 and 0.75. The flow impingement at the windward obstacle face caused local maxima in k , in particular at $y/H = 0.50$. The increased turbulent kinetic energy at $y/H = 0.25$ close to the front face of the cube is attributed to the horseshoe

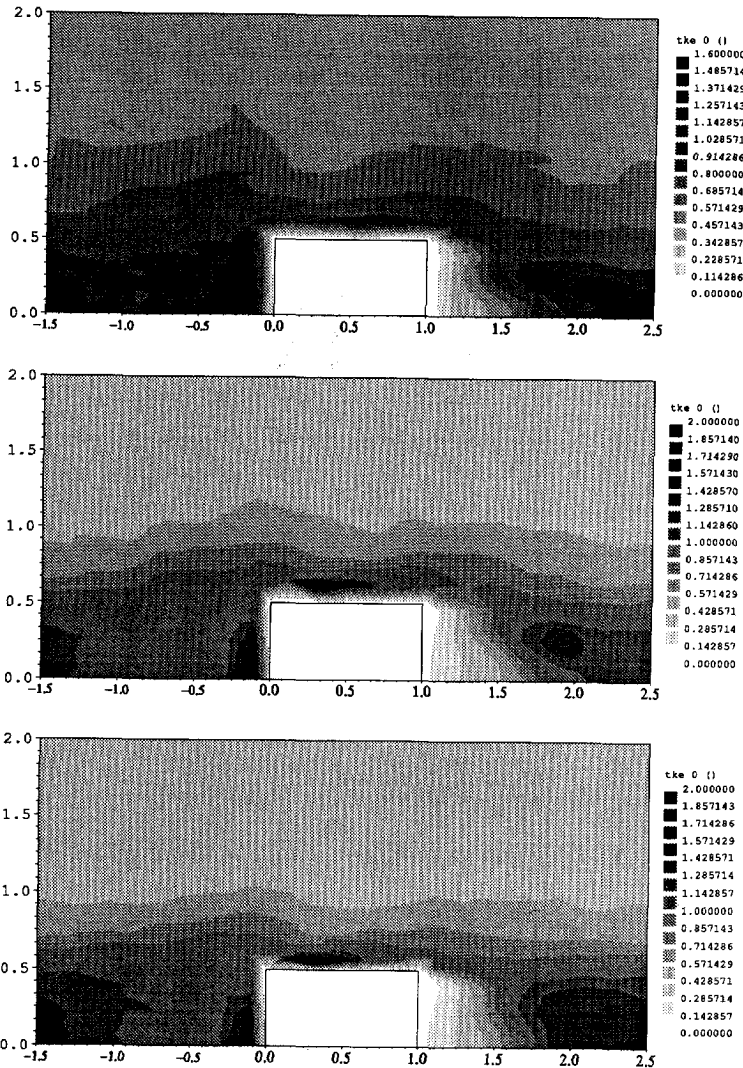


FIGURE 8.19: Grey-values plots of the turbulent kinetic energy $k_{2D} = 0.75 (\overline{u'^2} + \overline{w'^2})$ in the xz -plane for one half of a spatial period in the matrix for (a) $y/H = 0.25$, (b) $y/H = 0.50$ and (c) $y/H = 0.75$.

vortex. Similarly to the top face, the local maxima close to the sides were caused by the intermittent reattachment of the shear layers which were separated at the sharp leading side edges. Flow instabilities at these faces are amplified and initiate the vortex shedding behind the trailing side edges. The high turbulence levels directly behind these faces are, therefore, explained by the periodically fluctuating velocities.

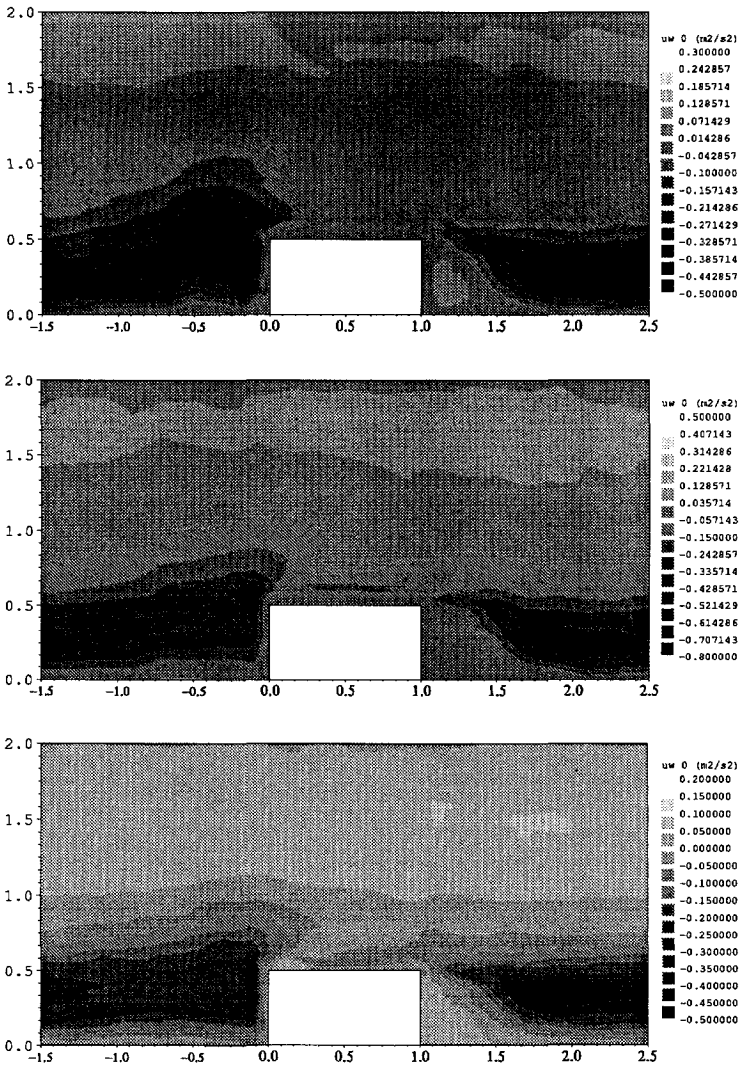


FIGURE 8.20: Grey-value plots of the Reynolds shear stresses $\overline{u'w'}$ in the xz -plane for one half of a spatial period in the matrix for (a) $y/H = 0.25$, (b) $y/H = 0.50$ and (c) $y/H = 0.75$.

Again the turbulent kinetic energy is rather low in the centre of the wake vortex indicating that the perturbations induced by the vortex shedding are damped. High k -concentrations were found at the symmetry-line at approximately $x/H = 2$ which is mainly due to the turbulent interaction between both side shear layers.

The Reynolds shear stress $\overline{u'w'}$ exhibits maxima at locations of large curvature of the mean streamlines. For example, at the obstacle windward face, the flow is

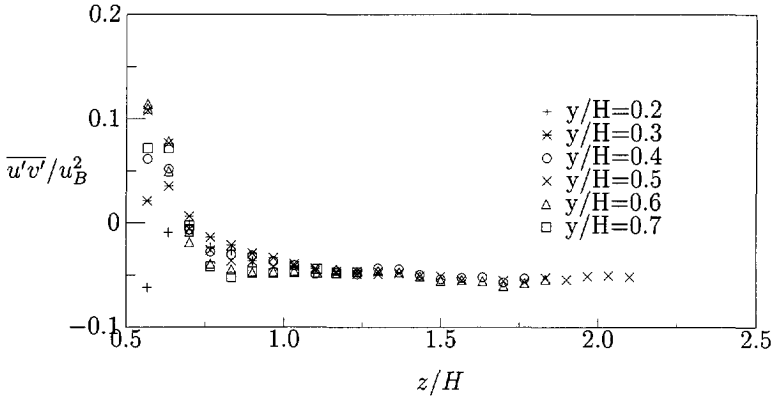


FIGURE 8.21: Profiles of the dimensionless Reynolds shear stresses $\overline{u'v'}/u_B^2$ at $x/H = 0.5$ parametric in height y/H .

directed around the cube causing large values in $\overline{u'w'}$, see Fig. 8.20 (the figures show iso-value plots of the Reynolds shear stress $\overline{u'w'}$ at $y/H = 0.25, 0.5$ and 0.75). The strong interaction between the wake vortex and the shear layer caused the maxima in the obstacle wake. The maxima shift towards the plane $y/H = 0$ where the flow reattaches and gradually move outwards for locations closer to the cube-front. Reynolds shear stresses $\overline{v'w'}$ are given in Fig. 8.21 for $x/H = 0.5$. These principle values were measured with the prism-cube when the measuring volume was reflected into the second quadrant, the side area of the cube (see Fig. 4.33 for the orientation in question). It is mentioned that these cross-terms can not be measured with conventional Laser Doppler Anemometry system since the measurement equipment will then block the flow. The values were again obtained only in the triangular shaped region due to the geometrical restrictions of the method. The shear stresses become maximum in the vicinity of the first horseshoe vortex which is expected due to the strong curvature of the flow in that region.

8.1.7 Vorticity fields

The grey-values representations of the normal component of the vorticity vector $\vec{\omega}$, $\omega_y = (\frac{\partial w}{\partial x} - \frac{\partial u}{\partial z})$, given in Fig. 8.22 show large vorticity concentrations in the near obstacle wake directly behind the lateral faces at locations where the velocity field exhibit strong spatial gradients. These maxima coincide with maxima in the turbulent quantities, represented as k . The vorticity concentrations at the side faces close to the leading edges are due to the rotating side vortices. The vorticity concentrations further downstream at the lateral faces are artifacts and were caused by the discontinuity in the velocity field: velocity measurements were performed up to 1 mm measured from the side edges.

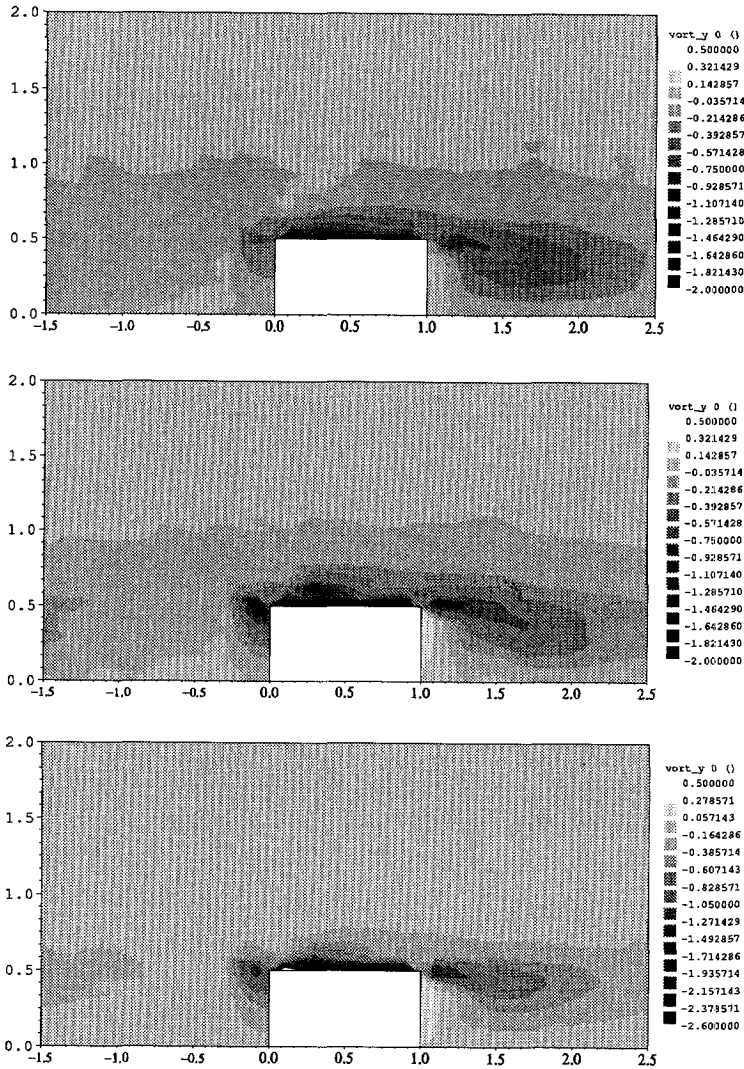


FIGURE 8.22: Grey-values plots of the vorticity $\omega_y = (\frac{\partial w}{\partial x} - \frac{\partial v}{\partial z})$ in the xz -plane for one half of a spatial period in the matrix for (a) $y/H = 0.25$, (b) $y/H = 0.50$ and (c) $y/H = 0.75$.

8.2 Convective heat transfer

Heat transfer was measured at one powered cube in the spatially periodic matrix flow, while all other cubes remained unpowered. The inlet temperature was used as a reference which, in this case, equals the adiabatic surface temperature [113, 114]. The heat transfer measurements were reproduced at different cubes in separate ex-

periments and the results were well within the 10% uncertainty limit. Heat transfer experiments were performed in the Reynolds number range of $2380 < Re_H < 5280$. The temperature of the copper core was kept at 75°C for all the heat transfer measurements. The inlet temperature varied between $19\text{--}21^\circ\text{C}$ (the real inlet temperature was measured for each experiment and used for further processing of the data).

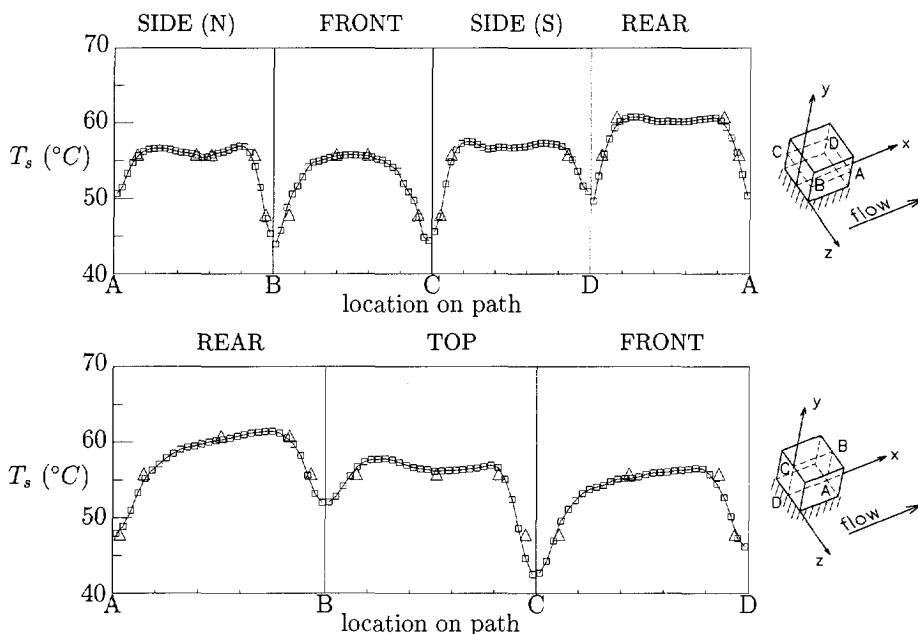


FIGURE 8.23: Comparison between liquid crystal (Δ) and infrared (lines with $\square$) surface temperature measurements along paths ABCDA (upper plot) and ABCD (lower plot) for $Re_H = 3854$.

The surface temperature distributions were validated with liquid crystal isotherm measurements. Surface temperature profiles are given in Fig. 8.23 for a Reynolds number of $Re_H = 3854$. These measurements are along the paths ABCDA and ABCD, which denote cross-sections at $y/H = 0.5$ and $z/H = 0$. The surface temperatures of both results coincide within the claimed experimental uncertainty of 0.4°C . Note that large spatial temperature gradients exist at a rather small surface (15 mm).

8.2.1 Average convective heat transfer

The surface-averaged heat transfer coefficients were derived from the surface-integrated convective heat flux and surface temperature. The cube-averaged heat transfer coefficients were derived in a similar way, but the convective heat flux and surface temperature are then an average over all the five faces of the cube exposed to air.

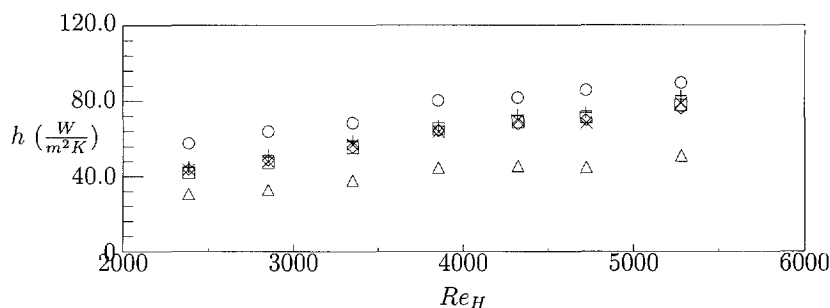


FIGURE 8.24: Surface-averaged heat transfer coefficient as a function of the Reynolds number Re_H : (+) side (north), (o) front, (x) side (south), (Δ) rear, ($\square$) top and ($\diamond$) cube average.

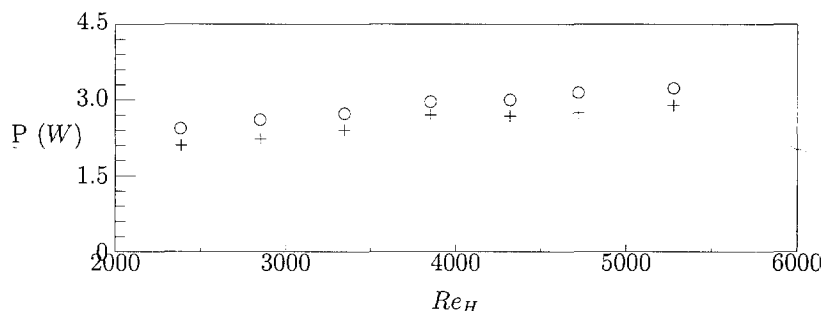


FIGURE 8.25: Total supplied power and the total heat through the five faces of the cube exposed to air flow as a function of the Reynolds number Re_H : (+) heat flow and (o) power supply.

flow. The results are shown in Fig. 8.24 as a function of the Reynolds number Re_H . Despite a small experimental uncertainty, the data follow an almost linear regression. A large heat transfer coefficient found at the front face is indicative for the impinging flow at this face. The side and top faces typically have the lowest averaged convective heat transfer coefficient which is due to the flow reversal, as shown later. The face-averaged heat transfer at the rear face is the lowest of all faces. The small deviations in the face-averaged heat transfer coefficients can be traced back to deviations in the power supply. The sum of the total convective and radiative heat transfer from the five faces exposed to the air flow is denoted as (+) in Fig. 8.25 as a function of the Reynolds number Re_H . The total power supplied to the heat source in the cube, which is product of the voltage drop and the current, is given in the same figure, indicated with (o). The difference between both values is the heat leakage through the sixth face, the interface between the wall-mounted cube and the channel wall, and consists of conduction losses through the base plate and lead wires. For all considered Reynolds numbers, this heat loss was approximately 10-15 %, where the smaller values pertain to the higher Reynolds numbers and vice versa.

8.2.2 Local convective heat transfer

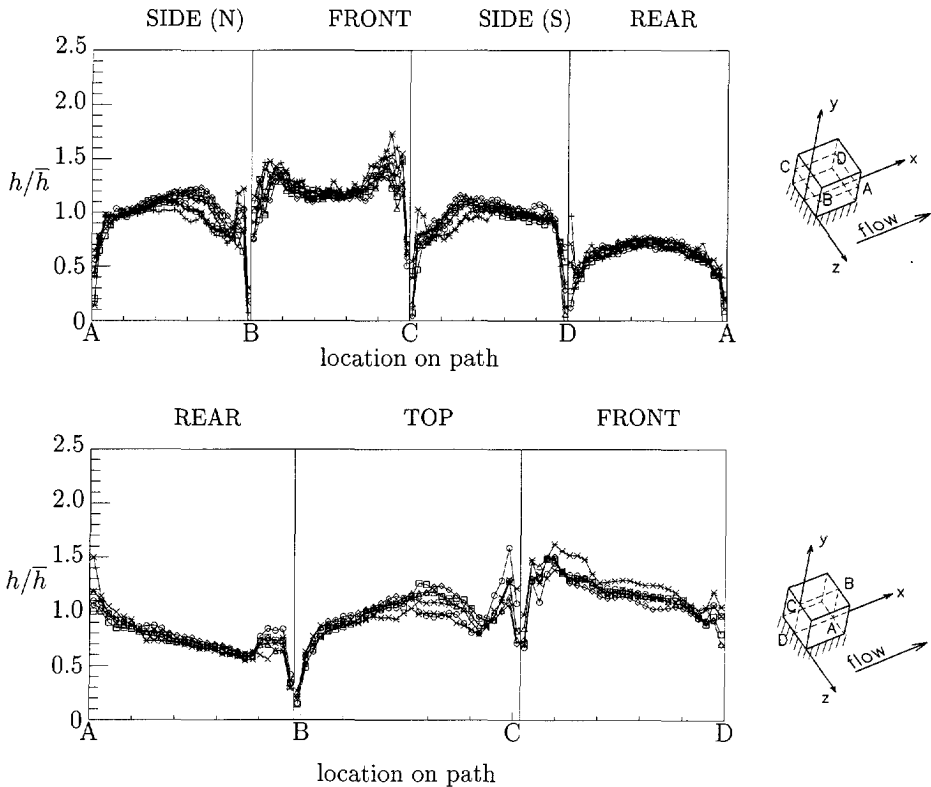


FIGURE 8.26: Heat transfer coefficient distributions along paths $ABCD$ and $ABCD$ as a function of the Reynolds number Re_H : (*) 2851, ($\circ$) 3348, ($\times$) 3854, ($\triangle$) 4321, ($\square$) 4722 and ($\diamond$) 5279.

Distributions of the heat transfer coefficient are shown along the paths $ABCD$ and $ABCD$ in Fig. 8.26 for different Reynolds numbers. The profiles are normalised with the cube-averaged heat transfer coefficient for the corresponding Reynolds number, as given in Fig. 8.24, and collapse within the experimental uncertainty. This suggests that the local heat transfer characteristics are similar throughout the range of Reynolds numbers considered ($2380 < Re_H < 5280$). Because of this similarity, the discussion can be focussed on one characteristic distribution of the local heat transfer coefficient, related to the characteristic flow field. Fig. 8.26 reveals that the heat transfer coefficient distributions show large spatial gradients at a relatively small surface which reflects a complexity of the flow structure around the matrix with flow recirculations and reattachment. Despite this large variation, the heat transfer displays a full symmetry indicating that local buoyancy effects are negligible. The flow is, therefore, classified as forced convection for all Reynolds numbers considered.

As mentioned earlier, the heat transfer at the windward face (FRONT) benefits

from the turbulent flow impingement and is maximum compared to the other faces. The peaks towards the side edges, see Fig. 8.26, are due to a local acceleration of the fluid, and thus causing a local increase in the heat transfer coefficient. Close to the mounting wall edge, the heat transfer coefficient shows a local minimum (the front face in the lower figure of Fig. 8.26) which is traced back to the presence of the horseshoe vortex. Heat convected away from the hot cube surface recirculates in the vortex causing the local vortex temperature to rise. This increased temperature level prevents a beneficial removal of heat at this location. A similar observation was made for the side faces (SIDES) of the cube. The side recirculations acted as insulation layers and, therefore, caused local minima close to the leading edge. On the contrary, the fresh air from the intermittent reattaching shear layer at approximately 5 mm downstream towards the trailing edge caused the local maxima in the heat transfer coefficient. Similar features were found at the top face, shown in the lower plot in Fig. 8.26 (TOP). The bound vortex, located in the vicinity of the top leading edge, caused the heat transfer to become minimum at about the centre of the vortex core. The subsequent maximum is due to the reattachment of the separated shear layer. The convective heat transfer from the leeward face is worst compared to the other faces which is mainly attributed to the wake vortex. The recirculating heat prevented also a beneficial cooling for this face.

Part IV

Conclusion

CHAPTER 9

Concluding discussion

The convective heat transfer from wall-mounted cubes placed in a turbulent channel flow has been studied in detail for four different test cases which included an isolated cube, a tandem of cubes, a row of cubes and a matrix of cubes. The research was mainly based on the experimental results obtained with Laser Doppler Anemometry and infrared thermography. In this chapter, the usefulness and accuracy of both measurement techniques are briefly discussed. Further, the major flow and convective heat transfer results of the four different test cases are discussed.

9.1 Infrared surface temperature measurements

Infrared thermography was used to measure the surface temperature distributions of wall-mounted cubes placed in a turbulent channel flow. A major difficulty in measuring surface temperature distributions of small three-dimensional bodies was maintaining a high accuracy and large depth-of-field without reducing the spatial resolution. This problem was solved by using an infrared image restoration technique with a Wiener filter which was based on the measured optical transfer function (OTF). For the application considered, it has been proven that the measured OTF was independent of the temperature of the scanned targets (temperatures in the range 30-70 °C). Further, a depth-of-field of at least $\pm 5\text{mm}$ was ensured by measurements of small out-of-focus step-edge responses. An in situ calibration was used to account for miscellaneous environmental contributions to the total intensity received by the camera.

Based on the surface temperature measurements from the independent liquid crystal technique, it was demonstrated that the temperature measurements which were not processed with the restoration filter showed systematic discrepancies up to 3 °C. The restored infrared temperatures coincided with the liquid crystal temperatures within the claimed experimental uncertainty of 0.4 °C. This result illustrates the necessity of image restoration in order to ensure a temperature accuracy of 0.4 °C for the entire surface of the cubes in question.

If infrared thermography is used without proper calibration and detailed analysis of the situation in which the infrared system is employed, substantial error may be encountered in the temperature reading. It is very important to validate measurements with a detailed accuracy analysis and preferably with additional measurements from a second independent measurement technique.

9.2 Laser Doppler Anemometry measurements

A standard single sample uncertainty analysis was performed to evaluate the accuracy of one velocity realisation with the laser probe in the standard orientation.

Additional contributions to the error, like misalignment and bias effects, were reported. From this analysis, it was concluded that the averaged velocities were accurate within 5% while the higher order moments were less accurate (between 7-10%). The results obtained with the prism cube had errors of comparable magnitude. However, the normal velocities obtained with the reconstruction of inclined velocity measurements had a somewhat larger uncertainty due to additional experimental error (10% and 20% for the averaged velocity and second order moments, respectively).

The claimed accuracy of the results was confirmed by several means, for example measurements in a laminar flow to verify the alignment of the system (absence of fringe gradients and chirps), comparison with literature results, comparison of similar results obtained from different techniques and so on.

Although LDA is a mature measurement technique widely used in flow dynamics research, it can never be considered as a black box. It is of major importance to evaluate all steps carefully, from alignment of the optics, to data acquisition and processing. Even if all steps are performed accurately, one should interpret the results with care.

9.3 Single cube

The time-averaged flow field around an isolated cube consisted of the horseshoe vortex, the top and the side vortices and the arch-shaped wake vortex. Some differences were detected as compared to the fully developed high-Reynolds number channel flow with a channel-height-to-cube-height ratio of 2, studied by Martinuzzi [101], which case is the closest to the present situation. Shear layer reattachment at the top and side faces of the cube was found for the present low-Reynolds number case while this was absent in the fully developed high-Reynolds number case from literature. Further, the flow reattachment at the channel floor downstream of the cube occurred at a larger distance than in the fully developed channel flow. The vortex shedding frequency was also different: the Strouhal number was $St = 0.095$ in the present case while it was $St = 0.145$ in the fully developed high Reynolds number flow.

The local flow structure had a significant effect on the local convective heat transfer from the cube. At the windward face, it was observed that the heat transfer coefficient is uniform about the stagnation point, commonly found for impinging flows. A local heat transfer minimum was found close to the base corner corresponding to the position of the horseshoe vortex core. Although this location coincided with an area of high wall shear stresses it was argued that the local increase in the fluid temperature prevented a beneficial removal of heat. Flow separation occurred at the leading side edges and flow instabilities caused significant vortex shedding behind these faces. Maxima of the heat transfer coefficient occurred in the proximity of the reattachment points, where the velocity fluctuations normal to the wall are large. Minima in h were found below the recirculation vortex cores, where the wall shear stress is high but where the local fluid temperature is also locally higher. The increased temperature of the vortex acted as an insulation layer and prevented

beneficial convective removal of heat. Similar features were found at the top face. The arch vortex in the wake of the cube caused rather uniform distributions of the local heat transfer coefficient.

Measurements of the static surface pressure were in agreement with the LDA-observations. A suction peak was found underneath the top and the side vortices while the reattachment region was characterised by a pressure maximum. This again illustrated the dissimilarity between surface heat transfer and the local flow structure.

A laminar developing channel flow was achieved by removing the tripping wire upstream of the cube such that a smooth entrance length remained. This situation enabled us to study the effect of the oncoming flow condition on the convective heat transfer. The vortices in the proximity of the isolated cube were generally larger in the laminar case. In particular the convective heat transfer at the top and side faces of the cube was significantly lower than for the turbulent case for a similar Reynolds number. Differences between the tripped and untripped results were negligible for $Re_H \leq 2200$. $Re_H \approx 2500$ was identified as a transition point for which the tripping wire became effective.

9.4 Row of cubes

The flow field study revealed that a horseshoe vortex structure originated directly upstream of the leading obstacle base. Flow separation from the sharp leading edges caused the forming of vortices at the top and side faces of the leading cube. Ordinary shear flow was found for the second and downstream cubes in the row. The complex vortex structure of the flow-field around the individual cubes caused a large non-uniformity in the convective heat transfer distributions. In particular, the flow separation and reattachment at the top and side faces of the first cube and the strong arch-shaped vortices between subsequent cubes gave rise to significant differences in the local heat transfer coefficients. It was shown that regions of intense flow separation are usually associated with a poor convective heat transfer and, in contrast, regions with flow reattachment exhibited the opposite behaviour.

A substantial variation in the averaged convective heat transfer was found along the array of cubes, in particular for the first three subsequent cubes. The leading cube appeared to have the lowest averaged heat transfer coefficient for the three highest Reynolds numbers studied. This was a consequence of the strong vortices at both the lateral and top faces. The second cube, however, is situated in the wake of the first cube and experiences a direct interaction with the horseshoe vortex and the separated shear layer which flows over the top of the leading cube. This caused the highest averaged heat transfer coefficient. Fully developed convective heat transfer was found starting from the third cube.

Buoyancy effects imposed a noticeable effect on the convective heat transfer only at the lowest Reynolds number studied, $Re = 795$. Especially in the corridors between subsequent cubes, the vertical fluid motion caused an asymmetric distribution of the local heat transfer coefficient.

9.5 Tandem of cubes

Aerodynamic interference between two cubes and the effect of neighbouring cubes was studied for in-line and off-axis tandem arrangements of cubes. It was shown for the in-line tandem that variation of S_x/H had only a significant effect on the mean flow and turbulence fields around the downstream cube, while the flow around the leading cube was hardly affected. The flow around the leading cube was more or less similar to that around the wall-mounted isolated cube. For $S_x/H = 2$, the separated shear layer reattached at the side edge of the downstream cube; it reattached in the inter-obstacle spacing at $x/H = 2.5$ for $S_x/H = 4$ and larger. Large similarities were found to exist in the flow structure around the upstream and the downstream cube for $S_x/H > 8$. The distinguishable flow features, for example, flow separation and reattachment, and recirculation led to marked differences in the distributions of h .

The frequency of vortex shedding appeared to become minimum for a critical distance of $S_x/H = 2.9$ which was dictated by the local flow structure. Although small, a minimum in the face-averaged heat transfer for the top and side faces was found for this critical streamwise distance. It was assumed that both the convective heat transfer and the dynamics of vortex shedding were a direct consequence of the flow structures which were dictated by the streamwise distance.

Further, the time-dependent convective heat transfer was measured with a fast-responding active heat flux sensor. The wake flow of the upstream cube was subjected to flow periodicities caused by vortex shedding. Dependent on S_x/H , the periodically oscillating shear layer reattached on the side faces of the downstream cube, causing the convective heat flux to vary periodically as well. This was demonstrated from power density spectra of the heat flux signal while the vortex shedding frequency was simultaneously measured with a hot-wire anemometer. These results were in good agreement with the LDA frequency measurements.

The flow structures detected around the cubes in the off-axis tandem arrangements were apparently similar to those measured around the cubes in the in-line tandem. Typically, the horseshoe, top, side and wake vortices were detected from LDA measurements and flow visualisations. Only for the tandems with small streamwise and spanwise distances, significant asymmetric flow structures were observed. These had comparable, but somewhat different consequences on the local convective heat transfer.

While large differences in the distributions of the local heat transfer coefficient existed between the different in-line and off-axis tandem arrangements, the cube-averaged heat transfer coefficients appeared to be insensitive to both the streamwise and spanwise distance between the cubes. This result suggests that the bulk velocity of the flow rather than the local flow structures dictates the overall cooling performance of the cubes.

9.6 Matrix of cubes

It was shown that the flow in the immediate proximity of the cube in the matrix configuration is characterised by a two-cell structure which was caused by the shear layer reattachment in the inter-obstacle space. A horseshoe type of vortex originates at the front face; an arc-shaped vortex exists in the wake. Flow separation at the sharp leading top and side edges caused flow recirculations to exist close to the edges and subsequent reattachment at the faces more downstream. Vortex shedding in the wake of the cube was detected from LDA-measurements. A typical dimensionless shedding frequency of $St = 0.109$ was found. Both the periodic fluctuations caused by vortex shedding and the large velocity gradients were responsible for large values of the turbulent kinetic energy in the free shear layers in the wake of the cube.

The pronounced differences in the flow structure around the cube caused large gradients in the distributions of the local heat transfer coefficient. Similarity of these distributions, scaled with the cube-averaged heat transfer coefficient, indicated that no Reynolds number effects existed in the range of Reynolds numbers investigated ($2380 < Re_H < 5280$). Regions of intense flow recirculations with subsequent flow separation showed relatively small values of the local heat transfer coefficient. In contrast, areas where the flow reattached were generally characterised by large values of the heat transfer coefficient.

9.7 Cube-averaged convective heat transfer

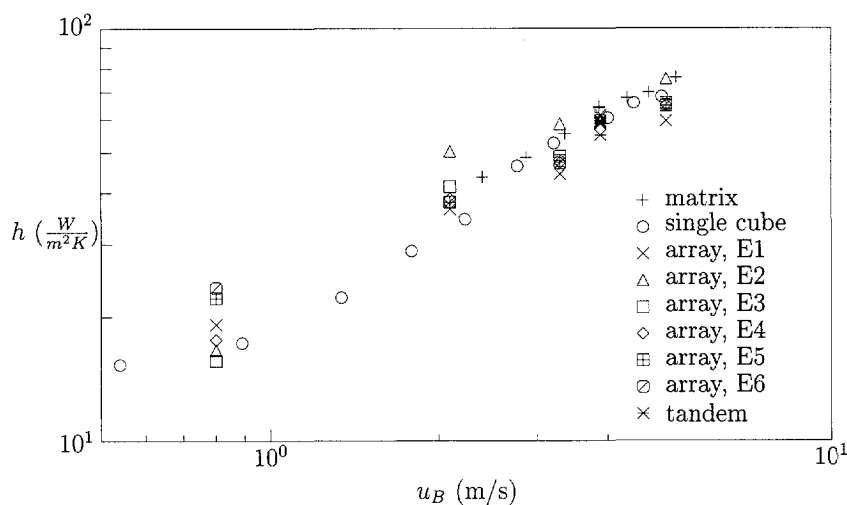


FIGURE 9.1: Cube-averaged heat transfer coefficient for all geometries studied as a function of the bulk velocity u_B .

For the tandem configurations, it was shown that the cube-averaged convective heat transfer was insensitive to both the streamwise and spanwise distance between

the cubes. A similar observation can be made for the other configurations studied. To do so, the cube-averaged heat transfer coefficients are plotted as a function of the bulk velocity in Fig. 9.1 for all different configurations considered in this thesis. The off-axis tandem results are omitted since the large number of data will reduce the ability of interpreting the figure. In particular for the larger bulk velocities, the heat transfer data collapse pretty well for all considered cases. In log-log coordinates, the cube-averaged heat transfer coefficient obeys a linear regression which suggests that the overall convective cooling characteristics for this turbulent region are dominated by the bulk flow through the channel. The results for the in-line row of cubes differ somewhat from the general trend. This deviation is largest for the upstream cubes in the row, while the deviations diminish for the cubes further downstream in the row. A sharp transition in the heat transfer coefficient is observed at $u_B = 2.5$ m/s which is attributed to the transition from a laminar developing to a turbulent developing channel flow as facilitated by the tripping wire upstream of the cubes.

It is noted that the results given in Fig. 9.1 were obtained for one channel-height-to-cube-height ratio, for one level of the free stream turbulence, one viscosity (that of air), and so on. It is, therefore, not known if the linear correlation is generally applicable for situations with different flow and fluid properties.

As mentioned before, the convective heat transfer characteristics in a spatially fully developed matrix flow was one of the test cases of the ERCOFTAC workshop on refined flow modelling, held at the Delft University of Technology on June 6-7, 1997. Bornoff and Parry [21] and Badie and Lasance [12] were the only participants who simulated the convective heat transfer in this matrix flow. They used the commercial code Flotherm [49, 50]. Although the velocity field around the cubes was wrongly predicted (with all the turbulence models incorporated in Flotherm) the cube-averaged convective heat transfer predictions corresponded remarkably well with the experimental data (deviations of maximum 10-20%). The explanation for this agreement is found in Fig. 9.1. Although large differences in the local heat transfer distributions were detected, the averaged heat transfer coefficient scales directly with the bulk flow.

CHAPTER 10

Conclusions and recommendations

In this last chapter, conclusions of the major findings of the research are drawn and possible directions for future research are highlighted.

10.1 Major conclusions

The major feature of the local heat transfer distribution for all four different test cases discussed in this thesis is a large spatial variation and steep gradients. Variations in h were up to 100% at a relatively small area of 15×15 mm. Further, marked differences were also observed between the face-averaged heat transfer coefficients. The large variation in h , i.e. the intensification and degradation of the local convective heat transfer, was caused by the vortex structure in the proximity of the cubes. A minimum in the surface heat transfer was typically found at locations where the fluid undergoes a recirculation, such as the top and side vortices. On the contrary, flow reattachment caused large convective heat transfer rates. Although the time-dependent heat transfer measurements were far from perfect, the results provided interesting additional information of the convective heat transfer dynamics involved.

The detailed accuracy analysis of the experimental techniques indicated that the acquired convective heat transfer and flow data appeared to be very reliable. The LDA measurements for the matrix configuration were in good agreement with numerical simulations which provided additional evidence of the carefully conducted experiments. In the development of advanced turbulence models and numerical solution methods, the need for accurate flow and heat transfer data is indispensable in order to make progress. The accurate results collected in this thesis are very suitable to validate further developments.

Although the configurations of wall-mounted cubes placed in a low-Reynolds number turbulent channel flow are considered to be representative test cases for realistic printed circuit boards (PCB), the experimental data presented in this thesis can not directly be translated to the practical heat transfer problems encountered on such boards. The study provides the design engineer at least sufficient background information how to solve these thermal problems and it makes the designer aware of the complexity of the convective heat transfer in question. An interesting result of the study is that for all considered configurations, the cube-averaged heat transfer coefficient was a linear function of the bulk velocity when plotted in log-log coordinates. Although this observation suggests that global parameters can be used to describe global convective heat transfer, such a relation will not provide information about local convective heat transfer phenomena which is of crucial importance to avoid local hot spots on electronic components. Further, the data acquired for the

four test cases can be used for benchmark purposes of commercial Computational Fluid Dynamics (CFD) codes, commonly used in the thermal design of PCBs.

In conclusion, the objectives of the research have been met but additional research is recommended to complement some of the aspects reported in this thesis.

10.2 Recommendations for future research

The time-dependent convective heat transfer is an issue which needs much more attention. The results reported in this thesis were a first attempt to analyse the dynamics in the convective heat transfer from wall-mounted cubes. A more advanced, a second generation sensor, is needed to accurately measure absolute values of the heat transfer coefficients. It was suggested that minimisation of the heat losses by means of isothermal conditions is a realistic option. This can be achieved by a guard heater, underneath the sensor. Another alternative is to determine the time-dependent heat losses from an experimental and numerical analysis. By a proper calibration of the sensor resistance versus the surface temperature, the actual sensor temperature is evaluated more accurately. Further, by embedding several thermocouples in the cube underneath the sensor, the conductive heat losses can be estimated. Numerical simulations, for example with Flotherm, can serve to estimate the three-dimensional conductive heat losses which can serve to guide the interpretation of the experimental results.

In order to elucidate the mechanisms which contribute to the degradation and the intensification of the local convective heat transfer, the time-dependent correlation between the flow structure and the surface heat flux needs to be investigated. Particle image velocimetry (PIV) is an almost mature technique to measure instantaneous vector fields. Synchronised simultaneous heat transfer measurements of the local surface heat flux can be performed with fast-response liquid crystal or infrared thermography. The simultaneous measurements should help in explaining the mechanisms that cause the observed dissimilarities between surface heat transfer and local flow structure in vortical flows.

Structure identification of the flow around the cube can be performed when instantaneous velocity data is available in planes. PIV is very suitable for providing data for this kind of analysis. With different methods based on flow topology and critical point theory, the vortex structure can then be identified. The structure identification based on the streakline patterns obtained from the oil-film visualisation only allowed a qualitative analysis. Since oil-film visualisations visualise the wall shear stress at the solid surface of the obstacle, and indirectly the local velocity field just above the surface, it is well suited for correlating the streakline structure with the local heat transfer coefficient. The only difficulty is to quantify the streakline patterns at the surface but at least a qualitative interpretation can be performed.

Direct measurement of the turbulent heat fluxes $\overline{u'\theta'}$ can be performed with Laser Doppler Anemometry and a cold wire positioned close to the measuring volume of the LDA-system. These results will also help in explaining the dissimilarities between the convective heat transfer from the cubes to the immediate flow. However, this is a very difficult measurement since the temperature-velocity correlation is

dependent on the local velocity and, therefore, on the location of the cold wire with respect to the measuring volume.

It is useful for engineering applications to relate the cube-averaged heat transfer coefficients to quantities which are easy to be measured in practical situations. One can think, for example, of the local static surface pressure in the channel or the free stream turbulence of the flow.

Another issue which deserves attention is how to deal with the definition of the local heat transfer coefficient. If the true local adiabatic reference temperature can be measured, temperature effects will be eliminated. In the case of epoxy cubes, strong temperature gradients exist at the cube surface due to the low thermal conductivity of epoxy. Since it is impossible to locally make the cube surface adiabatic, numerical simulations will help in estimating the order of magnitude of the effect. Further, the effect of the recirculation in the proximity of the cube can be investigated. Experimentally, one can think of measuring the response of a point source on the epoxy surface, for example by means of laser illumination. If the temperature distribution is in that case measured with infrared thermography, the temperature distribution in the epoxy layer can then be solved, giving the heat fluxes involved. In that way, the effect of a point source can be studied which may elucidate the concept of the adiabatic convective heat transfer.

This brief summation of recommendations for further research illustrates that there are still many issues to be addressed. The availability in the near future of highly accurate and more sophisticated measurement techniques as well as more advanced numerical methods and turbulence models facilitate the investigation of the time-dependent interaction between convective heat transfer and turbulence. Considering the complexity of the problem and the interesting transport phenomena involved, it is a big challenge to proceed with the investigation to address the questions which are still open.

APPENDIX A

Thermal conductivities

A.1 Thermal conductivity of the epoxy substrate

A transient and a steady-state method were developed to measure the thermal conductivity of the two-compound epoxy substrate. In addition to the thermal conductivity, the transient method provided also an estimate of the thermal diffusivity. The steady-state method provided the possibility to investigate the temperature dependency of the epoxy substrate.

A.1.1 Transient hot-wire method

The hot-wire method is based on the conductive heat transfer in a cylinder-symmetric infinite medium (cylinder) with a line source at the centreline. When the medium is considered to be isentropic and homogeneous, the Fourier equation for heat conduction expressed in cylindrical coordinates becomes [29]:

$$\frac{\partial T}{\partial t} = a \left(\frac{\partial^2 T}{\partial r^2} + \frac{1}{r} \frac{\partial T}{\partial r} \right), \quad (\text{A.1})$$

where $T(r, t)$ is the temperature at radius r and time t , a is the thermal diffusivity of the medium. The problem was solved with the initial and boundary conditions in terms of: $T = T_0$ for $t = 0$ and $r = \infty$ and the heat flux $Q = -2 \pi \lambda r \frac{\partial T}{\partial r} \bigg|_{r \rightarrow 0}$ for $t \geq 0$. The solution of this well-posed problem is given in Carslaw and Jaeger [29]. For relative large times, i.e. $\sqrt{4at} \gg 1$, the solution simplifies to [172]:

$$\theta(r, t) = A \ln(t) + B, \quad (\text{A.2})$$

where $\theta(r, t)$ is the temperature difference ($T(r, t) - T_0$). The constant A is the ratio of the heat production in the line source Q_l and the thermal conductivity λ :

$$A = \frac{Q_l}{4\pi\lambda}. \quad (\text{A.3})$$

B is a constant which is related to the thermal diffusivity of the material:

$$B = \frac{Q_l}{4\pi\lambda} \ln \left(\frac{4a}{R^2 c_E^*} \right), \quad (\text{A.4})$$

with $c_E^* = e^{c_E}$ and c_E is Euler's constant [172].

Design of the test-sample

The test-sample consisted of a cylinder of epoxy in which a high electrical resistance wire was embedded at $r = 0$. A sketch of the sample is given in Fig. A.1. Thermocouples were also embedded in the epoxy at two fixed radial positions R_1 and R_2 . It is noted that the two-compound liquid epoxy substrate was poured into a cylindrical mall with the dissipating wire and thermocouples already in position. After the solidification of the epoxy, it was ensured that no interface resistance existed between the wires and the surrounding epoxy. When a voltage drop is maintained across the wire, heat is diffused into the cylinder causing the temperature $T(r, t)$ to increase. The thermal conductivity was then derived from the temperature-time gradients at the fixed positions R_1 and R_2 .

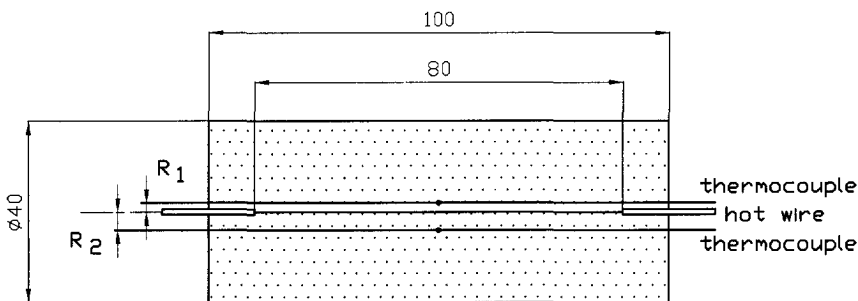


FIGURE A.1: Sketch of the cylindrical test-sample for the transient hot-wire method to measure the thermal conductivity of the epoxy substrate. Dimensions are given in mm.

The resistance wire had a diameter of 0.01 mm and a length of $l = 80$ mm. The power supply to the wire, the product of the voltage drop V and the electrical current I , gives then the heat per unit length: $Q_l = P/l = VI/l$. The cylindrical test-sample had a length of 100 mm and a radius of 20 mm. The two thermocouples were mounted in parallel to the centreline at distances of $R_1 = 1$ and $R_2 = 3$ mm. The temperature of the outer wall (at $r = 20$ mm) was also monitored with thermocouples to ensure that this surface temperature was not increased due to the power dissipation at the centre, and that the half-infinite approach was still valid. The whole cylinder mantle was thermally insulated with foam to minimise the heat losses to the environment.

The uncertainty in the measured thermal conductivity consisted of both experimental and structural errors. An estimate of the experimental uncertainty is based on the single sample uncertainty analysis, see [109, 110]. Structural errors were due to, for example, interface resistances, a not-infinite medium, heat losses through the leading wires, heat conduction at the lateral endings of the cylinder, and so on. It was verified that the dimensions as given in Fig. A.1 satisfied the requirements of an infinite medium. The temperature at the outer radius ($r = 20$ mm) of the test-sample did not increase during the experiment. The connecting leading wires,

to which the resistance wire was welded, were made of a Nickel alloy and had a low thermal conductivity to minimise conductive heat losses. Further, it had a high electrical conductivity to prevent power dissipation in the lead wires itself. The resistance wire (heat source) was fully embedded in the epoxy (the length of the wire was 8 cm while the length of the total epoxy cylinder was 10 cm, see Fig. A.1). The low thermal conductivity of epoxy prevented thus lateral heat losses. The thermal insulation at the lateral faces further minimised these conductive heat losses. From a detailed analysis, it was estimated that the heat losses were less than 1%. The thermocouples were welded such that the hot junction was in the centre and that both thermocouple wires were guided through the epoxy in opposite direction. This construction is shown in Fig. A.1. Since these thermocouple junctions were located at half of the cylinder length, a symmetric thermal situation inside the epoxy cylinder was guaranteed. The length of the cylinder was much larger than the radius R_1 and R_2 were the actual temperatures were measured and, therefore, the situation could be considered as one-dimensional, at least in a substantial part of the test-sample. Based on a detailed analysis, the accuracy of the thermal conductivity measured with the transient method was estimated to be 5%.

Results

A time series of the temperature measured with the thermocouple at R_1 is shown in Fig. A.2. The time is plotted on a logarithmic scale. The figure shows a linear

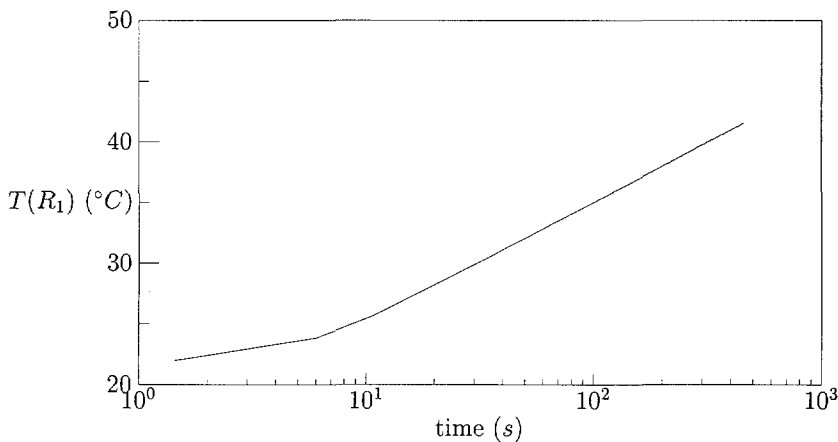


FIGURE A.2: Time series of the thermocouple temperature measurement at R_1 .

regression for times larger than approximately 20-30 seconds. This supports the requirement of long times for which the model is valid. A least-squares fit through the data in this linear region was performed to obtain the slope A . Experiments were performed for 6 different power supplies, resulting in 6 different time responses and, therefore, 6 different slopes. The point (0,0) is also added to the data points

since no temperature increase occurred for zero power supply. The 7 coefficients A are given in Fig. A.3 versus the power supply Q . A least-squares fit through

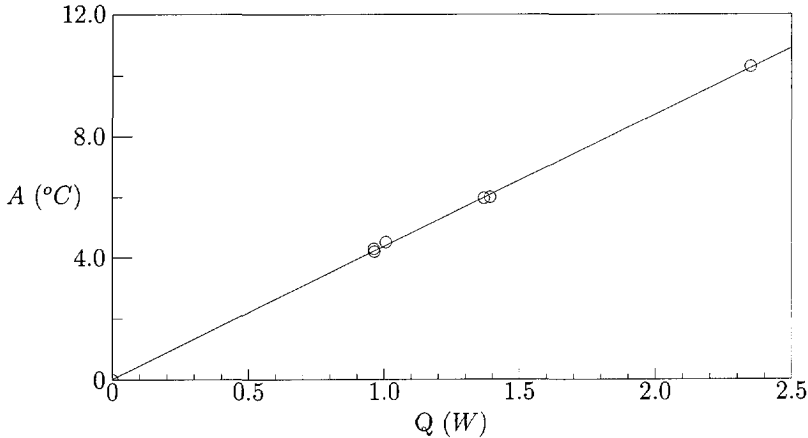


FIGURE A.3: Values of the slope A versus the supplied power.

these data points gave an accurate estimate of the gradient dQ/dA . The thermal conductivity λ is directly related to this gradient:

$$A = \frac{Q_l}{4\pi\lambda} \rightarrow \lambda = \frac{1}{4\pi l} \frac{dQ}{dA}. \quad (\text{A.5})$$

The slope dQ/dA was determined to be 4.361 which gave a thermal conductivity of $\lambda = 0.237$ W/mK. As mentioned, the accuracy of this value was 5%. It is noted that the transient method is only valid for homogeneous and isentropic material properties. The linearity of the time series, for example for the one shown in Fig. A.2, indicated that the material properties were constant in temperature. This is also supported by the linearity between A and the supplied power P , as shown in Fig. A.3. Although it was not possible to determine spatial dependencies (anisotropy) of the thermal conductivity with this method, it is assumed that possible anisotropy effects are of minor importance. The two-compound epoxy mixture was poured in a mall where it solidified very slowly. The slow process ensured sufficient time to diminish the internal stresses and to diminish possible preference orientations of long molecules.

Specific heat

Once the thermal conductivity is known, an estimate of the specific heat c_p was obtained from the intercept B (see eq. A.4). The thermal diffusivity a is expressed in terms of:

$$\frac{4a}{R^2 c_E^*} = e^{B \frac{4\pi\lambda}{Q_l}}. \quad (\text{A.6})$$

With the experimentally determined value for λ ($\lambda = 0.237$) and the value for B , the thermal diffusivity becomes $a = 1.206 \cdot 10^{-7} \text{ W/m}^2\text{K}$. From $a = \lambda/\rho c_p$, the specific heat was derived, i.e. $c_p = 1.65 \cdot 10^3 \text{ J/kgK}$. Although the value for the intercept B is somewhat difficult to determine accurately, the obtained values for a and c_p are in good agreement with tabulated properties of epoxy.

A.1.2 Steady state conduction

The second method explored to measure the thermal conductivity of the epoxy substrate was based on steady-state conduction in a two-dimensional annulus. The temperature distribution in the annulus was derived analytically from the cylinder-symmetric conduction problem with Dirichlet boundary conditions at the inner and outer radius. The temperature solution in the annulus reads [29]:

$$T(r) = T_2 + \frac{T_1 - T_2}{\ln(R_1/R_2)} \ln\left(\frac{r}{R_2}\right), \quad (\text{A.7})$$

where R_1 and R_2 denote the inner and outer radius, respectively. T_1 and T_2 are the temperature boundary conditions at R_1 and R_2 .

The conductive heat flux $\phi''(r)$ at any radius r in the annulus reads:

$$\phi''(r) = -\lambda \frac{\partial T}{\partial r} = -(T_1 - T_2) \frac{\lambda}{\ln(R_1/R_2)} \frac{1}{r}, \quad (\text{A.8})$$

where λ is the thermal conductivity. When the conductive heat flux is *a priori* known, an expression for the thermal conductivity of the material in the annulus results. If a line source with heat production per unit length Q_l is assumed at radius $r = 0$, the heat flux at $r = R_1$ reads in the assumption of zero heat losses:

$$\phi''(R_1) = \frac{Q_l}{2\pi R_1 l}. \quad (\text{A.9})$$

The thermal conductivity then results from:

$$\lambda = \frac{\phi''(R_1) \ln(R_1/R_2) R_1}{\Delta T}, \quad (\text{A.10})$$

where $\Delta T = T_2 - T_1$ is the temperature difference.

Test-sample

A test-sample was designed and constructed which consisted of an annulus of epoxy with copper annuli adjacent to both the inner and outer side of the epoxy material. A sketch of this test-sample is shown in Fig. A.4. The copper annuli were positioned in a mall such that an inner annulus of air resulted. The liquid epoxy mixture was poured into this vertical annulus where after the solidification process started. This vertical orientation prevented the occurrence of interface resistances between copper and epoxy since eventual shrinkage of the epoxy was compensated for with fresh epoxy (gravitation forces caused a constant pressure on the hardening epoxy, thus

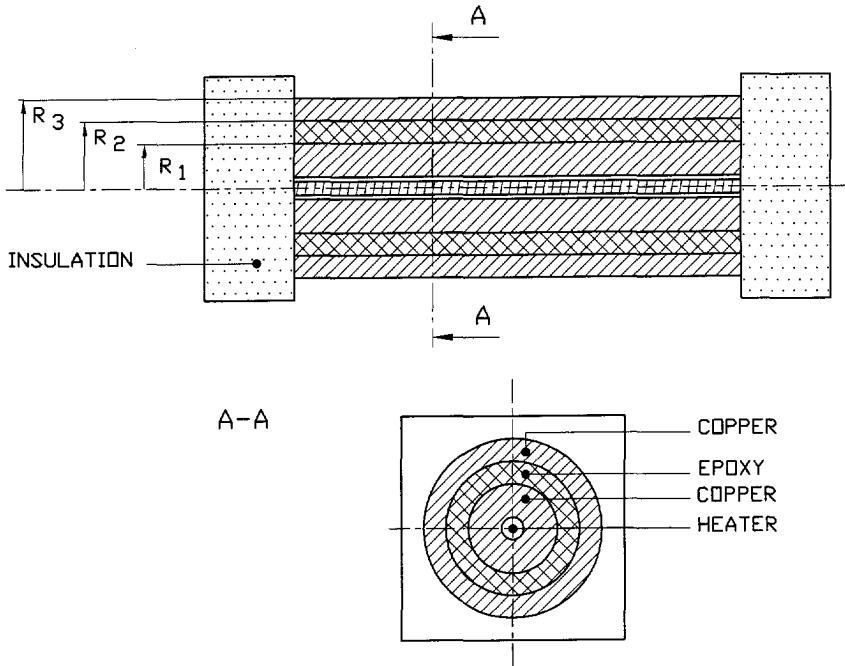


FIGURE A.4: Sketch of the test-sample with an annulus of epoxy used for the steady-state method to measure the thermal conductivity of the epoxy substrate.

enabling a good contact between the epoxy and copper). This construction ensured that the annulus width ($R_2 - R_1$) was constant within 0.2 % along the entire length of the test-sample and for the whole circumference of the annulus.

The length of the model was 10 cm. The three radii R_1 , R_2 and R_3 were 100, 150 and 200 mm, respectively. The epoxy annulus had a radius $R_2 - R_1 = 50$ mm. The axial edges were insulated with a low thermal conductivity foam (λ was less than 0.1 W/mK). This insulation and the sufficiently large aspect ratio (the ratio of length-to-width of the annulus was $100:5 = 20$) prevented severe heat losses in the axial direction. This resulted in a one-dimensional axis-symmetric situation (the temperature was only a function of the radius). This one-dimensionality was confirmed with experiments and is discussed later.

A heat source made from a spirally-winded resistance wire was placed at the centreline of the inner copper annulus. The connecting lead wires were again made from a Nickel alloy to minimise conductive heat losses and to prevent the power dissipation in the connecting pins. The high thermal conductivity of copper resulted in a uniform temperature of the inner annulus, even for relatively large heat fluxes. This temperature was monitored with 6 thermocouples distributed in the inner annulus. The average of these 6 temperatures gave the inner temperature at the radius R_1 : $T_1 = T(R_1)$. The standard deviation was about 0.1 °C. The entire sample was placed in a thermostated water bath. The temperature of the outer

copper annulus was dictated by the water temperature due to the good heat transfer coefficient of water. Again 6 thermocouples were used to measure the averaged copper temperature which represented the temperature at R_2 : $T_2 = T(R_2)$. By varying the water temperature (and thus $T(R_2)$) and the power supply to the heat source in the core of the sample, the temperature difference $\Delta T = T_2 - T_1$ was variable.

This one-dimensional model was derived to determine spatially- and temperature-independent thermal conductivities of the test material (epoxy). If the thermal conductivity is temperature-dependent, an averaged value is measured across the annulus radius which corresponds to the averaged temperature $(T_2 + T_1)/2$. If the temperature differences ΔT are kept relatively small, the temperature dependence of the thermal conductivity can then be examined by varying the temperature level $(T_2 + T_1)/2$. Experiments were conducted with temperature differences of 5, 10 and 15 °C at different temperature levels between 25 and 85 °C. While the results for the smaller temperature difference ($\Delta T = 5$ °C) gave a better temperature resolution, the results for the larger differences ($\Delta T = 10 - 15$ °C) were more accurate since the experimental uncertainty was then relatively smaller.

Results

Thermal conductivities, determined for temperature levels between 25 and 85 °C of the epoxy layer, are given in Fig. A.5 for a measurement series with fixed temperature differences of $\Delta T = 5, 10$ and 15 °C. Further, a measurement series

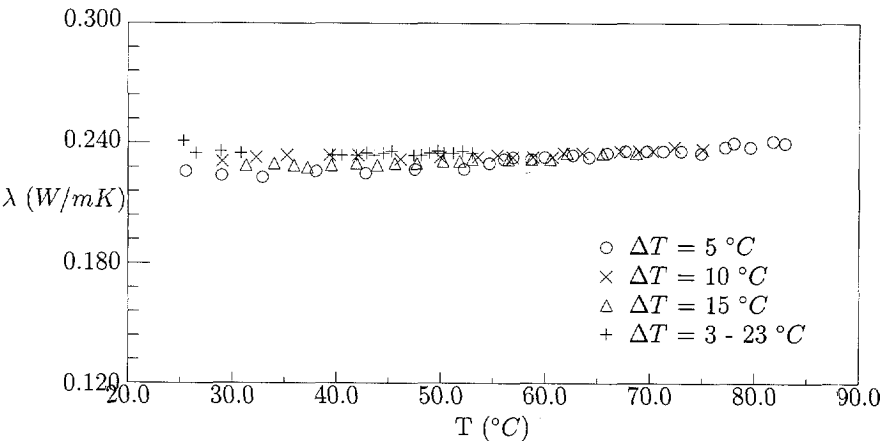


FIGURE A.5: The thermal conductivity versus the temperature level $(T_2 + T_1)/2$ for three fixed temperature differences: (o) $\Delta T = 5$ °C, ($\times$) $\Delta T = 10$ °C and (Δ) $\Delta T = 15$ °C. The symbols (+) refer to measurements with a variable ΔT .

for a variable ΔT is added to the figure. All the measurements collapse onto one line in the higher temperature range. For the lower temperature levels, the method

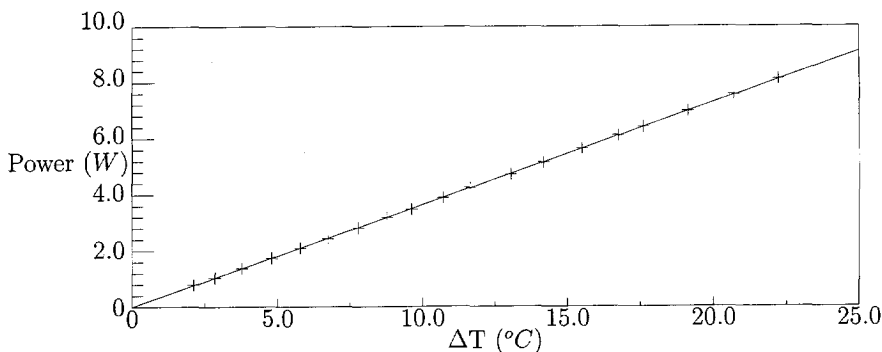


FIGURE A.6: The power supplied as a function of the temperature difference ΔT .

is less accurate which explains the larger scatter in this region. The measurements indicate that the thermal conductivity is a little temperature-dependent in the temperature range which is of most interest for our application: λ increases from about 0.23 W/mK at 55°C to about 0.24 W/mK at 80°C . This variation is smaller than the experimental uncertainty of 5% and is, therefore, ignored. The absence of a significant temperature dependence was already concluded from the results acquired with the transient method. The averaged thermal conductivity reads then $\lambda = 0.235 \text{ W/mK}$.

A gradual increase of the power supply resulted in a gradual increase in the temperature difference ΔT . The thermal conductivity then results from the gradient $dP/d\Delta T$:

$$\lambda = \frac{\ln(R_2/R_1)}{2\pi l} \frac{dP}{d\Delta T}. \quad (\text{A.11})$$

Again constant properties were assumed for this approach. The variation of the power supplied versus ΔT is given in Fig. A.6 showing a perfect linear increase. A least-squares linear regression through the data points gave a gradient of $dP/d\Delta T = 0.365 \text{ W/}^\circ\text{C}$ which corresponded to a thermal conductivity of $\lambda = 0.235 \text{ W/mK}$.

A.1.3 Concluding remarks

In conclusion, a transient and a steady-state method were explored to measure the thermal conductivity of the epoxy substrate. With the assumption of constant thermal properties of the epoxy material, a thermal conductivity of $\lambda = 0.237 \text{ W/mK}$ was measured with the transient technique. This assumption was confirmed by the measured logarithmic temperature profiles. A small temperature dependence was measured with the steady-state method. However, this variation was about 4% which is smaller than the 5% experimental uncertainty involved and, therefore, this dependency was not taken into account. The averaged thermal conductivity measured with the steady-state method was $\lambda = 0.235 \text{ W/mK}$. The thermal conductivities obtained with both methods are in very good agreement. It is, therefore, concluded that the epoxy substrate had a constant thermal conductivity of $\lambda =$

0.235 W/mK which was claimed to be accurate within 5%.

A.2 Thermal conductivity of the formaldehyde base plate

The base plate was fabricated from a low thermal conductivity formaldehyde. An accurate value of the thermal conductivity is required to estimate the heat losses through the base during the heat transfer measurements. Only a steady state conduction method was applied which was based on the so-called sandwich method.

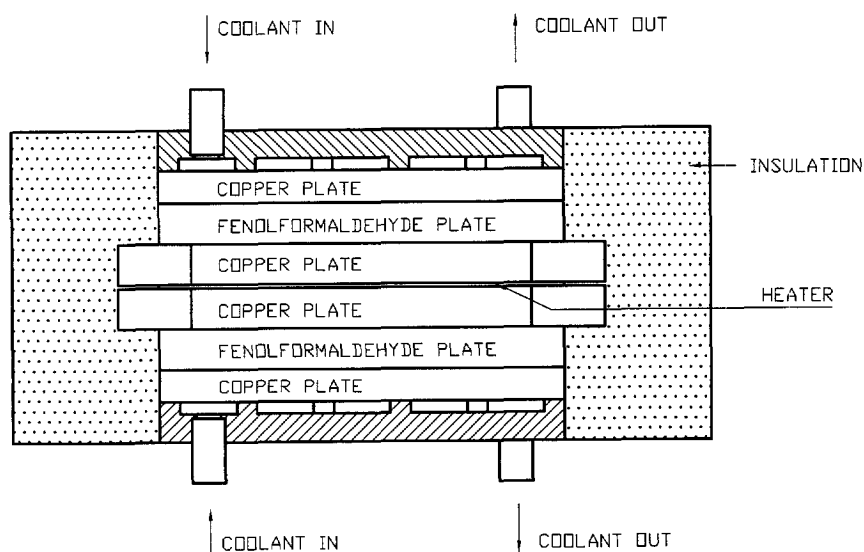


FIGURE A.7: Sketch of the steady-state sandwich set-up to measure the thermal conductivity of formaldehyde.

A.2.1 Experimental method

The formaldehyde material was machined to plates of exactly $100 \times 100 \times 10$ mm. The precise thickness of 10 mm was essential for an accurate measurement of the thermal conductivity. A copper plate was equipped with a commercial electrical resistance heater, operated between 5 and 100 Watts. The heat source was fully embedded in the copper inner plate of the sandwich. This inner plate had dimensions $120 \times 120 \times 10$ mm. The formaldehyde plates were placed on each side of the copper inner plate. Thermal heat compound ensured negligible thermal contact resistances between the copper and formaldehyde plate. Subsequently, two outer copper plates were placed at both outer sides of the formaldehyde material and again, thermal compound was used to minimise thermal resistances between the different materials. The outer plates consisted of machined channels through which water of adjustable temperature was recirculated. These heat exchangers enabled

that the outer temperature was kept at a prescribed level. The entire sandwich, constructed in this way, was pressed between two clamps. A sketch of the sandwich is given in Fig. A.7. Thermal insulation around all sides of the sandwich was applied to minimise heat losses to the environment. Thermocouples were embedded in both the copper inner and outer plates. Due to its large thermal conductivity, the copper plates were constant in temperature. The thermocouple measurements were averaged to obtain the averaged copper temperature.

The total heat ($P = VI$, where V is the voltage drop and I is the electric current) dissipated in the electrical heat source in the inner copper plate of the sandwich should equal the conductive heat through both sides of the inner core (Q_{cond}) plus some additional heat losses (Q_{loss}):

$$P = V \times I = Q_{cond} + Q_{loss}. \quad (\text{A.12})$$

Since a relative small thickness-to-width ratio was selected for the formaldehyde substrate (10:1), the situation was considered to be one-dimensional. The heat through one half of the sandwich is $Q_{cond}/2$ and equals:

$$Q_{cond}/2 = \lambda \frac{\Delta T}{d}, \quad (\text{A.13})$$

where ΔT is the one-dimensional temperature decay across the formaldehyde substrate and d is its thickness ($d = 10$ mm).

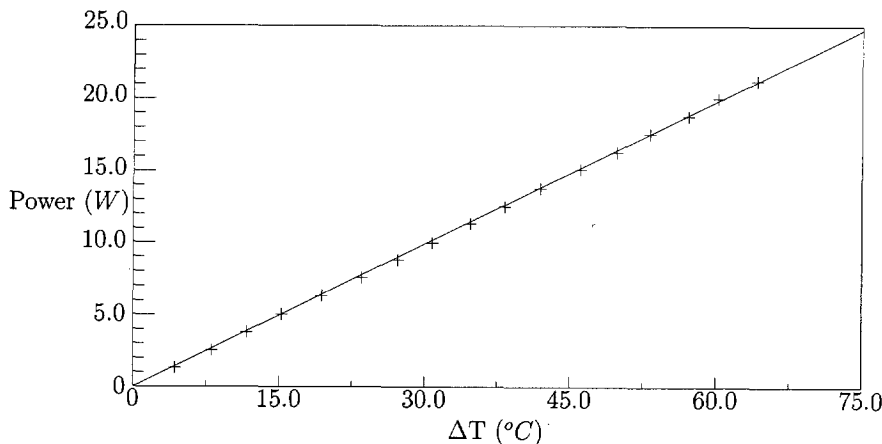


FIGURE A.8: The power supply as a function of the temperature difference ΔT across the formaldehyde plate.

A.2.2 Results

The temperature of the outer copper plates was fixed at a constant temperature between 19.5 and 20.0 °C. The power supply to the electrical heater was gradually increased which led to an increase in the temperature of the inner copper plate.

Since the temperature of the outer copper plate was kept at room temperature, the temperature decay across the formaldehyde plate increased also. The results, given in Fig. A.8 show a perfect linear increase in the temperature difference as a function of the power supply. The thermal conductivity was directly determined from its slope, according to:

$$\frac{dQ_{cond}/2}{d\Delta T} = \lambda/d. \quad (\text{A.14})$$

The slope was determined from a least-squares fit through the data which corresponded to a thermal conductivity of 0.33 W/mK. An accuracy of 10% was estimated from a single sample uncertainty analysis [109, 110] and included an estimate of the heat losses to the environment.

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SUMMARY

Experimental study of heat transfer in turbulent flows over wall-mounted cubes

Erwin R. Meinders

The turbulent heat transfer and the flow field characteristics of heated wall-mounted cubes placed in a channel flow are experimentally studied in this thesis. This investigation is of both fundamental scientific and practical interest. Sharp-edged obstacles on flat walls give rise to complex flow patterns with regions of flow separation, reattachment and recirculation. Further, periodic vortex shedding can occur in the wake of the obstacles because of flow instabilities. The fundamental interest lies in understanding the interaction between these large-scale vortex structures and the local surface heat transfer. Further, because of the complex convective heat transfer and turbulence involved, these configurations are suitable test cases for the benchmarking of turbulence models and numerical methods. From the application point of view, these geometries have a diagnostic value of practical relevance to the electronics industry, where local overheating is an important cause of technical failure of printed circuit boards.

Four different test cases were considered, namely a single cube, an in-line array of cubes, a spatially fully developed matrix flow and a tandem configuration of cubes with various streamwise and spanwise spacings between the cubes. The configurations were mounted on one of the walls of a low-Reynolds number closed windtunnel. Distributions of the time-averaged local convective heat transfer were determined from the internally controlled heat flux and the surface temperature distributions of special measurements cubes. Infrared and liquid crystal thermography were used to measure the surface temperature distributions. A high accuracy in the infrared temperature measurement (0.4°C) was achieved by the application of image restoration of the infrared images and an in situ calibration. The image restoration technique was based on the measured optical transfer function. The infrared measurements processed with the image restoration filter agreed very well with independent liquid crystal measurements. In addition to the time-averaged measurement method, some time-dependent heat transfer measurements were performed with a fast-responding heat flux sensor. Although the technique was far from perfect, some interesting results have been obtained for the tandem configuration of two cubes.

The three-dimensional flow structures around the cubes were deduced from oil-film surface patterns, smoke visualisation and detailed Laser Doppler Anemometry data. The two-dimensional LDA-system allowed the direct measurement of two velocity components. With two additional methods, information of the third component was obtained. A direct measurement of the normal velocity component was possible by integrating a prism in one of the cubes while the second method was based on a modification of the set-up to allow the reconstruction of the third velocity from inclined measurements.

A highly turbulent flow structure was typically observed for all configurations studied. For example, it was found that for the single cube or other configuration with a sufficiently large distance between the cubes, the flow was characterised by high vorticity recirculations at both the top and side faces, as well as the horseshoe vortex and the wake vortex. Flow instabilities caused the separated shear layers to fluctuate in a sinusoidal manner, the so called vortex shedding, which led to a highly turbulent wake flow. If cubes are located in the vicinity of other cubes, the flow structures are somewhat altered. This was clearly shown for a tandem of cubes for which the streamwise distance was varied. While the flow around the first cube was hardly affected by the downstream cubes, the flow around the latter was significantly influenced by the leading cube. Boundary layer flow was typically found for small streamwise distances while this generally changed to recirculative flow for the larger streamwise distances.

These, in nature, different flow types caused large variations in the local convective heat transfer for all configurations considered in this study. The steepest gradients in the distributions of the surface heat transfer were found at locations characterised by flow separation and reattachment. The resulting flow recirculations were typically accompanied by a minimum in the heat transfer coefficient. It was suggested that due to the local temperature rise of the air in the recirculation bubble, the bubble acted as an insulation layer and, therefore, prevented the removal of heat from the surface of the cubes. In contrast, the intermittent reattachment of the low-temperature shear flow was found to cause large heat transfer coefficients.

Despite the fact that a large variation in the local convective heat transfer existed for all configurations studied, dominated by the local vortex structure around the protrusions, the cube-averaged heat transfer appeared to scale with the bulk velocity of the flow. Although this is an interesting result for global thermal analysis, local hot spots still may occur which makes a detailed analysis of the flow features and related heat transfer characteristics indispensable.

SAMENVATTING

Experimenteel onderzoek van de warmteoverdracht in turbulente stromingen rond kubussen geplaatst op een wand

Erwin R. Meinders

De turbulente stromingsverschijnselen en de convectieve warmteoverdracht van kubussen geplaatst in een kanaalstroming zijn experimenteel onderzocht in dit verslag. Het beschreven onderzoek is zowel van fundamenteel wetenschappelijk als van praktisch belang. De stroming rond kubussen is gekarakteriseerd door een complex vortex systeem met grenslaag loslating, aanhechting en stromingsrecirculaties. Daarnaast kunnen instabiliteiten in de stroming aanleiding geven tot periodieke wervelafschudding. Het fundamentele karakter heeft vooral betrekking op het begrip van de koppeling tussen grootschalige vortex structuren en de lokale convectieve warmteoverdracht. Kubusvormige obstakels zijn verder uitermate geschikt voor het valideren van turbulentie modellen en numerieke oplosmethoden door de complexiteit van zowel de stroming als de lokale convectieve warmteoverdracht. Vanuit een praktisch oogpunt bezien hebben deze configuraties een diagnostische waarde met betrekking tot de koeling van elektronische printkaarten waarvoor lokale oververhitting nog steeds een belangrijke oorzaak van technisch falen is. Grondige kennis van de warmteverschijnselen van elektronische componenten en systemen is van vitaal belang om reeds tijdens het ontwerpstadium eventuele 'hot spots' te elimineren. De onderzochte configuraties van kubussen zijn geidealiseerde representanten van complexe printkaarten zoals die in computers en ander elektronische apparatuur te vinden zijn.

In deze studie zijn 4 configuraties onderzocht, namelijk een vrijstaande kubus, een rij kubussen, een tandem configuratie waarbij zowel de longitudinale als transversale afstand tussen de kubussen is gevarieerd en tenslotte een matrix van kubussen. De kubusconfiguraties zijn geplaatst in een kanaalstroming met een relatief laag Reynolds getal. De lokale tijdsgemiddelde warmteoverdracht is gemeten met speciaal daarvoor ontworpen meetkubussen. De verdeling van de oppervlakte temperatuur van deze kubussen is gemeten met infrarood thermografie en vloeibare kristallen (liquid crystals). Een hoge nauwkeurigheid, te weten $0.4\text{ }^{\circ}\text{C}$, is verkregen door de infrarood metingen te bewerken met een beeldreconstructie filter en verder door de metingen in situ te kalibreren. Het beeldreconstructie filter was gebaseerd op gemeten overdrachtsfuncties die geconstrueerd zijn uit responsies op perfecte thermische stapfuncties. De gemeten oppervlakte temperatuur is gebruikt als randvoorwaarde voor het bepalen van de lokale convectieve warmtefluxen door het oppervlak van de kubus. Daarnaast is deze temperatuur ook nodig om warmteoverdrachtscoëfficiënten te definiëren. De tijdsafhankelijke warmteoverdracht is gemeten met een zelf-ontwikkelde actieve sensor met een zeer snelle responsie. Ondanks dat de sensor nog in een ontwikkelfase verkeerde, zijn er toch interessante resultaten mee verkregen. De met deze sensor gemeten tijdsafhankelijke warmteflux door het zijvlak van een kubus geplaatst in het zog van een andere kubus bleek

een periodieke component te bevatten waarvan de frequentie overeen kwam met de frequentie van wervelafschudding, zoals gemeten met het LDA systeem.

Het driedimensionale tijdsgemiddelde en instantane stromingsveld is gemeten met verscheidene technieken, te weten Laser Doppler Anemometrie, 'Hot-wire' Anemometrie en stromingsvisualisaties (waaronder rook en olie-film visualisaties). Omdat het gebruikte LDA systeem alleen maar geschikt was om de twee snelheidscomponenten parallel aan de grondplaat te meten, zijn er twee methodes ontwikkeld om de derde component, loodrecht op het grondvlak, te meten. Een directe meting van deze snelheid was mogelijk door gebruik te maken van prisma geïntegreerd in één van de kubussen waarin de laser bundels 90 graden gespiegeld werden door een spiegellend diagonaalvlak (de hypotenusa van de kubus). In de tweede methode werd de loodrechte snelheidscomponent gereconstrueerd uit metingen onder verschillende hoeken.

De stroming rond een vrijstaande kubus, of een configuratie met een voldoende grote onderlinge afstand tussen de kubussen, bestaat typisch uit een hoefijzerachtige vortex ('horseshoe' vortex), die stroomopwaarts van de kubus ontstaat en door de hoofdstroming langs beide kanten wordt afgebogen. Verder bestaat de stroming uit een boogachtige vortex ('wake vortex') in het zog en stromingsrecirculaties op de top- en zijvlakken ('top' en 'side' vortex). Wervelafschudding van de zijvlakken veroorzaakt een oscillerende stroming in het zog van de kubus wat een verhoogde turbulentie intensiteit tot gevolg heeft. Enigszins afwijkende stromingsverschijnselen werden waargenomen indien de kubus in de nabijheid van een andere kubus was geplaatst. Dit is duidelijk te zien voor een tandem configuratie van twee op een lijn geplaatste kubussen waarvoor de longitudinale afstand was gevarieerd. Terwijl het stromingsveld rond de stroomopwaartse kubus nauwelijks door de stroomafwaartse kubus werd beïnvloed, blijkt het tegengestelde te gelden voor de stroomafwaartse kubus. De stroming langs de top- en zijvlakken van het desbetreffende object was een grenslaagstroming voor kleine onderlinge afstanden tussen de kubussen terwijl voor grotere separaties grenslaag loslating, en dus een recirculatieve stroming, plaatsvond.

De verschillende typen stromingen veroorzaakten grote verschillen in de lokale convectieve warmteoverdracht. De steilste gradienten zijn gevonden op locaties waar de stroming recirculeerde. De lokale warmteoverdrachtscoëfficiënt vertoonde lokale minima op plaatsen ongeveer ter hoogte van de kern van deze recirculaties. Dit verschijnsel werd deels verklaard door een temperatuurstijging van de wervel dat daardoor als een soort isolatie laag fungeerde. Daarentegen veroorzaakte het opnieuw aanhechten van de relatief koude luchtstroming maxima in de lokale warmteoverdrachtscoëfficiënt op plaatsen waar de stroming aanhechtte.

Ondanks het feit dat de lokale warmteoverdrachtscoëfficiënt grote verschillen vertoonde voor alle onderzochte configuraties, bleek dat de kubusgemiddelde warmteoverdrachtscoëfficiënt schaalte met de gemiddelde snelheid in het stromingskanaal. Hoewel dit een interessant resultaat is voor een globale thermische analyse van printkaarten, zal een nauwkeurige karakterisatie van het turbulente stromingsveld en de daaraan gerelateerde lokale convectieve warmteoverdracht onontbeerlijk zijn om lokale oververhitting van componenten te voorkomen.

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voor jou, liefje.

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1983-1989	Secondary school VWO, at 'Lyceum de Grundel', Hengelo (O)
1989-1994	Study Mechanical Engineering at the University of Twente, Enschede Graduation work at KSEPL, the exploration and production research laboratory of Shell, Rijswijk, during the period April 1993 - December 1993. The theoretical/experimental research involved the sound propagation in drilling fluids and, in relation with it, the development and testing of an acoustic detection system for the fast detection of possible gas influxes in drilling wells.
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